The Volatility Surface as a Field of Probability Laws

A working draft

Thibaut Delahaye

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Contents

Chapter 1

Introduction

[This chapter is written last. It will state the problem — finitely many noisy option prices, from which a surface of probability laws must be published — and the thesis that organizes every chapter: the separation of what follows from probability, what the market determines, and what the modeler must choose and disclose.]

Part I

Models

Chapter 2

The Log-Quantile-Density Model of the Volatility Smile

A volatility smile is a probability law expressed through the Black formula. This chapter constructs that law directly. The log-forward return is written as a monotone transport of a logistic variable, and the logarithm of the transport speed is expanded in Legendre polynomials of the percentile rank. Monotonicity, positive density, and unit mass are then automatic; a scalar shift imposes the martingale condition, and the sole remaining requirement is that the right endpoint speed be smaller than one. The endpoint speeds give the exact moment strip and the asymptotic Lee slopes. A single tail integral, the upper-share ledger, yields option prices, the density, the analytic calibration Jacobian, the variance-swap strike, and a complete test for calendar order. The construction separates what follows from probability, what is learned from quoted options, and what must be supplied by modeling convention. Theory, computation, and market examples are developed from the same representation.

2.1 A smile is a probability law

A listed option market gives finitely many prices. A smile model must turn them into a complete curve: a price and an implied volatility at every strike, including strikes that were not quoted. The completed curve will be differentiated for digitals and densities, integrated for variance contracts, and compared with curves at other expiries. It is therefore not enough to draw a smooth line through the quotes.

The obstruction is elementary. The second strike derivative of a call price is a probability density. A curve that fits every quote can nevertheless make this derivative negative between two quotes. It then assigns a negative price to a butterfly and a negative mass to an interval of terminal prices. No numerical routine has malfunctioned; the curve was allowed to leave the space of probability laws.

The useful question is thus not "which curve fits well?" but "in which coordinates is every fitted curve already a law?" The answer developed here uses quantiles. In rank coordinates, unit mass is automatic and positive density reduces to monotonicity. Taking the logarithm of the quantile speed removes even that constraint. The resulting log-quantile-density (LQD) family provides unconstrained coordinates for a broad class of mean-one laws with exponential log-return tails.

2.1.1 Normalized market

Fix one expiry T. Let F be its forward and D its discount factor. Under the T-forward measure define

$$X = \log \frac{S_T}{F}, \qquad Y = e^X = \frac{S_T}{F}, \qquad \mathbb{E}[Y] = 1. \tag{2.1}$$

The last equality is the martingale condition. Write $k = \log(K/F)$ for log-moneyness. Normalized call and put prices are

$$c(k) = \mathbb{E}[(Y - e^k)^+], \qquad P(k) = \mathbb{E}[(e^k - Y)^+], \qquad c(k) - P(k) = 1 - e^k. \tag{2.2}$$

A dollar price is DF times its normalized price.

Let τ denote variance time: calendar time, possibly adjusted for the desk's event-time convention. Total implied variance is $w(k) = \sigma(k)^2\tau$. The normalized Black call is

$$B(k, w) = \Phi(d_+) - e^k\Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \tag{2.3}$$

and $w(k)$ is the root of $B(k, w(k)) = c(k)$ [8]. Rates and carry have now disappeared from the single-slice problem.

2.1.2 The static constraint set

Use normalized strike $y = e^k$. If Y has density f_Y, differentiation of the payoff gives

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \qquad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \tag{2.4}$$

This is the Breeden–Litzenberger identity [10]. It identifies the geometry of the call curve with the law of Y: decreasing slope is a digital probability and convexity is positive density.

The identity can be read using traded portfolios. For a strike spacing δ, the symmetric butterfly

$$c(y - \delta) - 2c(y) + c(y + \delta)$$

is the price of a long call below, a long call above, and two short calls at the center. Dividing by δ^2 and shrinking the spacing recovers $f_Y(y)$. A negative second difference is therefore not merely undesirable curvature; it is a negative price for a nonnegative terminal payoff. The probability statement and the static-arbitrage statement are the same.

Definition 2.1 (Arbitrage-free slice)**.** An arbitrage-free slice is a call curve of the form (2.2) for a positive random variable Y with $\mathbb{E}[Y] = 1$. When Y has a density, this is equivalent to a decreasing convex function of strike y satisfying

$$(1 - y)^+ \leq c(y) \leq 1, \qquad -1 \leq c'(y) \leq 0, \qquad c(y) \longrightarrow 0 \quad (y \to \infty),$$

with $c'(y) \to -1$ as $y \downarrow 0$ when there is no mass at zero.

Each boundary condition has a probabilistic source. The upper bound follows from $(Y - y)^+ \leq Y$ and $\mathbb{E}[Y] = 1$. The lower bound is Jensen's inequality, $(\mathbb{E}[Y] - y)^+ \leq \mathbb{E}[(Y - y)^+]$. The slope is a survival probability and must lie between -1 and zero. Finally, calls vanish at infinite strike because the integrable random variable Y carries no mean mass arbitrarily far in its tail. A valid call curve is thus exactly a compact encoding of a mean-one law.

Convexity belongs to y, not to k. Indeed,

$$\frac{\partial^2 c}{\partial k^2} = e^k\{e^k f_Y(e^k) - \mathbb{P}(Y > e^k)\},$$

which is negative far in the left wing for any ordinary density. A convexity test performed in log-strike will therefore reject correct call curves.

The same constraint becomes less transparent in volatility coordinates. Differentiating $c(k) = B(k, w(k))$ twice yields

$$f_X(k) = g_{\mathrm{D}}(k)\frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_{\mathrm{D}}(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \tag{2.5}$$

Thus a volatility curve is butterfly-free precisely when $g_{\mathrm{D}}(k) \geq 0$ for all k [25]. The condition is global, nonlinear, and involves two derivatives. Direct volatility models must enforce it; the representation below makes it an identity.

This explains why good residuals alone are weak evidence. Quotes constrain only finitely many values of w. The function g_{D} also depends on the slope and curvature between them, and may become negative in a narrow unquoted interval. Increasing the strike grid reduces but does not remove the logical problem: the optimizer still moves in a space containing invalid curves. The quantile construction reverses the order of work. It first chooses a law and only then asks which smile the Black formula assigns to it.

Two checks fix the normalization in (2.5). If the smile is flat, $w' = w'' = 0$, so $g_{\mathrm{D}} \equiv 1$ and the remaining factor is exactly the Black density of X. At the money, the identities derived later in section 2.5 give $g_{\mathrm{D}}(0) = \sqrt{w_0} f_X(0)/\varphi(\sqrt{w_0}/2)$. Thus $g_{\mathrm{D}} \geq 0$ is not a penalty formula: it is positive density written in the volatility chart.

2.1.3 Notation ledger

The construction uses three horizontal axes. Strike y is the axis on which calls are convex; log-moneyness $k = \log y$ is the market axis; rank u and log-odds z are the model axes. For a rank $u \in (0,1)$, the *odds* are $u/(1-u)$ and the *log-odds* are

$$z = \log \frac{u}{1-u}.$$

The function taking rank to log-odds is called the *logit*: $\operatorname{logit}(u) = \log\{u/(1-u)\}$. It maps $(0,1)$ bijectively onto $\mathbb{R}$; its inverse is the logistic function $\Lambda(z) = 1/(1+e^{-z})$. Thus u and z record the same percentile on two different scales. The recurring notation is collected below.

$X,\ Y = e^X$	log and gross forward return	$u,\ z = \operatorname{logit} u$	rank and log-odds
$k,\ y = e^k$	log and normalized strike	$\Lambda,\ \rho$	logistic CDF and density
$c,\ P$	normalized call and put	$Q,\ q = Q'$	quantile and quantile density
$w = \sigma^2\tau$	total implied variance	g	smooth log-speed in rank
$x,\ m$	transport and martingale shift	G	upper-share ledger
L, R, a_n	polynomial coordinates	λ_-, λ_+	endpoint speeds
u_k, z_k	strike rank and log-odds	u_*, f_*, Δ	ATM rank, density, digital gap
β_L, β_R	Lee wing slopes	$N, Z_{\max}$	basis order and grid half-width

Table 2.1: Notation used throughout the chapter. A prime denotes an ordinary derivative of a one-variable function; ∂ denotes a partial derivative. The time τ is always variance time.

2.1.4 What the chapter separates

Three questions recur, and they should not be confused. *Validity* asks whether a parameter vector defines a probability law. *Information* asks which features of that law are determined by the quoted strip. *Convention* selects the remainder: basis order, regularization, tail coordinates, and the scope of calendar control. The LQD coordinates settle the first question structurally. The later sections measure the second and state the third explicitly.

The development is progressive. Section 2.2 constructs the coordinates. Sections 2.3 and 2.4 establish their probabilistic content. Section 2.5 derives prices and local smile quantities from one integral. Section 2.6 turns the representation into a fitter, section 2.7 into a computation, and section 2.8 into a surface. Section 2.9 then shows what the construction says on market and synthetic data. Proofs and implementation details are collected in the appendices.

2.2 From ranks to LQD coordinates

2.2.1 The speed of a quantile

Let $Q(u)$ be the quantile function of X, with rank $u \in (0,1)$. If U is uniform, then $Q(U)$ has the desired law. Hence probability one is built into the coordinate u, and the only remaining requirement is that Q increase.

Rank is a universal clock for distributions. The value $Q(0.01)$ is the first-percentile return whether the law is symmetric, skewed, or multimodal. Changing the law changes the returns assigned to the clock, not the clock itself. This is the main simplification: normalization of probability is handled once by the uniform variable rather than reimposed on every candidate density.

Write $q(u) = Q'(u)$ for the quantile density. Differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u)) = \frac{1}{q(u)}. \tag{2.6}$$

The function q is the speed with which return space is traversed as rank increases. Slow motion packs probability into a narrow return interval; fast motion thins it. Since Q is increasing exactly when $q > 0$, an arbitrary real function may be used for $\log q$.

An unbounded distribution forces q to diverge at both endpoints. Here a *tail* means the probability lying beyond a remote threshold. We say that X has exponential tails, with positive scales λ_- and λ_+, when

$$F_X(x) \sim C_- e^{x/\lambda_-} \quad (x \to -\infty), \qquad 1 - F_X(x) \sim C_+ e^{-x/\lambda_+} \quad (x \to +\infty), \tag{2.7}$$

for constants $C_-, C_+ > 0$. Their logarithms therefore decay linearly in $|x|$. By contrast, a Gaussian tail has log-probability proportional to $-x^2$: it decays quadratically on the log-probability scale and hence faster than every exponential tail. For the exponential class, the leading endpoint singularities of q are $1/u$ and $1/(1-u)$. Remove them explicitly:

$$\log q(u) = -\log u - \log(1-u) + g(u). \tag{2.8}$$

The remainder g is bounded for the exponential-tail class considered here.

The subtraction is not arbitrary. The left relation in (2.7) implies $Q(u) \sim \lambda_- \log u$ as $u \downarrow 0$ and $q(u) \sim \lambda_-/u$. The right endpoint is analogous. More precisely, if $X \sim N(0, s^2)$, then

$$F_X(x) \sim \frac{s}{|x|\sqrt{2\pi}} e^{-x^2/(2s^2)} \qquad (x \to -\infty),$$

with the symmetric formula on the right. Its quantile density is $q(u) = s/\varphi(\Phi^{-1}(u))$ and, by the normal tail equivalent,

$$q(u) \sim \frac{s}{u\sqrt{2\log(1/u)}} \qquad (u \downarrow 0).$$

Its leading $1/u$ singularity is removed, but the remainder still tends to $-\infty$. Exponential tails therefore lie inside the LQD chart, whereas Gaussian tails lie on a different boundary.

Now set

$$z = \operatorname{logit} u = \log \frac{u}{1-u}, \qquad u = \Lambda(z) = \frac{1}{1+e^{-z}}, \qquad \rho(z) = \Lambda(z)\Lambda(-z). \tag{2.9}$$

The function Λ is the logistic CDF and $\rho = \Lambda'$ its density. Because $\mathrm{d}u/\mathrm{d}z = u(1-u)$, the singularities in (2.8) cancel:

$$\frac{\mathrm{d}Q}{\mathrm{d}z} = q(u)u(1-u) = e^{g(u)}. \tag{2.10}$$

Thus the entire distribution is generated by integrating a positive, ordinary function on the log-odds line.

The coordinate also places every percentile at a finite location. The median is $z = 0$; the tenth and ninetieth percentiles are $z = \mp \log 9$; moving one unit in z multiplies the odds $u/(1-u)$ by e. Meanwhile $\rho(z) \sim e^{-|z|}$, so rank integrals become ordinary integrals on $\mathbb{R}$ with exponentially decaying weight. This is why the same coordinate is convenient both analytically and numerically.

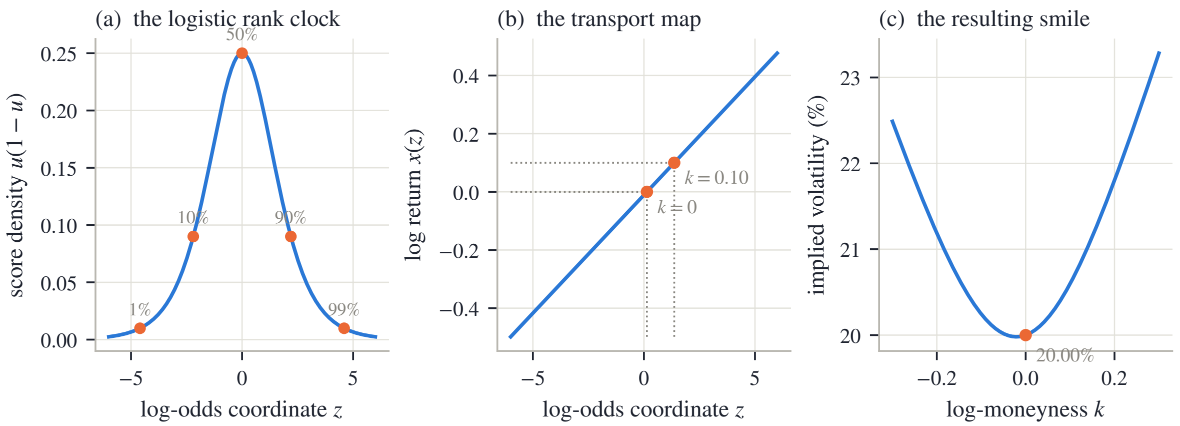

Figure 2.1: The transport construction for constant speed. Left: the logistic score and five percentile marks. Center: an affine map from log-odds to log-return, with two strike roots. Right: the implied-volatility smile priced from that law. The first two panels define the distribution; the last merely expresses it in market coordinates.

2.2.2 The LQD slice

Choose g to be a low-degree polynomial in rank. The shifted Legendre basis is convenient because it is orthogonal under the uniform rank measure. The constant and linear terms are written as an endpoint chord, leaving higher modes to describe the body.

Definition 2.2 (The LQD slice)**.** Fix $N \geq 2$ and $\theta = (L, R, a_2, \dots, a_N) \in \mathbb{R}^{N+1}$. Let Z have CDF Λ and density ρ. The LQD slice is the law of

$$\boxed{X = x(Z), \qquad x(z) = m + \int_0^z e^{g(\Lambda(t))}\,\mathrm{d}t, \qquad g(u) = (1-u)L + uR + \sum_{n=2}^{N} a_n P_n(1-2u),} \tag{2.11}$$

where P_n is the Legendre polynomial of degree n and m is chosen so that $\mathbb{E}[e^X] = 1$.

For every finite coefficient vector, $x'(z) = e^{g(\Lambda(z))} > 0$. Consequently

$$F_X(x(z)) = \Lambda(z), \qquad f_X(x(z)) = \frac{\rho(z)}{x'(z)} = \Lambda(z)\Lambda(-z)e^{-g(\Lambda(z))} > 0. \tag{2.12}$$

The quantile is $Q(u) = x(\operatorname{logit} u)$, and (2.10) recovers

$$q(u) = \frac{e^{g(u)}}{u(1-u)}. \tag{2.13}$$

Equations (2.11) and (2.13) are the transport and log-quantile-density descriptions of the same law.

Since a polynomial g is bounded on $[0,1]$, its exponential is bounded above and away from zero. Therefore $x(z)$ tends to $-\infty$ and $+\infty$ with z and is a diffeomorphism of the line. The construction cannot fold, stop, or reverse. Positivity of the density in (2.12) is thus a geometric fact about the coordinate map, not a pointwise inequality checked after the fit.

The two endpoint values are

$$g(0) = L + \sum_{n\geq 2} a_n, \qquad g(1) = R + \sum_{n\geq 2} (-1)^n a_n, \tag{2.14}$$

and their exponentials

$$\lambda_- = e^{g(0)}, \qquad \lambda_+ = e^{g(1)} \tag{2.15}$$

are the endpoint speeds. If $\bar{x}(z) = \int_0^z e^{g(\Lambda(t))} \mathrm{d}t$, then Lipschitz continuity of g and exponential convergence of Λ to its endpoints imply

$$\bar{x}(z) = b_- + \lambda_- z + O(e^z) \quad (z \to -\infty), \qquad \bar{x}(z) = b_+ + \lambda_+ z + O(e^{-z}) \quad (z \to +\infty). \tag{2.16}$$

The endpoint speeds will therefore govern both tails.

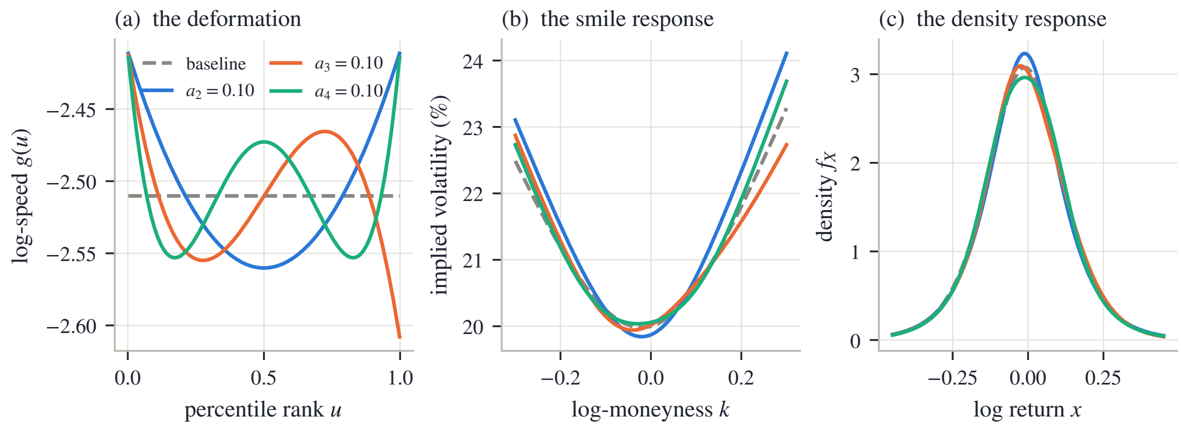

Figure 2.2: Three Legendre perturbations of the same reference law. Left: their action on log-speed g. Center: the response of implied volatility. Right: the response of the density. An even mode moves both endpoints in the same direction; an odd mode moves them in opposite directions. In all cases the density remains positive because the transport speed remains an exponential.

The modes have a direct interpretation. Raising g stretches returns near the affected ranks and therefore lowers density there by (2.6). Even modes act symmetrically; odd modes transfer shape from one side to the other; higher modes work on finer rank scales. The raw coefficients are not pure body controls, since (2.14) couples each of them to one or both tails. The endpoint chart introduced in section 2.6 removes that coupling without changing the family.

Legendre polynomials are a coordinate choice, not part of the probability argument. Rank integrals use uniform weight on $[0,1]$, under which Legendre modes are orthogonal. Their columns remain distinguishable in least-squares normal equations as order rises. Monomials span the same space but inherit Hilbert-matrix conditioning. Chebyshev modes would also be reasonable; the endpoint chord would remain useful because it names the two tail controls.

2.2.3 Martingale normalization and the finite-forward wall

Only a translation remains. Let

$$I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z) \, \mathrm{d}z.$$

Proposition 2.3 (Martingale normalization)**.** *If $I < \infty$, then $m = -\log I$ is the unique shift for which $\mathbb{E}[e^X] = 1$. Moreover, $I < \infty$ if and only if $\lambda_+ < 1$.*

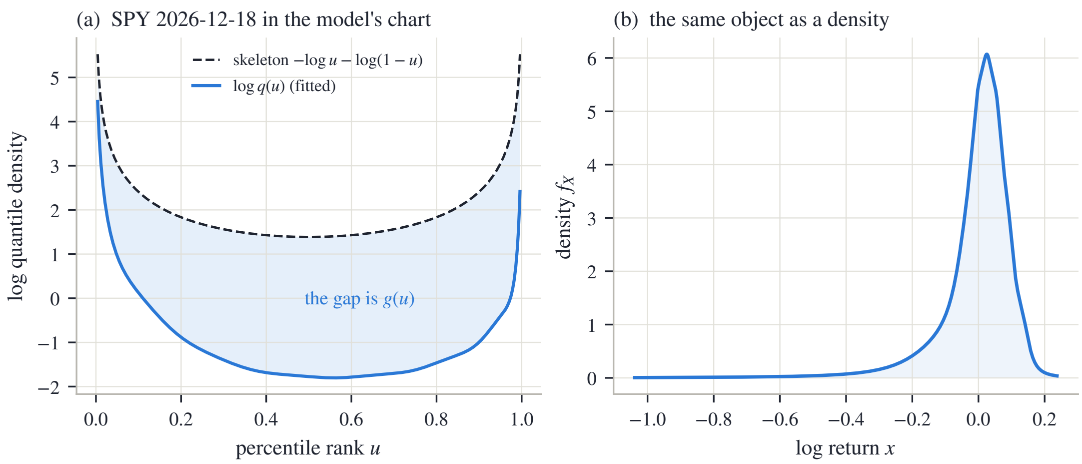

Figure 2.3: A fitted SPY December 2026 slice in the model's coordinates. Left: the universal boundary skeleton and the fitted $\log q$; their difference is g. Right: the density produced by the same transport. The quoted strikes occupy the central ranks; the continuation to the endpoints is extrapolation.

Proof. Since $X = m + \bar{x}(Z)$, $\mathbb{E}[e^X] = e^m I$, giving the unique shift. At the right endpoint, (2.16) and $\rho(z) \sim e^{-z}$ make the integrand asymptotic to a positive constant times $e^{(\lambda_+ - 1)z}$; it is integrable exactly when $\lambda_+ < 1$. At the left endpoint it behaves as $e^{(1+\lambda_-)z}$, which is integrable for every $\lambda_- > 0$. $\square$

The set of admissible raw parameters is therefore

$$\Theta_N = \{\theta \in \mathbb{R}^{N+1} : \lambda_+ < 1\}. \tag{2.17}$$

It is open and has a single boundary. A logistic coordinate for λ_+ will make that boundary unreachable.

2.2.4 The exactly solvable slice

Set $L = R = \log s$, $a_n = 0$, with $0 < s < 1$. Then $x(z) = m + sz$ and X is logistic with scale s. Its moment-generating function is the beta integral

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u}\right)^t \mathrm{d}u = \mathrm{B}(1+t, 1-t) = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \qquad |t| < 1. \tag{2.18}$$

Hence

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \qquad \mathrm{Var}(X) = \frac{\pi^2 s^2}{3}.$$

Matching a target ATM total variance w_0 gives the cold start

$$s = \frac{\sqrt{3w_0}}{\pi}. \tag{2.19}$$

This seed matches variance, not ATM price. As $s \downarrow 0$, $m = -\pi^2 s^2/6 + O(s^4)$. Moreover,

$$\mathbb{E}[Z^+] = \int_0^\infty z\rho(z)\,\mathrm{d}z = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} = \log 2,$$

where the logistic density is expanded as a geometric series on the positive half-line. Thus the LQD ATM call is $s \log 2 + O(s^2)$. Matching it to the small-variance Black price gives

$$\frac{w_{\text{implied}}}{w_0} \longrightarrow \frac{6(\log 2)^2}{\pi} \approx 0.918.$$

The seed is therefore about eight percent low in ATM variance, a known bias removed by calibration.

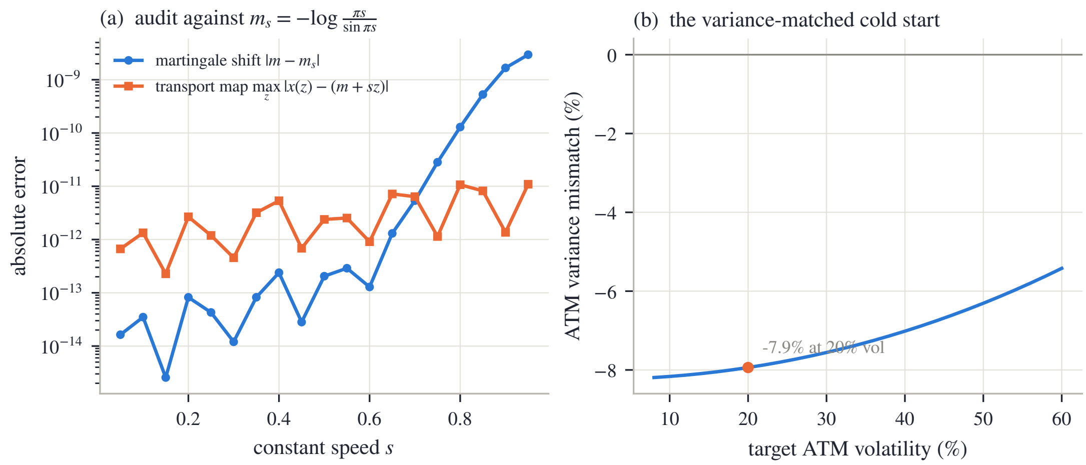

Figure 2.4: The solvable slice as an end-to-end numerical test. Left: errors in the computed transport and martingale shift against the closed forms from (2.18); the normalization becomes harder as s approaches the wall. Right: the variance-matched cold start and its derived small-s bias.

2.3 Validity and scope

The construction has two distinct claims. First, every admissible parameter vector defines an arbitrage-free slice. Second, increasing polynomial order fills out a specified class of laws. The first is exact at every order; the second determines the scope of the coordinates.

2.3.1 One-expiry validity

Proposition 2.4 (One-expiry no-arbitrage)**.** *Let $\theta \in \Theta_N$ and price calls from the LQD slice* (2.11)*. Then:*

(i) X has a strictly positive continuous density on $\mathbb{R}$, and $Y = e^X$ has a strictly positive density on $(0, \infty)$ with $\mathbb{E}[Y] = 1$;

(ii) as a function of strike y, the call is strictly decreasing and strictly convex, satisfies $(1-y)^+ \le c(y) \le 1$, has $c'(y) = -\mathbb{P}(Y > y) \in (-1, 0)$, and tends to zero as $y \to \infty$;

(iii) every nondegenerate finite-width butterfly has positive price, and put–call parity holds identically;

(iv) as a function of log-strike k, the call is concave sufficiently far in the money.

Proof. Equation (2.12) gives a positive continuous density; the shift of proposition 2.3 gives unit mean. Rank integration gives

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ \mathrm{d}u.$$

Each integrand is decreasing and convex in y, and differentiation yields $c'(y) = -\mathbb{P}(Y > y)$. Positivity of the density makes both monotonicity and convexity strict. Jensen gives $(1-y)^+ \le c(y)$, while $c(y) \le \mathbb{E}[Y] = 1$ and dominated convergence gives the right boundary. Butterfly positivity and parity follow immediately. Finally,

$$\frac{\partial^2 c}{\partial k^2} = e^k \{ f_X(k) - \mathbb{P}(Y > e^k) \},$$

which is negative as $k \to -\infty$ because $f_X(k) \to 0$ and the probability tends to one. □

The proposition is stronger than a dense-grid test. It covers every strike, including strikes between grid nodes and beyond the displayed smile, because the call is an expectation under a positive law. Calibration may move the coefficient vector arbitrarily inside Θ_N without approaching a hidden butterfly boundary. The numerical implementation must still preserve this continuous geometry; that separate question is treated by the certificates of section 2.7.2.

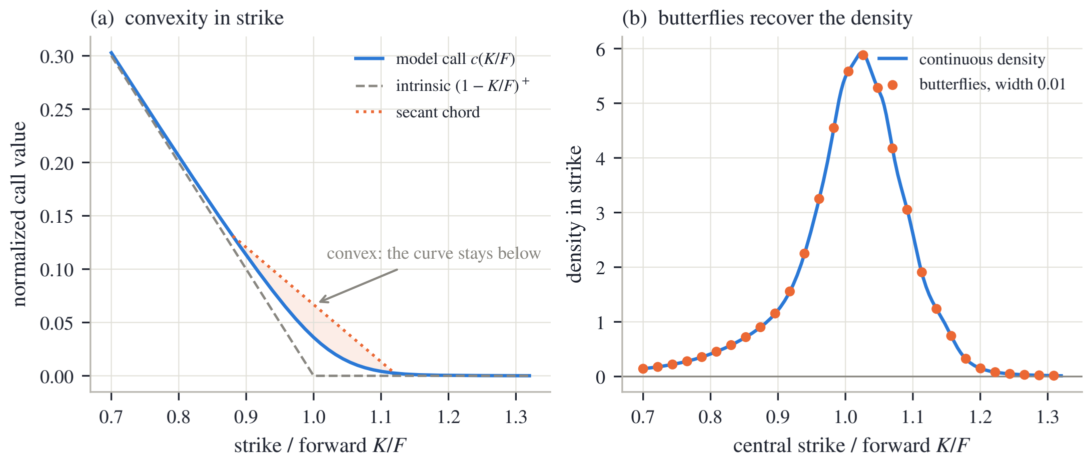

Figure 2.5: Validity on the fitted SPY December 2026 slice. Left: the call curve is convex in normalized strike and lies below its secant chord. Right: finite butterflies, divided by squared half-width, converge to the density. The geometry is produced by the law, not imposed during calibration.

2.3.2 Why the wall cannot be removed

Proposition 2.5 (Finite-forward wall)**.** *Within the LQD family the martingale normalizer exists if and only if $\lambda_+ < 1$. More generally, if a law satisfies $\mathbb{P}(X > x) \geq \kappa e^{-x}$ for some $\kappa > 0$ and all sufficiently large x, then $\mathbb{E}[e^X] = \infty$.*

Proof. The first statement is proposition 2.3. For the second, at any large x_0,

$$\mathbb{E}[e^X] \geq \int_{e^{x_0}}^{\infty} \mathbb{P}(X > \log t)\,\mathrm{d}t \geq \kappa \int_{e^{x_0}}^{\infty} \frac{\mathrm{d}t}{t} = \infty.$$

□

The boundary is therefore probabilistic, not parametric. A right log-return tail with scale at least one cannot support a finite forward. There is no left counterpart because e^X is bounded on the negative half-line.

2.3.3 Coverage of the coordinates

The natural target class can be described without reference to polynomials. For a differentiable quantile Q^* define

$$g^*(u) = \log Q^{*\prime}(u) + \log u + \log(1-u).$$

Continuity of g^* on $[0,1]$ is exactly the condition that the law have a positive continuous density and exponential tails with finite nonzero endpoint scales.

Proposition 2.6 (Universality, with exclusions)**.** *Let X^* be a mean-one law for which g^* extends continuously to $[0,1]$, with $e^{g^*(1)} < 1$. Then:*

(i) There exist admissible LQD slices with $g_N \to g^*$ uniformly. Their quantiles converge uniformly on compact subsets of $(0,1)$, their call curves converge uniformly over all strikes, and their gross-return laws converge in Wasserstein-p distance for every $1 \leq p < e^{-g^*(1)}$.

(ii) *No finite-order LQD slice has an atom, a zero-density interval, bounded support, mass at zero, or Gaussian-or-faster tails. These exclusions persist under uniformly convergent log-speeds. More precisely, let admissible g_N converge uniformly to g_∞. If $g_\infty(1) < 0$, the limiting law remains in the class of part (i). If $g_\infty(1) = 0$, weak convergence can fail because mass escapes to $-\infty$; if a weak limit nevertheless exists, it has full support and a strictly positive continuous density, but sits at the finite-forward boundary. Its moments of order $r > 1$ diverge, and its mean may be smaller than one.*

Here the one-dimensional Wasserstein metric has the concrete quantile form

$$W_p(\mu, \nu)^p = \int_0^1 |Q_\mu(u) - Q_\nu(u)|^p \,\mathrm{d}u.$$

The proof is in appendix 2.A. Its approximation step is Weierstrass: polynomials approximate the continuous g^* uniformly. Since $g^*(1) < 0$, sufficiently accurate approximants remain inside the open admissible set. The tail margin supplies a common integrable envelope for normalizers, calls, and moments.

Uniform convergence of g is the correct strength for this result. It controls the transport speed at every rank, including both endpoints. Local approximation on the quoted ranks alone would say nothing about normalization or tail moments. Once a strict margin $g^*(1) < 0$ is fixed, the same exponential envelope controls all approximating normalizers and supplies uniform integrability of gross returns. This is what upgrades pointwise shape approximation to uniform convergence of call prices.

The exclusions follow from the same geometry. A positive continuous transport speed makes the quantile strictly increasing and unbounded. An atom would require a quantile jump; a gap, a flat interval; bounded support, a finite endpoint; and a Gaussian tail, $g(u) \to -\infty$ at an endpoint. None is available to a finite polynomial, nor to a uniform limit of bounded log-speeds.

Multimodality is not excluded. By (2.6), density modes are created where the transport slows and are separated where it accelerates. A positive speed can vary many times without ever becoming zero. Thus the family may represent a smooth double-hump density, but not a literal gap of zero density between the humps. Basis order determines how sharply the speed can vary, as the synthetic example in section 2.9 demonstrates.

Thus the LQD family is not a coordinate system for every probability law. It is a coordinate system for a flexible exponential-tail class. That class is large enough to approximate smooth unimodal and multimodal smile laws, while its boundary and exclusions remain explicit.

2.4 Tails, moments, and wings

The endpoint speeds introduced in (2.15) have an exact probabilistic meaning. This section follows the chain

$$\text{endpoint speed} \longrightarrow \text{tail scale} \longrightarrow \text{finite moments} \longrightarrow \text{asymptotic smile slope.}$$

2.4.1 Endpoint speeds are tail scales

Proposition 2.7 (Exponential tails)**.** *For every admissible LQD slice there are constants $K_\pm > 0$ such that*

$$\mathbb{P}(X > x) \sim K_+ e^{-x/\lambda_+} \quad (x \to +\infty), \qquad \mathbb{P}(X < x) \sim K_- e^{x/\lambda_-} \quad (x \to -\infty). \tag{2.20}$$

Equivalently, $\mathbb{P}(Y > y) \sim K_+ y^{-1/\lambda_+}$ and $\mathbb{P}(Y < y) \sim K_- y^{1/\lambda_-}$.

Proof. On the right, (2.16) gives $x(z) = m + b_+ + \lambda_+ z + O(e^{-z})$. Its inverse satisfies

$$z(x) = \frac{x - m - b_+}{\lambda_+} + O(e^{-z(x)}).$$

Since $1 - \Lambda(z) \sim e^{-z}$,

$$\mathbb{P}(X > x) = 1 - \Lambda(z(x)) \sim e^{(m+b_+)/\lambda_+} e^{-x/\lambda_+}.$$

The left tail is identical after reversing signs. □

The gross return therefore has power tails, while the log-return has exponential tails. The two endpoint coefficients control extrapolation in a directly readable unit: log-return distance per e-fold reduction in tail probability.

2.4.2 The moment strip

Every expectation may be written either in rank or log-odds:

$$\mathbb{E}[h(X)] = \int_0^1 h(Q(u))\,\mathrm{d}u = \int_{-\infty}^{\infty} h(x(z))\rho(z)\,\mathrm{d}z. \tag{2.21}$$

Proposition 2.8 (Moment strip)**.** *For an admissible LQD slice,*

$$\mathbb{E}[e^{rX}] < \infty \quad\Longleftrightarrow\quad -\frac{1}{\lambda_-} < r < \frac{1}{\lambda_+}. \tag{2.22}$$

Consequently the last finite positive moment of Y beyond the first and the last finite inverse moment are

$$r_+^* = \frac{1}{\lambda_+} - 1, \qquad r_-^* = \frac{1}{\lambda_-}.$$

Proof. In (2.21), the right-tail integrand for $h(x) = e^{rx}$ is asymptotic to $e^{(r\lambda_+ - 1)z}$; the left-tail integrand is asymptotic to $e^{(r\lambda_- + 1)z}$. Integrability gives exactly (2.22), with divergence at both boundary values. □

The martingale condition is the point $r = 1$. It lies inside the strip exactly when $\lambda_+ < 1$, so normalization, the finite-forward wall, and the first moment are three forms of the same fact.

The asymmetry of the strip is financially important. A large λ_- may eliminate low-order inverse moments of Y without threatening the forward; negative log-returns contribute at most one to Y. A comparable right-tail scale would destroy the first positive moment and hence the forward itself. For the long NVDA example, $\lambda_- > 1$, so even $\mathbb{E}[Y^{-1}]$ diverges, while $\mathbb{E}[Y] = 1$ remains exact.

2.4.3 Closed Lee slopes

Lee's moment formula [34] relates the critical moments to the asymptotic slopes of total implied variance:

$$\beta_R = \limsup_{k\to+\infty} \frac{w(k)}{k} = \Psi(r_+^*), \qquad \beta_L = \limsup_{k\to-\infty} \frac{w(k)}{|k|} = \Psi(r_-^*), \tag{2.23}$$

where

$$\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r), \qquad r \geq 0.$$

Proposition 2.9 (Speed-to-slope maps)**.** *The LQD wing slopes are*

$$\beta_R = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \qquad \beta_L = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \tag{2.24}$$

Both maps increase with their endpoint speed. For small speeds, $\beta_\pm \sim \lambda_\pm/2$; as $\lambda_+ \uparrow 1$, $\beta_R \uparrow 2$. For LQD slices the limsups in (2.23) are ordinary limits.

Proof. Substitute $r_+^* = 1/\lambda_+ - 1$ and $r_-^* = 1/\lambda_-$ into Ψ and rationalize. For example, with $s = \sqrt{1-\lambda_+}$,

$$\Psi(1/\lambda_+ - 1) = 2 - \frac{4s}{1+s} = \frac{2\lambda_+}{(1+s)^2}.$$

The left formula is obtained with $t = \sqrt{1+\lambda_-}$. Monotonicity and limits follow from the final expressions. By proposition 2.7, the return tails are regularly varying; the Benaim–Friz refinement of Lee's formula therefore gives convergence rather than only a limsup [4]. $\square$

For completeness, the left algebra is

$$\Psi(1/\lambda_-) = 2 - \frac{4(\sqrt{1+\lambda_-} - 1)}{\lambda_-} = \frac{2\lambda_-}{(1+\sqrt{1+\lambda_-})^2}.$$

The formulas expose two useful scales. When λ is small, the wing slope is about half the endpoint speed. On the right, the map reaches Lee's model-free ceiling 2 exactly when the first moment reaches its boundary. The tail coordinate and the no-arbitrage wing bound meet at the same point.

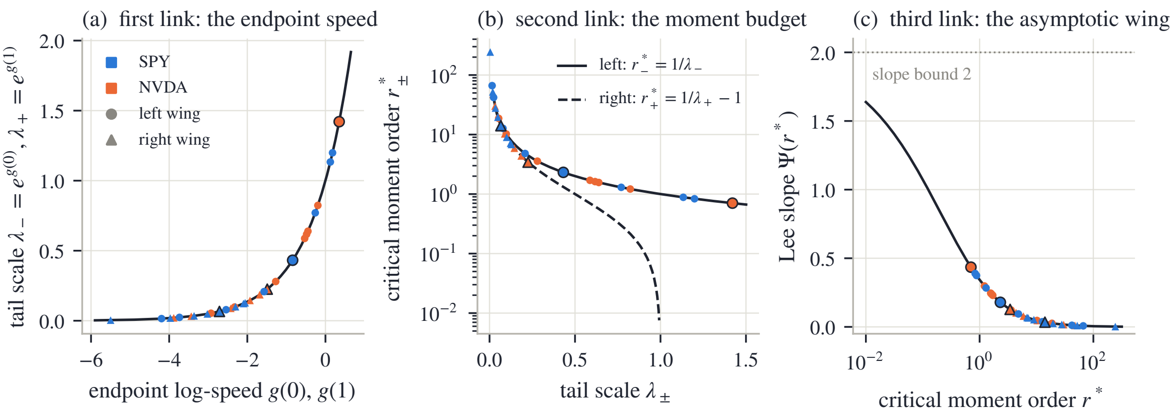

Figure 2.6: The tail chain. Left: endpoint speed determines the last finite moment; the right map reaches the finite-forward wall at $\lambda_+ = 1$. Center: Lee's moment-to-slope function. Right: the closed compositions in (2.24). Markers show the fitted SPY December 2026 and NVDA December 2027 slices.

The marked market slices occupy only a small part of the admissible range. For SPY December 2026 the endpoint speeds are $\lambda_- = 0.432$ and $\lambda_+ = 0.066$, giving slopes $\beta_L = 0.179$ and $\beta_R = 0.034$. The long NVDA slice has $\lambda_- = 1.421$ and $\lambda_+ = 0.225$, with slopes 0.435 and 0.128. The heavier single-name tails appear as larger endpoint speeds before the smile is ever plotted.

2.4.4 The limit and the traded wing

The numbers $\beta_\pm$ describe limits, not quoted strikes. If $w(k) = \beta|k| + b + o(1)$, then $w(k)/|k| = \beta + b/|k| + o(1/|k|)$ and may approach its limit from either side. The sign of the intercept b is not fixed.

No finite option strip identifies an asymptotic decay rate by itself. In a fit, $\lambda_\pm$ are selected jointly by the outer quotes, basis order, regularization, and endpoint coordinates. The model makes this choice low-dimensional and interpretable, but it does not turn extrapolation into observation.

The finite-strike discrepancy is large enough to matter. On the SPY slice in fig. 2.7, the effective right slope is 1.69 times its limit at ten delta and 1.05 times at one delta. On the left, the corresponding ratios are 0.70 and 0.70, both below one. A limiting slope can therefore overstate one traded wing and understate the other on the same fitted law.

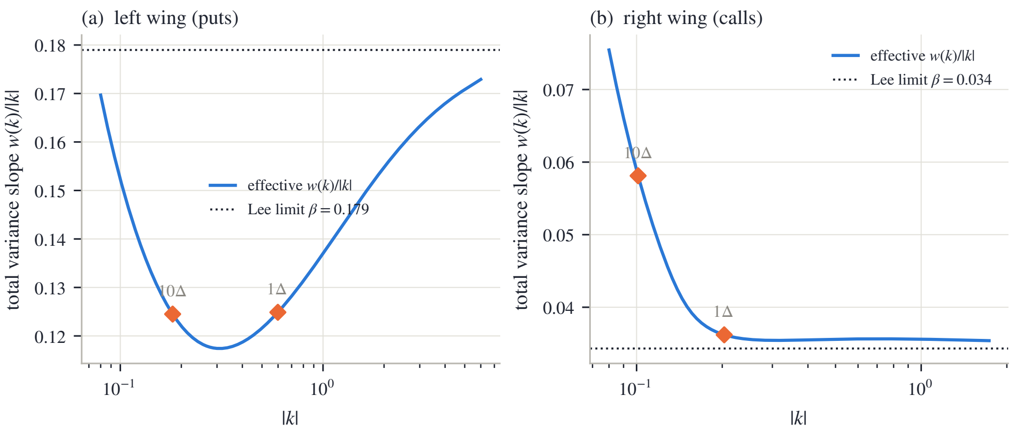

Figure 2.7: Effective variance slopes for the fitted SPY December 2026 slice. The left wing lies below its Lee limit at the marked deltas, while the right wing lies above. The asymptotic slopes are boundary conditions for extrapolation; they are not substitutes for prices at finite strikes.

2.5 The ledger calculus

Once the transport x is known, a single tail integral contains nearly every quantity needed by the fitter.

2.5.1 The upper-share ledger

Definition 2.10 (Upper-share ledger)**.** For an admissible slice define

$$G(z) = \int_z^\infty e^{x(t)}\rho(t)\,\mathrm{d}t = \mathbb{E}[Y\mathbf{1}\{Z > z\}]. \tag{2.25}$$

The name is literal: $G(z)$ is the fraction of the unit forward carried by outcomes above log-odds level z. It satisfies

$$G(-\infty) = 1, \qquad G(+\infty) = 0, \qquad G'(z) = -e^{x(z)}\rho(z) < 0. \tag{2.26}$$

For strike k, let z_k be the unique root and u_k its rank:

$$x(z_k) = k, \qquad u_k = \Lambda(z_k) = F_X(k). \tag{2.27}$$

Proposition 2.11 (Ledger pricing)**.** *For every admissible slice,*

$$c(k) = G(z_k) - e^k(1 - u_k), \tag{2.28}$$

$$P(k) = e^k u_k - (1 - G(z_k)), \tag{2.29}$$

$$c'(k) = -e^k(1 - u_k), \tag{2.30}$$

$$c''(k) = -e^k(1 - u_k) + e^k\frac{\rho(z_k)}{x'(z_k)}, \tag{2.31}$$

$$f_X(k) = e^{-k}\{c''(k) - c'(k)\} = \frac{\rho(z_k)}{x'(z_k)}. \tag{2.32}$$

Proof. Because x increases, exercise is the event $Z > z_k$. Splitting asset and cash legs gives

$$c(k) = \mathbb{E}[Y\mathbf{1}\{Z > z_k\}] - e^k\mathbb{P}(Z > z_k) = G(z_k) - e^k(1 - u_k),$$

and the put follows on the complementary event. Differentiating the call, the moving-root term is

$$\{G'(z_k) + e^k\rho(z_k)\}\frac{\mathrm{d}z_k}{\mathrm{d}k} = 0$$

because $G'(z_k) = -e^{x(z_k)}\rho(z_k) = -e^k\rho(z_k)$. This yields (2.30). Since $\mathrm{d}z_k/\mathrm{d}k = 1/x'(z_k)$, a second derivative gives (2.31); subtracting (2.30) gives (2.32). □

Pricing is thus the difference between the share delivered on exercise and the strike paid on the same event. The digital probability is $1-u_k$, and the density is one division of two arrays already present in the transport. No price differentiation is required in the implementation.

The same rank change of variables prices any integrable terminal payoff:

$$\mathbb{E}[H(Y)] = \int_{-\infty}^{\infty} H(e^{x(z)})\rho(z)\,\mathrm{d}z. \tag{2.33}$$

The ledger is the specialization in which $H(y) = y\mathbf{1}\{y > e^{x(z_0)}\}$ and the threshold z_0 is allowed to vary. Calls are unusually cheap because their payoff splits into this asset tail and an ordinary probability tail. Digitals retain only the probability tail; power and log contracts become moments of the transport. The whole payoff calculus therefore remains on one fixed rank measure.

The constant-speed slice gives a concrete first calculation. Choose its scale so that the six-month ATM volatility is exactly twenty percent. The solution is $s = 0.081250$ and the closed-form shift is $m = -0.010883$. For the call with $k = 0.10$, the strike rank is $u_k = 79.65\%$. The upper-share entry is $G(z_k) = 0.247221$, while the cash entry is $e^k(1-u_k) = 0.224876$. Hence

$$c(0.10) = 0.247221 - 0.224876 = 0.022345,$$

corresponding to 20.60% implied volatility. The symmetric law produces a smile because its exponential return tails carry more remote-strike value than the lognormal matched at the money.

The fitted law also gives the log-contract variance directly:

$$w_{\mathrm{vs}} = -2\mathbb{E}[\log Y] = -2\int_{-\infty}^{\infty} x(z)\rho(z)\,\mathrm{d}z. \tag{2.34}$$

This is the direct-law form of the familiar strip replication: the classical construction synthesizes $-\log Y$ from calls and puts over all strikes [15, 38], while here the law is already in hand and the same payoff is integrated against its rank measure. Under the usual continuous-path, continuous-monitoring identity this is the fair variance-swap strike; jumps and discrete sampling take their standard separate corrections.

2.5.2 The ATM microscope

At the money, the smile is determined locally by three quantities from the law. Let z_* solve $x(z_*) = 0$, and write

$$c_0 = G(z_*) - (1-u_*), \qquad u_* = \Lambda(z_*), \qquad f_* = \frac{\rho(z_*)}{x'(z_*)}. \tag{2.35}$$

The ATM Black variance is

$$w_0 = 4\left[\Phi^{-1}\left(\frac{1+c_0}{2}\right)\right]^2, \qquad d = \frac{\sqrt{w_0}}{2},$$

and define the digital mismatch

$$\Delta = u_* - \Phi(d). \tag{2.36}$$

Proposition 2.12 (Exact ATM handles)**.** *The total-variance smile $B(k, w(k)) = c(k)$ satisfies*

$$w(0) = w_0, \qquad\qquad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta, \tag{2.37}$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right) w'(0)^2. \tag{2.38}$$

In volatility units, $\sigma(k) = \sqrt{w(k)/\tau}$*,*

$$\sigma(0) = \sqrt{w_0/\tau}, \qquad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \tag{2.39}$$

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4}\left(\frac{\Delta}{\varphi(d)}\right)^2\right]. \tag{2.40}$$

The full differentiation is given in section 2.A.2. The first derivative contains the main idea. At $(k, w) = (0, w_0)$,

$$\partial_k B = -(1 - \Phi(d)), \qquad \partial_w B = \frac{\varphi(d)}{2\sqrt{w_0}}, \qquad c'(0) = -(1 - u_*).$$

Implicit differentiation therefore gives

$$w'(0) = \frac{c'(0) - \partial_k B}{\partial_w B} = \frac{2\sqrt{w_0}}{\varphi(d)}\{u_* - \Phi(d)\}. \tag{2.41}$$

The model digital is $1 - u_*$; the matched flat-Black digital is $1 - \Phi(d)$. Hence Δ is Black digital minus model digital. A negative equity skew is exactly a model digital richer than the flat-Black digital. Level, digital, and density determine level, skew, and curvature.

Three sign checks make the interpretation concrete. For the matched Black law, $X \sim \mathcal{N}(-w_0/2, w_0)$, so $u_* = \mathbb{P}(X \le 0) = \Phi(d)$ and the skew vanishes. If $\sigma'(0) < 0$, then $u_* < \Phi(d)$ and the model digital exceeds the Black digital. Conversely, an upward skew is exactly a cheaper model digital. The comparison is with $\Phi(d)$, not with one half: the martingale shift may place the forward rank on either side of the median even when the skew is small.

The curvature formula adds one new local fact: f_*. Holding ATM price and digital fixed, a larger density at the forward raises $w''(0)$. This matches the geometry of a peaked distribution: more mass close to the forward makes nearby options cheap relative to options a little farther away, and the smile bends upward more rapidly. The other terms in (2.38) are the Black-coordinate correction and a quadratic skew contribution. Price level, first CDF value, and first density value form a hierarchy for the three smile handles.

The identities also run in reverse. A target ATM level fixes c_0; a target skew fixes u_* through (2.37); and a target curvature fixes f_* through (2.38). These are local constraints on the law rather than finite differences of a plotted smile.

2.5.3 Envelope cancellation

Calibration requires derivatives with respect to every parameter. The call depends on a parameter both through G and through the moving root z_k. The same cancellation used for (2.30) removes the latter.

Theorem 2.13 (Envelope cancellation)**.** *For every coordinate* θ_j*,*

$$\frac{\partial c(k;\theta)}{\partial \theta_j} = \left.\frac{\partial G(z;\theta)}{\partial \theta_j}\right|_{z=z_k(\theta)}. \tag{2.42}$$

Proof. Differentiate (2.28):

$$\frac{\partial c}{\partial \theta_j} = \left.\frac{\partial G}{\partial \theta_j}\right|_{z_k} + \{G'(z_k) + e^k \rho(z_k)\}\frac{\partial z_k}{\partial \theta_j}.$$

The bracket vanishes by (2.26) and $x(z_k) = k$. □

This is an envelope theorem: the exercise boundary is the point where the payoff integrand vanishes, so its displacement has no first-order value.

Because g is affine in the coefficients, all columns are obtained by two cumulative passes. With $\bar{x} = x - m$ and $I = \int e^{\bar{x}}\rho$, let $b_j(u) = \partial g(u)/\partial\theta_j$. Then

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z e^{g(\Lambda(t))} b_j(\Lambda(t))\,\mathrm{d}t, \tag{2.43}$$

$$\frac{\partial m}{\partial \theta_j} = -\frac{1}{I}\int_{-\infty}^{\infty} e^{\bar{x}(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j}\rho(z)\,\mathrm{d}z, \tag{2.44}$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j}\right)\rho(t)\,\mathrm{d}t. \tag{2.45}$$

The basis columns are fixed functions,

$$b_L(u) = 1 - u, \qquad b_R(u) = u, \qquad b_{a_n}(u) = P_n(1 - 2u).$$

The first pass differentiates the transport speed and integrates it. The normalization dot product removes the component that would change $\mathbb{E}[e^X]$. The reverse pass differentiates the upper-share tail. Every stage follows the direction of the corresponding primal build, so the derivative is obtained by array operations rather than a separate root-and-price routine for each coefficient.

The variance-swap row is obtained from the same sensitivity of x:

$$\frac{\partial w_{\mathrm{vs}}}{\partial \theta_j} = -2\int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j}\right)\rho(z)\,\mathrm{d}z. \tag{2.46}$$

Both analytic and finite-difference Jacobians scale as $O(Pn_{\mathrm{grid}})$ for $P = N + 1$ parameters, but the analytic route avoids $P + 1$ complete rebuilds. In the reference implementation it reaches the same optimum, with costs agreeing to 1.7×10^{-10} relative and parameters to 3.7×10^{-7}, in 1.41–1.76× less time. The worst relative disagreement with an independent central difference is 3.7×10^{-6}.

The theorem is exact for the continuous curves. Numerically, x and G are interpolated separately, so between nodes the price interpolant is not algebraically identical to the integral of the quantile interpolant. The analytic derivative inherits this fourth-order interpolation error. The finite-difference comparison in fig. 2.8 measures the discrete discrepancy instead of assuming the continuous cancellation transfers without loss.

It is worth counting the shared work. One build stores (x, x', G, G') on a fixed grid. A price is one root and two ledger entries; a density is one division; the log-contract variance is one weighted sum; the Jacobian reads ledger sensitivities at the same roots; and calendar order, in section 2.8, will compare two such arrays. These are not five approximations to the law; they are five queries of the same law.

2.6 Calibration

Validity has already been settled by the coordinates. Calibration may therefore concentrate on information: selecting, among admissible laws, one that represents the quoted strip without inventing unsupported detail.

2.6.1 A one-expiry objective

Suppose a slice contains quotes (k_i, σ_i) with $w_i = \sigma_i^2\tau$. The reference fit minimizes

$$\sum_i \left(\frac{c(k_i;\theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta}\right)^2 + \alpha \sum_{n\geq 4} n^{2\nu} a_n^2. \tag{2.47}$$

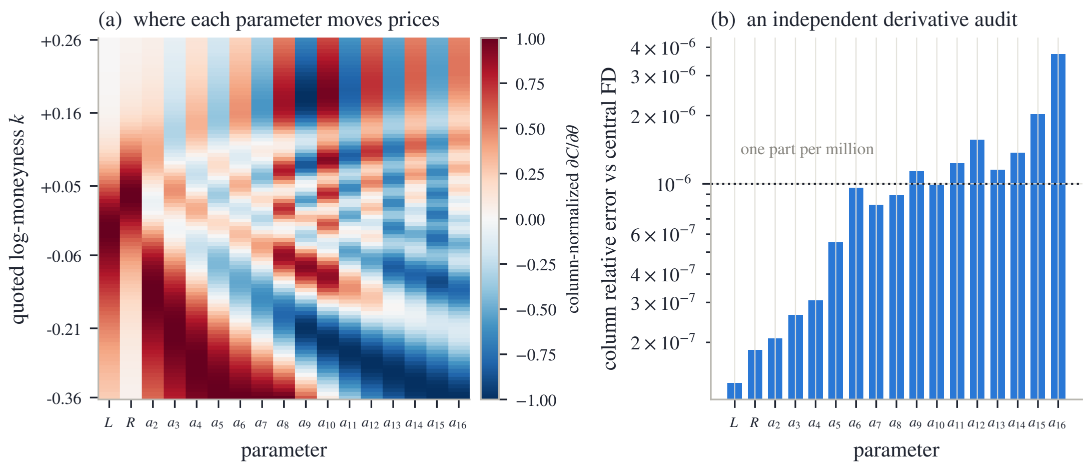

Figure 2.8: The analytic Jacobian on the fitted SPY December 2026 slice. Left: the normalized price-sensitivity pattern of every parameter across strikes. Right: columnwise disagreement with an independent central finite difference. The comparison tests the entire sensitivity pass by a separate route.

Each element of (2.47) answers one practical problem. The residual is computed in price, so implied-volatility inversion stays outside the optimization loop, where it cannot fail on an exploratory trial point. Division by the Black vega $\partial_\sigma B = \varphi(d_+)\sqrt{\tau}$ makes a central residual a volatility error — the unit in which quotes are judged — while the floor η lets the metric degrade to a scaled price error where vega vanishes, instead of assigning unbounded weight to a negligible option. The $n^{2\nu}$ ridge makes fine modes progressively more expensive but leaves a_2 and a_3, the main curvature and asymmetry modes, unpenalized.

The reference objective also contains a smooth row $\log(1 + e^{50(\lambda_+ - 0.90)})$. The admissibility chart already prevents the wall from being reached; this row is instead an explicit prior against a nearly exhausted right-tail moment budget.

A quoted option is an interval, not necessarily a point. Three targets are useful. A *mid* fit uses (2.47) directly. A *band* fit uses the outside-band hinge together with a weak mid anchor. If $[p_i^{\mathrm{lo}}, p_i^{\mathrm{hi}}]$ is the price band and ϕ_i the vega normalizer, its two residuals are

$$\frac{(c_i - p_i^{\mathrm{hi}})^+ + (p_i^{\mathrm{lo}} - c_i)^+}{\phi_i}, \qquad \sqrt{0.05}\,\frac{c_i - p_i^{\mathrm{mid}}}{\phi_i}. \tag{2.48}$$

The squared hinge is continuously differentiable at a band edge, and the weak mid row — its weight deliberately small — selects one point when a wide band leaves a flat valley of equally admissible laws. A *haircut* fit first shrinks the volatility band by h:

$$\sigma_i^{\mathrm{lo}} = \min(\sigma_i^{\mathrm{bid}} + h, \sigma_i^{\mathrm{mid}}), \qquad \sigma_i^{\mathrm{hi}} = \max(\sigma_i^{\mathrm{mid}}, \sigma_i^{\mathrm{ask}} - h). \tag{2.49}$$

If the original spread is narrower than $2h$, the band collapses to the mid instead of changing orientation. This occurs for every SPY quote in the frozen sample; the wider NVDA spreads retain live bands.

The two underliers illustrate both regimes. The SPY December node has median spread 4 vol bp, far inside the default $2h = 100$ vol-bp width, and none of its haircut bands survives. Its figures therefore show raw bid–ask intervals around a mid fit. On the featured NVDA nodes, 47% and 52% of the bands remain live. The same rule becomes a mid target on a liquid index and a genuine interval target on the wider single-name book.

The informational division can be read by location. Between liquid quotes a small change in the law moves observed prices and is strongly penalized: the strip determines the call curve there, and with it the ATM level, skew, curvature, and low-order moments. Near the edge of the strip, data and regularization share control; beyond the outer quote, the endpoint speeds and the ridge supply most of the continuation. The curve itself is smooth — what changes gradually is the source of information, a transition made visible by plotting quote locations with the density as in fig. 2.3. Tail coordinates, ridge, and basis order should be reported with a fit because they make these choices.

2.6.2 Order and starting values

At order N the model has $N+1$ parameters. The reference implementation uses the effective order

$$N_{\text{eff}} = \max\left(\min\left(N, \left\lfloor \frac{n_q}{2} \right\rfloor - 1\right), \min(N, 6)\right), \tag{2.50}$$

where n_q is the quote count. The cap keeps the parameter count well below the number of observations; the floor leaves thin strips enough shape to form a usable smile. The rule is an identification convention, not a validity condition. Changing it changes inter-quote density detail while every resulting slice remains arbitrage-free.

Away from the floor, (2.50) requires roughly two quotes per body degree of freedom. This is not a theorem about optimal smoothing; it is a guard against inferring one oscillatory direction from each noisy observation. On the shortest NVDA slice, 17 quotes give $N_{\text{eff}} = 7$. The fit remains visibly less certain there rather than borrowing high-order detail from the basis.

Cold starts use the constant-speed value (2.19). In a maturity sweep, the previous expiry supplies a warm start. A multi-start check on the featured SPY and NVDA nodes used 10 randomized starts per node and found 1 basin, with worst fit-quality spread 0.00 vol bp. This supports the starting rule on those slices; event-dominated books may still require a broader search.

2.6.3 Coordinates for optimization

Three charts describe the same polynomial family.

1. The *raw chart* is $(L, R, a_2, \ldots, a_N)$. It is simple, but (2.14) couples body modes to both tails.

2. The *endpoint chart* is $(\log \lambda_-, \log \lambda_+, a_2, \ldots, a_N)$. Solving (2.14) for (L, R) is linear. Equivalently, each body basis function is replaced by an endpoint-vanishing combination. Body moves then hold both tail scales fixed.

3. The *logistic chart*, the reference default, writes $\lambda_+ = \operatorname{expit} \xi$ with $\xi \in \mathbb{R}$. Every finite coordinate vector is admissible, and

$$\frac{\mathrm{d} \log \lambda_+}{\mathrm{d}\xi} = 1 - \lambda_+. \tag{2.51}$$

   Steps shrink smoothly as the wall is approached.

The endpoint-vanishing body functions are explicit:

$$\widetilde{P}_n(u) = P_n(1-2u) - (1-u) - (-1)^n u, \qquad n \geq 2. \tag{2.52}$$

They satisfy $\widetilde{P}_n(0) = \widetilde{P}_n(1) = 0$. A tail policy can therefore set (λ_-, λ_+) while calibration adjusts the body in a subspace that cannot move either endpoint.

The objective is identical in all three charts. Fits in the frozen sample agree across charts to 2.2×10^{-7} in the recovered raw parameters. In floating point, $\operatorname{expit} \xi$ eventually rounds to one, so the implementation also enforces $\lambda_+ \leq 1 - 10^{-6}$.

2.6.4 Calibration sequence

The complete one-slice calculation is short enough to state as an algorithm.

1. Convert strikes and prices to (k_i, c_i) using the node's forward and discount factor. Map bid and ask volatilities through Black once to construct price bands and compute the quote vegas used for normalization.

2. Choose the effective order from (2.50) and select mid, band, or haircut residuals. This fixes the data part of the objective before optimization begins.

3. Construct a cold start from the ATM variance using (2.19), or map the previous expiry into the endpoint chart for a warm start. In either case, convert to logistic coordinates so all finite trial vectors are admissible.

4. At each trust-region step, build (x, x', G, G') on the fixed grid of section 2.7, price every quote by (2.28), and obtain the whole residual Jacobian from (2.43)–(2.45). Failed exponent-budget trials return a large residual rather than an undefined price.

5. After convergence, rebuild the accepted parameters on the dense grid. Evaluate ATM handles, tail moments, Lee slopes, the log-contract variance, and the numerical certificates (section 2.7.2) from this one final object.

6. If the slice belongs to a maturity sequence, add calendar rows against the accepted nearer slice and repeat. Report any residual common-support gap in normalized price units.

This order matters. Market conventions are fixed before the solve; probability validity holds during the solve; diagnostics are computed after the solve from the same law that produced the quoted prices. No separate post-processing smoother is inserted between calibration and publication.

2.7 Numerical realization

The theorems of the preceding sections concern the continuous objects x and G; the fitter computes with discrete realizations of them. One grid build produces every array used above, analytic corrections handle what lies beyond the grid, and two certificates verify that discretization has preserved what the theory guarantees.

2.7.1 One grid build

All integrals use a symmetric log-odds grid $[-Z_{\max}, Z_{\max}]$, with $Z_{\max} = 40$. Optimization uses 2001 nodes; the accepted slice is rebuilt on 8001 nodes. A build performs four operations: evaluate g, integrate e^g cumulatively to form $\bar{x}$, compute the martingale shift, and integrate $e^x \rho$ backward to form G. Exact nodal derivatives $x' = e^g$ and $G' = -e^x \rho$ accompany the values. Cubic Hermite interpolation then gives fourth-order off-grid evaluation and a Newton-polished strike root.

The missing logistic tails are integrated analytically. To leading order,

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}} \rho \, \mathrm{d}z \simeq \frac{e^{\bar{x}(Z_{\max}) - Z_{\max}}}{1 - \lambda_+}, \qquad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}} \rho \, \mathrm{d}z \simeq \frac{e^{\bar{x}(-Z_{\max}) - Z_{\max}}}{1 + \lambda_-}. \tag{2.53}$$

For a strike beyond the right return grid, continue the root along the tail asymptote,

$$z_R(k) = Z_{\max} + \frac{k - x(Z_{\max})}{\lambda_+}.$$

Both the ledger and cash leg are then single exponentials. Their difference is

$$c(k) \simeq e^{k - z_R(k)} \frac{\lambda_+}{1 - \lambda_+}. \tag{2.54}$$

The left-wing continuation follows from parity.

To derive the factor in (2.54), restart the right-tail correction at $z_R(k)$, where $x = k$. The asset leg is approximately $e^{k-z_R(k)}/(1-\lambda_+)$ and the cash leg is $e^{k-z_R(k)}$. Subtraction leaves $\lambda_+/(1-\lambda_+)$. The continuation is therefore the tail form of the same asset-minus-cash identity (2.28), not an independent wing model.

Three floating-point rules are essential. First, compute $\rho(z) = \operatorname{expit}(z)\operatorname{expit}(-z)$ rather than $u(1-u)$, since u rounds to one for large positive z. Compute the cash leg as $\exp\{k - \log(1+e^{z_k})\}$ using a stable log-sum-exp. Second, reject a trial if an interior value of g or $\bar{x}$ exceeds the exponent budget, even when the endpoints are admissible. Third, evaluate the Lee function at large r in the rationalized form

$$\Psi(r) = 2 - \frac{4}{\sqrt{1 + 1/r} + 1}, \tag{2.55}$$

which avoids subtracting nearly equal large numbers. The full build and sensitivity pseudocode appear in appendix 2.B.

2.7.2 Certificates

The continuous theorems concern x and G; numerical prices use their interpolants. Two inexpensive certificates connect the two levels. On each Hermite segment, the Fritsch–Carlson inequalities require both endpoint derivatives to lie between zero and three times the secant slope [22]. Applying them to x and $-G$ certifies monotone interpolation; segments below round-off are declared flat to a stated tolerance. Across the traded strike span, the analytic derivatives of $w(k)$ are also inserted in (2.5) on a dense grid and required to satisfy $g_{\mathrm{D}} \geq -10^{-4}$.

An independent randomized battery prices 27 admissible slices across several orders and stress classes at sub-grid strikes. Its worst errors in call bounds, butterflies, digitals, and agreement with a four-times-finer grid are respectively 1.1×10^{-11}, 8.2×10^{-13}, 1.5×10^{-12}, and 2.3×10^{-9} in normalized price. The exact constant-speed test of fig. 2.4 independently checks transport, normalization, tail correction, and root evaluation. These numerical tests do not create validity; they verify that the discrete realization has preserved it to the reported tolerance.

The two-grid rebuild supplies a final resolution check. Parameters fitted on 2001 nodes and refitted on 8001 nodes agree to 1.4×10^{-6}. The accepted vector is rebuilt once on the dense grid, so prices, densities, ATM handles, and calendar ledgers all read the same final discrete object.

These diagnostics answer different questions and should not be collapsed into one flag. Quote residuals measure agreement with the strip; effective order and ridge describe the asserted inter-quote structure; endpoint speeds describe the extrapolation and its moment budget; monotonicity and belly margins certify that discretization preserved validity; calendar gaps describe compatibility across expiries. A fit can be excellent on one axis and weak on another.

2.8 Calendar order

Butterfly freedom is a property of one marginal law. A surface also requires later expiries to dominate earlier ones in convex order. Under deterministic rates and carry, this comparison has a simple form in normalized units.

Lemma 2.14 (Calendar ordering at equal log-moneyness)**.** *Let $F_{0,t}$ be the deterministic forward curve and suppose $M_t = S_t/F_{0,t}$ is a positive martingale. For $T_1 < T_2$ and every k, the normalized calls satisfy*

$$c_1(k) \leq c_2(k). \tag{2.56}$$

Proof. Since $M_{T_i} = S_{T_i}/F_i = Y_i$, conditional Jensen gives

$$\begin{aligned} c_2(k) &= \mathbb{E}[(M_{T_2} - e^k)^+] \\ &\geq \mathbb{E}[(\mathbb{E}[M_{T_2} \mid \mathcal{F}_{T_1}] - e^k)^+] = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k). \end{aligned}$$

□

Equal k means strikes $K_i = F_i e^k$, which are generally different dollar strikes. The comparison is a forward-scaled calendar relation, not the usual listed equal-strike spread. In probability language, the laws of Y_i have equal means and increase in convex order.

The scaling is essential. A violation at k is monetized by shorting $1/(D_1F_1)$ calls at $K_1 = F_1e^k$ and buying $1/(D_2F_2)$ calls at $K_2 = F_2e^k$. After normalization both payoffs are convex functions of the common martingale M_t. Without deterministic carry, one must first specify the numeraire and the objects being compared; equal log-moneyness alone is not a model-free calendar relation.

2.8.1 Calls and ledgers are conjugate

Lemma 2.15 (Ledger–call conjugacy)**.** *For any admissible slice,*

$$\begin{aligned} c(k) &= \max_{z\in[-\infty,\infty]} \{G(z) - e^k(1 - \Lambda(z))\}, \\ G(z) &= \min_{k\in[-\infty,\infty]} \{c(k) + e^k(1 - \Lambda(z))\}. \end{aligned} \tag{2.57}$$

The optimizers are $z = z_k$ and $k = x(z)$, respectively.

Proof. For fixed k,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k)\rho(t)\,\mathrm{d}t.$$

The integrand is negative before z_k and positive after it. Starting the integral at z_k therefore maximizes it and gives the call price. The first identity implies $c(k) + e^k(1 - \Lambda(z)) \geq G(z)$ for every k, with equality at $k = x(z)$, proving the second. □

Geometrically, the first identity scans all possible exercise ranks. Before z_k, the integral includes outcomes where the payoff is negative; after z_k, it discards outcomes where the payoff is positive. The strike root is the unique point of tangency between call and ledger descriptions. The second identity reconstructs the ledger from the full call curve, so no information is lost in changing representation.

Theorem 2.16 (Calendar order is ledger order)**.** *For two admissible mean-one slices,*

$$c_1(k) \leq c_2(k) \text{ for all } k \iff G_1(z) \leq G_2(z) \text{ for all } z. \tag{2.58}$$

Proof. If $G_1 \leq G_2$, the maximum representation in (2.57) gives $c_1 \leq c_2$. If $c_1 \leq c_2$, the minimum representation gives $G_1 \leq G_2$. Both implications are pointwise. □

Thus the continuum of calendar inequalities is equivalent to ordering two arrays already produced by the slice builds. This is also the classical convex-order criterion: by the results of Strassen and Kellerer, a family of equal-mean laws increasing in convex order can be realized as the marginals of a martingale [33, 46].

This is an existence statement. Ordered ledgers make the marginal laws compatible with some martingale dynamics; they neither select that dynamics nor determine forward smiles. The calendar problem treated here is exactly the static marginal problem.

2.8.2 A support-confined control

The theorem is an exact diagnostic. A finite calibration requires a chosen enforcement rule. When fitting a farther expiry after a nearer one, the reference objective adds

$$\sqrt{w_{\mathrm{cal}}}\,\zeta_m\{c_{\mathrm{near}}(k_m) - c_{\mathrm{far}}(k_m)\}^+ \tag{2.59}$$

on a grid of strikes in the intersection of their quoted spans. The weights ζ_m taper to zero with a $\cos^2$ profile over the outer fifteen percent of that span. The number of nodes is $\max(49, 4N+1)$.

The rows are imposed in price space, not ledger space. A ledger value at one rank integrates the entire upper tail, so constraining it couples the quoted region of one expiry to the unquoted continuation of another: the ledger is ideal for diagnosis precisely because it sees the whole law, and unsuitable as a constraint for the same reason. In a reproduced stress comparison, a full-grid ledger floor moved the worst quote error from 10.6 to 1095.0 vol bp; fixed-strike call rows, each a specified payoff in the common quoted region, localize the control. The weight is finite, and any remaining violation is reported as a price gap and as a fraction of the local bid–ask width. Calendar order is thus controlled on common quoted support; outside it, the ledger comparison remains a diagnostic of the extrapolations.

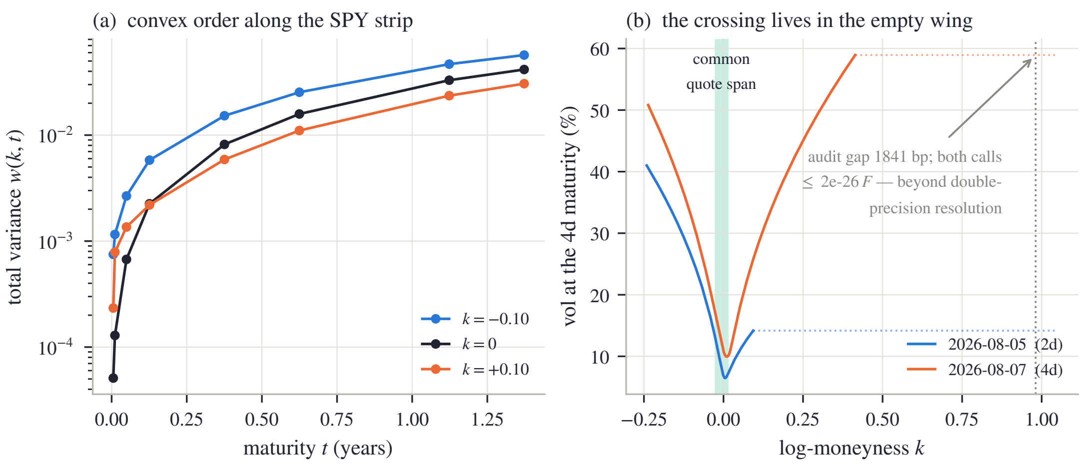

Figure 2.9: Calendar order in the frozen SPY sample. Left: total variance is ordered through maturity at representative log-moneyness values. Right: the two shortest expiries are ordered on their common quoted span, while a volatility-space crossing appears only in a numerically empty far wing. The price-space gap there is at round-off scale.

For the two shortest SPY expiries, a full-grid volatility audit flags a 1840.6 vol-bp crossing at $k = +0.981$, far outside the common quote span $[-0.028, +0.016]$. The corresponding calls are 5.4×10^{-107} and 1.6×10^{-26} of forward. A direct price audit finds a worst violation of only 2.2×10^{-15}. The crossing is a disagreement between extrapolations in a region where Black inversion is ill-conditioned, not an economically available calendar trade.

2.9 Examples and the model contract

All market examples use one frozen after-hours snapshot of SPY and NVDA listed options from the 2026-08-03 US session. Each underlier has eight expiries, for 16 fitted nodes in total. The common recipe is order $N = 16$ subject to (2.50), the logistic chart, the haircut target (2.49), ridge $\alpha = 10^{-6}$, and the calendar rows (2.59). No node is tuned separately. Across the sample the median quote error is 5.9 vol bp and the worst is 24.6 vol bp.

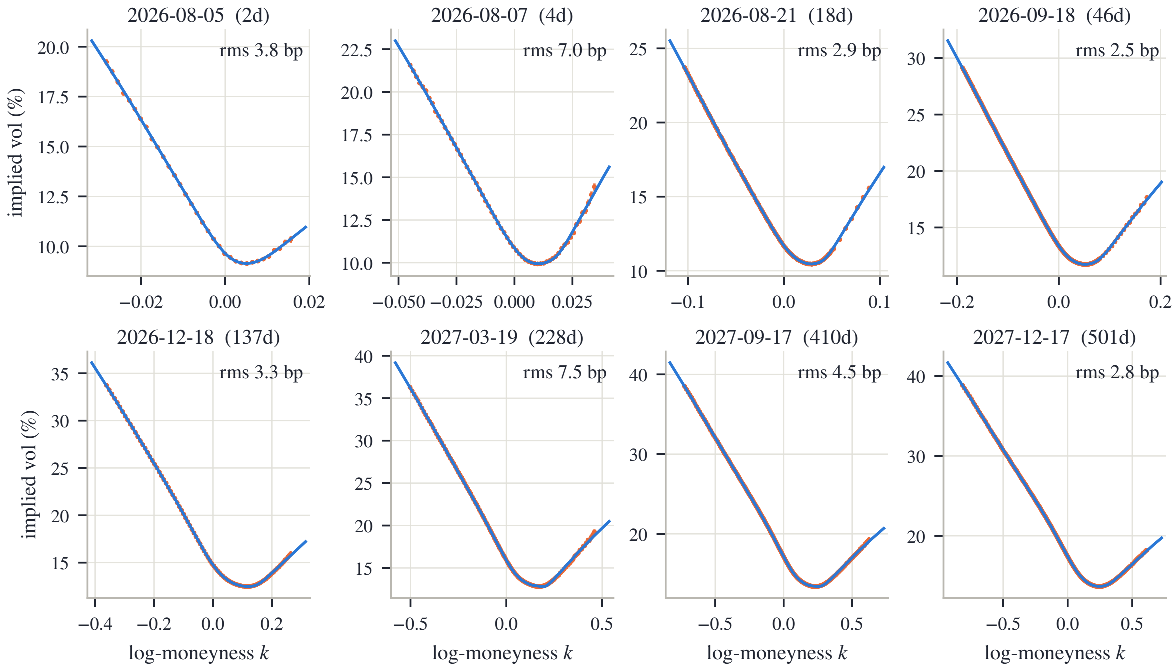

Figure 2.10: Eight SPY expiries fitted with one recipe. Each panel shows raw bid–ask quotes, mids, the fitted LQD slice, and rms error. The short end is steeper and more localized; the long end broadens smoothly. Beyond the last quote, each curve is determined by the fitted exponential-tail convention.

2.9.1 SPY: a term structure and one slice

The gallery separates recurring market shape from the completion chosen by the model. Through the quoted region the slices follow the term structure of level and skew; outside it the two endpoint speeds continue the curve. The boundary is most visible at the short end, where quoted spans are narrow in log-moneyness: the sharp local skew is supported by quotes, while the remote upward turn is the exponential-tail completion. At longer maturities the span widens and a larger fraction of the displayed smile is directly informed by prices. The median error across the eight slices is 3.6 vol bp; the worst is 7.5 vol bp.

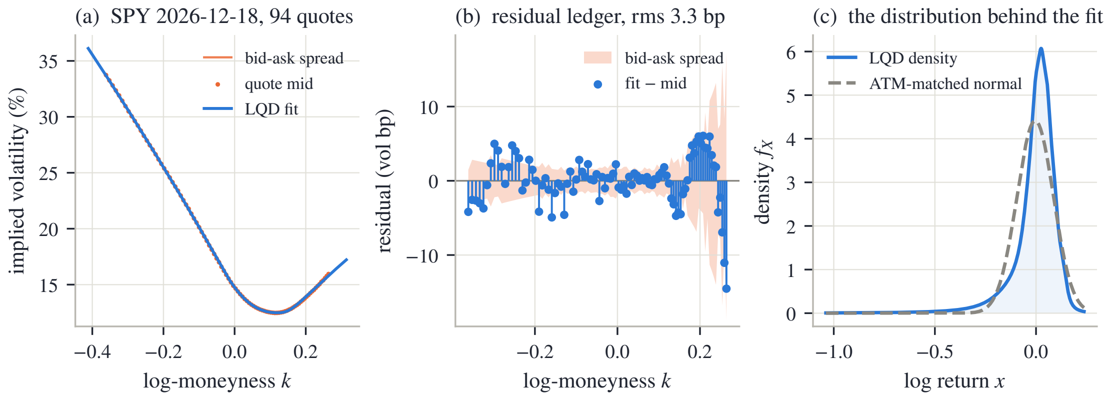

Figure 2.11: The SPY December 2026 slice. Left: quotes and fitted volatility. Center: signed residuals in vol bp. Right: the fitted density against an ATM-matched normal. The density view makes the negative skew visible as a transfer of mass toward the left tail.

The December 2026 slice contains 94 quotes and fits them to 3.3 vol bp rms. Its exact ATM quantities from proposition 2.12 are volatility 14.76%, skew −0.430, and curvature 4.40. The forward rank is 41.31%%, below the matched Black rank; hence $\Delta < 0$, in agreement with the negative skew. The log-contract variance corresponds to 19.84% annualized volatility.

The three panels answer different questions. The first tests the nonlinear map from law to implied volatility. The second asks whether residuals retain systematic strike structure. The third removes the Black coordinate and shows the fitted object itself: a left-heavy density. The ATM-matched normal has the same central price level but not the same digital or tail allocation, and therefore cannot reproduce the skew.

The ledger also makes an individual option transparent. For the listed December 800 call, $k = 0.0409$ and the strike rank is 65.27%%. The upper-share and cash entries are 0.378366 and 0.361834, so

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532.$$

This is \$12.70 per share and corresponds to 13.37% implied volatility.

2.9.2 NVDA and the order guard

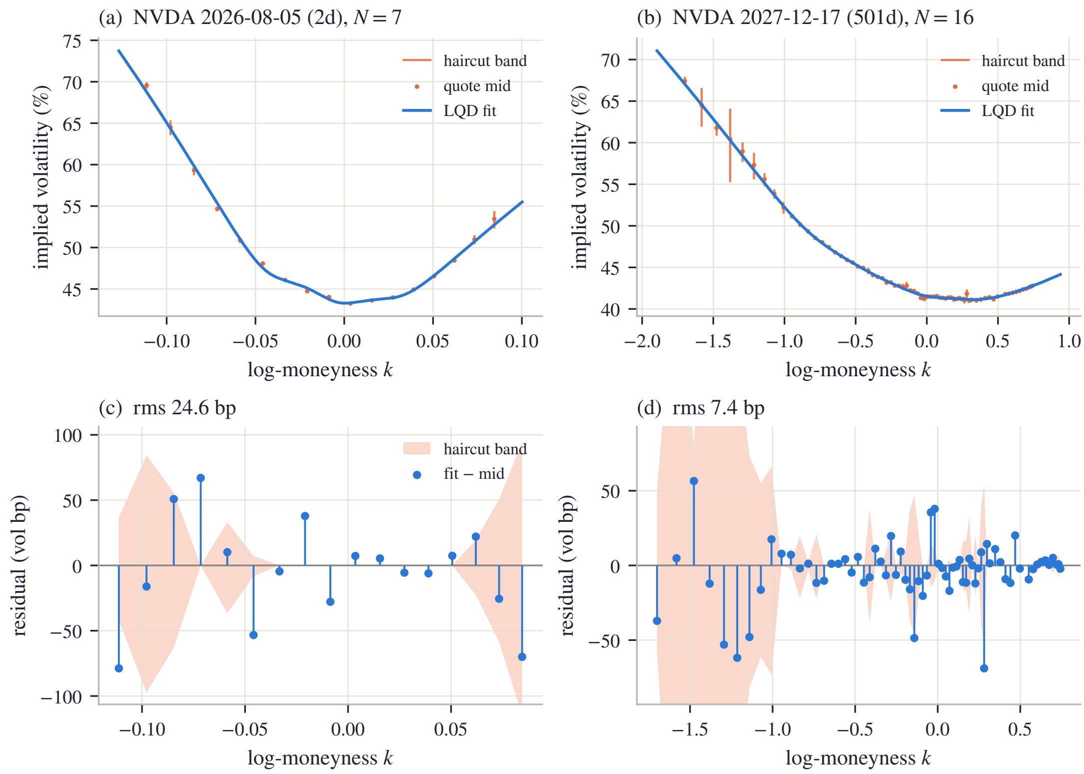

Figure 2.12: Two NVDA slices. Left: the shortest expiry, with 17 quotes and effective order $N_{\text{eff}} = 7$. Right: the December 2027 expiry at full order. Wider NVDA spreads leave live haircut bands and permit visibly larger residual variation than SPY.

The shortest node fits at 24.6 vol bp rms; the long node, with 69 quotes, at 7.4 vol bp. The latter has heavier fitted tails than SPY. In particular its left endpoint speed exceeds one, which is admissible: only the right endpoint is constrained by the finite-forward wall. The consequence is a last finite inverse-moment order 0.70 below one.

The shortest node is also the clearest example of the order guard. A thin strip cannot distinguish sixteen smooth modes, even though an optimizer can assign values to them. Reducing N_{eff} converts that lack of information into visible residual uncertainty instead of invisible oscillation between quotes. The long node has enough observations to use the full order and separate body curvature from tail continuation.

2.9.3 What basis order asserts

The synthetic benchmark is a martingale-normalized mixture of two lognormal components. It has a genuinely bimodal density but only gentle shoulder structure in implied volatility.

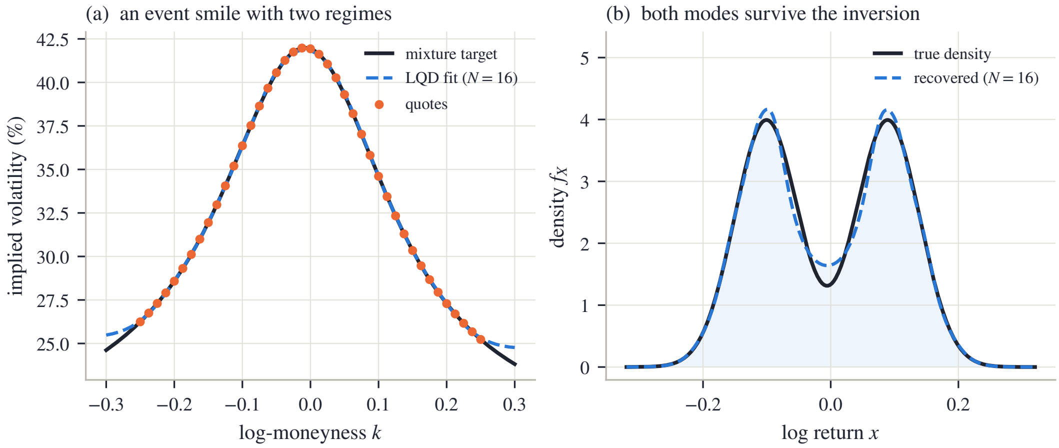

Figure 2.13: The double-hump benchmark at $N = 16$. Left: the fitted smile matches the synthetic quotes to 4.8 vol bp rms. Right: the fitted density recovers both modes, the trough, and the asymmetry of the target.

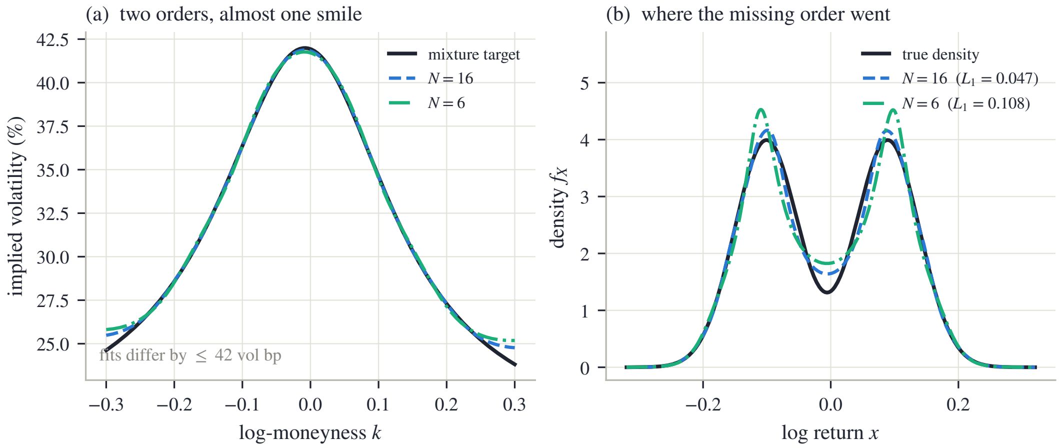

Figure 2.14: The same target at orders 6 and 16. Left: the implied-volatility curves nearly coincide over the quoted region. Right: the densities do not; the lower-order fit displaces the modes and fills part of the trough. Basis order controls how much inter-quote structure the model is allowed to assert.

At $N = 16$ the fitted mode locations are -0.099 and 0.088, against true locations -0.101 and 0.089. At $N = 6$ both modes remain, but their shape is distorted. The L^1 density error rises from 0.047 to 0.108, even though the two implied-volatility curves differ by at most 41.8 vol bp on the comparison grid. Option quotes can be nearly indifferent between laws that price density-sensitive structures differently.

The lesson is not that high order is always preferable. At $N = 16$ the synthetic target is known to contain fine structure, so extra modes recover a feature known to be real. In market data the truth is not supplied. High order then grants the model permission to assert finer inter-quote density shape. Digitals and narrow butterflies are sensitive to that permission; broad vanilla prices may not be. Basis order should be chosen for the downstream instruments, not solely for the in-sample volatility residual.

2.9.4 Position among existing models

The construction joins several classical ideas. Breeden–Litzenberger identify option prices with a density [10]; Lee and Benaim–Friz connect moments with wing slopes [4, 34]; quantile modeling supplies the rank coordinate [26, 39]; and Petersen–Müller use the same log-quantile-density transform to place density-valued data in an unconstrained function space [41]. The metalog family is a related direct quantile expansion [32].

SVI and SSVI work in volatility coordinates and characterize admissible parameter regions [23, 25]; arbitrage-free smoothing works directly with constrained prices [20]. LQD instead makes the probability law primary. Its contribution is the combined architecture: structural one-slice validity, readable tail coordinates, and one ledger for pricing, differentiation, density, variance, and calendar order.

2.9.5 The contract

Two things remain: how to read a fitted slice, and what precisely is claimed.

A fitted slice should be read in the same order as the theory. First inspect quote residuals over the retained span. Then read the three ATM handles as local level, digital, and density information. Next inspect $g(u)$ and the density over the ranks occupied by quotes. Only after that should one read $\lambda_\pm$, the moment strip, and Lee slopes, since these describe the extrapolated endpoints. Finally compare the ledger with neighboring expiries. This sequence prevents a tail convention or a calendar diagnostic from being mistaken for a directly observed quote.

Structural	**Numerically controlled**	**Outside the claim**
Positive density, unit mean, and butterfly-free calls for every admissible slice	Interpolation monotonicity, traded-range density, and randomized price audits	Atoms, mass at zero, density gaps, bounded support, and Gaussian-or-faster tails
Exact exponential tail scales, moment strip, and Lee limits	Tail quadrature, exponent budgets, grid convergence, and finite-strike wing prices	Tail parameters as observations, or asymptotic slopes as finite-strike quotes
Exact ATM handles, analytic Jacobian, and ledger characterization of calendar order	Quote fit, basis-order control, multi-start checks, and calendar order on common quoted support	Dynamics, jump decomposition, realized-variance corrections, and calendar structure outside the controlled support

Table 2.2: The model contract: theorem, numerical control, and excluded scope.

The central distinction is now visible in one place. Probability supplies validity. Quotes supply information over a finite strip. Basis, ridge, tails, and calendar scope supply convention elsewhere. The value of the LQD representation is not that convention disappears, but that it can no longer be mistaken for a theorem or for market information.

2.A Proofs deferred from the main text

2.A.1 Derivation of the volatility-space density factor

Let $c(k) = B(k, w(k))$. For general (k, w), the identity $e^k \varphi(d_-) = \varphi(d_+)$ gives

$$\partial_k B = -e^k \Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

The second partials required by the chain rule are

$$\partial_{kk} B = -e^k \Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}},$$

$$\partial_{kw}B = \frac{d_+\varphi(d_+)}{2w},$$
$$\partial_{ww}B = -\frac{d_+\varphi(d_+)}{2\sqrt{w}}\,\partial_w d_+ - \frac{\varphi(d_+)}{4w^{3/2}}, \qquad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}.$$

Differentiate $c(k)$ twice and subtract the first derivative:

$$c'' - c' = (\partial_{kk}B - \partial_k B) + (2\partial_{kw}B - \partial_w B)w' + \partial_{ww}B(w')^2 + \partial_w B w''.$$

Factor $\varphi(d_+)/\sqrt{w}$. Since $d_+ = -k/\sqrt{w} + \sqrt{w}/2$,

$$2\partial_{kw}B - \partial_w B = -\frac{k\varphi(d_+)}{w^{3/2}},$$

and a direct multiplication gives

$$d_+\partial_w d_+ = -\frac{k^2}{2w^2} + \frac{1}{8}.$$

Therefore

$$\begin{aligned} c'' - c' &= \frac{\varphi(d_+)}{\sqrt{w}}\left[1 - \frac{kw'}{w} + \left(\frac{k^2}{4w^2} - \frac{1}{4w} - \frac{1}{16}\right)(w')^2 + \frac{w''}{2}\right] \\ &= \frac{\varphi(d_+)}{\sqrt{w}}\left[\left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}\right]. \end{aligned}$$

Finally, $f_X(k) = e^{-k}(c'' - c')$ and $e^{-k}\varphi(d_+) = \varphi(d_-)$, which proves (2.5).

2.A.2 Proof of the ATM formulas

The smile is defined by $B(k, w(k)) = c(k)$. Put $w_0 = w(0)$ and $d = \sqrt{w_0}/2$. The identity $e^k\varphi(d_-) = \varphi(d_+)$ follows from $d_+^2 - d_-^2 = -2k$ and gives

$$\partial_k B = -e^k\Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiating once more and evaluating at $(0, w_0)$ gives

$$\begin{aligned} \partial_k B &= -(1 - \Phi(d)), & \partial_w B &= \frac{\varphi(d)}{2\sqrt{w_0}}, \\ \partial_{kk}B &= -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, & \partial_{kw}B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \\ \partial_{ww}B &= -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}. \end{aligned} \tag{2.60}$$

By (2.30)–(2.32),

$$c'(0) = -(1 - u_*), \qquad c''(0) = f_* - (1 - u_*).$$

One derivative of the defining identity gives $\partial_k B + \partial_w B\, w'(0) = c'(0)$, hence

$$w'(0) = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)}\Delta.$$

Two derivatives give

$$\partial_{kk}B + 2\partial_{kw}B\, w' + \partial_{ww}B\,(w')^2 + \partial_w B\, w'' = c''.$$

Solving at the money and using (2.60),

$$\begin{aligned} w''(0) &= \frac{2\sqrt{w_0}}{\varphi(d)}\left[f_* + \Delta - \frac{\varphi(d)}{\sqrt{w_0}} - \frac{\varphi(d)}{2\sqrt{w_0}}w'(0)\right. \\ &\qquad\qquad \left. + \left(\frac{\varphi(d)}{4w_0^{3/2}} + \frac{\varphi(d)}{16\sqrt{w_0}}\right) w'(0)^2\right] \\ &= \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8}\right) w'(0)^2. \end{aligned}$$

The terms linear in Δ cancel because $w'(0) = 2\sqrt{w_0}\Delta/\varphi(d)$.

Finally, $\sigma = \sqrt{w/\tau}$ gives

$$\sigma' = \frac{w'}{2\sqrt{\tau w}}, \qquad \sigma'' = \frac{w''}{2\sqrt{\tau w}} - \frac{(w')^2}{4\sqrt{\tau}\, w^{3/2}}.$$

Substitution yields (2.39)–(2.40).

2.A.3 Proof of universality and the closure dichotomy

Let g^* satisfy proposition 2.6, write $\lambda_+^* = e^{g^*(1)} < 1$, and fix $\bar\lambda \in (\lambda_+^*, 1)$.

Polynomial approximation. By Weierstrass there are polynomials p_N with $\varepsilon_N = \|p_N - g^*\|_\infty \to 0$. Every polynomial of degree at most N is represented by the endpoint chord and $P_2, \dots, P_N$. For all large N,

$$p_N(1) \le g^*(1) + \varepsilon_N < \log\bar\lambda < 0,$$

so the approximating slice is admissible.

Transport convergence and a common envelope. Let $\bar x_N$ be the corresponding unshifted transport. On every finite z-interval,

$$|\bar x_N(z) - \bar x^*(z)| \le (e^{\varepsilon_N} - 1)e^{\|g^*\|_\infty}|z|,$$

and therefore $\bar x_N \to \bar x^*$ locally uniformly. Continuity of g^* at the right endpoint gives z_0 such that, for all large N, $e^{p_N(\Lambda(z))} \le \bar\lambda$ when $z \ge z_0$. Hence a constant C_1, independent of N, satisfies

$$e^{\bar x_N(z)} \le C_1 e^{\bar\lambda z} \quad (z \ge 0), \qquad e^{\bar x_N(z)} \le C_1 \quad (z \le 0). \tag{2.61}$$

Normalizers, shifts, and quantiles. The functions $e^{\bar x_N}\rho$ converge pointwise and, by (2.61), are dominated by integrable multiples of $e^{(\bar\lambda-1)z}$ on the right and e^z on the left. Dominated convergence gives $I_N \to I^*$, so $m_N = -\log I_N \to m^*$. Thus $x_N \to x^*$ locally uniformly and $Q_N \to Q^*$ uniformly on compact subsets of $(0,1)$.

Uniform convergence of calls. Couple all laws with one uniform rank and let $Y_N = e^{Q_N(U)}$, $Y^* = e^{Q^*(U)}$. Since positive-part payoffs are one-Lipschitz,

$$\sup_k |c_N(k) - c^*(k)| \le \mathbb{E}|Y_N - Y^*|.$$

The integrand converges pointwise. The envelope above and boundedness of m_N give a constant C_2 such that $e^{Q_N(u)} \le C_2\{1 + (1-u)^{-\bar\lambda}\}$. This is integrable, so the calls converge uniformly.

Wasserstein convergence. The quantile coupling is optimal on the line:

$$W_p(Y_N, Y^*)^p = \int_0^1 |e^{Q_N(u)} - e^{Q^*(u)}|^p \, du.$$

For $1 \le p < 1/\lambda_+^*$ choose $\bar{\lambda} \in (\lambda_+^*, 1/p)$. The integrand is dominated by a multiple of $1+(1-u)^{-p\bar{\lambda}}$, proving the claim by dominated convergence.

Finite-order exclusions. At finite order, x'_N is positive and continuous. The quantile is a strictly increasing homeomorphism from $(0,1)$ onto $\mathbb{R}$; it can neither jump, remain flat, nor terminate. This excludes atoms, density gaps, bounded support, and mass at zero. A Gaussian-or-faster tail would require $g(u) \to -\infty$ at an endpoint, which no polynomial can do.

Uniform closure away from the wall. Suppose admissible g_N converge uniformly to g_∞. Necessarily $g_\infty(1) \le 0$. If $g_\infty(1) < 0$, the preceding dominated-convergence argument applies with g_∞ as target. The limit retains a positive continuous density and exponential tails.

The boundary branch. If $g_\infty(1) = 0$, weak convergence may fail. For $g_N \equiv \log(1 - 1/N)$, the solvable slice gives

$$I_N = \frac{\pi(1 - 1/N)}{\sin(\pi/N)} \longrightarrow \infty, \qquad m_N = -\log I_N \longrightarrow -\infty.$$

Every fixed-rank quantile tends to $-\infty$, so mass escapes.

Assume instead that the laws converge weakly. The shifts are bounded above because every normalizer has a positive central contribution. They are bounded below because $m_N \to -\infty$ along a subsequence would contradict tightness. Every convergent subsequence of shifts then yields

$$Q_\infty(u) = m_\infty + \int_0^{\operatorname{logit} u} e^{g_\infty(\Lambda(t))} \, dt.$$

Weak convergence identifies this quantile uniquely, so the full shift sequence converges. The limiting speed is continuous and bounded away from zero; hence the limit has full support and a strictly positive continuous density, and all geometric exclusions remain.

The right endpoint speed is one. For each $\delta > 0$ it is eventually at least $1-\delta$, so the argument of proposition 2.8 gives $\mathbb{E}[e^{rX}] = \infty$ for every $r > 1$. Fatou also gives $\mathbb{E}[e^X] \le \liminf_N \mathbb{E}[e^{X_N}] = 1$, with equality not guaranteed. This is the boundary law in proposition 2.6.

2.B Numerical implementation

This appendix records the load-bearing parts of a reference implementation. The pseudocode follows the notation of the chapter; tail-sensitivity terms and error handling are described after the listings.

2.B.1 Slice build

Listing 2.1: Build the transport and ledger on one fixed grid.

```
def build(theta, grid):
    L, R, a = theta.L, theta.R, theta.a
    g = (1 - grid.u)*L + grid.u*R + a @ grid.Pmat
    if not finite(g) or max(g) > 700:
        raise InteriorOverflow

    dx = exp(g)                         # x'(z)
```

```
    xbar = cumulative_simpson(dx, anchor=grid.mid)
    if max(xbar) > 700:
        raise InteriorOverflow

    lam_m, lam_p = exp(g[0]), exp(g[-1])
    if lam_p >= 1 - 1e-6:
        raise InadmissibleTail

    mass = exp(xbar) * grid.rho
    I = integrate(mass) \
        + exp(xbar[-1] - grid.zmax)/(1 - lam_p) \
        + exp(xbar[0]  - grid.zmax)/(1 + lam_m)
    m = -log(I)
    x = m + xbar

    dG = -exp(x) * grid.rho              # G'(z)
    G = reverse_integrate(-dG) \
        + exp(x[-1] - grid.zmax)/(1 - lam_p)
    return Slice(x=x, dx=dx, G=G, dG=dG,
                 m=m, lam=(lam_m, lam_p), theta=theta)
```

The cached density must be computed as `expit(z)*expit(-z)`. Forming `u*(1-u)` loses the right tail when `u` rounds to one. Both x and G carry their exact nodal derivatives, so a cubic Hermite representation needs no global spline solve.

The median is the natural anchor for the cumulative transport. It fixes $\bar{x}(0) = 0$ and lets a single forward cumulative sum build the positive half-line while the same stencil, reversed, builds the negative half-line. The additive freedom is then carried entirely by m. Simpson quadrature is fourth order on the uniform grid; the Hermite interpolant has the same local order because both values and exact nodal derivatives are available.

The grid does not truncate the probability law. From (2.16), on the right

$$e^{\bar{x}(z)}\rho(z) = e^{b_+ + (\lambda_+ - 1)z}\{1 + O(e^{-z})\}.$$

Integrating its leading term from $Z_{\max}$ gives the first correction in (2.53); the relative error is $O(e^{-Z_{\max}})$. The left correction follows from $e^{\bar{x}(z)}\rho(z) \sim e^{b_- + (\lambda_- + 1)z}$. The factor $1 - \lambda_+$ in the right denominator is also a numerical warning: tail mass becomes increasingly sensitive as the finite-forward wall is approached.

2.B.2 Call evaluation

Listing 2.2: Evaluate a normalized call.

```
def call_price(slc, k, grid):
    if k > slc.x[-1]:
        z = grid.zmax + (k - slc.x[-1]) / slc.lam[1]
        return exp(k - z) * slc.lam[1] / (1 - slc.lam[1])
    if k < slc.x[0]:
        return 1 - exp(k) + put_asymptote_left(slc, k, grid)

    z = hermite_root(slc.x, slc.dx, k)
    share = hermite_eval(slc.G, slc.dG, z)
    cash = exp(k - logaddexp(0, z))        # e^k(1-Lambda(z))
    return share - cash
```

The log-space cash expression is essential. The algebraically simpler `exp(k)*(1-expit(z))` becomes zero after rank saturation and can create a spurious upward step in the far call wing.

The two branches beyond the return grid are part of the same law. On the right, (2.54) continues the exact power tail. On the left, put asymptotics are computed first and call parity is applied. This avoids subtracting two quantities close to one when a deep in-the-money call is evaluated directly.

2.B.3 Sensitivity pass

Listing 2.3: All raw-coordinate Jacobian columns on one build.

```
def sensitivities(slc, grid):
    # rows: dg/dL, dg/dR, dg/da_2, ..., dg/da_N
    B = vstack([1-grid.u, grid.u, grid.Pmat])
    dxbar = cumulative_simpson(slc.dx * B,
                               anchor=grid.mid, axis=1)
    weight = exp(slc.x) * grid.rho
    dm = -integrate(weight * dxbar, axis=1)
    dx = dm[:, None] + dxbar

    integrand = weight[None, :] * dx
    dG = reverse_integrate(integrand, axis=1)
    dG_slope = -integrand
    dvs = -2 * integrate(dx * grid.rho, axis=1)
    return dG, dG_slope, dvs

def price_jacobian_row(slc, sens, k):
    z = hermite_root(slc.x, slc.dx, k)
    return hermite_eval(sens.dG, sens.dG_slope, z)
```

The derivatives of the two tail corrections in (2.53) must also be included. They use the endpoint sensitivities

$$\partial\lambda_-/\partial\theta = \lambda_-(1,0,1,1,\ldots), \qquad \partial\lambda_+/\partial\theta = \lambda_+(0,1,1,-1,\ldots).$$

Derivatives in the endpoint or logistic chart follow by multiplication with the corresponding chart Jacobian.

The analytic pass performs one cumulative integral for each basis column, one normalization dot product, and one reverse cumulative integral. Its work is $O(Pn_{\text{grid}})$ and its storage may be either $O(Pn_{\text{grid}})$, when all columns are retained, or $O(n_{\text{grid}})$ per streamed column. Finite differences have the same formal order but repeat normalization, ledger construction, and every strike root $P+1$ times. Envelope cancellation removes the roots from the analytic derivative and is therefore responsible for most of the measured constant-factor gain.

2.B.4 Interpolation certificate

Listing 2.4: Fritsch–Carlson margin for a uniform-grid Hermite interpolant.

```
def monotone_margin(y, d, dz, flat_tol=1e-9):
    sec = diff(y) / dz
    d0, d1 = d[:-1], d[1:]
    active = ~((abs(sec) < flat_tol) &
               (abs(d0) < flat_tol) &
               (abs(d1) < flat_tol))
    margin = minimum.reduce([d0, d1,
                             3*sec-d0, 3*sec-d1])
    return min(margin[active])
```

An increasing interpolant is certified when the returned margin is positive. Apply the same routine to $-G$. Segments excluded by `flat_tol` are recorded as flat to tolerance rather than strictly monotone.

For a cubic Hermite segment, let s be its secant slope and d_0, d_1 its endpoint derivatives. The sufficient Fritsch–Carlson box is

$$0 \le d_0 \le 3s, \qquad 0 \le d_1 \le 3s.$$

The routine returns the minimum distance to the four faces of this box. A positive number is therefore not merely a pass flag but a margin. For smooth functions sampled on a fine grid, $d_i/s = 1 + O(h^2)$, so the certificate should clear the boundary comfortably. A small margin is a useful indication of under-resolution even before monotonicity fails.

Monotonicity of x and G is not by itself a convexity certificate for the interpolated call. The separate belly test uses the exact factorization (2.5): analytic w, w', w'' are evaluated at 801 points across the retained strike span and g_{D} must exceed -10^{-4}. Since a flat smile has $g_{\mathrm{D}} = 1$, the tolerance permits a negative density component no larger than 10^{-4} of the local Black density factor. The threshold is therefore dimensionless and interpretable.

2.B.5 Floating-point failure catalogue

Four failures deserve explicit tests.

1. *Rank saturation.* For z near 37, double precision rounds $\operatorname{expit} z$ to one. Any formula that subsequently forms $1 - u$ loses the right-tail probability. Use $\operatorname{expit}(-z)$ directly and keep the cash leg in log space.
2. *Interior overflow.* Admissible endpoints do not bound a polynomial between them. Large Legendre coefficients can make g exceed the exponent range in the body while $g(0)$ and $g(1)$ remain moderate. The build must inspect the full grid and reject the trial before exponentiation.
3. *Near-wall cancellation.* Normalization and right-tail prices contain $1/(1-\lambda_+)$. The logistic chart slows parameter steps near the wall, while the hard guard leaves a finite numerical margin.
4. *Lee cancellation.* In the textbook expression for $\Psi(r)$, $\sqrt{r^2 + r} - r$ subtracts nearly equal quantities for large r. Use the rationalized form (2.55), with the limiting value zero handled explicitly as $r \to \infty$.

Cached, parameter-independent grid arrays should be read-only. An accidental in-place modification of ranks, logistic weights, or basis values otherwise corrupts every later slice without necessarily producing an exception.

2.B.6 Validation stack

The solvable slice tests the most code with the least ambiguity. Across $s \in [0.05, 0.95]$, compare the computed transport with $m + sz$, the shift with $-\log(\pi s/\sin \pi s)$, and the martingale integral with one. The observed worst errors are 1.1×10^{-11}, 3.0×10^{-9}, and 3.3×10^{-16}. The shift error grows toward the wall because the normalizing integrand decays at rate $1 - s$.

The randomized battery serves a different purpose. It samples several basis orders, ordinary slices, near-wall slices, and slices with large interior modes. Strikes are placed between grid nodes, and convexity is checked in normalized strike rather than log-strike. A four-times-finer independent build provides a reference price. Together with the interpolation and belly certificates, this separates algebraic validity, interpolation validity, and grid accuracy.

2.B.7 Reference defaults and timing

The exact timings depend on hardware and caches. The portable facts are the $O(Pn_{\mathrm{grid}})$ analytic complexity and the removal of repeated builds. The reported medians and interquartile ranges come from a single run that alternates analytic and finite-difference solves after warm-up. Alternation reduces drift from cache state and background load; medians avoid a best-of-n claim. A timing table should always travel with its grid size, quote count, order, solver tolerances, and machine context.

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(2.50)
Grid	$[-40, 40]$	Nodes	8001 (fit: 2001)
Wall guard	$\lambda_+ \leq 1 - 10^{-6}$	Exponent budget	700
Ridge (α, ν)	$(10^{-6}, 1)$, $n \geq 4$	Vega floor η	10^{-4}
Tail barrier	center 0.90, slope 50	Haircut h	0.005 vol
Solver	trust-region LSQ	Tolerances	10^{-10}
Chart	logistic	Calendar nodes	$\max(49, 4N + 1)$
Belly check	801 points	Cold start	$\sqrt{3w_0}/\pi$

Table 2.3: Reference implementation defaults.

Configuration	Order	Median	IQR	vs. finite difference
Analytic Jacobian	6	15.9 ms	0.5 ms	1.41×
Finite difference	6	22.5 ms	2.7 ms	—
Analytic Jacobian	12	26.0 ms	2.1 ms	1.68×
Finite difference	12	43.8 ms	3.2 ms	—
Analytic Jacobian	16	28.9 ms	0.5 ms	1.76×
Finite difference	16	50.9 ms	3.9 ms	—

Table 2.4: Single-slice calibration times from one interleaved run. The last column is the median finite-difference time divided by the median analytic time for the same configuration.

For a new implementation, the shortest reliable porting sequence is:

1. reproduce the constant-speed normalizer and transport;
2. implement the stable cash leg and tail corrections;
3. keep (x, x', G, G') as one immutable slice object;
4. add the envelope sensitivity pass;
5. run monotonicity, Durrleman, and randomized sub-grid checks on every release.

2.C Data and reproducibility

Every empirical number and figure is generated from one frozen JSON snapshot, captured after the 2026-08-03 US session from a delayed consolidated US listed-options feed. The file contains SPY and NVDA chains for eight expiries each, with spots 757.67 and 206.80, together with forwards, discount factors, prepared quotes, fitted parameters, and display curves. Maturities run from 2 calendar days to 1.3726 years.

The snapshot is self-contained. For each node it records the normalized chain, forward and discount provenance, retained quote set, fitted raw parameters, and diagnostic curves. Reproduction therefore requires neither credentials nor a market-data connection. The capture timestamp, reference date, and session are part of the manifest, so a later book cannot silently replace the one used for the chapter.

The manifest records the common fitting recipe: order 16 under (2.50), logistic chart, haircut $h = 0.005$, equal quote weights, ridge $\alpha = 10^{-6}$ with $\nu = 1$, and calendar rows active. All 16 nodes were fitted; none was tuned individually. The shortest NVDA node has the largest residual, consistent with its sparse and steep quote strip. SPY bands collapse to mids under the haircut rule; NVDA retains interval targets. These states are stored per quote rather than reconstructed from rounded display values.

The figure generator reads only this snapshot and fixed synthetic specifications. Real nodes are rebuilt from the frozen coefficients rather than refitted, and the rebuilt curves agree with the stored curves to 0.000 vol bp in the worst case. The script emits both the 14 figure PDFs and the LaTeX macro file supplying every empirical value in the chapter. Randomized audits use fixed seeds. Timing values follow a fixed interleaved protocol but vary with the machine.

The synthetic double-hump target is a fixed two-component lognormal mixture, martingale-normalized and sampled on a realistic strike ladder. The constant-speed benchmark requires no data and is evaluated directly from (2.18). Regeneration consists of running the figure generator and then compiling the present manuscript. Refetching the market feed would create a different dataset and is not part of reproduction.

One family-level diagnostic is retained with the data. The two shortest SPY expiries cross in implied-volatility space only far outside their common quote span, at call prices near numerical zero. Their direct normalized call prices remain ordered to 2.2×10^{-15} on the audited grid. Keeping both measurements makes the support-confinement example of section 2.8 reproducible without interpreting the picture by eye.

Chapter 3

Families in the Volatility Chart: SVI-JW and Superposition

Chapter 2 built a probability law first and read the smile from it. This chapter does what the industry does instead: it writes the smile down directly, as a formula for total implied variance. The gain is legibility. The SVI formula draws one expiry's smile with five parameters, and its jump-wing version speaks the desk's own vocabulary of level, skew, and wings. The price is that a formula for a curve is not automatically a probability law, so validity must now be checked rather than inherited. The chapter develops SVI slowly: the geometry of the curve, why its wing slopes are capped strictly inside Lee's bound of two, why passing the cheap checks does not yet make a slice a law, how the trader coordinates work and exactly where they fail, and how to calibrate so that the cheap conditions hold at every step. It then builds the Multi-Core Sigmoid, which adds local detail to the body of a smile without touching its wings, and closes by fitting both families and Chapter 2's model to the same frozen quotes, so that each family's strengths and blind spots show in the same three numbers.

3.1 Working in the volatility chart

Chapter 2 built a smile in two steps. First choose a probability law for the return; then read prices, and hence implied volatilities, off that law. The order of the steps did real work. Because every curve the model could produce came from a genuine law, validity never needed checking. The cost was indirection: the model's coordinates (L, R, a_n) — the endpoint speeds and polynomial coefficients of its rank-space construction — describe how probability is transported, and none of them is a number a trading desk quotes.

This chapter works in the opposite order, which is also the industry's order. Two definitions from Chapter 2 are used on every page that follows, so we restate them once. The *log-moneyness* of a strike K against the forward F is $k = \log(K/F)$: the money is $k = 0$, and negative k is the put side. The *total implied variance* at that strike is $w(k) = \sigma(k)^2\tau$, the squared implied volatility times the variance time τ. One expiry's smile is then a single function $w(k)$, and the direct route is to write that function down as an explicit formula with a few parameters. Working this way — on the picture whose axes are strike and variance, rather than behind it on the law — is what we call working *in the volatility chart.*

The direct route has a real cost, and it is exactly the one section 2.1 identified when it introduced the function g_{D}. Most functions $w(k)$ are not the smile of any probability law. A model that writes $w(k)$ freely therefore moves through a space that contains invalid curves, and validity — which Chapter 2 got for free — must now be policed. Part of the policing can be done by cheap inequalities that the optimizer respects while it runs. Part of it cannot, and needs a check on the finished curve. Learning which part is which is most of this chapter.

Why pay that cost? Because the direct route serves two audiences at once. An optimizer wants a formula that evaluates in one line and differentiates in closed form. A desk wants parameters it can talk about: where the at-the-money volatility sits, how steep the skew is, how heavy each wing is. The stochastic-volatility-inspired (SVI) family of Gatheral [23] answers the optimizer with a tilted hyperbola,

$$\boxed{w(k) = a_S + b_S\Big\{\rho_S(k - m_S) + \sqrt{(k - m_S)^2 + s_S^2}\Big\},} \tag{3.1}$$

five parameters under the structural conditions $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$. (The subscript S is there to keep the book's notation unambiguous: bare ρ is Chapter 2's logistic density and bare m its martingale shift, so the SVI parameters carry their own mark.) The desk is answered by a second coordinate system on the same curve — the jump-wing handles of section 3.5, five numbers chosen to match the desk's questions. The two coordinate systems describe the same family, and theorem 3.5 will say exactly when one converts to the other.

Before any geometry, we need to be precise about what "valid" means here, because different validity conditions will be supplied by different mechanisms. Five graded conditions recur through the chapter; table 3.1 collects them once, and the sections that follow fill the table in. The first three are inequalities on the parameters, cheap enough to build into the fit itself. The fourth is the full butterfly condition $g_{\mathrm{D}} \geq 0$ of (2.5) — recall that $g_{\mathrm{D}}(k)$ is, up to a positive factor, the probability density the smile implies at strike k, so a valid smile needs $g_{\mathrm{D}} \geq 0$ everywhere. No cheap inequality implies it: section 3.4 exhibits a slice that passes every screen and still fails it. The fifth condition couples different expiries and is not new to this chapter: it is the calendar coupling of section 2.8, which applies to these families unchanged.

The table names two symbols and two mechanisms that are still ahead of us, so read it as a map of the chapter rather than a summary of things known: β_L and β_R are the smile's two wing slopes, defined in section 3.2; a *hinge penalty* is a cheap fence built into the fit, explained in section 3.4; and the *structural chart* is a choice of optimizer coordinates, built in section 3.6. We will return to this table as each row is earned.

Condition	Mathematical statement	Supplied by
Structural	$b_S > 0$, $\lvert\rho_S\rvert < 1$, $s_S > 0$	the optimization chart (section 3.6)
Positive floor	$\min_k w(k) \geq 0$	a hinge penalty; automatic under the structural chart
Wing-admissible	$\max(\beta_L, \beta_R) \leq \beta_{\max} < 2$	a hinge penalty; automatic under the structural chart
Butterfly-free	$g_{\mathrm{D}}(k) \geq 0$ for all k	*not* implied by the rows above; checked on a dense grid (section 3.4)
Calendar-ordered	$w_{\tau_1}(k) \leq w_{\tau_2}(k)$, $\tau_1 < \tau_2$	the coupling of section 2.8, applied model-agnostically

Table 3.1: Five grades of validity for a slice written directly in the volatility chart. In Chapter 2 the first four held by construction. Here each must be attributed to the mechanism that supplies it — and the fourth row is the one no cheap mechanism reaches.

The chapter now proceeds in order. Section 3.2 works out the geometry of the hyperbola (3.1): its vertex, its wings, and a rigidity that will matter later. Section 3.3 explains why the wing slopes must be capped, and why the cap sits strictly inside Lee's bound. Section 3.4 shows what the caps do *not* protect — the middle of the smile — and states the chapter's division of labor between fences and checks. Section 3.5 develops the trader coordinates and proves exactly where they can be inverted. Section 3.6 turns to calibration: what the least-squares objective integrates against, and how to choose coordinates so the cheap conditions hold at every step. Section 3.7 then builds the Multi-Core Sigmoid, which adds capacity in the body of the smile without touching the wings. Section 3.8 closes by fitting LQD, SVI, and MCS to the same frozen quotes under one protocol.

3.2 One hyperbola, five parameters

We have a formula, (3.1), and a claim that its five parameters are legible. This section earns the claim: every geometric feature of the curve — where it bottoms, how it curves there, how steep each wing is — comes out in closed form.

Start by computing, because the formula is friendlier than it looks. Take

$$a_S = 0.01, \qquad b_S = 0.10, \qquad \rho_S = -0.4, \qquad m_S = 0, \qquad s_S = 0.2,$$

with a quarter year of variance time, $\tau = 0.25$. At the money, $k = 0$, the square root is $\sqrt{0^2 + 0.2^2} = 0.2$, so

$$w(0) = 0.01 + 0.10\,(-0.4 \cdot 0 + 0.2) = 0.030,$$

and the implied volatility is $\sqrt{0.030/0.25} = \sqrt{0.120} \approx 34.6\%$. Further out on the call side, at $k = 0.3$: the square root is $\sqrt{0.09 + 0.04} \approx 0.361$, so $w(0.3) = 0.01 + 0.10\,(-0.12 + 0.361) \approx 0.0341$, an implied volatility of about 36.9%. And at $k = -0.3$: $w(-0.3) = 0.01 + 0.10\,(0.12 + 0.361) \approx 0.0581$, about 48.2%. The left side came out much higher than the right. That is the equity put skew, and it entered through the sign of ρ_S: a negative tilt makes the put side steep and the call side shallow. Everything else in this section is this computation done once and for all, in general.

For the general statement, three shorthands keep the algebra readable:

$$x_S = k - m_S, \qquad r_S = \sqrt{x_S^2 + s_S^2}, \qquad q_S = \sqrt{1 - \rho_S^2} \tag{3.2}$$

(the subscripts keep Chapter 2's transport x and the moment order r of (2.23) untouched).

Proposition 3.1 (Geometry of one slice)**.** *Under $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$, the slice (3.1) is strictly convex with a single minimum:*

$$w'(k) = b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \qquad\qquad w''(k) = \frac{b_S\, s_S^2}{r_S^3} > 0, \tag{3.3}$$

$$k^* = m_S - \frac{s_S \rho_S}{q_S}, \qquad\qquad w^* = w(k^*) = a_S + b_S\, s_S\, q_S. \tag{3.4}$$

Its far field is two straight rays:

$$w(k) = \beta_L |k| + O(1)\ (k \to -\infty), \qquad w(k) = \beta_R\, k + O(1)\ (k \to +\infty), \tag{3.5}$$

with

$$\beta_L = b_S(1 - \rho_S), \qquad \beta_R = b_S(1 + \rho_S).$$

The curvature at the vertex is $\kappa^* = w''(k^*) = b_S\, q_S^3 / s_S$.

Proof. Differentiate (3.1) once and twice in k, using $r_S' = x_S/r_S$:

$$w' = b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \qquad w'' = b_S\,\frac{r_S - x_S \cdot (x_S/r_S)}{r_S^2} = b_S\,\frac{r_S^2 - x_S^2}{r_S^3} = \frac{b_S\, s_S^2}{r_S^3},$$

which is (3.3); it is positive because $b_S, s_S > 0$, so the curve is strictly convex and any stationary point is the unique global minimum. To find that point, set $w' = 0$, i.e. $x_S/r_S = -\rho_S$. Square both sides and use $r_S^2 = x_S^2 + s_S^2$:

$$x_S^2 = \rho_S^2\,(x_S^2 + s_S^2) \quad\Longrightarrow\quad x_S^2\,(1 - \rho_S^2) = \rho_S^2 s_S^2 \quad\Longrightarrow\quad |x_S| = \frac{s_S |\rho_S|}{q_S}.$$

Since $r_S > 0$, the equation $x_S/r_S = -\rho_S$ forces x_S to have the sign opposite to ρ_S, so $x_S = -s_S\rho_S/q_S$ and hence $k^* = m_S - s_S\rho_S/q_S$. At this point

$$r_S = \sqrt{\frac{s_S^2\rho_S^2}{q_S^2} + s_S^2} = s_S\,\frac{\sqrt{\rho_S^2 + q_S^2}}{q_S} = \frac{s_S}{q_S},$$

because $\rho_S^2 + q_S^2 = 1$. Substituting into (3.1),

$$w^* = a_S + b_S\Big(-\frac{s_S\rho_S^2}{q_S} + \frac{s_S}{q_S}\Big) = a_S + b_S\,s_S\,\frac{1-\rho_S^2}{q_S} = a_S + b_S\,s_S\,q_S,$$

which is (3.4), and plugging $r_S = s_S/q_S$ into (3.3) gives $\kappa^* = b_S s_S^2\,(q_S/s_S)^3 = b_S q_S^3/s_S$. Finally, in either wing $|x_S| \to \infty$ and $r_S = |x_S|\sqrt{1 + s_S^2/x_S^2} = |x_S| + O(1/|x_S|)$; substituting $r_S \approx |x_S|$ into (3.1) leaves $w \approx a_S + b_S(\rho_S x_S + |x_S|)$, whose slope in k is $b_S(\rho_S + 1) = \beta_R$ on the right and whose slope in $|k|$ is $b_S(1-\rho_S) = \beta_L$ on the left — which is (3.5). □

In words: the curve is a rounded hinge. Far from the vertex it is two straight rays, of slope β_L on the left and β_R on the right. The tilt ρ_S divides a fixed total steepness $\beta_L + \beta_R = 2b_S$ between the two sides — our worked slice had $\beta_L = 0.10 \times 1.4 = 0.14$ and $\beta_R = 0.06$, steep left, shallow right. Near the vertex, s_S decides how sharply one ray turns into the other, m_S slides the turn left or right, and a_S lifts the whole curve. (For the worked slice, (3.4) puts the vertex at $k^* = 0 - 0.2 \cdot (-0.4)/\sqrt{0.84} \approx 0.087$, with $w^* = 0.01 + 0.10 \cdot 0.2 \cdot 0.917 \approx 0.028$ — slightly right of the money, slightly below $w(0)$, as a left-steep curve requires.)

Figure 3.1 sweeps the three shape parameters one at a time around a common slice, and each panel isolates one of the readings above. In panel (a), only ρ_S varies: watch the left and right rays trade steepness while their total stays fixed — the vertex drifts, but the outer geometry only tilts. In panel (b), only s_S varies: the rays do not move at all, and the curves differ only in how tightly they round the corner between them. In panel (c), only b_S varies: both rays open or close together, like scissors. Every member, in every panel, is one convex curve with one minimum.

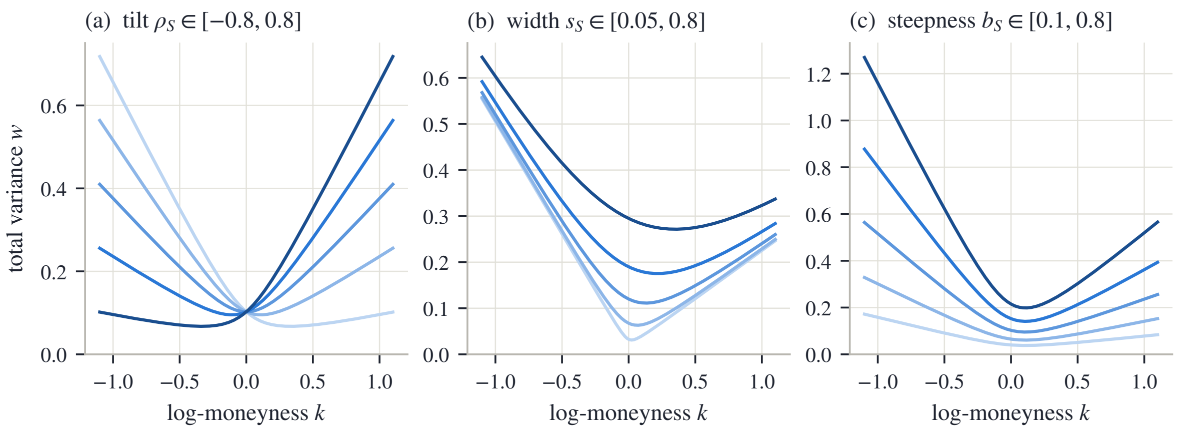

Figure 3.1: The reach of the family (3.1), one parameter at a time around a common slice. (a) The tilt ρ_S redistributes a fixed total steepness between the wings. (b) The width s_S rounds the vertex without moving the rays. (c) The steepness b_S opens or closes both wings together. Every member is one convex curve with one minimum.

We now have the whole geometry. Two consequences of it organize the rest of the chapter.

First, the tails collapse to two numbers. As $|k|$ grows, $w(k)/|k|$ tends to β_L on the put side and β_R on the call side, and no other feature of the slice survives to infinity. Whatever the market's tails are worth, this family records them in (β_L, β_R) alone. That is exactly the pair that Lee's moment bound constrains, and section 3.3 develops the correspondence.

Second, strict convexity is a rigidity. One hyperbola has one belly and one monotone turn of the slope, from $-\beta_L$ up to $+\beta_R$. This excludes a shape that markets do produce: the W, a central trough with a raised shoulder on each side.

Proposition 3.2 (A convex slice cannot carry a W)**.** *A convex function on an interval has no interior strict local maximum. Hence a total-variance curve with a central trough flanked by two shoulders — an interior local maximum on each side — is not representable by any globally convex slice, in particular by* (3.1)*.*

Proof. For a convex function the derivative (or, where it fails to exist, the one-sided derivatives) is non-decreasing. At a strict interior local maximum the derivative would have to pass from positive, on the way up, to negative, on the way down — a decrease, contradicting monotonicity. By proposition 3.1, $w'' > 0$ everywhere for the family (3.1), so the family is convex and the exclusion applies to it. ☐

Whether this rigidity helps or hurts depends on the smile in front of you. On a liquid index chain it helps: it forbids the wiggles that quote noise would otherwise carve into the fitted curve. Ahead of a scheduled event, or in a single stock (a *name*, in desk speech) whose holders are split between two scenarios, it is a ceiling — the market's smile genuinely turns more than once, and the family cannot follow it. Section 3.7 removes the ceiling without giving up the wings.

3.3 Why the wing slopes are capped

We know from section 3.2 that the family's tails collapse to the two slopes (β_L, β_R). Table 3.1 promised a cap on those slopes. This section explains where the cap comes from, why it is set strictly *inside* the theoretical bound rather than at it, and what a fitted slope does and does not say about the return distribution.

The starting point is Lee's moment formula, met in Chapter 2 as (2.23). In plain terms: the steeper a wing of total variance, the fatter the corresponding tail of the return law, and "fatter" is measured by which moments still exist. Recall the actors. Y is the gross forward return of Chapter 2 — the underlying's final level divided by its forward price, the random variable behind every smile in this book. Its moments $\mathbb{E}[Y^r]$ make sense for *any* real exponent r, not just integers, and for a fat-tailed law they stop being finite beyond some critical exponent. Write $1 + r_+^*$ for that critical exponent: $\mathbb{E}[Y^{1+r}]$ is finite for $r < r_+^*$ and infinite for $r > r_+^*$. (The subscript $+$ marks the right tail; a mirror image r_-^* governs the left.) Lee's formula ties the right wing to this critical exponent: $\beta_R = \Psi(r_+^*)$, where

$$\Psi(r) = 2 - 4\big(\sqrt{r^2 + r} - r\big).$$

The function Ψ decreases: more finite moments mean a shallower wing. As the law loses its moments, $r_+^* \downarrow 0$ and $\Psi \uparrow 2$. So the number 2 is a wall: a wing of total variance steeper than 2 is the tail of no probability law at all. For LQD the correspondence between tails and moments was a closed formula in the endpoint speeds (proposition 2.9). For a family whose wings are exactly straight lines it is even simpler — a checkable inequality on two parameters, $\max(\beta_L, \beta_R) \le 2$.

That inequality looks like the whole story. It is not, and the cleanest way to see why is to follow the bound into the function that actually tests for arbitrage. Recall the Durrleman function of (2.5) — up to a positive factor it is the probability density the smile implies, so a valid smile needs $g_{\mathrm{D}} \ge 0$:

$$g_{\mathrm{D}}(k) = \Big(1 - \frac{k\,w'}{2w}\Big)^2 - \frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2}.$$

It mixes w, w', and w''. Deep in the wings of the family (3.1), however, all three are controlled by the slope alone, and the mixture settles to a limit.

Proposition 3.3 (The wing slope sets the sign of the tail density)**.** *For a slice* (3.1) *with* $w > 0$,

$$\lim_{k\to-\infty} g_{\mathrm{D}}(k) = \frac{4-\beta_L^2}{16}, \qquad \lim_{k\to+\infty} g_{\mathrm{D}}(k) = \frac{4-\beta_R^2}{16}. \tag{3.6}$$

Hence for $\beta < 2$ *the corresponding tail of* g_{D} *is eventually positive and for* $\beta > 2$ *eventually negative. At* $\beta_R = 2$ *the limit vanishes and the sign falls to the next order: the right wing is* $w(k) = 2k + (a_S - 2m_S) + O(k^{-1})$, *and*

$$g_{\mathrm{D}}(k) = \frac{(a_S - 2m_S) - 2}{4k} + O(k^{-2}), \tag{3.7}$$

so the boundary slice carries negative tail density whenever $a_S - 2m_S < 2$*. The mirror statement holds on the left with the intercept* $a_S + 2m_S$*.*

Proof. Work on the right wing; the left is symmetric. There, by proposition 3.1, $w = \beta k + O(1)$ with $\beta = \beta_R$, $w' \to \beta$, and $w'' \to 0$. Take the three terms of g_{D} in turn:

$$\frac{k\,w'}{2w} \longrightarrow \frac{k\,\beta}{2\,\beta k} = \frac{1}{2}, \qquad \text{so} \qquad \Big(1 - \frac{k\,w'}{2w}\Big)^2 \longrightarrow \frac{1}{4};$$

$$\frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) \longrightarrow \frac{\beta^2}{4}\Big(0 + \frac{1}{4}\Big) = \frac{\beta^2}{16},$$

because $w \to \infty$ kills the $1/w$; and $w''/2 \to 0$. Adding the three limits gives $\frac{1}{4} - \beta^2/16 = (4-\beta^2)/16$, which is (3.6).

For the boundary case set $b_S(1+\rho_S) = 2$ and expand one order deeper. With $x_S = k - m_S$,

$$r_S = x_S\sqrt{1 + s_S^2/x_S^2} = x_S + \frac{s_S^2}{2x_S} + O(x_S^{-3}),$$

so

$$w = a_S + b_S(\rho_S x_S + r_S) = a_S + 2\,(k - m_S) + \frac{b_S s_S^2}{2k} + O(k^{-2}) = 2k + (a_S - 2m_S) + O(k^{-1}),$$

and differentiating, $w' = 2 + O(k^{-2})$. The intercept $a_S - 2m_S$ recurs below; we carry it written out. First term: dividing numerator and denominator by $4k$,

$$\frac{k\,w'}{2w} = \frac{2k + O(k^{-1})}{4k + 2(a_S - 2m_S) + O(k^{-1})} = \frac{1}{2}\Big(1 - \frac{a_S - 2m_S}{2k}\Big) + O(k^{-2}) = \frac{1}{2} - \frac{a_S - 2m_S}{4k} + O(k^{-2}),$$

so

$$\Big(1 - \frac{k\,w'}{2w}\Big)^2 = \Big(\frac{1}{2} + \frac{a_S - 2m_S}{4k}\Big)^2 + O(k^{-2}) = \frac{1}{4} + \frac{a_S - 2m_S}{4k} + O(k^{-2}).$$

Second term: $(w')^2/4 = 1 + O(k^{-2})$ and $1/w = 1/(2k) + O(k^{-2})$, so

$$\frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) = \frac{1}{4} + \frac{1}{2k} + O(k^{-2}).$$

Third term: $w'' = O(k^{-3})$. Subtracting,

$$g_{\mathrm{D}}(k) = \frac{a_S - 2m_S}{4k} - \frac{1}{2k} + O(k^{-2}) = \frac{(a_S - 2m_S) - 2}{4k} + O(k^{-2}),$$

which is (3.7). □

The proposition says two things, and the second is the one that is easy to miss. Below the bound, the tail of the implied density is eventually positive: the wing condition genuinely protects the far tail. *At* the bound, it protects nothing. The limit is zero, and the sign is decided by the $1/k$ term — that is, by the intercept, which no condition of the form $\beta \le 2$ even looks at.

A concrete slice makes the boundary failure explicit, and it is worth computing along. Take

$$(a_S, b_S, \rho_S, m_S, s_S) = (0.04,\ 2,\ 0,\ 0,\ 0.2).$$

With $\rho_S = 0$ and $m_S = 0$ the curve is symmetric, and its minimum, at $k = 0$, is $w^* = 0.04 + 2 \cdot 0.2 = 0.44$ — comfortably positive, so the floor condition holds. Both wings have slope $b_S(1 \pm 0) = 2$, so a constraint written as $\beta \le 2$ is satisfied (with equality) and raises no objection. Yet the intercept is $a_S - 2m_S = 0.04$, far below 2, so (3.7) makes the tail density negative at every large strike — at $k = 10$ the predicted leading term is $(0.04 - 2)/40 \approx -0.049$, and the exact value on this slice is $g_{\mathrm{D}}(10) = -0.0485$. A constraint that uses Lee's constant as its right-hand side therefore admits curves that are not the smile of any law.

The reference implementation draws the consequence as a convention, stated once: the enforced cap sits strictly inside the bound,

$$\max(\beta_L, \beta_R)\ \le\ \beta_{\max}, \qquad \beta_{\max} = 1.95 < 2. \tag{3.8}$$

Is the buffer expensive? To answer, invert Lee's formula and ask which laws a slope just under 2 still allows. Set $\beta = \Psi(r)$ and solve for r. Writing $\sqrt{r^2 + r} - r = (2 - \beta)/4$, move r across and square:

$$r^2 + r = \Big(r + \frac{2-\beta}{4}\Big)^2 = r^2 + 2r\,\frac{2-\beta}{4} + \Big(\frac{2-\beta}{4}\Big)^2 \quad \Longrightarrow \quad r\Big(1 - \frac{2-\beta}{2}\Big) = \frac{(2-\beta)^2}{16},$$

and since $1 - (2-\beta)/2 = \beta/2$, this gives the *moment budget* at slope β:

$$r^* = \Psi^{-1}(\beta) = \frac{(2-\beta)^2}{8\beta}, \tag{3.9}$$

the number of moments beyond the first that a wing of slope β still permits. At the cap, $r^* = (0.05)^2/(8 \times 1.95) = 1.6 \times 10^{-4}$: the only laws the buffer excludes are those whose entire moment budget beyond the mean is smaller than about two parts in ten thousand. The cap costs essentially nothing in distributional scope, and it restores the implication the boundary slice broke: wing-admissible now implies that the tail density is eventually positive.

The map (3.9) also tells us when a fitted wing may be read as a moment estimate, because its sensitivity is extreme at both ends of the range. Expand the quotient:

$$r^* = \frac{4 - 4\beta + \beta^2}{8\beta} = \frac{1}{2\beta} - \frac{1}{2} + \frac{\beta}{8}, \qquad \text{so} \qquad \frac{\mathrm{d}r^*}{\mathrm{d}\beta} = -\frac{1}{2\beta^2} + \frac{1}{8} = -\frac{4 - \beta^2}{8\beta^2}.$$

For small β the derivative behaves as $-1/(2\beta^2)$: a shallow wing pins essentially no bound on the moments, and a small change in the fitted slope moves the inferred moment violently. As $\beta \to 2$ the budget collapses to zero. A fitted β is a usable moment reading only in the middle of the range — and only for a slice that actually is a law, which is the subject of the next section.

Figure 3.2 draws the three parts of the story. In panel (a), the tail limit (3.6) is plotted against the slope β: a downward parabola that crosses zero exactly at $\beta = 2$ (dotted vertical line), with the working cap $\beta_{\max} = 1.95$ marked dashed just inside it. Left of the crossing the far tail is positive; right of it, negative; at it, undecided. Panel (b) shows the boundary slice of the worked example: its Durrleman function (orange) is negative at every plotted strike, creeping up toward zero from below exactly as the $1/k$ law (3.7) prescribes — while the same slice with its wings brought back to the cap (blue) is positive and settles at the limit $(4 - \beta_{\max}^2)/16$ (dotted

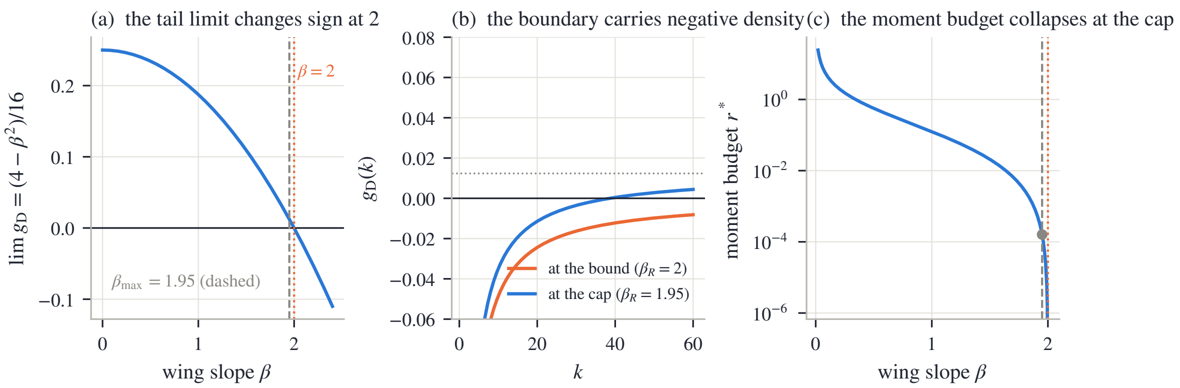

Figure 3.2: Lee's bound inside the density factor. (a) The tail limit (3.6) against the wing slope: a downward parabola crossing zero exactly at $\beta = 2$; the dashed line is the working cap $\beta_{\max} = 1.95$. (b) The boundary slice of the text: at $\beta_R = 2$ (orange) the tail density is negative at every plotted strike, approaching zero from below as (3.7) prescribes; the same slice with its wings at the cap (blue) turns positive and settles at the limit $(4 - \beta^2_{\max})/16$ (dotted). (c) The moment budget (3.9) on a log scale: it explodes as $\beta \to 0$ and collapses at the cap (marked point, $r^* = 1.6 \times 10^{-4}$).

horizontal). Panel (c) plots the moment budget (3.9) on a log scale: it explodes as $\beta \to 0$ — shallow wings say nothing about moments — and collapses at the cap, where the marked point sits at $r^* = 1.6 \times 10^{-4}$.

We now know that the floor guards the vertex and the cap guards the tails. The next question is what guards everything in between — and the answer, for now, is nothing.

3.4 What the inequalities do not check: the belly

The floor condition of table 3.1 guards the vertex, and the cap (3.8) guards the wings. This section shows that the region between them — the belly of the smile — is guarded by neither, on a concrete slice that passes every cheap test and still fails to be a law.

The root of the problem is that the word "convex" hides two different second derivatives. The family (3.1) has $w'' > 0$ by construction: the *variance curve* is convex in log-moneyness. What no-arbitrage requires is different: the *option price* must be convex in the strike. A butterfly spread — buy one option at $K - \delta$, sell two at K, buy one at $K + \delta$ — has a nonnegative payoff, so its price must be nonnegative, and that price is (up to scale) the second difference of the call price in strike. Price convexity is what the sign of g_D tests, and g_D mixes w, w', and w''. In the wings the two notions of convexity agree — proposition 3.3 showed that the sign of g_D out there is decided by the wing slope. In the interior they need not agree.

That the gap is real, and not a defect of some particular fitting code, is shown by a classical example due to Vogt, reproduced as Example 3.1 by Gatheral and Jacquier [25]:

$$(a_S, b_S, \rho_S, m_S, s_S) = (-0.0410,\ 0.1331,\ 0.3060,\ 0.3586,\ 0.4153).$$

Score this slice against table 3.1, row by row. The structural conditions hold: $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$. The floor holds: the minimum total variance, from (3.4), is $w^* = 0.0116$, safely positive. The wings hold with an enormous margin: the larger slope is $\max(\beta_L, \beta_R) = 0.174$, an order of magnitude below the cap, and by (3.6) both tails of g_D settle near 0.25 — clean, positive tail density on both sides. Everywhere the cheap inequalities look, this slice is impeccable.

Yet it is not a law. In the interior, g_D dips to -0.033 near $k = 0.88$ — squarely between the two clean tails. The implied density is negative there, which means the slice prices some butterfly below zero. Figure 3.3 shows both halves of the verdict. In panel (a), the total-variance curve itself: smooth, convex, floor marked, nothing visibly wrong — this is the point; the defect is invisible in the chart everyone looks at. In panel (b), the same slice's Durrleman function: it approaches the positive limit (3.6) on both sides, exactly as the wing analysis promises, and turns negative in between. The wing conditions fence the tails; the belly escapes them.

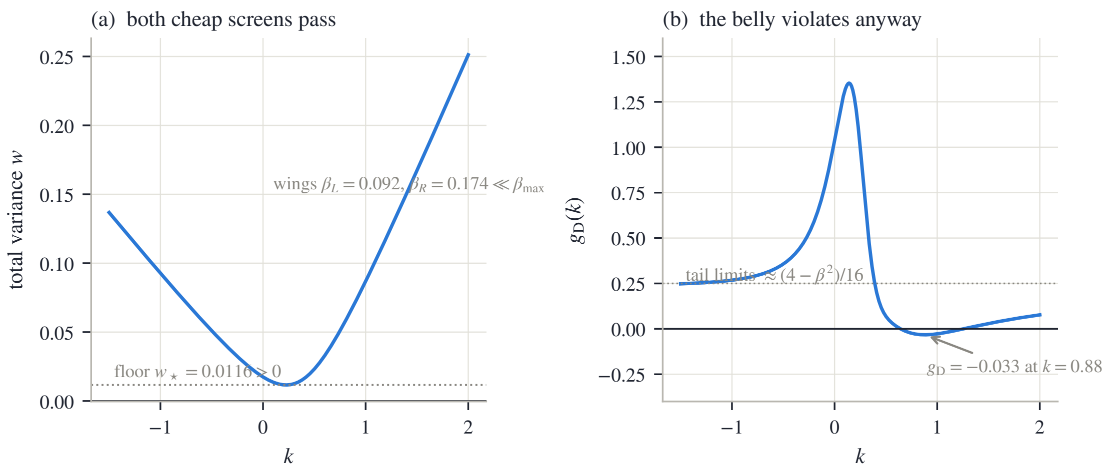

Figure 3.3: The Vogt slice. (a) Total variance: the floor is positive and both wing slopes are far below the cap — every inequality of table 3.1 up to wing-admissibility holds strictly. (b) Its Durrleman function approaches the positive tail limits (3.6) on both sides yet turns negative in the interior. The wing conditions fence the tails; the belly escapes them.

Could a better inequality close the gap? In principle, yes: the exact butterfly-free domain of the five SVI parameters has been characterized completely by Martini and Mingone [36]. But the characterization is not one short inequality — it involves a parameter rescaling, root finding, and a numerical minimization — and it is specific to this five-parameter family, with no analogue for the superposition family of section 3.7. A family-by-family exact domain is not a strategy a multi-family fitter can build on.

The reference implementation adopts a different division of labor for every family in the volatility chart, and it is the chapter's second convention. *During* the fit, the floor and the cap are enforced inside the least-squares residual — the stacked vector of per-quote misfits that the optimizer drives toward zero — by appending two cheap penalty rows to it:

$$W\ \max\big(-(a_S + b_S s_S q_S),\, 0\big), \qquad W\ \max\big(b_S(1+|\rho_S|) - \beta_{\max},\, 0\big), \tag{3.10}$$

with W a multiplier large enough to dominate any violation. Read the two rows: the first is active only when the floor $w^* = a_S + b_S s_S q_S$ of (3.4) goes negative, the second only when the larger wing $b_S(1+|\rho_S|)$ exceeds the cap. A row of this shape is called a *hinge*: it is exactly zero whenever its inequality holds, so a clean fit is identical with or without it, and it only pushes back when the optimizer strays. *After* the fit, the condition the hinges cannot see is checked where it lives: g_D is evaluated on a dense grid of strikes across the traded range — always from the model's own derivatives (3.3), never by differencing prices, which would inject curvature noise at exactly the scale being measured — and the slice is accepted only if the sampled minimum clears a small tolerance. This is the same posture as the numerical certificates of section 2.7.2: an analytic guarantee where one exists, and where it does not, a computed check on the finished object, with its scope (the sampled grid on the traded range, not the whole line) stated rather than implied. The minimum of g_D over that grid is called the fit's *validity margin*; the comparison of section 3.8 reports it for every fit of every family.

One could try to push the butterfly condition into the optimizer too, as a hinge on sampled values of g_D over a grid — and the superposition family of section 3.7, whose localized bump corrections (its *kernels*, built there) can bend the density in ways a hyperbola cannot, carries exactly such a row (appendix 3.A). It helps, but it settles nothing: a finite-weight hinge is traded against the data term like any other residual row, so it discourages violations without excluding them, and section 3.8 shows fits that fail the final check with that hinge active. In the other direction, trusting the cheap fences alone would publish the Vogt slice. So the division stands: fence cheaply during the fit to keep the optimizer in the region where the check usually passes — and let the check decide.

3.5 Trader coordinates: the jump-wing handles

The raw quintuple $(a_S, b_S, \rho_S, m_S, s_S)$ is an excellent computational coordinate system and a poor market language. No single coordinate is "the level" or "the skew," and each of b_S, ρ_S, s_S moves a wing and the interior at once. The jump-wing (JW) coordinates of Gatheral and Jacquier [25] re-read the same curve through five quantities chosen to match the desk's questions. (The name is inherited, not descriptive: the coordinates arose from reading wing behavior in stochastic-volatility models with jumps; nothing in this section requires jumps.) This section defines them, shows how to read them without mixing up their units, and then answers the harder question: given five quoted handles, can the hyperbola be reconstructed — and when it cannot, what information was lost?

A useful way to remember the five handles: three of them describe the *interior* of the smile and two describe its *tails*.

Definition 3.4 (Jump-wing handles)**.** At variance time τ, with $w_0 = w(0)$ the ATM total variance, the jump-wing coordinates of a slice are

$$\underbrace{v_J = \frac{w_0}{\tau}, \qquad \psi_J = \left(\sqrt{w}\right)'(0), \qquad \tilde{v}_J = \frac{\min_k w(k)}{\tau},}_{\text{interior}} \qquad \underbrace{p_J = \frac{\beta_L}{\sqrt{w_0}}, \qquad c_J = \frac{\beta_R}{\sqrt{w_0}}}_{\text{tails}}. \tag{3.11}$$

In words: v_J is the ATM variance per year, ψ_J is the ATM slope (of total volatility $\sqrt{w}$, not of w — this matters below), $\tilde{v}_J$ is the smile's minimum variance per year, and p_J, c_J are the put and call wing slopes, each divided by $\sqrt{w_0}$.

For the family (3.1) all five have closed forms. Four are immediate from proposition 3.1; the ATM slope takes two lines. Since $(\sqrt{w})' = w'/(2\sqrt{w})$, and at $k = 0$ the shorthands (3.2) read $x_S = -m_S$ and $r_S = \sqrt{m_S^2 + s_S^2}$, formula (3.3) gives $w'(0) = b_S(\rho_S - m_S/\sqrt{m_S^2 + s_S^2})$, hence

$$v_J = \frac{w_0}{\tau}, \quad \psi_J = \frac{b_S}{2\sqrt{w_0}}\Big(\rho_S - \frac{m_S}{\sqrt{m_S^2 + s_S^2}}\Big), \quad p_J = \frac{b_S(1-\rho_S)}{\sqrt{w_0}}, \quad c_J = \frac{b_S(1+\rho_S)}{\sqrt{w_0}}, \quad \tilde{v}_J = \frac{w^*}{\tau}, \tag{3.12}$$

with w^* from (3.4).

Reading the handles in their own units

Each handle lives in its own units, and misreading them is the most common JW error, so we spell the dictionary out. Write $I(k) = \sqrt{w(k)/\tau}$ for the implied-volatility curve. Then $I(0) = \sqrt{w_0/\tau} = \sqrt{v_J}$; differentiating, $I'(0) = w'(0)/(2\sqrt{\tau w_0}) = (\sqrt{w})'(0)/\sqrt{\tau} = \psi_J/\sqrt{\tau}$; and inverting the definitions of the wing handles, $\beta_L = p_J\sqrt{w_0} = p_J\sqrt{v_J\tau}$ and likewise for β_R:

$$I(0) = \sqrt{v_J}, \qquad I'(0) = \frac{\psi_J}{\sqrt{\tau}}, \qquad \min_k I(k) = \sqrt{\tilde{v}_J}, \qquad \beta_L = p_J\sqrt{v_J\tau}, \qquad \beta_R = c_J\sqrt{v_J\tau}. \tag{3.13}$$

So: v_J and $\tilde{v}_J$ are variances, and what an implied-volatility chart shows is their square roots. ψ_J is the ATM slope of total *volatility*, so the tangent drawn on the IV chart has slope $\psi_J/\sqrt{\tau}$. And p_J, c_J are *normalized* total-variance wing slopes — they are not slopes of the IV curve at all.

Figure 3.4 annotates a fitted slice in exactly these units. The data is the SPY December 2026 expiry of the frozen snapshot — the fixed market capture, described in appendix 2.C, from which every empirical example in this book is drawn — and this particular (asset, expiry) pair, a *node* in the book's vocabulary, will recur through the chapter. The fit misses the quotes by 7.5 vol bp on average (a *vol bp* is one hundredth of a volatility percentage point, the book's unit of fit error), and carries $\sqrt{v_J} = 14.89\%$, tangent slope -0.430, floor $\sqrt{\tilde{v}_J} = 12.41\%$, and wing handles $(p_J, c_J) = (1.384, 0.724)$. Panel (a) is the implied-volatility chart, where the three *interior* handles are read: the orange dots are the mid quotes, the blue curve is the SVI fit, the two dotted horizontals mark the level $\sqrt{v_J}$ and the floor $\sqrt{\tilde{v}_J}$, and the dashed line through the money is the ATM tangent, of slope $\psi_J/\sqrt{\tau}$ — negative, as an equity skew requires. Panel (b) switches to the total-variance chart, the only place the two *tail* handles can be read honestly: the two dashed rays are the asymptotes, of slopes $\beta_L = p_J\sqrt{v_J\tau}$ and $\beta_R = c_J\sqrt{v_J\tau}$. Plotting all five handles on one IV axis would give the wing handles units they do not have.

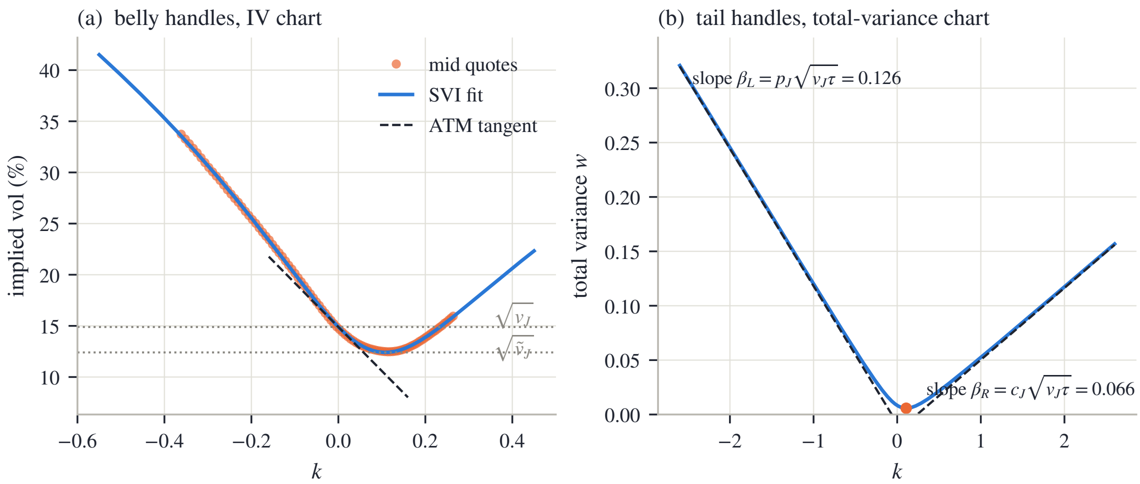

Figure 3.4: The five handles in their honest units, on the SVI fit of the SPY December 2026 quotes (rms 7.5 vol bp). (a) The three interior handles are read on the IV chart: the level $\sqrt{v_J}$, the minimum $\sqrt{\tilde{v}_J}$, and the ATM tangent of slope $\psi_J/\sqrt{\tau}$. (b) The two tail handles are read on the total-variance chart, as the slopes $\beta_L = p_J\sqrt{v_J\tau}$ and $\beta_R = c_J\sqrt{v_J\tau}$ of the asymptotes.

One caution before inverting. The tail handles are *normalized* indicators: by (3.13), the actual asymptotic slopes — the objects Lee's bound and the moment budget (3.9) read — carry a factor $\sqrt{v_J\tau}$ of the supposedly interior level. Raise v_J with (p_J, c_J) held fixed and both actual wings steepen, moving the inferred moments. JW is a quoting convention adapted to the desk, not a decomposition of the slice into independent interior and tail degrees of freedom.

3.5.1 Inverting the handles

Reading (3.12) off a fitted slice is mere evaluation. The interesting direction is the reverse: a desk quotes five handles — can we reconstruct the hyperbola? We build the inverse in three steps and watch where it can break.

Step 1: the tail handles fix the scale and the tilt. From (3.12), $p_J\sqrt{w_0} = b_S(1-\rho_S)$ and $c_J\sqrt{w_0} = b_S(1+\rho_S)$. Adding and subtracting,

$$b_S = \frac{\sqrt{w_0}}{2}(p_J + c_J), \qquad \rho_S = \frac{c_J - p_J}{c_J + p_J}, \qquad w_0 = v_J\tau. \tag{3.14}$$

Step 2: the ATM slope fixes where the vertex sits. The combination of m_S and s_S that appears in ψ_J is the *normalized vertex displacement* $\chi := m_S/\sqrt{m_S^2 + s_S^2} \in (-1, 1)$. Solve (3.12) for it: $\psi_J = \frac{b_S}{2\sqrt{w_0}}(\rho_S - \chi)$ and $b_S/\sqrt{w_0} = (p_J + c_J)/2$ give

$$\chi = \rho_S - \frac{4\psi_J}{p_J + c_J}. \tag{3.15}$$

Step 3: the gap between level and floor sets the width. Evaluate (3.1) at the money and at the vertex:

$$w_0 = a_S + b_S\Big(\sqrt{m_S^2 + s_S^2} - \rho_S m_S\Big), \qquad w^* = a_S + b_S\, s_S\sqrt{1-\rho_S^2}. \tag{3.16}$$

By the definition of χ, $m_S = \chi s_S/\sqrt{1-\chi^2}$ and hence $\sqrt{m_S^2 + s_S^2} = s_S/\sqrt{1-\chi^2}$. Subtract the second identity from the first: the unknown a_S cancels, and

$$w_0 - w^* = b_S\, s_S\Big(\frac{1-\rho_S\chi}{\sqrt{1-\chi^2}} - \sqrt{1-\rho_S^2}\Big) = b_S\, s_S\, \mathcal{D},$$

where we name the bracket

$$\mathcal{D}(\rho_S, \chi) = \frac{1-\rho_S\chi}{\sqrt{1-\chi^2}} - \sqrt{1-\rho_S^2}. \tag{3.17}$$

Since $w_0 - w^* = (v_J - \tilde{v}_J)\tau$, the width and the remaining two parameters follow:

$$s_S = \frac{(v_J - \tilde{v}_J)\,\tau}{b_S\,\mathcal{D}}, \qquad m_S = \frac{\chi\, s_S}{\sqrt{1-\chi^2}}, \qquad a_S = \tilde{v}_J\tau - b_S\, s_S\sqrt{1-\rho_S^2}. \tag{3.18}$$

The whole inverse therefore stands or falls with the sign of the denominator $\mathcal{D}$, and that sign is a disguised Cauchy–Schwarz inequality. Consider the two unit vectors $(\rho_S, \sqrt{1-\rho_S^2})$ and $(\chi, \sqrt{1-\chi^2})$ — two points on the upper half of the unit circle. Their dot product is at most 1:

$$\rho_S\chi + \sqrt{1-\rho_S^2}\sqrt{1-\chi^2} \;\le\; 1, \quad \text{i.e.} \quad \mathcal{D} \;\ge\; 0, \tag{3.19}$$

(multiply the definition (3.17) through by the positive $\sqrt{1-\chi^2}$ to see the two statements are the same), with equality exactly when the two vectors coincide — that is, when $\chi = \rho_S$, which by (3.15) means $\psi_J = 0$. This single inequality gives both the domain of the coordinate system and its one singularity.

Theorem 3.5 (The image of the jump-wing map)**.** *Fix $\tau > 0$ and restrict (3.1) to $b_S > 0$, $|\rho_S| < 1$, $s_S > 0$, $w_0 > 0$.*

(i) A JW point has a unique inverse, given by (3.14)–(3.18), exactly on

$$v_J > 0, \qquad p_J > 0,\; c_J > 0, \qquad -\frac{p_J}{2} < \psi_J < \frac{c_J}{2}, \qquad \psi_J \neq 0, \qquad \tilde{v}_J < v_J. \tag{3.20}$$

(ii) The image also contains the stratum *$\psi_J = 0$, $\tilde{v}_J = v_J$ (with $v_J, p_J, c_J > 0$) — a lower-dimensional sheet of handle points — every point of which is attained by infinitely many slices. There are no other image points.*

Proof. If $p_J, c_J > 0$ then (3.14) gives $b_S > 0$ and $\rho_S \in (-1, 1)$; conversely every admissible slice has $p_J, c_J > 0$ because $b_S > 0$ and $|\rho_S| < 1$ make both $\beta_L, \beta_R > 0$.

Next, the band on ψ_J is exactly the condition $|\chi| < 1$, which keeps m_S finite in (3.18). To see the equivalence, note from (3.14) that

$$\rho_S - 1 = \frac{(c_J - p_J) - (c_J + p_J)}{c_J + p_J} = \frac{-2p_J}{p_J + c_J}, \qquad \rho_S + 1 = \frac{2c_J}{p_J + c_J},$$

so, by (3.15),

$$|\chi| < 1 \iff \rho_S - 1 < \frac{4\psi_J}{p_J + c_J} < \rho_S + 1$$
$$\iff -\frac{2p_J}{p_J + c_J} < \frac{4\psi_J}{p_J + c_J} < \frac{2c_J}{p_J + c_J} \iff -\frac{p_J}{2} < \psi_J < \frac{c_J}{2}.$$

By (3.19), $\mathcal{D} > 0$ exactly when $\chi \neq \rho_S$, i.e. $\psi_J \neq 0$; and then $s_S > 0$ in (3.18) requires $v_J > \tilde{v}_J$. So on the domain (3.20) the three steps produce a unique admissible slice, and each failure of (3.20) breaks a requirement: $p_J \le 0$ or $c_J \le 0$ forces $b_S \le 0$ or $|\rho_S| \ge 1$; ψ_J outside the band sends $|\chi| \ge 1$; $\psi_J = 0$ annihilates $\mathcal{D}$; $\tilde{v}_J \ge v_J$ makes $s_S \le 0$.

For the stratum: $\psi_J = 0$ gives $\chi = \rho_S$, i.e. $m_S/\sqrt{m_S^2 + s_S^2} = \rho_S$, which solves to $m_S = s_S\rho_S/\sqrt{1-\rho_S^2}$; substituting into (3.4),

$$k^* = m_S - \frac{s_S\rho_S}{q_S} = \frac{s_S\rho_S}{\sqrt{1-\rho_S^2}} - \frac{s_S\rho_S}{\sqrt{1-\rho_S^2}} = 0$$

— the smile bottoms exactly at the money — so the floor is attained there and $\tilde{v}_J = v_J$ automatically. Conversely, fix (v_J, p_J, c_J) on the stratum. Then (3.14) sets (b_S, ρ_S), and for *every* width $s_S > 0$, the slice with $m_S = \rho_S s_S/\sqrt{1-\rho_S^2}$ and $a_S = v_J\tau - b_S s_S\sqrt{1-\rho_S^2}$ reproduces the same five handles: one handle set, a one-parameter family of slices. □

Remark 3.6 (The blind spot is the interior curvature). The free width on the stratum is not an abstract degeneracy; it is a specific piece of missing information. On the stratum the vertex sits at the money, the handles pin the level and both tails — and what they omit is exactly how sharply the curve turns through its bottom. By (3.4) the vertex curvature is $\kappa^* = b_S q_S^3/s_S$, which runs through every positive value as the free width s_S does.

Nor is the failure confined to the stratum itself; it is felt nearby. Differentiate (3.17) in χ (quotient rule, then simplify the numerator):

$$\frac{\partial\mathcal{D}}{\partial\chi} = \frac{-\rho_S(1-\chi^2) + \chi(1-\rho_S\chi)}{(1-\chi^2)^{3/2}} = \frac{\chi - \rho_S}{(1-\chi^2)^{3/2}},$$

which vanishes at $\chi = \rho_S$: the denominator is flat there, not just zero. Differentiate once more by the product rule,

$$\frac{\partial^2\mathcal{D}}{\partial\chi^2} = (1-\chi^2)^{-3/2} + 3\chi\,(\chi-\rho_S)\,(1-\chi^2)^{-5/2},$$

and evaluate at $\chi = \rho_S$, where the second term drops: $\partial^2\mathcal{D}/\partial\chi^2 = (1-\rho_S^2)^{-3/2}$. So by Taylor expansion, and then substituting $\chi - \rho_S = -4\psi_J/(p_J + c_J)$ from (3.15),

$$\mathcal{D} \sim \frac{(\chi-\rho_S)^2}{2\,(1-\rho_S^2)^{3/2}} = \frac{8\,\psi_J^2}{(p_J+c_J)^2(1-\rho_S^2)^{3/2}}, \qquad \psi_J \to 0. \tag{3.21}$$

The denominator closes *quadratically* in ψ_J. Three consequences, in increasing order of practical pain: near the stratum, a finite width forces the gap $v_J - \tilde{v}_J$ to be of order ψ_J^2, so the two handles carry nearly the same number; quote noise in the handles is amplified by $1/\psi_J^2$ in the recovered width; and a direct evaluation of (3.17) subtracts two nearly equal numbers, losing precision exactly where precision is scarce. An implementation can remove that last cancellation by evaluating an algebraically identical form — but no rearrangement can restore a curvature the handles never carried. And the stratum is not exotic: ATM sitting at the smile's minimum is where a desk would expect the coordinates to be simplest, and fitted index slices pass close to it.

Figure 3.5 draws the two halves of the remark. In panel (a), the denominator $\mathcal{D}$ is plotted against ψ_J at fixed (p_J, c_J) (blue): it closes at $\psi_J = 0$, and the dashed curve — the quadratic asymptote (3.21) — hugs it near the origin, confirming that the closing is quadratic, not linear. In panel (b), three slices share one handle quintuple $(v_J, 0, p_J, c_J, v_J)$ on the stratum: same level, same asymptotes, visibly different turns through the bottom, one per width s_S. The script verifies that the three slices agree on all five handles to 1.1×10^{-17} — machine precision — so nothing about the handles distinguishes curves the eye tells apart at a glance.

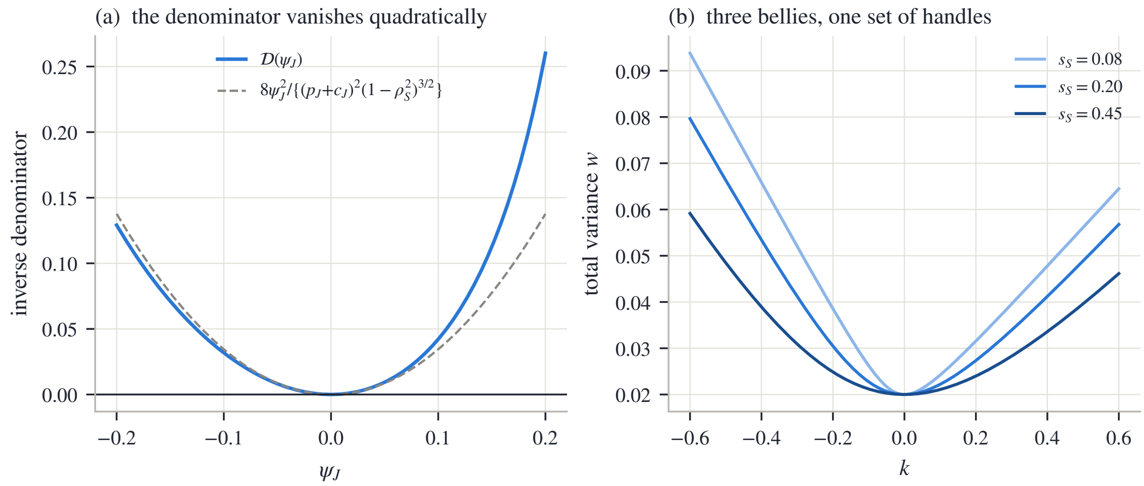

Figure 3.5: Where the jump-wing chart tears. (a) The inverse denominator $\mathcal{D}$ against ψ_J at fixed (p_J, c_J): it closes quadratically at $\psi_J = 0$, following (3.21) (dashed). (b) Three slices with identical handles $(v_J, 0, p_J, c_J, v_J)$ — same level, same asymptotes — differing only in width s_S: the five handles lose the interior curvature (handle agreement across the three slices: 1.1×10^{-17}).

We close the section with a practical dividend. The validity screens of table 3.1 translate directly into handle vocabulary, so a quoted handle set can be sanity-checked before any conversion: the floor condition is $\tilde{v}_J \geq 0$; the domain condition is $-p_J/2 < \psi_J < c_J/2$; and since $\max(\beta_L, \beta_R) = \max(p_J, c_J)\sqrt{w_0}$ by (3.13), the wing cap (3.8) reads

$$\max(p_J, c_J)\,\sqrt{v_J \tau} \;\leq\; \beta_{\max}. \tag{3.22}$$

None of the three certifies the interior, for the reason section 3.4 established: the belly answers to no cheap inequality, in any coordinate system.

3.6 Calibration in the chart

We can now write a slice, read it, and screen it. This section fits one to quotes. Chapter 2 already settled the anatomy of a slice objective — units, band targets, regularization (section 2.6) — and most of it carries over unchanged. Two choices genuinely change when the model lives in the volatility chart, and they are this section's subject: what the objective integrates against, and what coordinates the optimizer moves in.

First, the objective itself. Because the model produces $w(k)$ directly, no price inversion stands between model and data, and the residual can live in implied volatility: with quotes (k_i, σ_i) and weights μ_i, the data term is

$$\sum_i \mu_i \left(I(k_i;\theta) - \sigma_i\right)^2, \qquad I(k;\theta) = \sqrt{w(k;\theta)/\tau}, \tag{3.23}$$

optionally replaced by the band or haircut targets of (2.48) and (2.49) — Chapter 2's variants that fit to the quoted bid–ask spread, or to that spread shrunk by a declared margin, instead of to the mid — evaluated in volatility space.

3.6.1 The measure a fit integrates against

Start with a concrete ladder. A liquid expiry might list thirty strikes within a few percent of the money and three in the far put wing. Fit with equal weights and the wing contributes three rows out of thirty-three: less than a tenth of the objective decides the entire region where the tail lives. Nobody chose that allocation — it happened.

To say precisely what happened, idealize (3.23) as a continuum problem: minimize $\int (I - \sigma)^2 \, \mathrm{d}\mu(k)$ for some measure μ on the strike axis — "measure" in the ordinary sense of a distribution of weight along the axis, saying which strike ranges the fit cares about. A finite quote set turns the integral into a weighted sum, so the weight vector is a *quadrature rule* for μ — a recipe assigning each sample point its share of the integral: the weights *are* the measure. Equal weights make μ the counting measure of the exchange's strike listing — dense at the money, sparse in the wings. That listing was designed by a listings committee for trading convenience, not for inference, and a fit that adopts it inherits an accidental prior. It even fails the most basic sanity check: add more ATM listings — more information — and the fitted wings get *worse*, because the new near-duplicate rows outvote the isolated wing quotes by sheer count.

The repair is to *declare* the measure instead of inheriting it. A natural declaration weights strike ranges by the time value they carry: $\mathrm{d}\mu = \mathrm{TV}(k)\,\mathrm{d}k$, where $\mathrm{TV}(k)$ is the normalized price of the out-of-the-money option at k — the part of the premium that is genuinely optionality rather than intrinsic value, and so a gauge of how much economic content a strike carries. To discretize it on a quote ladder, give each quote the piece of axis nearer to it than to any neighbor: quote i's cell C_i runs half the gap to each adjacent quote (one-sided at the ends), with width $|C_i|$. The scheme

$$\mu_i \;\propto\; \mathrm{TV}_i\,|C_i| \tag{3.24}$$

then weights each quote by time value times the stretch of axis it represents.

Proposition 3.7 (Weights as a quadrature)**.** *For any continuous integrand f,*

$$\sum_i \mathrm{TV}_i\,|C_i|\,f(k_i) \quad \textit{is a Riemann sum of} \quad \int \mathrm{TV}(k)\,f(k)\,\mathrm{d}k$$

over the partition $\{C_i\}$, converging under refinement. In particular the weighted loss with (3.24) discretizes the continuum time-value objective; on a uniform strike grid all $|C_i|$ coincide and the scheme reduces to pure time-value weighting.

Proof. The cells partition the quoted interval and $k_i \in C_i$, so the sum is a Riemann sum over that partition; convergence under refinement is the standard property of Riemann sums. The uniform case is arithmetic. □

Two conventions keep the quadrature usable on real ladders, each in one sentence. The cell of an isolated far-wing quote can be arbitrarily wide, so the width factor is capped (the reference implementation uses ten times the mean spacing): beyond the cap the scheme deliberately under-weights the deep wing rather than let one quote carry a tenth of the objective. And the weights are normalized to unit mean, so switching schemes redistributes the data term without rescaling it against the regularization. Having said all this, the comparisons in this chapter still use equal weights throughout — deliberately. When three families are raced on the same ladder, the simplest shared convention is the fair one, and the accidental measure, once *recognized* as a measure, is no longer accidental. The lesson here is not that one scheme is right; it is that a weight vector *is* a measure, and should be chosen as one rather than inherited from a listings committee.

3.6.2 Two charts

Now the coordinates. Recall from section 2.6.3 what an optimization *chart* is: a smooth map from an unconstrained vector $\theta \in \mathbb{R}^n$ onto the model's parameters, built so that every θ lands in the allowed set — constraints eliminated by construction instead of policed by the solver. (This is the word "chart" in its second sense; the *volatility* chart of section 3.1 is the picture the model draws in, and an optimization chart is a coordinate system on the family.)

The historical chart for SVI maps $\theta \in \mathbb{R}^5$ to

$$a_S = \theta_1, \quad b_S = \mathrm{softplus}(\theta_2), \quad \rho_S = \tanh(\theta_3), \quad m_S = \theta_4, \quad s_S = e^{\theta_5}, \tag{3.25}$$

with $\mathrm{softplus}(\theta) = \log(1 + e^{\theta})$. Each lift does one job: softplus and the exponential keep b_S and s_S positive, tanh keeps ρ_S in $(-1, 1)$, and a_S, m_S need no lift. Every finite θ is therefore a structurally valid hyperbola — the first row of table 3.1, and nothing more. The floor can still go negative and a wing can still exceed the cap; those are fenced by the hinge rows (3.10), which the optimizer can trade against the data term while it runs.

The structural chart pushes the same idea one level up: parameterize the slice by the quantities the inequalities are *about*. Those are the two wing slopes, the vertex location, the floor, and the vertex curvature — five numbers, and the inversion displayed below in (3.27) will show that they determine the hyperbola completely. Lift each so that every finite coordinate is admissible:

$$\beta_L = \beta_{\max}\Lambda(\theta_1), \quad \beta_R = \beta_{\max}\Lambda(\theta_2), \quad k^* = \theta_3, \quad w^* = \mathrm{softplus}(\theta_4), \quad \kappa^* = e^{\theta_5}, \tag{3.26}$$

where Λ is the logistic function of section 2.1, $\Lambda(\theta) = 1/(1 + e^{-\theta})$, which maps the whole line onto $(0, 1)$ — so each wing lands strictly between 0 and $\beta_{\max}$, and the floor is strictly positive, before the optimizer has taken a single step.

The raw parameters are recovered by inverting proposition 3.1, one line per parameter. Adding and subtracting $\beta_L = b_S(1 - \rho_S)$ and $\beta_R = b_S(1 + \rho_S)$ gives b_S and ρ_S; solving $\kappa^* = b_S q_S^3 / s_S$ for s_S gives the width; the vertex location (3.4) then gives m_S, and the floor gives a_S:

$$\begin{gathered} b_S = \frac{\beta_L + \beta_R}{2}, \qquad \rho_S = \frac{\beta_R - \beta_L}{\beta_R + \beta_L}, \qquad s_S = \frac{b_S(1 - \rho_S^2)^{3/2}}{\kappa^*}, \\ m_S = k^* + \frac{s_S \rho_S}{\sqrt{1 - \rho_S^2}}, \qquad a_S = w^* - b_S s_S \sqrt{1 - \rho_S^2}. \end{gathered} \tag{3.27}$$

Under this *structural chart*, every finite iterate has $w(k) \geq w^* > 0$ and both wings strictly inside $(0, \beta_{\max})$: the second and third rows of table 3.1 hold by construction, and the hinge rows (3.10) are identically zero along the entire optimization path. What the chart does *not* deliver is the fourth row. The Vogt slice of section 3.4 satisfies floor and cap, so it corresponds to a perfectly ordinary θ in either chart, and the dense-grid check remains the deciding instance.

The two charts describe the same family, so a well-converged fit should not depend on which one the solver walked through — and it does not: on the SPY December quotes the two charts reach the same slice to 0.00 vol bp (fig. 3.6). What changes is the geometry the solver works in. A hinge row has a kink where the violation starts, exactly where an optimizer that needs derivatives is most likely to stall; the structural chart removes those kinks from the path entirely, because no iterate can reach them.

Figure 3.6 shows both halves. In panel (a), the wing lift $\beta = \beta_{\max}\Lambda(\theta)$: the whole horizontal axis (every finite coordinate) maps into the band between 0 and the dashed cap, itself strictly below the dotted Lee bound at 2 — admissibility is a property of the chart, not a constraint on the step. In panel (b), the five structural coordinates are drawn on the SPY December fit: the marked vertex at $(k^*, w^*) = (0.108, 0.0058)$, the osculating parabola — the parabola matching the curve's value, slope, and curvature $\kappa^* = 0.397$ at the vertex — hugging it at the bottom, and

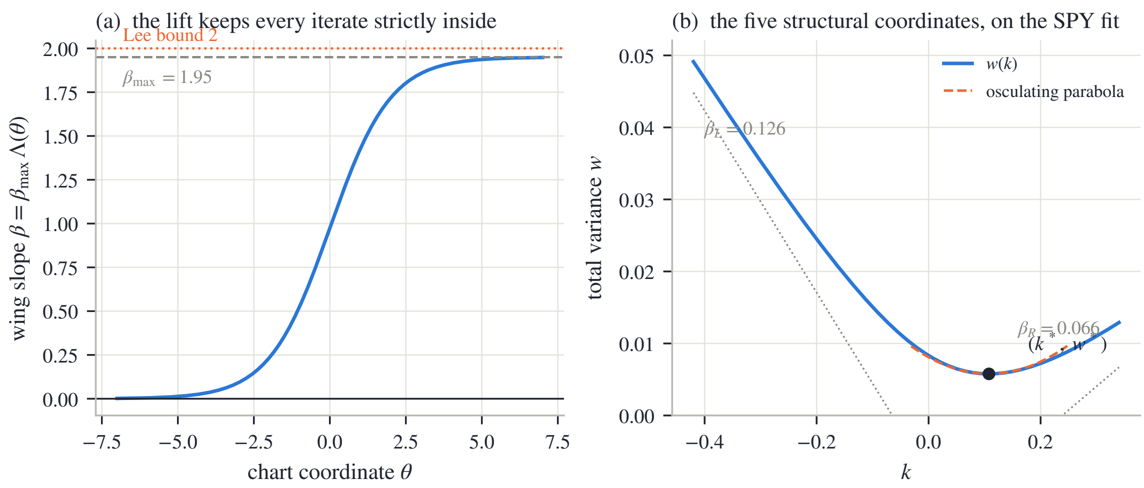

Figure 3.6: The structural chart. (a) The wing lift $\beta = \beta_{\max}\Lambda(\theta)$: every finite coordinate lands strictly between 0 and $\beta_{\max}$, itself strictly below Lee's bound — admissibility is a property of the chart, not a constraint on the step. (b) The five structural coordinates read on the SPY December fit: vertex $(k^*, w^*) = (0.108, 0.0058)$, osculating parabola of curvature $\kappa^* = 0.397$, and the two asymptote slopes.

the two asymptote slopes. The optimizer moves exactly the quantities the inequalities constrain — which is the whole idea.

Both charts are consumed by the same least-squares machinery as Chapter 2's, with analytic derivatives assembled by the chain rule; the assembly is bookkeeping rather than mathematics, and appendix 3.A records it. We now have everything needed to fit one hyperbola well. What remains is the shape it cannot take.

3.7 Superposition: capacity without touching the wings

Proposition 3.2 left a gap: some smiles — ahead of a binary event, or in a name whose holders are split between two scenarios — carry a central trough with a shoulder on each side, and no globally convex slice can follow them. We now fill the gap. But the design problem is sharper than "add flexibility." The wings are the *solved* part of the curve: Lee's bound caps them, the outermost quotes pin them, and section 3.3 showed that the tail density reads them alone. What is missing is detail in the *body*, and whatever supplies it must not move the wings at all.

The Multi-Core Sigmoid (MCS) meets this specification by superposition: a convex base curve that owns the wings, plus localized corrections that are silent — in value, in slope, *and* in curvature — in both tails. The whole construction rests on one elementary mechanism, and we build it first.

3.7.1 The base

The family works in a standardized moneyness

$$\zeta = \frac{k}{\sigma_{\text{ref}}\sqrt{\tau}}, \tag{3.28}$$

where σ_{ref} is a reference volatility fixed before the fit (the quoted volatility nearest the money). The scaling makes ζ count moneyness in standard deviations, so parameters mean roughly the same thing at every expiry: with $\sigma_{\text{ref}} = 20\%$ and $\tau = 0.25$, one ζ-unit is $0.20 \times 0.5 = 0.10$ of log-moneyness; at $\tau = 1$ it is 0.20. Displacements along this axis are written $\bar{\zeta}$: the base below measures its displacement from its center ζ_0, a kernel from its own center ζ_r, and $\bar{\zeta}$ always denotes the distance from the center at hand. (The symbol ζ is new to this chapter's ledger; Chapter 2's z is the log-odds of rank, an unrelated object.)

The convexity primitive of the family is the log-cosh:

$$\mathcal{A}_\kappa(\bar\zeta) = \frac{4}{\kappa^2}\log\cosh\frac{\kappa\bar\zeta}{2}, \qquad \mathcal{A}'_\kappa(\bar\zeta) = \frac{2}{\kappa}\tanh\frac{\kappa\bar\zeta}{2}, \qquad \mathcal{A}''_\kappa(\bar\zeta) = \operatorname{sech}^2\frac{\kappa\bar\zeta}{2}, \tag{3.29}$$

a smooth, even, convex softening of the absolute value, with κ setting how fast the softening straightens out. Two facts about it carry the whole section. Near the origin, $\log\cosh$ of a small argument is half its square, so $\mathcal{A}_\kappa(\bar\zeta) = \bar\zeta^2/2 + O(\bar\zeta^4)$: a rounded vertex, of unit curvature whatever κ. Far out it is *asymptotically affine* — straight line plus a vanishing remainder. To see this, write

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{|x|}}{2}\big(1 + e^{-2|x|}\big) \quad\Longrightarrow\quad \log\cosh x = |x| - \log 2 + \log\big(1 + e^{-2|x|}\big),$$

and the last term dies exponentially; scaling by $4/\kappa^2$ at argument $\kappa\bar\zeta/2$,

$$\mathcal{A}_\kappa(\bar\zeta) = \frac{2}{\kappa}|\bar\zeta| - \frac{4}{\kappa^2}\log 2 + o(1). \tag{3.30}$$

The base models annualized variance $V(\zeta) = w/\tau$ as level plus tilt plus one rounded hinge:

$$V_{\text{base}}(\zeta) = V_0 + V_1(\zeta - \zeta_0) + V_2\,\mathcal{A}_\kappa(\zeta - \zeta_0), \qquad \kappa = \begin{cases}\kappa_P, & \zeta < \zeta_0,\\ \kappa_C, & \zeta \ge \zeta_0,\end{cases} \tag{3.31}$$

six parameters: level V_0, slope V_1, and convexity amplitude $V_2 \ge 0$ at the center ζ_0, with separate put-side and call-side steepnesses κ_P, κ_C. Switching the steepness at ζ_0 costs no smoothness: $\mathcal{A}_\kappa$ and $\mathcal{A}'_\kappa$ both vanish at the origin, and $\mathcal{A}''_\kappa(0) = 1$ for *every* κ, so value, slope, and curvature all match across the switch — the curve is C^2, and the asymmetry lies only in how fast each side straightens.

The wings of the base follow from (3.30). As $\zeta \to +\infty$ the hinge contributes slope $2V_2/\kappa_C$, so V_{base} has asymptotic slope $V_1 + 2V_2/\kappa_C$; on the left, $V_1 - 2V_2/\kappa_P$, typically negative — the put wing rises as ζ falls. To convert to the total-variance slopes that Lee's bound reads, two bookkeeping steps: the map $w(k) = \tau V(k/(\sigma_{\text{ref}}\sqrt{\tau}))$ contributes a chain-rule factor $\tau/(\sigma_{\text{ref}}\sqrt{\tau}) = \sqrt{\tau}/\sigma_{\text{ref}}$; and on the put side the convention of (3.5) measures the slope in $|k|$, which flips the sign of the falling left slope:

$$\beta_P = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(\frac{2V_2}{\kappa_P} - V_1\Big), \qquad \beta_C = \frac{\sqrt{\tau}}{\sigma_{\text{ref}}}\Big(V_1 + \frac{2V_2}{\kappa_C}\Big). \tag{3.32}$$

These six numbers set the wings. The design goal for everything added next is that nothing moves them.

3.7.2 A kernel that vanishes to second order

Here is the mechanism. A centered second difference — value at the left, minus twice the value at the center, plus value at the right — annihilates affine functions exactly: for any straight line $L(\bar\zeta) = \text{const} + \text{slope}\cdot\bar\zeta$,

$$L(\bar\zeta - \ell) - 2L(\bar\zeta) + L(\bar\zeta + \ell) = \big(-\ell\cdot\text{slope}\big) + \big(+\ell\cdot\text{slope}\big) = 0 \qquad \text{identically.}$$

Now take the second difference of a function that is *almost* affine far out — such as $\mathcal{A}_\kappa$, by (3.30). Far out, the second difference sees an affine function plus an exponentially small remainder, and returns the remainder alone: a bump that dies in both tails. Near the center, where $\mathcal{A}_\kappa$ is genuinely curved, the second difference is not zero — that is the bump itself.

The correction kernel is exactly this, normalized: with $\bar\zeta = \zeta - \zeta_r$,

$$\mathcal{B}_{\zeta_r,\ell_r,\kappa_r}(\zeta) = \frac{\mathcal{A}_{\kappa_r}(\bar\zeta - \ell_r) - 2\mathcal{A}_{\kappa_r}(\bar\zeta) + \mathcal{A}_{\kappa_r}(\bar\zeta + \ell_r)}{2\,\mathcal{A}_{\kappa_r}(\ell_r)}, \tag{3.33}$$

centered at ζ_r, of half-width ℓ_r and steepness κ_r. The normalizer makes the peak height exactly one — at $\bar\zeta = 0$ the numerator is $2\mathcal{A}_{\kappa_r}(\ell_r)$, because $\mathcal{A}$ is even with $\mathcal{A}(0) = 0$ — so a kernel's amplitude will read directly as a variance displacement. Its derivatives are the same second difference applied to $\mathcal{A}'$ and $\mathcal{A}''$, over the same normalizer.

Lemma 3.8 (Zero wings)**.** *$\mathcal{B} \to 0$, $\mathcal{B}' \to 0$, and $\mathcal{B}'' \to 0$ as $\zeta \to \pm\infty$.*

Proof. By (3.30), $\mathcal{A}_\kappa$ is affine plus $o(1)$ in each tail; a centered second difference annihilates the affine part exactly (the computation displayed above), so what remains tends to zero. The same argument covers the derivatives: $\mathcal{A}'_\kappa$ tends to the constants $\pm 2/\kappa$ in the tails — affine of zero slope — so its second difference also tends to zero, and $\mathcal{A}''_\kappa$ itself decays. □

The lemma is the entire design, and notice that it does not depend on the log-cosh: *any* asymptotically affine primitive yields a tail-silent kernel by the same second difference. The log-cosh is a convenience — the model and its derivatives share one set of formulas (3.29), the unit peak makes amplitudes readable, and the half-width ℓ_r and steepness κ_r are separate dials where a Gaussian bump has only one width. One more reading will be used below: the kernel's curvature at its own center. Apply the second difference to $\mathcal{A}''$ from (3.29) at $\bar\zeta = 0$, using that $\mathcal{A}''$ is even with $\mathcal{A}''(0) = 1$:

$$\mathcal{B}''(\zeta_r) = \frac{\mathcal{A}''_{\kappa_r}(-\ell_r) - 2\mathcal{A}''_{\kappa_r}(0) + \mathcal{A}''_{\kappa_r}(\ell_r)}{2\,\mathcal{A}_{\kappa_r}(\ell_r)} = \frac{\operatorname{sech}^2(\kappa_r\ell_r/2) - 1}{\mathcal{A}_{\kappa_r}(\ell_r)} < 0,$$

negative because $\operatorname{sech}^2 < 1$ away from zero. So a *positive* multiple of $\mathcal{B}$ is locally concave — it builds a shoulder — and a negative multiple digs a notch.

3.7.3 The model, and what the superposition preserves

The MCS slice is the base plus M signed kernels:

$$V_M(\zeta) = V_{\mathrm{base}}(\zeta) + \sum_{r=1}^{M} \gamma_r\,\mathcal{B}_{\zeta_r,\ell_r,\kappa_r}(\zeta), \qquad w(k) = \tau\,V_M\Big(\frac{k}{\sigma_{\mathrm{ref}}\sqrt{\tau}}\Big), \tag{3.34}$$

with $6 + 4M$ parameters (6 for the base; center, half-width, steepness, and amplitude γ_r per kernel). By lemma 3.8, every member of the family has the same asymptotic wings as its base, whatever M: expressiveness in the body and control of the tails are decoupled by construction.

Figure 3.7 shows the mechanism end to end, on a synthetic target built to be globally admissible — a hyperbolic total-variance base with two Gaussian shoulders, asymptotically linear wings, and $\min g_{\mathrm{D}} = 0.277$ verified from analytic derivatives on a dense grid out to $|k| = 12$ (appendix 3.A records the construction and why a target must be certified this carefully). Panel (a) shows the primitive $\mathcal{A}_\kappa$ merging with its affine asymptote (3.30): within a couple of units the two are indistinguishable, and only the rounded vertex tells them apart — this exponential merging is what the kernel will inherit as tail silence. Panel (b) shows one kernel (3.33): a unit-height bump (blue) whose slope (orange) and curvature both die visibly in the tails — silent to all three orders, as lemma 3.8 proved. Panel (c) runs the fit on the W-shaped target (orange dots). The convex base alone ($M = 0$) cannot reach the two shoulders — proposition 3.2 forbids it — and misses by 118.2 vol bp, leaving a residual that is not noise but two localized features. Adding two kernels (centers dotted) seats both shoulders and cuts the miss to 0.7 vol bp, while the fitted slice remains butterfly-free: $\min g_{\mathrm{D}} = 0.420$ on the wide diagnostic grid.

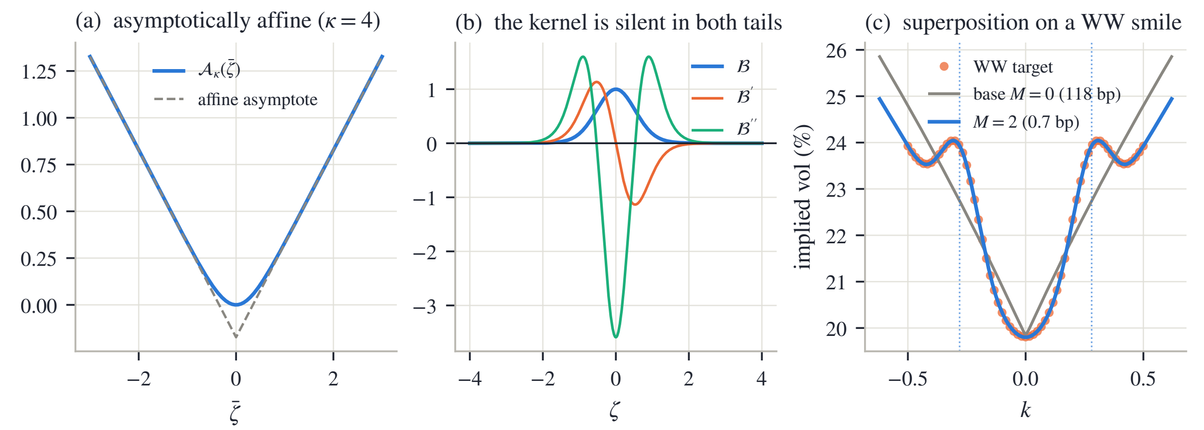

Figure 3.7: Superposition in three steps. (a) The log-cosh primitive merges with its affine asymptote (3.30); only the rounded vertex differs. (b) The centered second difference (3.33): a unit-height bump whose value, slope, and curvature all decay — the correction is silent in the wings to all three orders (lemma 3.8). (c) A W-shaped target (dots): the convex base misses both shoulders by 118.2 vol bp exactly as proposition 3.2 requires; two signed kernels (centers dotted) seat them, reducing the miss to 0.7 vol bp while the wings — and the fitted slice's admissibility — are preserved.

Two boundaries of the construction, each stated once. *Wing-neutral is weaker than wing-admissible*: the kernels preserve the base's slopes (3.32), but nothing in the family forces those slopes under the cap — the base is screened like any other slice. And *the family is not arbitrage-free by construction*: a signed kernel can drive variance negative or break price convexity locally, so the dense-grid check of section 3.4 decides for MCS exactly as it does for SVI — and since two overlapping kernels can trade amplitude without changing the curve, the fitted parameters are means to an end, and only the fitted curve is comparable across fits.

3.7.4 Capacity and its governance

The number of cores M is the family's capacity dial. Counting degrees of freedom already bounds it: $6 + 4M$ parameters against n_q quotes requires

$$M \;\leq\; \left\lfloor \frac{n_q - 6}{4} \right\rfloor \tag{3.35}$$

for the fit to be determined at all — the analogue, for this family, of the LQD order guard (2.50). A thin chain with fifteen quotes, for instance, admits at most $\lfloor (15 - 6)/4 \rfloor = 2$ cores.

But identification is the weaker constraint, and the stronger one is worth stating in words. A kernel is a local feature detector. On a smile that has genuine local features — an event trough, a second mode — it finds them. On a smile that has none, it finds quote noise instead: the error against the quotes falls while the error against the underlying smile rises, and the manufactured curvature shows up exactly where section 3.4 taught us to look — as a negative dip of g_{D} in a sparsely quoted region, where no data resists it. Section 3.8 measures this on the frozen snapshot: at $M = 2$ on liquid index chains, MCS fits the mid quotes better than SVI on every node and fails the validity check on most of them; at $M = 0$ the margins are uniformly clean and the fit quality is SVI's. The capacity is real, and so is its price.

The reference implementation therefore treats M as a governed dial — capped by (3.35), ridged in amplitude, and reserved for smiles that are genuinely non-convex — rather than as a default. The calibration is layered to match the model's structure: fit the base first; seed kernels one at a time on the largest remaining residual features; then refine all $6 + 4M$ parameters jointly (appendix 3.A records the settings).

3.8 Three families on the same quotes

The chapter closes by putting the three families side by side on the same data: the frozen snapshot of appendix 2.C, all 16 nodes, under one protocol. Each family is calibrated by the reference implementation to the same mid quotes with equal weights. LQD runs at polynomial order 16 — its capacity dial, cut back by the guard (2.50) on thin chains — in its own constraint-eliminating coordinates (the logistic chart of section 2.6.3); SVI runs in the structural chart of section 3.6.2; MCS runs at $M = 2$. Each fit reports the same three quantities: the rms distance to the quotes, the validity margin $\min g_D$ over a dense grid on the traded range, and its wing slopes. Every Durrleman function is evaluated from the family's own analytic structure; appendix 3.A records the details. The point of the exercise is not a ranking. It is to watch each family's contract — what it guarantees, what it must have checked — show up in numbers.

3.8.1 The December node and the SPY gallery

Figure 3.8 shows the SPY December 2026 slice, the node the chapter has used throughout. In panel (a), the orange dots are the mid quotes with their bid–ask band, and the three fitted curves — LQD in green, SVI in blue, MCS in violet — overlie so closely that at chart scale they read as one curve inside the band. Whatever separates the families, it is not visible here. Panel (b) is where the information lives: it plots each family's *residuals*, quote by quote, against the mids. Now the three separate cleanly. LQD tracks the quotes at 3.4 vol bp rms with a residual that looks like noise — no structure left. SVI carries 7.5 vol bp, and its residual is not noise: it is one slow wave across the strike range, the visible signature of a five-parameter curve absorbing a whole quoted skew — the cost of parsimony made concrete. MCS at $M = 2$ sits between, at 5.6 vol bp: its two kernels have spent themselves on the two largest features of the SVI-shaped wave.

The three fits also agree on the wings to within a few hundredths of slope — $\beta_R = 0.034$, 0.066, and 0.054 respectively — and it is worth pausing on why this is expected rather than reassuring. Section 2.4 made the point for LQD: no finite strip of quotes identifies an asymptotic slope by itself; a fitted wing is the model's convention continuing the last quoted evidence. Here, three different conventions continue similar evidence in similar ways. The agreement measures the evidence, not the truth of any family's tails.

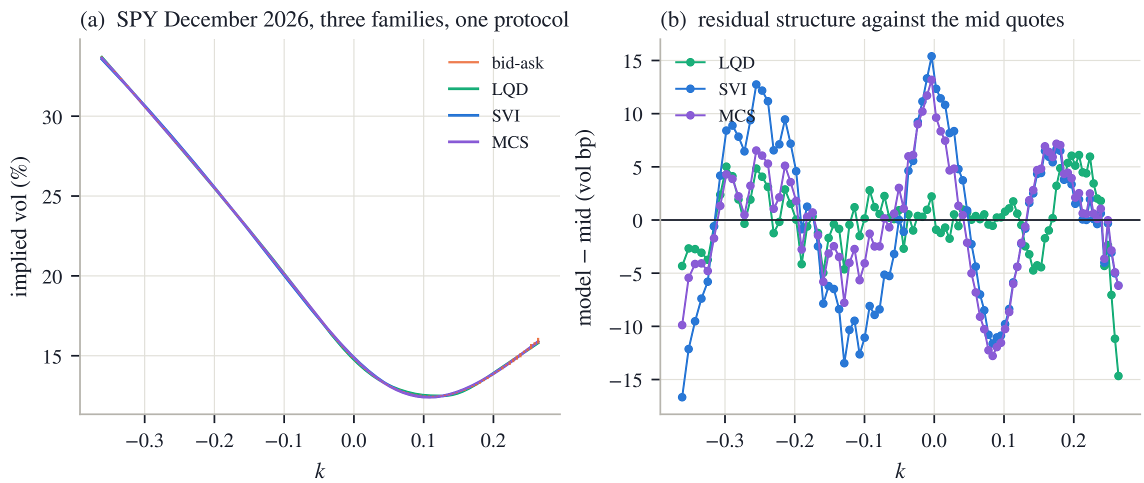

Figure 3.8: Three families, one protocol, on SPY December 2026. (a) The fitted curves overlie at chart scale inside the bid–ask band. (b) The residuals against the mid quotes separate them: LQD (3.4 bp rms, 17 parameters), MCS (5.6 bp, 14), SVI (7.5 bp, 5). More coordinates buy a whiter residual.

Figure 3.9 extends the reading to all eight SPY expiries. Panel (a) plots fit quality by expiry, and the ordering is stable everywhere: LQD, then MCS, then SVI, with SVI's gap widening at the long end, where the skew's shape departs furthest from a hyperbola. More parameters fit better, uniformly — no surprise. Panel (b) is the panel to study: the validity margin $\min g_{\mathrm{D}}$ by expiry, with the zero line marked. The picture inverts. LQD's margin is positive on every node — proposition 2.4 made that structural, so this panel is simply the numerical trace of a theorem. SVI's margin is positive on every node as well, and it cannot be by theorem — the Vogt slice of section 3.4 is proof that the family contains invalid members. It is by rigidity: five parameters fitted to a liquid convex smile have no capacity left over to bend the density negative. MCS at $M = 2$ fails the check on most of the SPY expiries — even though its own optimizer carries a sampled hinge on g_{D} (appendix 3.A), the concrete form of section 3.4's warning that a finite-weight fence discourages violations without excluding them. Its kernels, finding no genuine local features on a liquid index smile, chase quote noise, and the manufactured curvature surfaces as a negative dip of g_{D} — typically in the thinly quoted region, where no data resists it.

Across all 16 nodes (both names), the medians read:

	median rms (bp)	worst rms (bp)	median $\min g_{\mathrm{D}}$	nodes with margin < 0
LQD (order 16, guarded)	7.3	34.0	0.184	0 of 16
SVI (structural chart)	19.0	68.0	0.173	0 of 16
MCS, $M = 2$	11.8	38.5	−0.026	10 of 16
MCS, $M = 0$ (base only)	18.1	—	0.179	0 of 16

Read the two MCS rows against each other: they isolate the dial. Withdraw the kernels and the margins return to uniformly positive while the fit quality falls back to SVI's level. Capacity is neither free nor fatal — it is a choice, and on these liquid chains the correct choice is zero.

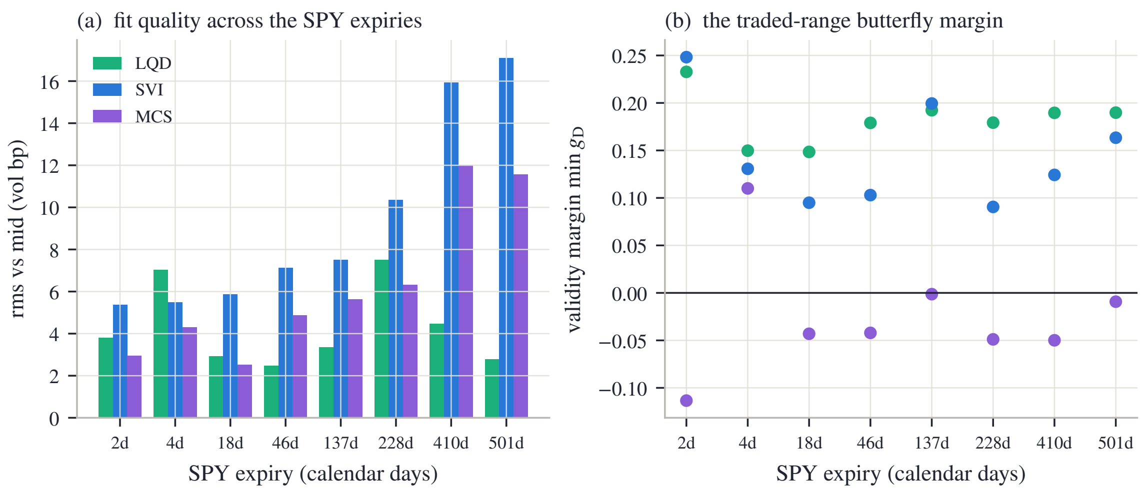

Figure 3.9: The SPY gallery under one protocol. (a) Fit quality by expiry: more parameters fit better, uniformly. (b) The validity margin by expiry: LQD is positive by construction, SVI by rigidity, and MCS at $M = 2$ — its kernels chasing noise on a smile with no genuine local features — fails the dense-grid check on most nodes. The two panels together are the chapter's thesis in one figure: in the volatility chart, fit quality and validity are separate accounts.

3.8.2 Expressiveness measured on a legal target

The gallery shows what excess capacity does where it is not needed. The complementary question — what rigidity costs where capacity *is* needed — has to be asked carefully. It would prove nothing to fit a W-shaped curve that no law can produce; the target must itself be a genuine law, so that an arbitrage-aware family is allowed to reproduce it.

A mixture of two lognormals supplies one. Take component weights 0.65 and 0.35 and component volatilities 14% and 42% — a calm scenario and a stressed one — with a quarter year to expiry. Give the stressed component a forward of 0.85; the martingale identity (the weighted forwards must average to one) then fixes the calm component's forward:

$$0.65\, f_1 + 0.35 \times 0.85 = 1 \quad \Longrightarrow \quad f_1 = \frac{1 - 0.2975}{0.65} = 1.081.$$

This law is genuine by construction — a mixture of laws is a law — and because its density is known exactly, its own Durrleman minimum can be computed rather than estimated: $\min g_{\mathrm{D}} = 0.369$ on the quoted range. Bimodal, but strictly admissible.

Figure 3.10 fits the three families to this smile. In panel (a), the target smile (orange dots) shows the two-mode signature, and the three fitted curves now visibly separate — unlike on the index node, chart scale is enough. LQD (green) follows the two modes at quote scale. SVI (blue) cuts through them with its single belly. MCS (violet) recovers most of the shape. Panel (b) makes the comparison quantitative with the signed errors. LQD fits at 3.6 vol bp with margin 0.371. SVI misses by 166.4 vol bp — a structured, two-lobed error of about two volatility points, which is proposition 3.2 in numbers: one belly cannot seat two modes. Note what SVI's failure is *not*: its margin is 0.431, perfectly valid. Rigidity fails at expressiveness, not at arbitrage. MCS halves SVI's miss to 87.3 vol bp, but overdrives its kernels into a small violation (−0.066): on this target its capacity is genuinely needed — and still needs its check. No family wins all three columns, which is the result.

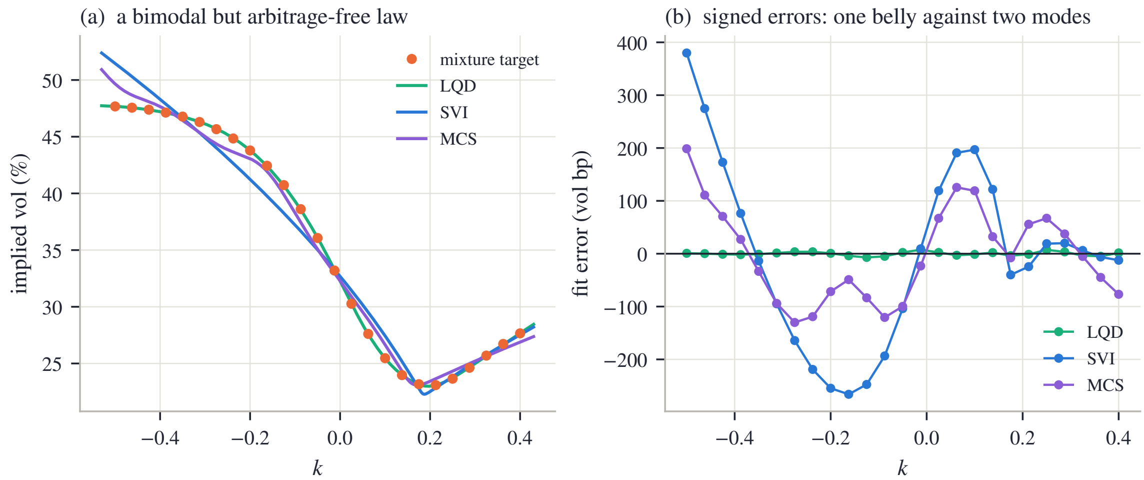

Figure 3.10: The rigidity question asked legally: a bimodal but strictly admissible mixture law ($\min g_{\mathrm{D}} = 0.369$). (a) The smile and the three fits. (b) Signed errors: SVI's single belly leaves a ±two-vol-point structured miss; MCS halves it; LQD follows the modes at quote scale.

3.8.3 The chapter's contract

The node, the gallery, and the mixture say one thing three ways, and it is the chapter's summary. Working in the volatility chart does not remove the constraint that a smile be a probability law. It moves the constraint out of the coordinates and into the process: some of it becomes inequalities a chart can absorb (the floor, the cap), some of it becomes a check no chart absorbs (the butterfly grid), and some of it becomes capacity that must be rationed because nothing structural rations it (the cores). What the chart buys in exchange is legibility: coordinates a desk can quote, wings that are two readable numbers, corrections that are local by construction.

	LQD (Ch. 2)	SVI	MCS
Validity	structural: every admissible vector is a law (proposition 2.4)	policed: floor and cap structural in the chart; butterfly checked on a grid	policed: wings neutral by lemma 3.8; butterfly checked on a grid
Tails	exponential class, slopes from endpoint speeds (proposition 2.9)	exactly linear wings; slopes capped at $\beta_{\max} < 2$	base-owned wings (3.32); admissibility a separate check
Interior capacity	polynomial order, guarded by (2.50)	one belly, one turn (proposition 3.2)	M kernels, guarded by (3.35); governed, not default
Interpretability	endpoint speeds and ledger functionals	JW handles on the domain of theorem 3.5; curvature lost on the stratum	curve-level only: kernels are non-unique by design
What convention supplies	order, ridge, tail chart	$\beta_{\max}$, chart, weights	M, ridge, weights

Table 3.2: What each family guarantees, what must be checked, and what the modeler chooses. The families do not dominate one another; they allocate the same difficulty differently.

Chapter 2 and this chapter are therefore not competitors but two poles of one design axis — validity built into the coordinates at the price of legibility, or coordinates built for the reader at the price of policing. The choice of chart allocates the difficulty; it does not remove it. The next chapter moves to a third position on the same axis: local volatility, where the modeled object is neither a law nor a curve but the field consistent with all of them at once.

3.A Comparison protocol and numerical notes

Protocol

Every number and figure in this chapter is produced by one deterministic script from the frozen snapshot of appendix 2.C and the fixed synthetic constructions below; nothing is typed by hand. The three-family comparison of section 3.8 calibrates each family to the same mid quotes of each node with equal unit-mean weights:

- **LQD**: order 16 reduced by the guard (2.50), logistic tail chart, ridge $\alpha = 10^{-6}$ with $\nu = 1$ — the recipe of section 2.6 with the mid objective.
- **SVI**: the structural chart (3.26) with $\beta_{\max} = 1.95$ and hinge multiplier $W = 10^3$ (the rows are structurally zero in this chart; the multiplier matters only for the historical chart (3.25)).
- **MCS**: $M = 2$ under the cap (3.35), amplitude ridge 10^{-2}, and a one-sided hinge on sampled g_{D} over a grid extending two ζ-units past the quoted range, weighted double on the put side — a soft fence with the same status as (3.10): zero on any admissible fit. Kernel centers may roam half a unit past the quoted range; half-widths are confined to $[0.15, 1.5]$ and steepnesses to $[1, 12]$ in ζ-units. The base is fitted first, kernels are seeded greedily on the largest residuals with a masked neighborhood so two kernels cannot seed one feature, and a joint refinement follows.

The validity margin is $\min g_{\mathrm{D}}$ over 801 equally spaced points spanning exactly the node's quoted log-moneyness range.

Evaluating g_{D} per family

Each family's Durrleman function is computed from its own analytic structure, never by differencing prices — a finite difference of reconstructed prices carries curvature noise at exactly the scale the diagnostic measures. For SVI, (w, w', w'') are the closed forms (3.3). For MCS, the ζ-space derivatives of (3.34) are analytic (second differences of (3.29)) and convert by $w' = \sqrt{\tau}\, V'/\sigma_{\mathrm{ref}}$, $w'' = V''/\sigma_{\mathrm{ref}}^2$. For LQD no derivatives are needed at all: (2.5) states $f_X(k) = g_{\mathrm{D}}(k)\, \varphi(d_-)/\sqrt{w}$, so

$$g_{\mathrm{D}}(k) = \frac{f_X(k)\sqrt{w(k)}}{\varphi\big(d_-(k, w(k))\big)}, \qquad d_- = -\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2},$$

with f_X the slice's explicit density — the identity that defined g_{D} is run backwards. For the mixture target of fig. 3.10 the same identity is used with the exact mixture density, which is how the target's own margin is known rather than estimated.

The residual Jacobian, in layers

The least-squares solvers of both charts consume an analytic Jacobian, and it assembles by the chain rule in three layers. The raw-parameter partials of the hyperbola follow from (3.1) with (3.2):

$$\frac{\partial w}{\partial a_S} = 1, \quad \frac{\partial w}{\partial b_S} = \rho_S x_S + r_S, \quad \frac{\partial w}{\partial \rho_S} = b_S x_S, \quad \frac{\partial w}{\partial m_S} = -b_S\Big(\rho_S + \frac{x_S}{r_S}\Big), \quad \frac{\partial w}{\partial s_S} = \frac{b_S s_S}{r_S}. \tag{3.36}$$

The data rows live in volatility, so each is scaled by

$$\frac{\partial I}{\partial w} = \frac{1}{2\sqrt{w\tau}} = \frac{1}{2\tau I}. \tag{3.37}$$

Finally the lift: for (3.25) the three nontrivial factors are

$$\frac{\mathrm{d}b_S}{\mathrm{d}\theta_2} = 1 - e^{-b_S}, \qquad \frac{\mathrm{d}\rho_S}{\mathrm{d}\theta_3} = 1 - \rho_S^2, \qquad \frac{\mathrm{d}s_S}{\mathrm{d}\theta_5} = s_S, \tag{3.38}$$

while the structural chart pushes the same raw-space rows through the closed-form 5×5 matrix $\partial(a_S, b_S, \rho_S, m_S, s_S)/\partial\theta$ obtained by differentiating (3.27) along (3.26). Hinge rows contribute their linear part where the violation is strictly positive and a zero row otherwise. Nothing here is deep; the point of recording the layers is that a chart is a change of variables like any other, and its cost is one matrix product per iterate rather than a constrained solver.

Synthetic constructions

The W-shaped target of fig. 3.7 is built in total variance so that its wings are asymptotically linear and its derivatives are exact:

$$w(k) = 0.008 + 0.012\sqrt{k^2 + 0.15^2} + 0.0025\big(e^{-\{(k-0.28)/0.11\}^2} + e^{-\{(k+0.28)/0.11\}^2}\big), \qquad \tau = 0.25,$$

sampled at 61 equally spaced quotes on $[-0.5, 0.5]$. Gaussians vanish with all derivatives, so both wings carry the asymptotic slope 0.012 — far under Lee's bound — and the tail limit of g_{D} is $(4 - 0.012^2)/16 = 0.250$; the script verifies $\min g_{\mathrm{D}} = 0.277 > 0$ on a dense grid to $|k| = 12$ from the analytic derivatives before any figure is written. A target built carelessly in volatility space with linear volatility tails would violate Lee far outside the plotted window: total variance would grow quadratically. Certify a synthetic target the same way as a fit, globally, or it certifies nothing.

The mixture target is the two-lognormal law of section 3.8: weights $(0.65, 0.35)$, volatilities $(0.14, 0.42)$, second component forward 0.85 and the first fixed by the martingale identity $0.65\, f_1 + 0.35 \cdot 0.85 = 1$ (so $f_1 = 1.0808$), $\tau = 0.25$, quoted at 25 strikes on $k \in [-0.5, 0.4]$ through exact Black inversion of the mixture call.

Floating-point strictness

The structural chart's guarantee — wings strictly inside $(0, \beta_{\max})$, floor strictly positive — is a statement about real numbers, and float64 does not honor it automatically. Two boundary effects deserve record. A logistic lift evaluated at a moderately large coordinate rounds to exactly 1, parking a wing exactly at the cap; the implementation clips the lift one unit in the last place inside 1. And when one wing saturates against an ordinary other, the quotient for ρ_S in (3.27) can round to ± 1, making $1 - \rho_S^2$ an exact zero and m_S undefined; the implementation computes it by the algebraically identical product form

$$1 - \rho_S^2 = \frac{4\,\beta_L \beta_R}{(\beta_L + \beta_R)^2},$$

which never rounds to zero for admissible wings. A chart whose strictness silently fails at the boundary reproduces exactly the failure mode it was built to remove.

Chapter 4

Local Volatility

Chapters 2 and 3 gave each expiry its own probability law and used a calendar condition to keep neighboring expiries consistent. This chapter asks for more: one model that carries every expiry of the surface at once. The Dupire equation provides it. Its coefficient — the local volatility — turns out to be a ratio of the two quantities the earlier chapters already policed: the calendar increment on top, the butterfly factor below. So the field exists exactly where the surface is valid. Computing the field, however, splits into a wrong way and a right way. Read as a formula, the ratio asks for two derivatives of noisy quotes, and fails no matter how good the data gets. Read as a pricing model, it poses a well-behaved fitting problem: a sheet of positive numbers on a grid, priced by a stable marching scheme, differentiated exactly, and fitted to the quotes like every other model in this book. The chapter builds that fit slowly, shows why the quotes alone cannot pin the sheet down, and ends by fitting the book's frozen SPY and NVDA chains with one sheet each.

4.1 One diffusion for the whole surface

Chapters 2 and 3 both ended in the same place. Each expiry of a fitted surface carries its own probability law for the return: Chapter 2 built the law first and read the smile from it, Chapter 3 wrote the smile directly and then checked that it described a law. Between neighboring expiries one further condition held, the calendar order of section 2.8: at the same log-moneyness, the later normalized call is worth at least the earlier one. A surface, so far, is a stack of laws, related two at a time.

This chapter asks for more. Is there one random process — one model of the underlying's path — whose distribution at each expiry is exactly the law that expiry's smile encodes? If there is, the surface stops being a stack and becomes a single object: one dynamics, and every European price of every maturity comes out of it.

Half of the answer is already known. Recall two terms from Chapter 2. Calendar order across all adjacent pairs is *convex order* of the normalized returns — later laws are more spread out. And a *martingale* is a process with no drift: its expected future value, given everything known today, is today's value. A theorem of Kellerer [33] says these two notions meet exactly: if the family of laws is in convex order, then *some* martingale has these laws as its distributions at the given dates. But the theorem is pure existence. It does not name the martingale, it gives it no structure a computation could use, and it does not say what extra information any particular choice smuggles in.

Dupire's observation [18] answers all three objections at once, by restricting attention to one concrete class of processes. Work in the normalized units the book has used since section 2.8: $Y_\tau = S_\tau / F_{0,\tau}$ is the underlying divided by its forward. Dividing by the forward is what strips the drift: the forward is the expected future level under the pricing law (with deterministic carry, as section 2.8 assumed), so the ratio starts at 1 and has no tendency to grow or shrink —

it is a positive martingale worth 1 at time zero. And τ is variance time, the maturity variable the book has quoted variance against since Chapter 2, so that a slice's total implied variance is $w = \sigma^2\tau$. The reader knows the Black–Scholes dynamics $\mathrm{d}Y = \sigma Y \mathrm{d}W$, with one constant volatility. Now let the volatility depend on where the process is and on when it is there:

$$\mathrm{d}Y_\tau = \sqrt{v(\tau, Y_\tau)}\, Y_\tau \,\mathrm{d}W_\tau, \qquad Y_0 = 1, \tag{4.1}$$

where $v(\tau, y) \geq 0$ is a fixed, deterministic function of two variables. The function v is called the *local variance*; its square root $\sigma_{\text{loc}} = \sqrt{v}$ is the *local volatility*. It is a field: a whole surface of variances, one for every (time, level) pair, laid out in advance. When the process visits level y at time τ, it diffuses at exactly the rate the field prescribes there. Nothing else is random: the field is the entire model.

Within this class, the vague question above has an exact answer, in both directions. Given the field, one linear equation — derived in the next section — produces the marginal law of every maturity at once. Given the surface, a pointwise formula recovers the field, wherever the implied density is positive. A valid surface is therefore not merely *consistent with* some model. It *is* the coefficient field of one canonical model, written in different coordinates. This chapter is about that change of coordinates: how to compute it, why the obvious way of computing it fails on real data, and what a market's finitely many quotes do and do not say about the field.

Before starting, be clear about what the object is not, on two counts. First, the diffusion (4.1) is a *representative*, not a discovery. A theorem of Gyöngy [28] says that any well-behaved martingale with the same marginal laws — however wild its own volatility, random or path-dependent — is *mimicked* by (4.1): take $v(\tau, y)$ to be the average instantaneous variance of that process over the paths that sit at y at time τ, and the diffusion reproduces every one of its marginals. Many different dynamics share one field. Anything that depends only on the marginal laws — every European price, every quantity of Chapters 2–3 — is therefore safe to compute in the model. Anything that depends on the joint law across dates — above all, how the smile moves when the spot moves — is not determined by the surface at all. That question is postponed to Chapter 9, and no computation in this chapter will touch it.

Second, the class (4.1) contains no jumps. Every path is continuous, so a genuine gap — an earnings night, a scheduled announcement — can only be imitated by a brief burst of very large diffusive variance. The honest treatment of scheduled events is a change of clock, and it waits for Chapter 8. Throughout this chapter, rates, dividends and borrow — the cost of shorting the stock — live inside the forward as usual (Chapter 6 makes that precise), $y = e^k$ is the normalized strike, $c(\tau, y)$ is the normalized call, and the field v is variance per unit of variance time.

The plan follows the two directions of the correspondence. Section 4.2 derives the forward equation, shows that a nonnegative field makes the whole surface arbitrage-free automatically, and turns the equation around into the extraction formula — whose two ingredients will be exactly the two validity conditions of the static chapters. Section 4.3 shows why the extraction formula must never be applied to market quotes. Sections 4.4 and 4.5 build the alternative: a field with finitely many positive parameters, and a pricing scheme that keeps the theory's guarantees on a finite grid. Section 4.6 asks what the quotes actually determine about the field, and sections 4.7 and 4.8 assemble the calibration and run it on the book's frozen data.

4.2 The equation the surface must obey

4.2.1 Forward: from the field to every marginal

Fix a bounded measurable field $v \geq 0$ and let Y solve (4.1). We now derive the equation that the normalized call satisfies as a function of maturity. It is the computational engine of the whole chapter: solve it once, marching left to right in maturity, and every expiry's smile comes out along the way.

Proposition 4.1 (The forward Dupire equation)**.** *Suppose Y_τ has a density $f(\tau,\cdot)$ such that $vy^2 f$ and $y\,\partial_y(vy^2 f)$ are integrable and vanish at $y=0$ and as $y\to\infty$. Then the normalized call $c(\tau,y)=\mathbb{E}[(Y_\tau-y)^+]$ satisfies*

$$\boxed{\partial_\tau c(\tau,y)=\tfrac12\,v(\tau,y)\,y^2\,\partial_{yy}c(\tau,y),\qquad c(0,y)=(1-y)^+.} \tag{4.2}$$

Proof. The proof has three steps: find the equation the density obeys, integrate that equation against the call payoff, and move the derivatives across.

Step 1: the density. Apply Itô's formula to $\Xi(Y_\tau)$, for a smooth function Ξ — recall that Itô adds a second-order term to the ordinary chain rule, because the squared increment $(\mathrm{d}Y)^2$ accumulates at the rate $v\,Y^2\,\mathrm{d}\tau$ here:

$$\mathrm{d}\,\Xi(Y_\tau)=\Xi'(Y_\tau)\,\sqrt{v}\,Y_\tau\,\mathrm{d}W_\tau+\tfrac12\,v(\tau,Y_\tau)\,Y_\tau^2\,\Xi''(Y_\tau)\,\mathrm{d}\tau.$$

Take expectations. The $\mathrm{d}W$ term has mean zero, and writing the remaining expectations against the density f,

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int_0^\infty \Xi(y)\,f(\tau,y)\,\mathrm{d}y=\tfrac12\int_0^\infty v(\tau,y)\,y^2\,\Xi''(y)\,f(\tau,y)\,\mathrm{d}y.$$

Let Ξ vanish outside a bounded set away from 0, and integrate the right side by parts twice, moving both derivatives off Ξ and onto $vy^2 f$ (no boundary terms, since Ξ vanishes near the ends):

$$\int v\,y^2 f\,\Xi''\,\mathrm{d}y=-\int\partial_y(v\,y^2 f)\,\Xi'\,\mathrm{d}y=\int\partial_{yy}(v\,y^2 f)\,\Xi\,\mathrm{d}y.$$

This holds for every such Ξ; and if the two integrands differed anywhere, choosing a Ξ concentrated near that point would break the equality. So they agree:

$$\partial_\tau f=\tfrac12\,\partial_{yy}(v\,y^2 f).$$

This is the *Fokker–Planck equation* — the equation every diffusion's density obeys; the two displays above are its entire derivation.

Step 2: integrate against the payoff. The call is $c(\tau,y_0)=\int_0^\infty(y-y_0)^+ f(\tau,y)\,\mathrm{d}y$; differentiating under the integral sign (which the decay hypothesis licenses),

$$\partial_\tau c(\tau,y_0)=\int_0^\infty(y-y_0)^+\,\partial_\tau f(\tau,y)\,\mathrm{d}y=\tfrac12\int_0^\infty(y-y_0)^+\,\partial_{yy}(v\,y^2 f)\,\mathrm{d}y.$$

Step 3: move the derivatives onto the payoff. Integrate by parts once. The payoff's derivative in y is the step function $\mathbf{1}_{\{y>y_0\}}$ — slope 0 below the strike, 1 above — and the boundary term $(y-y_0)^+\,\partial_y(vy^2 f)$ vanishes at both ends by the decay hypothesis:

$$\tfrac12\int_0^\infty(y-y_0)^+\,\partial_{yy}(vy^2 f)\,\mathrm{d}y=-\tfrac12\int_{y_0}^\infty\partial_y(v\,y^2 f)\,\mathrm{d}y.$$

The remaining integral is elementary: it integrates an exact derivative, and $vy^2 f$ vanishes at the far end, so

$$-\tfrac12\int_{y_0}^\infty\partial_y(v\,y^2 f)\,\mathrm{d}y=-\tfrac12\Big[v\,y^2 f\Big]_{y_0}^\infty=\tfrac12\,v(\tau,y_0)\,y_0^2\,f(\tau,y_0).$$

Finally, recall the density identity of section 2.1: differentiating the call twice in strike returns the density, $f(\tau,y_0)=\partial_{yy}c(\tau,y_0)$. Substituting gives (4.2). The initial condition is $Y_0=1$, whose call is $(1-y)^+$. □

Read (4.2) as a physics problem. It is a heat equation in the *strike* variable: the call price plays the temperature, the payoff $(1-y)^+$ is the initial temperature profile, maturity plays the role of time, and $\frac12 vy^2$ is the conductivity of the material at position y and time τ. One field, one linear march, and every marginal law of the surface comes out at once. The analogy is not decoration; it is the source of every structural property this chapter uses. Heat flow spreads bumps and never sharpens them, and "spreading", translated into option language, is exactly the no-arbitrage direction. The next proposition makes that precise, and it is the reason this chapter can *promise* validity rather than police it. (One name in it deserves its reminder: a *butterfly* is the three-strike spread whose payoff is a tent, and its price is a second difference of call prices — so ruling out negative butterfly prices is exactly the condition $\partial_{yy}c \geq 0$, a nonnegative implied density.)

Proposition 4.2 (Positive local variance propagates no-arbitrage)**.** *Let c solve (4.2) with $v \geq 0$ bounded. Then for every $\tau > 0$: (i) $\partial_{yy}c(\tau,\cdot) \geq 0$ — each slice is butterfly-free; (ii) $\partial_\tau c(\tau, y) \geq 0$ — the normalized call is non-decreasing in maturity at fixed y, which is the calendar order of section 2.8.*

Proof sketch. Differentiate (4.2) twice in y: the function $f = \partial_{yy}c$ — by Step 3's density identity, the same f as in proposition 4.1 — satisfies $\partial_\tau f = \frac{1}{2}\,\partial_{yy}(vy^2 f)$ — the Fokker–Planck equation of Step 1 again — with initial datum $\partial_{yy}(1-y)^+$, the unit point mass at $y = 1$: at $\tau = 0$ all the probability sits at $Y = 1$. Equations of this type preserve nonnegativity. Probabilistically there is nothing to prove: $f(\tau,\cdot)$ *is* the density of the diffusion (4.1) started at 1, and densities are nonnegative. Analytically, the same fact is the parabolic maximum principle, which we quote from PDE theory. With $f \geq 0$ established, part (ii) is the equation (4.2) itself read as a sign statement: $\partial_\tau c = \frac{1}{2}vy^2 f \geq 0$. □

Pause on what this proposition buys, because it is the strategy of the entire chapter. In Chapters 2 and 3, validity was work, done slice by slice: Chapter 2's LQD chart (its log-quantile-density coordinates, in which every parameter choice is automatically a law) made butterfly freedom an identity within one expiry, the volatility chart had to police $g_{\mathrm{D}} \geq 0$, and calendar order across expiries was in both cases a separate condition. Here, one sign condition on one function of two variables delivers both families of conditions, across the whole surface, automatically. The rest of the chapter is organized around protecting that gain. Choose a parametrization in which $v > 0$ is trivial to enforce (section 4.4). Choose a discretization that *inherits* the proposition on a finite grid, rather than merely approximating it (section 4.5). Then measure what little the finite grid leaks, instead of assuming it leaks nothing (section 4.8).

4.2.2 Backward: from the surface to the field

We now know how a field produces a surface. Read the equation the other way. If the surface is given — so that both derivatives of c are known — then (4.2) can simply be solved for the coefficient:

$$v(\tau, y) = \frac{\partial_\tau c(\tau, y)}{\frac{1}{2}\, y^2\, \partial_{yy} c(\tau, y)}, \tag{4.3}$$

wherever the density in the denominator is positive. This is the uniqueness half of the correspondence: within the class (4.1), at most one field is consistent with a given surface, and (4.3) computes it pointwise. The formula becomes most instructive when translated into the coordinates of Chapter 3 — the volatility chart, where a surface is the total-variance function $w(k, \tau)$.

Proposition 4.3 (The field in the volatility chart)**.** *Write the surface in implied form, $c = B(k, w(k, \tau))$ with $w(\cdot, \tau)$ the total-variance slice. Then (4.3) reads*

$$v(\tau, e^k) = \frac{\partial_\tau w(k, \tau)}{g_{\mathrm{D}}(k)}, \tag{4.4}$$

with g_{D} the Durrleman factor of (2.5), evaluated on the slice at maturity τ [24].

Proof. Compute the numerator and the denominator of (4.3) in turn.

Numerator. At fixed k, the call is the Black function of the slice's total variance, so by the chain rule $\partial_\tau c = \partial_w B \cdot \partial_\tau w$. The derivative $\partial_w B$ is the vega of the Black formula in total-variance units — the sensitivity of the price to the variance the formula is fed — and it is worth computing once in full. The normalized call is $B = \Phi(d_+) - e^k \Phi(d_-)$ with $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$. The two arguments satisfy

$$d_+ - d_- = \sqrt{w}, \qquad d_+ + d_- = -\frac{2k}{\sqrt{w}}, \qquad \text{so} \qquad d_+^2 - d_-^2 = (d_+ + d_-)(d_+ - d_-) = -2k.$$

Because $\varphi(d) = e^{-d^2/2}/\sqrt{2\pi}$, the ratio of the two normal densities is

$$\frac{\varphi(d_+)}{\varphi(d_-)} = \exp\Big(-\tfrac{d_+^2 - d_-^2}{2}\Big) = e^k, \qquad \text{i.e.} \qquad \varphi(d_+) = e^k \varphi(d_-).$$

Now differentiate B in w and use this identity:

$$\partial_w B = \varphi(d_+)\,\partial_w d_+ - e^k \varphi(d_-)\,\partial_w d_- = \varphi(d_+)\,\partial_w (d_+ - d_-) = \varphi(d_+)\,\partial_w \sqrt{w} = \frac{e^k \varphi(d_-)}{2\sqrt{w}}.$$

Denominator. Two facts convert $\partial_{yy} c$ into chart language. First, the density of Y transforms into the density of the log return $X = \log Y$ by the usual change of variables: with $y = e^k$, $f(\tau, y) = f_X(k)/y$ (the factor $1/y$ is $\mathrm{d}y = y\,\mathrm{d}k$). Second — this is (2.5), quoted from Chapter 2 — the log-return density of a slice with total variance w is

$$f_X(k) = g_{\mathrm{D}}(k)\,\frac{\varphi(d_-)}{\sqrt{w}},$$

where g_{D} is the Durrleman function: up to the positive factor $\varphi(d_-)/\sqrt{w}$, it *is* the probability density the smile implies, so $g_{\mathrm{D}} \geq 0$ is exactly the butterfly condition Chapter 3 policed. The denominator of (4.3) is then

$$\tfrac{1}{2}\,y^2\,\partial_{yy} c = \tfrac{1}{2}\,y^2\,f(\tau, y) = \tfrac{1}{2}\,y\,f_X(k) = \tfrac{1}{2}\,e^k\,g_{\mathrm{D}}(k)\,\frac{\varphi(d_-)}{\sqrt{w}}.$$

Divide numerator by denominator: the factors e^k, $\varphi(d_-)$ and $2\sqrt{w}$ all cancel, leaving $v = \partial_\tau w / g_{\mathrm{D}}$. □

The reader has met both halves of this ratio before, as the two families of static validity. The numerator is the calendar increment: it is nonnegative exactly when the slices around maturity τ are calendar-ordered in the sense of section 2.8. The denominator is the butterfly factor: it is nonnegative exactly when the slice at τ implies a genuine (nonnegative) density. So (4.4) is more than a formula; it is a compatibility statement. *A local-volatility field exists precisely where the surface is valid*, and each mode of static arbitrage has its own place to fail: a calendar violation makes the numerator negative, a butterfly violation the denominator. This chapter therefore takes as its input hypothesis what Chapters 2–3 produced as output: slices that are butterfly-free and calendar-ordered, in the normalized units of section 2.8. Where the hypothesis fails, no field exists, and (4.4) says so by changing sign.

Figure 4.1 evaluates the ratio on the book's frozen SPY surface — the snapshot's eight fitted slices, with total variance interpolated linearly in τ between expiries. Walk its three panels left to right. Panel (a) is the numerator: the calendar increment $\partial_\tau w$, piecewise constant in τ because the interpolation between expiries is linear, and largest toward the short-dated put side, where the smile grows fastest with maturity. Panel (b) is the denominator: each slice's Durrleman factor, the implied-density factor of the chart. Panel (c) is their ratio, displayed as local volatility $\sqrt{v}$: a steep put-side wall, a quiet call side, sharp structure at the short end — the

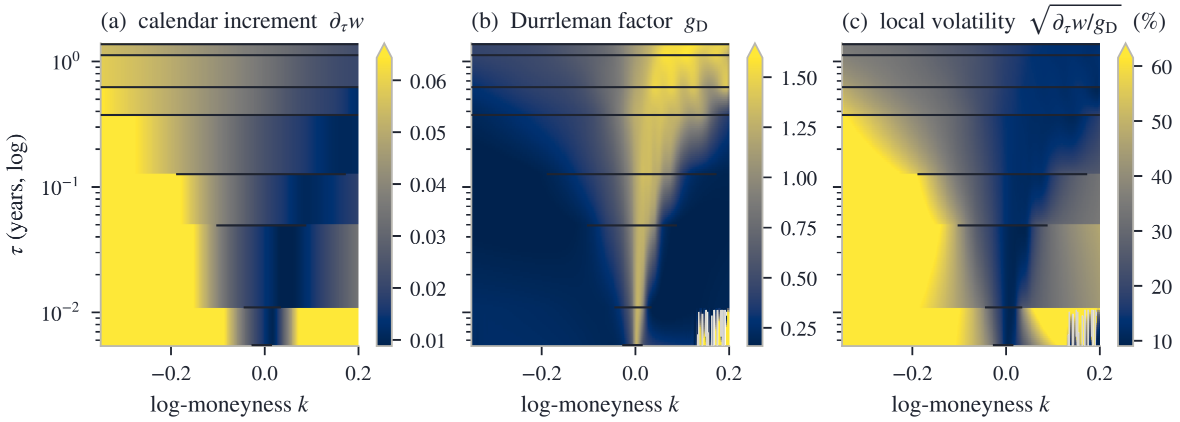

Figure 4.1: The field as a ratio, on the frozen SPY surface (stored per-expiry fits of the snapshot, total variance interpolated linearly in τ between the eight expiries; horizontal bars mark each expiry's quoted span). (a) The calendar increment $\partial_\tau w$. (b) The Durrleman factor of each slice. (c) Their ratio, displayed as local volatility; grey cells have a negative numerator or denominator and are inadmissible.

field inherits the smile's features in recognizable form, and where the data speaks the extracted values range from 9.0% to 61.5%. The grey cells are the diagnostic part of the picture: there one of the two signs fails and no local variance exists. They amount to 0.5% of the displayed mesh, and every one of them sits *outside* the common quoted support of the two expiries that bracket it — restricted to the quoted region the share is 0.00% — in territory owned by extrapolation conventions rather than by prices.

One caution closes the section. The surface in this figure is fitted and smooth, and evaluating the ratio on a fitted surface is a legitimate operation — it is how the figure was drawn. What must never be done is to apply the formula to the market's raw quotes. The next section shows why.

4.3 The wrong direction

Formula (4.3) suggests an obvious procedure: interpolate the quoted smile, differentiate it twice in strike and once in maturity, divide. On a continuum of exact prices, this is a theorem. On a real chain — a few dozen quotes per expiry, each carrying bid–ask noise — it fails. The failure is structural, not bad luck, and it is worth computing exactly how large it is.

Start with how curvature is estimated from data. From three quotes at equally spaced strikes, the standard estimate is the second difference,

$$\partial_{kk} w \;\approx\; \frac{w(k-\Delta k) - 2\,w(k) + w(k+\Delta k)}{\Delta k^2},$$

which recovers the curvature of smooth data as $\Delta k \to 0$. Now let each quote err. Suppose the middle quote reads too low by ε and its two neighbors too high by ε — the alternating pattern a bid–ask bounce produces naturally, one print near the bid, the next near the ask. The three errors pass through the formula as

$$\frac{(+\varepsilon) - 2\,(-\varepsilon) + (+\varepsilon)}{\Delta k^2} = \frac{4\varepsilon}{\Delta k^2},$$

an error in the estimated curvature of 4ε divided by the *square* of the strike spacing. Put numbers in. On a smile at 23% volatility with three months to run, a ripple of 50 volatility basis points (a basis point is 0.01 volatility percent) moves total variance by about $\varepsilon = 2\sigma\tau\,\delta\sigma \approx 5\times10^{-4}$. Sent through strikes $\Delta k = 0.02$ apart, the curvature error is $4\times(5\times10^{-4})/(0.02)^2 = 5$ — several times larger than the true curvature, which for the smile at hand is of order one. And the dependence on the spacing is inverse-square: halving Δk multiplies the error by four. More quotes — a finer, ostensibly better chain — makes the estimate *worse*, without bound. There is no stable limit to converge to. Numerical differentiation of noisy data is the textbook ill-posed operation.

The place this amplified noise lands could not be worse chosen. By (4.4), the curvature sits inside g_{D} — the factor that carries the *sign* of the implied density. Recall its expression from (2.5), with primes denoting strike derivatives of the slice:

$$g_{\mathrm{D}}(k) = \Big(1 - \frac{k\,w'}{2w}\Big)^2 - \frac{(w')^2}{4}\Big(\frac{1}{w} + \frac{1}{4}\Big) + \frac{w''}{2}.$$

The estimated curvature enters additively, through the last term: a curvature error of 5 is a g_{D} error of 2.5, dwarfing the factor's healthy values (panel (b) of fig. 4.1 showed them). The first time the noise pushes the estimated density through zero, the extraction returns a negative variance. That is not a large error; it is a non-object. No diffusion has negative variance, and no pricing equation, simulation or risk system downstream can accept one.

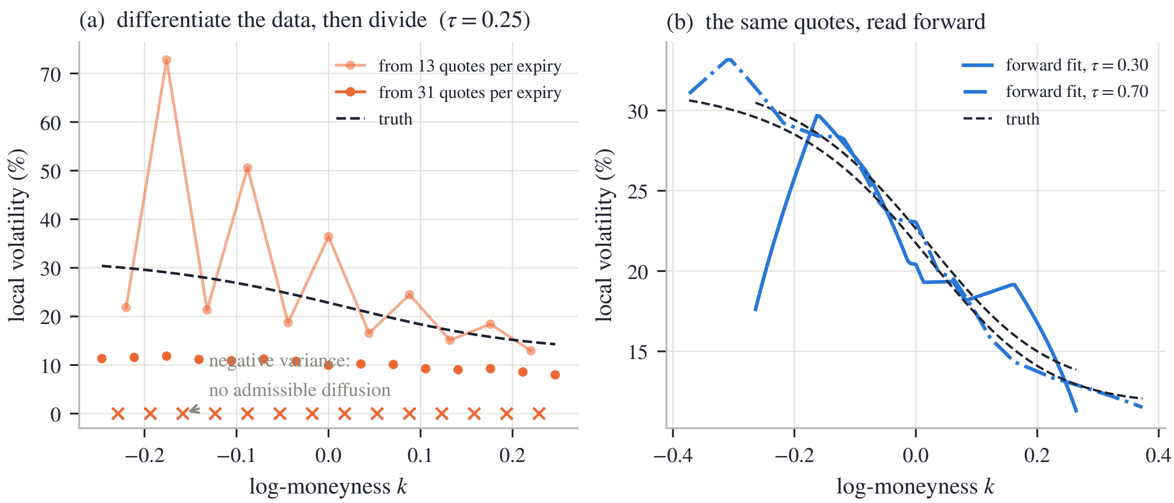

Figure 4.2: The same synthetic market, read both ways. Quotes come from a known smooth field (dashed) plus a deterministic alternating ±50 bp volatility ripple — a stylized bid–ask bounce. (a) The extraction (4.4) from the noisy quotes at $\tau = 0.25$, from 13 and from 31 quotes per expiry; crosses mark strikes where it returns a negative variance or density. (b) The same 31-quote noisy chains handed to this chapter's forward calibration: admissible by construction, with reprice residuals of rms 49 bp — the scale of the ripple itself.

Figure 4.2 performs the experiment on a synthetic market whose answer is known. The smooth field and the quote recipe are recorded in appendix 4.A; the "noise" is a deterministic alternating ripple of ±50 bp of volatility, a stylized bid–ask bounce, so that every number in the figure is reproducible. Panel (a) is the extraction at $\tau = 0.25$. The dashed curve is the true local volatility, whose level at the money is 23%. From 13 quotes per expiry, the extracted curve sawtooths between 13% and 73% — already unusable. From 31 quotes of the *same* market — the same ripple, only finer strike spacing — it does worse than oscillate: at 14 strikes, marked by crosses, it fails outright, reporting a negative variance or a negative density. That is the inverse-square law at work. Refining the grid, which improves every well-posed computation, drives this one to report, at nearly half the strikes, that no diffusion exists.

Panel (b) previews the repair, run on the harder case: the same 31-quote noisy chains are handed to the calibration this chapter builds. The result is admissible by construction — every iterate of the search is a genuine positive field — and it tracks the truth through the quoted region, staying within 12.9 volatility points of it along the plotted cross-sections, while the quotes reprice to 49 bp rms, the scale of the ripple itself. The noise ends up in the residual, where it belongs, instead of in a derivative. One honest caveat is visible in the panel and explained properly in section 4.6: at the edges of the shortest expiry's quoted span, where a single rippled quote is the only witness, the fitted field wanders by as much as 22 volatility points. Noise rejection comes from dense data and declared smoothing, not from magic.

The repair itself, due to Andreasen and Huge [2], is a change of role, not of formula. Stop treating the field as the output of a differentiation. Declare it the unknown. Parametrize it; price it through the forward equation (4.2), which differentiates only the smooth numerical solution and never the data; and ask an optimizer for the positive field whose prices meet the quotes. Positivity is imposed on the *parameters*, so every iterate of the search is an admissible diffusion before any quote is consulted. Readers of the earlier chapters will recognize the design: it is constraint elimination again, the idea that built the LQD chart (section 2.6.3) and the structural SVI chart (section 3.6.2), now one dimension up. Choose the coordinates in which the constraint set is trivial, and let the pricing map do the hard work.

Three questions remain, and they organize the rest of the chapter. What parametrization makes positivity of a two-variable field trivial (section 4.4)? What discretization of (4.2) keeps proposition 4.2 true on a finite grid, rather than approximately true (section 4.5)? And what do the quotes actually determine about the field, once the pointwise formula is off the table (section 4.6)?

4.4 A sheet with finitely many parameters

4.4.1 Positivity as a box

The unknown is now a positive function of two variables — an infinite-dimensional object with a one-sided constraint, which sounds harder than anything fitted so far in this book. The finite version is built from the simplest ingredient available: straight-line interpolation. Picture the smallest example that shows everything. Take two maturities and three strikes — a grid of six points — and write a positive number on each. Split each of the two rectangular cells along a diagonal, giving four triangles. Inside each triangle, define the field as the unique plane through its three corner values. The result is a continuous surface made of flat triangular facets, passing through the six numbers — and those six numbers are the model's only parameters. The general object just replaces six by a few hundred.

Two names before the formal definition. A *tensor grid* is a rectangular grid: pick a list of maturities and a list of strikes, and take every combination as a vertex. The *hat function* of a vertex is the tent that rises to 1 there, slopes to 0 at every neighboring vertex, and stays 0 beyond; on each triangle it is the affine function with those corner values. Its values at a point are also called the point's *barycentric weights* — the three nonnegative numbers, summing to one, that locate the point inside its triangle.

Definition 4.4 (P_1 local-variance sheet)**.** Fix vertices (τ_i, y_j), $i = 1..n_\tau$, $j = 1..n_y$, on a tensor grid, triangulated by splitting each rectangular cell along one diagonal (remark 4.6). The local variance is the continuous piecewise-affine interpolant

$$v_\theta(\tau, y) = \sum_{\ell=1}^{n_\theta} \theta_\ell \, \mathcal{H}_\ell(\tau, y), \qquad n_\theta = n_\tau n_y, \tag{4.5}$$

where $\mathcal{H}_\ell$ is the hat function of vertex ℓ — barycentric on each triangle, equal to 1 at its own vertex and 0 at every other — and the parameters $\theta_\ell = v(\tau_i, y_j) > 0$ are the *nodal local variances*, indexed flat as θ_ℓ or by row and column as $\theta_{i,j}$, whichever is convenient.

With the names in place, read (4.5) as plainly as it deserves: the sum simply interpolates the nodal values. At any point of the grid, only the three hats of the surrounding triangle are nonzero; their values there are the point's barycentric weights; and v_θ at that point is the corresponding weighted average of the three corner parameters.

Why affine pieces, and why triangles? Because a weighted average with nonnegative weights cannot leave the range of the numbers it averages. That one-line fact is the model's entire validity theory:

Proposition 4.5 (Nodal bounds are surface bounds)**.** *If $v_{\mathrm{lo}} \le \theta_\ell \le v_{\mathrm{hi}}$ for all ℓ, then $v_{\mathrm{lo}} \le v_\theta(\tau, y) \le v_{\mathrm{hi}}$ everywhere on the triangulation's hull.*

Proof. At any point of the hull, the only nonzero hat functions are the three belonging to the surrounding triangle, and their values there are nonnegative and sum to one — they are the point's barycentric weights. Hence

$$v_\theta(\tau, y) = \sum_\ell \theta_\ell \, \mathcal{H}_\ell(\tau, y) \;\le\; v_{\mathrm{hi}} \sum_\ell \mathcal{H}_\ell(\tau, y) = v_{\mathrm{hi}},$$

and the lower bound is the same computation with v_{lo}. □

Positivity of a surface has been reduced to positivity of finitely many numbers: keep every θ_ℓ inside the box $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$, and the whole field stays inside it too, everywhere on the hull — the region the triangles cover. A box is the one constraint that least-squares solvers handle natively — clip each coordinate — with no penalty terms and no feasibility questions. This is the promised chart, one dimension up: as with the LQD chart of Chapter 2 and the structural chart of section 3.6.2, the coordinates are chosen so that every point of the parameter set is a valid object, and the optimizer never learns that a constraint existed. Combined with proposition 4.2, the chain of guarantees is complete before any data arrives:

$$\text{box} \implies \text{positive field} \implies \text{butterfly-free, calendar-ordered surface.}$$

Every iterate of every fit in this chapter is an admissible diffusion.

Figure 4.3 shows the object this chapter will calibrate: the SPY sheet of section 4.8, with 9 maturity rows, 24 strike columns, and therefore 216 vertices, drawn with the very triangulation the pricing uses. Read it as a list of numbers, because that is all it is. Every dot is one calibration parameter; every triangle is a region where local variance is an affine function of (τ, y). The tall put-side wall, the sharp structure at the short end, and the flat call-side plain are nothing but nodal values. Notice also that the strike columns crowd toward the money at short maturities; that is deliberate grid design, and section 4.6 explains it.

Figure 4.4 takes the construction apart on a small demonstration grid. In panel (a), the triangulated vertex grid; the shaded star of triangles around one vertex is the support of that vertex's hat function — the entire region of the surface its parameter can influence. A nodal variance is a local dial: turn it, and only the prices of options whose diffusion passes through that star can move. (The cell diagonals in the panel do not all point the same way; that is real, and remark 4.6 explains it.) In panel (b), a cross-section along one vertex row: each hat rises to 1 at its own vertex, is 0 at every other, and the hats sum to 1 identically — the partition of unity behind the weighted-average argument of proposition 4.5.

Remark 4.6 (Which diagonal, and does it matter)*.* On a tensor grid every cell is a rectangle, and a rectangle's four corners lie on one circle — exactly the tie case of the standard (Delaunay) criterion for well-shaped triangles, which prefers triangulations in which no vertex falls strictly inside another triangle's circumscribed circle. The triangulation routine therefore breaks the tie cell by cell, deterministically but by no designed rule, and the figures draw the resulting patchwork faithfully. The choice is second-order small: the two candidate interpolants of a cell agree at all four corners and along all four edges, and differ inside by at most half the cell's

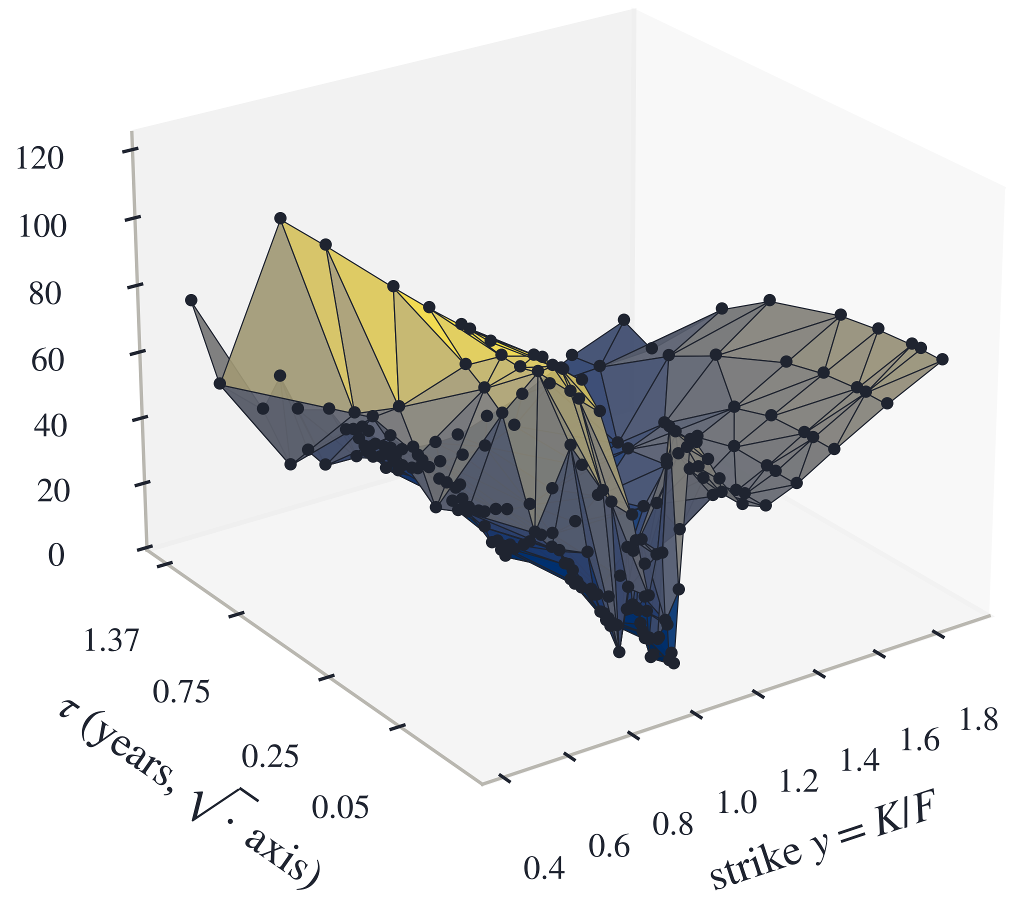

Figure 4.3: The model, drawn: the calibrated SPY local-volatility sheet of section 4.8 (9 maturity rows $\times$ 24 strike columns, 216 vertices), with the triangulation the pricing basis actually uses. Every dot is one calibration parameter; every triangle is a region where local variance is affine in (τ, y).

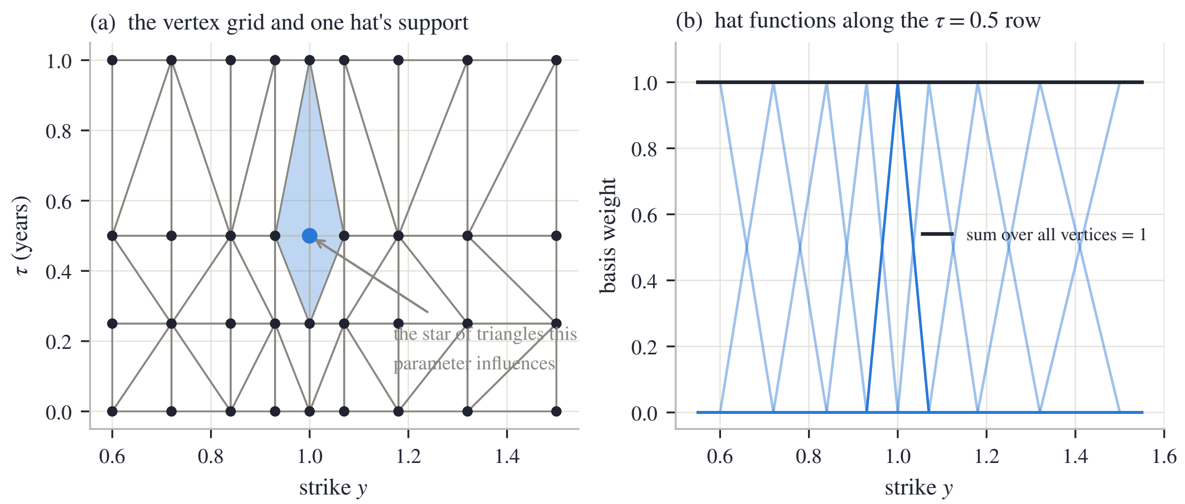

Figure 4.4: The anatomy of the sheet, on a small demonstration grid. (a) The triangulated vertex grid; the shaded star of triangles is the support of one vertex's hat function. The cell diagonals are not uniformly oriented; that is real (remark 4.6). (b) A strike cross-section along a vertex row: each hat is 1 at its own vertex, 0 at every other, affine between, and they sum to 1 identically.

twist, $(\theta_{i,j} + \theta_{i+1,j+1} - \theta_{i,j+1} - \theta_{i+1,j})/2$, at the center — zero whenever the four values are locally bilinear. It is nonetheless a convention, so it is pinned: the triangulation is computed once, cached, and the same triangles price every iterate.

4.4.2 Beyond the hull: the wing contract

The grid must end somewhere, and pricing does not. Below the lowest strike column, the sheet continues local variance *linearly*, with slope a times the slope of the first interior cell; above the highest column it is clamped flat. Three settings of a are meaningful. The default, $a = 0$, clamps the put wing flat as well. A fixed $a > 0$ keeps an unquoted put wing rising. And a is handed to the optimizer as a free parameter in exactly one circumstance: when a variance-swap quote is present. A variance swap pays the realized variance of the underlying to expiry, and its fair price integrates the whole smile, deep wings included — it is the one instrument in the problem that genuinely reads the deep-put tail, as Chapter 5 will show.

Remark 4.7 (The extrapolation sits outside the positivity guarantee). Proposition 4.5 stops at the hull, and for a reason: outside it, interpolation becomes extrapolation, and extrapolation weights are signed — an extrapolated value is no longer an average of nodal values. With $a > 0$ and a first cell whose variance decreases toward the boundary, the linear continuation can reach zero before $y = 0$ (pricing floors it there, so the continued curve kinks rather than turning negative), and a rising continuation can exceed v_{hi} freely. Every guarantee in this chapter that leans on the box is therefore scoped to the hull and to the flat-clamped wings. The wing is a disclosed convention — the same standing as the tail conventions of Chapters 2–3 — and the chapter's contract table records it.

Remark 4.8 (Lee's bound, inherited). A bounded field bounds the smile, and the bound is worth deriving because Chapter 3 had to legislate what comes out here for free. The first step is that the marched call is *monotone in the field*: raising local variance anywhere can only raise call prices. To see it, let c_v and $c_{v+\delta v}$ solve (4.2) for fields v and $v + \delta v$ with $\delta v \geq 0$, and subtract the two equations. Their difference $s = c_{v+\delta v} - c_v$ starts at $s(0, \cdot) = 0$ and satisfies

$$\partial_\tau s = \tfrac{1}{2}\,(v + \delta v)\, y^2\, \partial_{yy} s + \tfrac{1}{2}\,\delta v\, y^2\, \partial_{yy} c_v$$

(add and subtract $\frac{1}{2}(v+\delta v)y^2\partial_{yy}c_v$ to check). The source term is nonnegative — $\delta v \geq 0$ by assumption, $\partial_{yy}c_v \geq 0$ by proposition 4.2 — and a heat equation driven by a nonnegative source from a zero start stays nonnegative (the maximum principle again). So $s \geq 0$. Now compare with the constant field v_{hi}. With flat-clamped wings, $v \leq v_{\mathrm{hi}}$ holds on the entire strike line, so every call price is dominated by its price in the Black–Scholes model with variance rate v_{hi}; and since the Black price increases in total variance (vega is positive), the domination transfers to implied variance:

$$w(k,\tau) \;\leq\; v_{\mathrm{hi}}\,\tau \qquad \text{for every } k.$$

Total implied variance bounded in strike means its wing slope decays to zero: the surface sits strictly inside Lee's cap of 2 automatically, where Chapter 3 had to impose a cap by hand (section 3.3). A steep released continuation (a free) escapes v_{hi}, and with it this bound.

4.5 Pricing on the lattice

4.5.1 The implicit march and its M-matrix

The forward equation must now be solved numerically, and this section chooses the scheme with one criterion above all others: it must inherit proposition 4.2, not merely approximate it.

First the setting. The equation is solved on a strike grid over $[0, y_{\max}]$, with the two end values pinned: $c(\cdot,0)=1$ (a call struck at zero delivers the forward, worth 1 in normalized units) and $c(\cdot,y_{\max})=0$ (far enough out, the call is worthless). Pinned end values are called *Dirichlet* boundaries. The march then steps maturity forward in increments $\Delta\tau$, and at each step it must decide at which time level to evaluate the strike derivative. The *fully implicit* choice — evaluate at the new level — is the one this chapter defends.

On the grid, the second derivative becomes a weighted second difference, and the weights are worth deriving because their signs carry the whole theory. The grid need not be uniform; write $\Delta y_i^- = y_i - y_{i-1}$ and $\Delta y_i^+ = y_{i+1} - y_i$ for the gaps around node i. Taylor-expand the two neighbors of a smooth function c around y_i:

$$c_{i\pm1} = c_i \pm \Delta y_i^{\pm}\, c'(y_i) + \tfrac{1}{2}\,(\Delta y_i^{\pm})^2\, c''(y_i) + O(\Delta y^3).$$

Multiply the "+" expansion by Δy_i^-, the "−" expansion by Δy_i^+, and add: the first-derivative terms cancel, leaving

$$\Delta y_i^-\, c_{i+1} + \Delta y_i^+\, c_{i-1} - (\Delta y_i^- + \Delta y_i^+)\, c_i = \tfrac{1}{2}\,\Delta y_i^+ \Delta y_i^- (\Delta y_i^+ + \Delta y_i^-)\, c''(y_i) + O(\Delta y^4),$$

and solving for c'' gives the three-point weights. Multiplying by $\frac{1}{2}v_i y_i^2$, the discrete generator $\mathcal{L}$ encoding $\frac{1}{2}vy^2\partial_{yy}$ has the stencil

$$\mathcal{L}_{i,i\mp1} = \frac{v_i\, y_i^2}{(\Delta y_i^- + \Delta y_i^+)\, \Delta y_i^{\mp}} \;\geq 0, \qquad \mathcal{L}_{ii} = -(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}), \tag{4.6}$$

which reduces to the familiar $(1,-2,1)/\Delta y^2$ pattern on a uniform grid. Note the two signs that will do all the work: the off-diagonal entries are nonnegative, and each diagonal is exactly minus the sum of its row's off-diagonals.

An implicit step then solves a linear system for the vector U^{n+1} of calls at the new maturity, the operator evaluated at the new level:

$$\mathcal{M}^{n+1}U^{n+1} = U^n + (\text{boundary terms}), \qquad \mathcal{M}^{n+1} = I - \Delta\tau\, \mathcal{L}^{n+1}. \tag{4.7}$$

Because the stencil couples only neighbors, $\mathcal{M}$ is tridiagonal, and each step is one tridiagonal solve — linear cost in the number of nodes. The boundary terms carry the pinned values 1 and 0 through the stencil's end columns, and two later proofs lean on them, so write them out once as a vector b^{n+1}: only the row whose lower neighbor is the boundary node $y = 0$ has a nonzero entry,

$$b_i^{n+1} = \Delta\tau\, \mathcal{L}_{i,i-1}^{n+1} \times 1 \quad \text{(the pinned call value at } y = 0\text{)}, \qquad b_i^{n+1} = 0 \ \text{ at every other row}$$

— the far boundary would contribute through its own stencil arm, but its pinned value is 0.

Now the promised discipline. Proposition 4.2 was proved for the continuous equation. Does the discrete march *inherit* it — no invented negative densities, no broken calendar order — or merely approximate it, with errors of unknown sign? The answer runs through one classical notion from linear algebra. A matrix with positive diagonal, nonpositive off-diagonal entries and enough diagonal dominance is called an *M-matrix*, and the property that sign pattern buys is exactly the one needed here: the inverse of an M-matrix has no negative entries.

Proposition 4.9 (The step matrix is an M-matrix)**.** *For any $v \geq 0$ and any $\Delta\tau > 0$, the matrix $\mathcal{M} = I - \Delta\tau\mathcal{L}$ of (4.7) has positive diagonal, non-positive off-diagonals, and each diagonal entry exceeds the absolute sum of its row's off-diagonals by at least 1 (exactly 1 at fully interior rows). Hence $\mathcal{M}$ is a strictly diagonally dominant nonsingular M-matrix, and $\mathcal{M}^{-1} \geq 0$ elementwise.*

Proof. The sign pattern reads off (4.6): the off-diagonals of $\mathcal{M}$ are $-\Delta\tau\mathcal{L}_{i,i\mp1} \leq 0$; the diagonal is $1 + \Delta\tau(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}) \geq 1$; and the dominance gap is the 1 contributed by the identity at interior rows, more at rows whose stencil arm reaches a boundary column (that neighbor is not among the unknowns). This is the defining pattern of a nonsingular M-matrix, and such matrices are inverse-positive [7]. A self-contained argument fits in four lines. To solve $\mathcal{M}u = b$ with $b \geq 0$, split $\mathcal{M} = D - \mathcal{N}$ into its diagonal $D > 0$ and off-diagonal part $\mathcal{N} \geq 0$, and iterate Jacobi's method from zero:

$$u^{(m+1)} = D^{-1}(\mathcal{N}u^{(m)} + b), \qquad u^{(0)} = 0.$$

Every iterate is nonnegative, being assembled from nonnegative pieces; the iteration converges, because every row of $D^{-1}\mathcal{N}$ sums to less than one (that is diagonal dominance), so each update multiplies the largest error entry by a factor below one; and the limit solves $\mathcal{M}u = b$. So $b \geq 0$ forces $u \geq 0$, for every such b — which is the statement $\mathcal{M}^{-1} \geq 0$, column by column. □

Inverse positivity is the discrete counterpart of heat flow never turning a nonnegative temperature profile negative, and it translates directly into the property numerical analysts call *monotonicity* — order in, order out — and into stability in the strongest everyday sense: the largest absolute value in the solution can never grow.

Corollary 4.10 (Discrete maximum principle, unconditionally)**.** *For any $\Delta\tau > 0$ the scheme (4.7) is* monotone *— $U^n \geq V^n$ with the same boundary data implies $U^{n+1} \geq V^{n+1}$ — and every step keeps the values between the extremes of the previous step and the boundary data: the march is unconditionally ℓ^∞-stable (the maximum absolute value never grows), with no step-size restriction, and cannot create new extrema.*

Proof. Monotonicity is inverse positivity applied to a difference: if $U^n \geq V^n$, the two systems (4.7) subtract to $\mathcal{M}^{n+1}(U^{n+1} - V^{n+1}) = U^n - V^n \geq 0$, and multiplying by $\mathcal{M}^{-1} \geq 0$ preserves the sign. For the upper bound, let U_* be the larger of $\max_i U_i^n$ and the boundary values 1 and 0, and test the constant vector $U_*\mathbf{1}$, where $\mathbf{1}$ is the all-ones vector (a vector here, not the indicator of proposition 4.1's proof). At a fully interior row, the row of $\mathcal{M}$ sums to 1 (because the stencil's row sums to zero), so $(\mathcal{M}\,U_*\mathbf{1})_i = U_* \geq U_i^n$. At a row next to a boundary, the arm that would reach the boundary column is missing from $\mathcal{M}$, leaving the larger value $(\mathcal{M}\,U_*\mathbf{1})_i = U_* + \Delta\tau\,\mathcal{L}_{i,\mathrm{bd}}\,U_*$, with $\mathcal{L}_{i,\mathrm{bd}}$ the missing arm's weight; the right-hand side there is $U_i^n + b_i^{n+1}$ with $b_i^{n+1} \leq \Delta\tau\,\mathcal{L}_{i,\mathrm{bd}} \times 1$. Since U_* dominates both U_i^n and the boundary value 1, the left side dominates the right at every row: $\mathcal{M}(U_*\mathbf{1}) \geq U^n + b^{n+1}$ entrywise. Multiplying by $\mathcal{M}^{-1} \geq 0$ gives $U_*\mathbf{1} \geq U^{n+1}$. The lower bound is symmetric. □

Remark 4.11 (What the M-matrix argument does and does not prove). Proposition 4.9 and corollary 4.10 control the call *values*: order preservation, boundedness, no invented extrema. They do not by themselves give nonnegative second differences — a bounded monotone vector can still be locally concave — so "the scheme cannot manufacture negative densities" would need a separate convexity-preservation proof for the actual stencil and boundary treatment, and this chapter does not claim one. What it does instead is measure: the scheme is monotone and stable by theorem, and the butterfly and calendar cleanliness of the *output* is checked on the price grid with every fit. Section 4.8 reports the measured minima for the chapter's two calibrated sheets; they sit at the scale of solver rounding.

4.5.2 Why not Crank–Nicolson

The implicit march is only first-order accurate in the time step, and the standard second-order upgrade is Crank–Nicolson: evaluate the operator half at the old level and half at the new, replacing (4.7) by

$$(I - \tfrac{\Delta\tau}{2}\mathcal{L})U^{n+1} = (I + \tfrac{\Delta\tau}{2}\mathcal{L})U^n.$$

Look at what the upgrade costs. The implicit factor on the left is still an M-matrix — that part survives. But monotonicity now also needs the *explicit* factor on the right to map nonnegative vectors to nonnegative vectors, and its diagonal entries are $1 - \frac{\Delta\tau}{2}(\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1})$: nonnegative only when the step is small enough,

$$\Delta\tau \;\le\; \frac{2}{\mathcal{L}_{i,i-1} + \mathcal{L}_{i,i+1}} \;=\; \frac{2\,\Delta y^2}{v_i\, y_i^2} \quad \text{(uniform grid).} \tag{4.8}$$

This is a step-size restriction of the Courant–Friedrichs–Lewy (CFL) type — the generic fate of schemes with an explicit part — and it tightens with the *square* of the strike step. Put numbers in: at a 40% local volatility on a $\Delta y = 0.005$ lattice — the resolution section 4.6 will show short-dated smiles require — the bound is 0.0003 years, more than thirty times smaller than the standard 0.01 marching step. Monotone Crank–Nicolson on realistic lattices is simply not on offer. When the bound is violated, the scheme remains second-order *accurate* but is no longer *monotone*, and the payoff kink at $y = 1$ seeds the oscillation that fig. 4.5 displays.

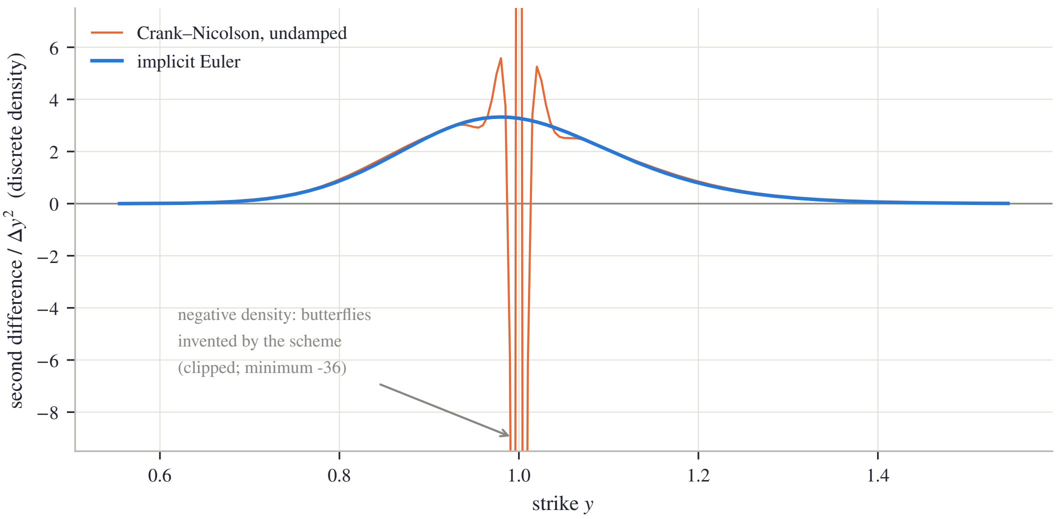

Figure 4.5: The same flat 40% surface, the same $\Delta y = 0.005$ lattice, marched to $\tau = 0.1$ in ten steps by the same solver — only the scheme differs. (a) Implicit Euler: the discrete density is a clean bell, minimum 0.0012 in the plotted window. (b) Undamped Crank–Nicolson: the kink oscillation swings the discrete density to -36.

Figure 4.5 shows what breaking monotonicity does to the object we actually care about: the discrete density, i.e. the second differences of the marched prices. Both panels march the same flat 40% surface on the same $\Delta y = 0.005$ lattice to $\tau = 0.1$ in ten steps; only the scheme differs. In panel (a), implicit Euler: the second differences form a clean bell — a discrete Gaussian, as they should — whose minimum over the plotted window is 0.0012: nonnegative, exactly as corollary 4.10 guarantees. In panel (b), undamped Crank–Nicolson: the oscillation seeded by the payoff kink swings the discrete density down to -36 — a negative density, which is butterfly arbitrage, manufactured by the scheme that is, in the textbook ordering, the more accurate of the two. Damped start-up steps — a few implicit steps before switching to Crank–Nicolson — are the standard repair [27]. The reference implementation declines the trade altogether: it pays one order of time accuracy and keeps corollary 4.10 unconditionally.

4.5.3 Reading out, and the operator's blind spot

The observation operator is plain: the normalized call at a quote's (τ, y) is read off the marched solution by interpolation in strike, and quoted expiries are forced to be lattice time-nodes, so no interpolation in maturity is ever needed.

One principle from inverse problems completes the section, because it dictates part of the chapter's protocol. An optimizer facing a *fixed* discretization will happily bend its parameters to cancel the operator's own discretization error. The residual measured on the fitting operator therefore flatters the fit, and validating a reconstruction with the operator that produced it has a name in the inverse-problems literature: the *inverse crime*. The chapter's protocol answers it by repricing every fit on a *refined* operator — half the strike step, four times the time steps — and reporting both numbers. The gap between them is not noise; it is the measured size of the compensation, and section 4.8 will show it is the dominant error term at the short end.

4.6 What the quotes determine

Before writing an objective function, a chapter should ask whether the problem it is about to pose has one answer. This one does not, in three distinct ways — one about resolution, one about averages, one about entire regions of the sheet — and each failure dictates one piece of the calibration's design. The section works through the three in order, then shows the sharpest of them as an experiment.

The length scale. Everything interesting about a smile at maturity τ happens within a few multiples of $\sigma\sqrt{\tau}$ of the money — the diffusive width, the distance the underlying typically moves by expiry. Work out its size and the design problem appears: a two-day smile at 20% volatility is about 1.5% wide, while a one-year smile at the same volatility is 20% wide — more than ten times wider. Both live on the same grids. Three placement rules follow. First, *vertex placement*: strike columns are spaced by probability, not by strike. Fix a ladder of probabilities (the desk's "deltas"), and place a column at the strike the return ends beyond with each given probability — under a normal approximation, the strike at probability u is $y = \exp\big(\sigma\sqrt{\tau}\,\Phi^{-1}(u)\big)$. Equal steps of probability make strike steps that are tight near the money and wide in the wings, so parameter density automatically follows where optionality lives; the exact ladder is in appendix 4.A. Second, *vertex coverage*: an axis laid out for the longest expiry undersamples the shortest, whose entire smile can fall between two adjacent columns; so every expiry is guaranteed several columns within its own traded range, spread evenly across it — this is what crowds the short-dated money in fig. 4.3. Third, *lattice resolution*: the strike step must resolve the narrowest smile, and it must also be at least as fine as the quote spacing — a lattice coarser than the quotes cannot represent them faithfully, and the fitted smile develops spurious wiggles at the quote spacing (an *alias* of the data, in signal-processing language) until the lattice out-resolves it. The exact placement rules used in this chapter are collected in appendix 4.A.

An averaging fact, and the floor. At the money, implied and local variance are related as average to integrand. If the field near the money depends only on time, $v(\tau, y) \approx v(\tau, 1)$, the model is Black–Scholes with a time-dependent volatility, and the first-course identity applies exactly:

$$w(0, \tau) = \int_0^\tau v(s, 1)\,\mathrm{d}s.$$

In general the identity holds to leading order near the money — a fact we quote rather than prove [24]. Its consequence for calibration is simple: an average can only be matched by an integrand that crosses it. (One reading convention for the numbers that follow: volatilities are quoted in percent, and the field v is their *square*.) Whenever the ATM *term structure* — implied variance read across maturities at the money — is *increasing*, early local variance must sit strictly *below* the shortest implied variance: if the one-week implied is 12% and the one-month is 18%, the field's first days must run below $(0.12)^2$ in variance units. A parameter floor set carelessly — say at a round 5% volatility, $v_{\mathrm{lo}} = 0.0025$ — therefore makes some low-volatility short-dated term structures literally unfittable, with the optimizer pinned to the box and the misfit looking like a mystery. The box of proposition 4.5 must be generous and *adaptive*: the cap a fixed multiple of the largest implied volatility in view, the floor a fraction of the smallest *ATM* implied — keyed to ATM quotes deliberately, so that one noisy deep-wing print cannot drag down the floor where it does real stabilizing work.

Rows the quotes cannot see. The averaging fact has a sharper corollary at the front of the grid. The quotes at the first expiry τ_1 constrain — through the integral above — only the *total* of local variance accumulated over $[0, \tau_1]$. A vertex row strictly inside that interval is individually invisible: the optimizer can ring one sub-front row up and the next one down, leave every integral unchanged, and pay nothing in fit. Under noise it will do exactly that — this is how the rippled fit of fig. 4.2 spent its freedom at the shortest expiry, wandering 22 volatility points where a single quote was the only witness. The cure is not more data, because no vanilla — an ordinary call or put — expires inside $[0, \tau_1]$. The cure is a declared prior: tie each sub-front row to the first identified row — the *front tie* of section 4.7 — turning an invisible subspace into a stated constant continuation.

Two sheets, one quote set. The three paragraphs above are instances of one geometry, and it is best seen as an experiment. A fit like this chapter's carries a couple of hundred nodal parameters. The quotes outnumber them — nearly five to one on the frozen SPY chain of section 4.8, between two and three on NVDA — but the counting misleads. The quotes cluster near the money of a handful of expiries, and each vanilla reads the field only through a smoothing integral along the diffusion's paths. Whole regions of the sheet — deep wings, sub-front rows, maturities beyond the last expiry — are therefore weakly determined no matter what the count says.

Figure 4.6 stages the experiment on the synthetic market. In panel (a), a strike cross-section of the calibrated sheet at $\tau = 0.5$ (solid), and the same sheet with one deliberate change: its deep-put column at $y = 0.45$, below every quoted strike, is pushed down by 10 full volatility points (dashed). Inside the shaded band — the quoted region — the two curves are indistinguishable; the change lives entirely in unquoted territory. In panel (b), the cost of the change where the market can see it: the absolute reprice difference at every quote of every expiry. The largest is 22.6 bp, the rms across the quote set is 2.5 bp, and the effect concentrates in the longest expiry's lowest strikes — the only quotes whose diffusion reaches the moved column at all. A thousand basis points of surface, twenty basis points of quotes.

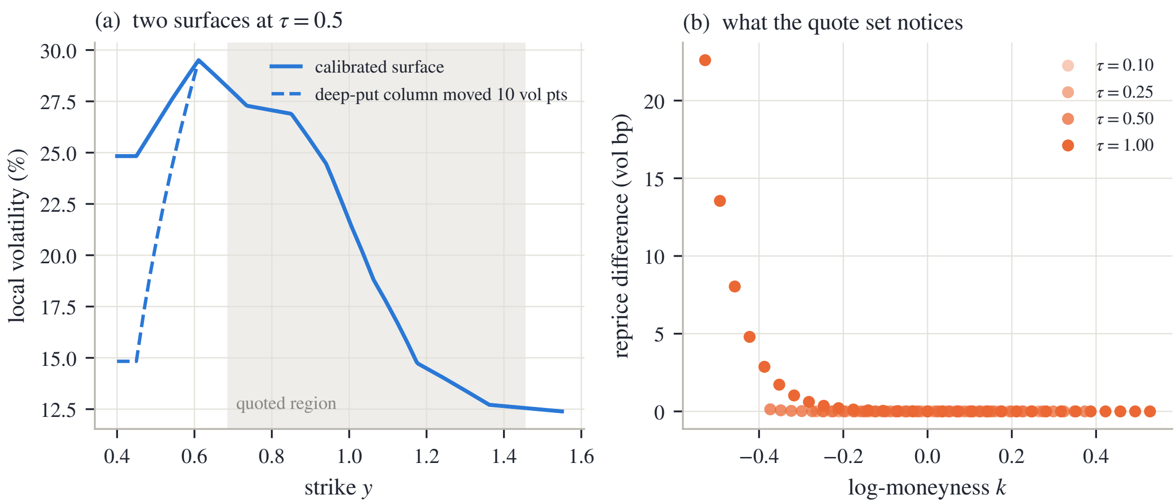

Figure 4.6: What sparse quotes do not see. (a) The calibrated synthetic sheet at $\tau = 0.5$ (solid) and the same sheet with its unquoted deep-put column pushed down by 10 volatility points (dashed); the shaded band is the quoted region. (b) The absolute reprice change at every quote of every expiry: at most 22.6 bp, concentrated where the longest expiry's lowest strikes feel the moved column.

What a calibration returns, then, is one *representative* of an equivalence class of sheets the data cannot tell apart, and every device in the next section's objective is a declared choice of representative. The roughness term selects the smoothest member near a reference. The box clips the unidentified extremes. The front tie pins the rows below the first expiry. The seed decides which basin a non-convex search lands in. None of this is hidden tuning; it is the "convention" column of the book's thesis, operating inside a calibration. And it is why section 4.8 reports quote error and surface error as two separate numbers: the first says how well the representative prices the market; only the second says how far the representative can be trusted *as the field.*

4.7 The objective and its derivatives

4.7.1 The objective

The pieces now assemble into a bound-constrained least-squares problem over the nodal variances:

$$\min_{\theta \in [v_{\text{lo}}, v_{\text{hi}}]^{n_\theta}} \; \tfrac{1}{2}\Big(\sum_q r_q(\theta)^2 \;+\; \lambda \left\|\Gamma(\theta - \theta_{\text{ref}})\right\|^2 \;+\; \lambda_0 \left\|\theta_{1,\cdot} - \theta_{2,\cdot}\right\|^2 \Big). \tag{4.9}$$

Its three blocks, in order: a data term, a roughness penalty, and a tie between the first two vertex rows — $\theta_{1,\cdot}$ is the $\tau = 0$ row, $\theta_{2,\cdot}$ the row at the first quoted expiry. Each block was argued for before it was written down; we take them in turn.

The *data block* compares the marched call c_θ at quote q's coordinates with the quoted one, in volatility units: $r_q = (c_\theta(\tau_q, y_q) - c_q)/\eta_q$, where η_q is one volatility percent of the quote's vega, floored. Recall what vega is: the derivative of the Black price with respect to implied volatility — the exchange rate between price units and volatility units. Dividing each price error by it makes every residual read as a volatility error, so a deep wing's tiny price errors and the money's large ones compete on the same scale; the floor (the same device as Chapter 2's vega floor η; this chapter's constant is in appendix 4.A) keeps the scale from vanishing where vega does. This is the residual design of section 2.6 — solve in price space, measure in volatility space — carried over unchanged, and when band edges rather than mids are trusted, the band objective of that section — residuals measured to the nearest bid–ask band edge, zero inside the band — applies verbatim.

The *roughness block* $\lambda\|\Gamma(\theta - \theta_{\mathrm{ref}})\|^2$ is Tikhonov regularization — the standard name for adding a penalty so that, among sheets fitting the data equally well, the optimizer returns the smoothest. The operator Γ takes second divided differences along strike within each maturity row and along time within each strike column — the discrete curvature of the sheet — normalized by the local vertex spacing, so that a tightly spaced money column is not penalized more than a wide wing one (on a uniform grid it reduces to the $(1, -2, 1)$ stencil). The penalty pulls toward a reference sheet θ_{ref} — flat, at the median ATM variance, in this chapter's protocol — and it is the declared answer to section 4.6: of the equivalence class the data allows, return the smoothest member near the reference.

The *front tie*, at weight λ_0, pins the $\tau = 0$ row to the first quoted row, closing the invisible sub-front subspace of section 4.6.

Two further row types plug into the same stack when their instruments are present, and are only named here. A variance-swap quote adds one scalar residual per quoted expiry, constraining the deep-put tail integral that vanillas cannot see — Chapter 5 develops the integral, and it is exactly the instrument that justifies releasing the wing slope a of remark 4.7. And prior structures from a previous fit enter as reference terms in the same Γ-and-θ_{ref} position — Chapter 10 develops when and how much to trust them.

4.7.2 Every derivative from the same factorization

A least-squares solver runs on the Jacobian: the matrix of derivatives of every residual with respect to every parameter. For a model priced by a PDE march, the naive route is finite differences — perturb θ_ℓ, re-march the surface, subtract — which costs n_θ extra marches per iteration, hundreds here. The sheet's structure gives something better: *exact* derivatives, for the price of extra right-hand sides through a factorization already computed. The key is linearity. By (4.5) the field depends linearly on θ, so the discrete generator does too, and its derivative in one parameter is itself a stencil:

$$\frac{\partial \mathcal{L}}{\partial \theta_\ell} = \mathcal{L}[\mathcal{H}_\ell],$$

the stencil (4.6) with the field v replaced by the single hat function $\mathcal{H}_\ell$ — zero outside vertex ℓ's star of triangles.

Proposition 4.12 (Tangent: same factor, extra right-hand sides)**.** *The sensitivities $s_\ell^n = \partial U^n / \partial \theta_\ell$ obey*

$$\mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]\, U^{n+1}, \qquad s_\ell^0 = 0, \tag{4.10}$$

where $\mathcal{L}[\mathcal{H}_\ell]\, U^{n+1}$ applies the boundary-inclusive stencil to the solution extended by its fixed boundary values. This is the discretization of the continuous sensitivity equation $\partial_\tau s_\ell = \frac{1}{2} v y^2 \partial_{yy} s_\ell + \frac{1}{2} \mathcal{H}_\ell\, y^2 \partial_{yy} c$.

Proof. Write one step over the interior unknowns as $\mathcal{M}^{n+1} U^{n+1} = U^n + b^{n+1}$, with b^{n+1} the boundary vector displayed after (4.7) — it carries the Dirichlet values through the stencil's boundary columns, and therefore depends on θ through the field at boundary-adjacent nodes. Differentiate the whole equation with respect to θ_ℓ, using the product rule on the left:

$$\left(\partial_{\theta_\ell} \mathcal{M}^{n+1}\right) U^{n+1} + \mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \partial_{\theta_\ell} b^{n+1}.$$

By linearity, $\partial_{\theta_\ell} \mathcal{M}^{n+1} = -\Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]$ exactly — no approximation is involved. Move it to the right-hand side:

$$\mathcal{M}^{n+1} s_\ell^{n+1} = s_\ell^n + \Delta\tau\, \mathcal{L}[\mathcal{H}_\ell]\, U^{n+1} + \partial_{\theta_\ell} b^{n+1},$$

and the last two terms combine into the boundary-inclusive product of the statement, because the boundary *values* themselves (1 and 0) do not move with θ — only the stencil weights through which they enter do. The start is $s_\ell^0 = 0$: the payoff does not depend on the field. $\square$

Read (4.10) twice. As a computation: the sensitivity system has the *same* matrix $\mathcal{M}^{n+1}$ as the value system — only the right-hand side differs — so every sensitivity marches behind the same tridiagonal factorization as the value solve. The entire n_θ-column Jacobian costs one factorization plus n_θ extra back-substitutions per step; the columns form a block of right-hand sides that a solver processes together. As probability: the source term says that vertex ℓ influences a price exactly through the diffusion's visits to ℓ's star of triangles, weighted by the convexity $\partial_{yy}c$ it finds there.

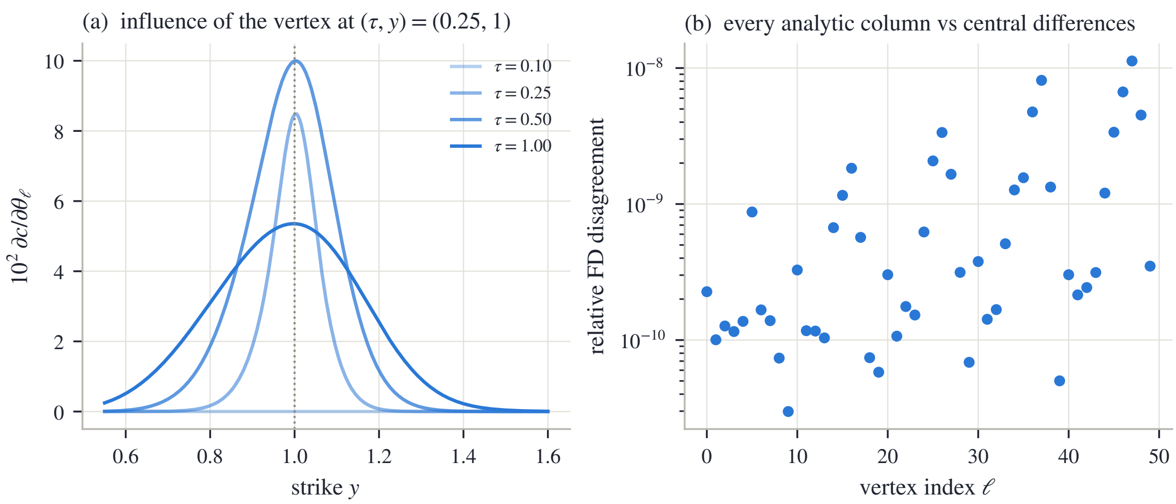

Figure 4.7: The tangent system, seen and audited. (a) Sensitivity of the priced call curve to the single vertex at $(\tau, y) = (0.25, 1)$, across the four quoted expiries. (b) Every analytic column of the Jacobian against a central finite difference of the full march, over all 50 vertices: worst relative disagreement 1.1×10^{-8}.

Figure 4.7 shows both readings on the synthetic market. In panel (a), the sensitivity of the priced call curve to the single vertex at $(\tau, y) = (0.25, 1)$, plotted across the four quoted expiries. At $\tau = 0.10$ the curve is identically zero: an earlier expiry cannot feel a later vertex, because the march has not reached it yet. At $\tau = 0.25$, the vertex's own maturity, the sensitivity is a sharp cone centered on its strike. At the two later expiries the cone spreads and flattens: the diffusion gradually forgets where it collected its variance. In panel (b), the audit: every analytic Jacobian column is compared against a central finite difference of the full march, over all 50 vertices, and the worst relative disagreement is 1.1×10^{-8}. The derivative is exact, not approximate.

The optimization itself is then standard machinery used plainly: a trust-region reflective least-squares iteration on (4.9) — a method of the Gauss–Newton family (linearize the residuals at the current iterate, solve the resulting linear least squares, repeat) that handles box bounds natively, shrinking its steps rather than leaving the box [9] — with the analytic Jacobian of proposition 4.12 supplied at every iterate and a flat seed at the reference sheet. The tangent route's Jacobian cost grows linearly with n_θ; the classical *adjoint* alternative — run the march backward once through its transposed steps, collecting the whole gradient in a single sweep, at a cost independent of the parameter count — is derived in appendix 4.A for the reader who will need it at larger grids.

4.8 Examples and the contract

All fits in this section use one protocol — the grids of section 4.6, the objective of section 4.7, every constant recorded in appendix 4.A — applied unchanged to a synthetic market whose answer is known and to the book's frozen SPY and NVDA chains. No fit is tuned separately.

4.8.1 The round trip, and its two separate numbers

The first test is a market whose answer is known. The synthetic field of appendix 4.A is priced through an *independently implemented* dense scheme — different grids, different code, so the test cannot flatter the chapter's march by validating it against itself — and quoted cleanly at four expiries: 124 quotes in all. The calibration starts from a flat seed. Following section 4.6, the result is reported as two numbers, not one. The *quote* number: the fitted sheet reprices the quotes to 0.6 bp rms, 1.8 bp at worst — in the space the optimizer sees, the fit is essentially exact. The *surface* number: the fitted field agrees with the true field to 0.78 volatility points rms over the quote-covered region, 3.10 points at worst — more than a hundred times the quote error.

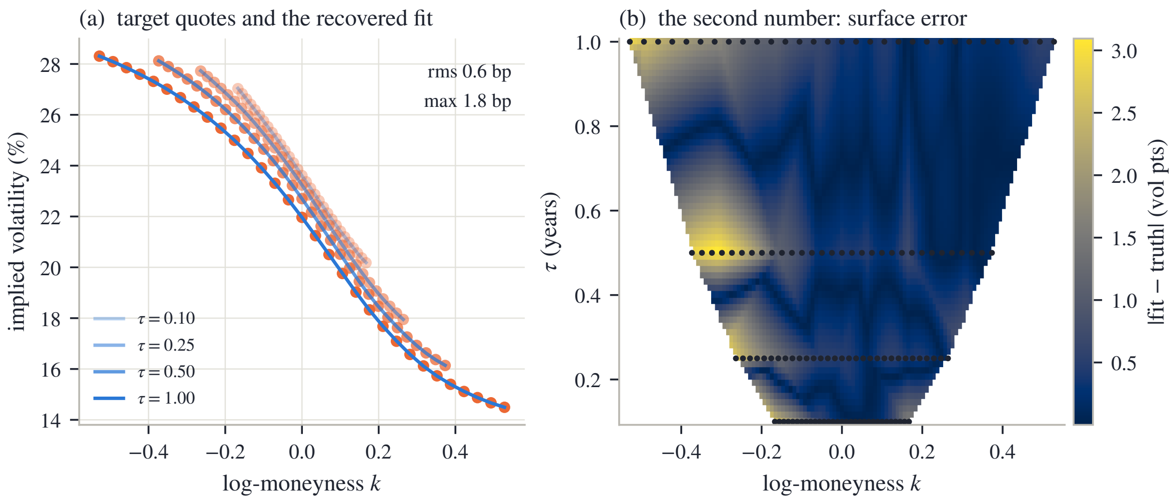

Figure 4.8: The synthetic round trip. (a) Target quotes (dots) and the recovered fit (lines) at the four expiries: rms 0.6 bp, max 1.8 bp. (b) |fitted − true| local volatility over the quote-covered region, quote positions dotted.

Figure 4.8 shows where each number lives. In panel (a), the four quoted smiles (dots) and the recovered fit (lines): the curves pass through the data, and the two numbers printed above — rms 0.6 bp, worst 1.8 bp — are all there is to say. In panel (b), the map of |fitted − true| local volatility over the quote-covered region, with the quote positions dotted. The error is not spread evenly: it concentrates between the expiries and toward the edges of each quoted span — exactly where fig. 4.6 said the quotes do not look, and largest where the nearest quote is farthest. Both numbers are correct at once: the quotes are matched almost perfectly, and the field between them is known only to a fraction of a volatility point.

4.8.2 The frozen chains

The real test fits one sheet per underlier to every mid quote of the snapshot's eight expiries at once: 1026 quotes against 216 nodal variances for SPY, 506 against 207 for NVDA. Figure 4.3 already showed the calibrated SPY sheet itself; fig. 4.9 shows what it prices. Its four panels are four expiries of the same surface, each priced by the *one* calibrated diffusion, against the frozen quotes (dots are mids, whiskers the bid–ask band; each panel's annotation gives that expiry's rms on the fitting operator). Read them as a progression. In the first panel, the two-day smile: narrow, steep, and carrying visibly the most residual of the four — the next figure isolates why. In the last, a year-and-a-half LEAP (market shorthand for a long-dated listed option): wide, gentle, and essentially exact. The two middle panels bridge the regimes, and the entire gallery comes from one list of positive numbers.

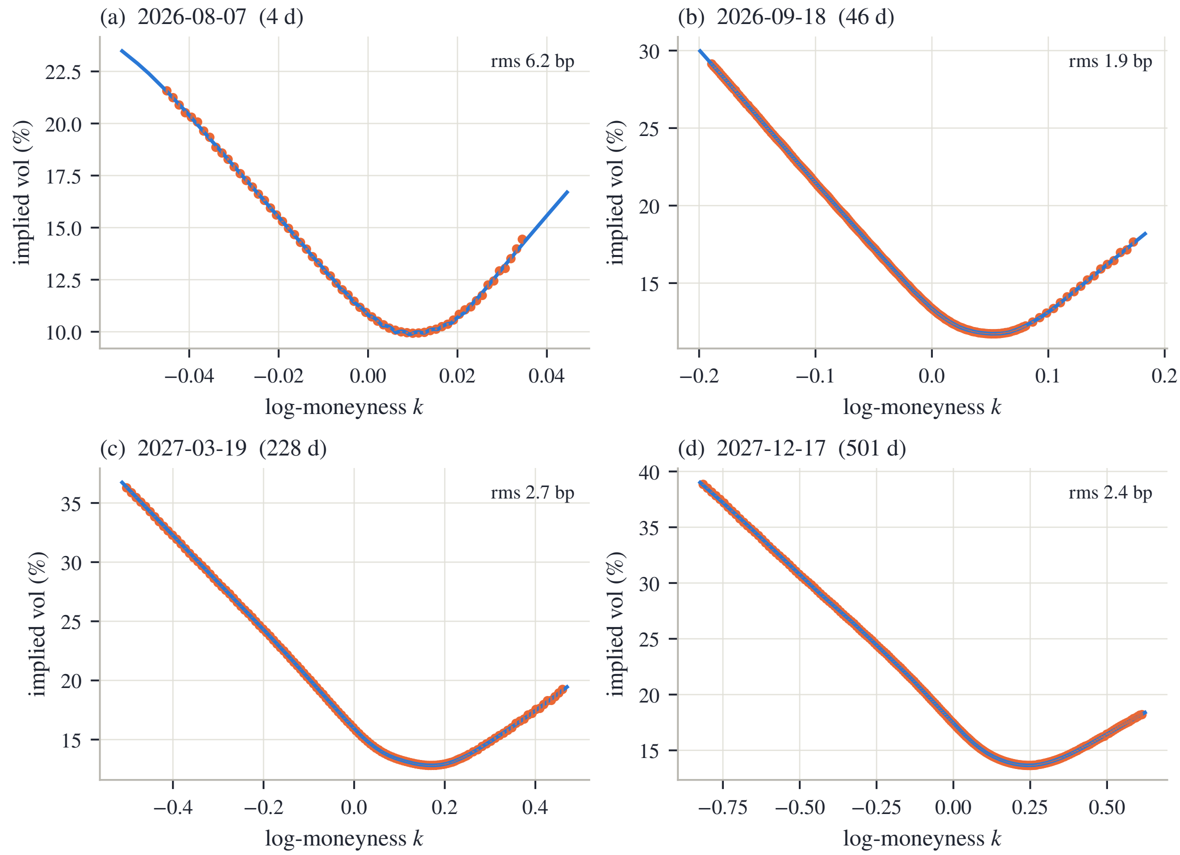

Figure 4.9: Four SPY expiries priced by the single calibrated sheet, against the frozen quotes (dots: mids; whiskers: bid–ask). Panel titles give the expiry and its distance; annotations the per-expiry rms on the fitting operator.

Figure 4.10 reports the chapter's two metrics side by side, per expiry and on a log scale: SPY in panel (a), NVDA in panel (b). The grey bars are the rms on the fitting operator — across all quotes, 2.7 bp for SPY and 16.1 bp for NVDA, whose wider spreads and harder short-dated smiles carry more residual everywhere. The blue bars are the same fitted sheets repriced on the refined operator of section 4.5: 13.6 and 46.1 bp, with the worst single expiry — in both names the one-day front — at 28.2 and 116.9 bp. The gap between the paired bars is the operator's blind spot made quantitative: the optimizer partially cancels the coarse march's discretization error at the expiries it is fitted on, the in-operator residual flatters accordingly, and the refined number is the one to quote. Where the pairs agree — the long end — the fit is genuinely converged. Where they diverge — day-scale maturities — the remaining error is discretization, not disagreement with the quotes.

The measured cleanliness promised by remark 4.11 completes the report. Across every expiry of both calibrated sheets, the minimum divided second difference of the marched prices is -8.5×10^{-13} (SPY) and -2.1×10^{-13} (NVDA), and the minimum adjacent-expiry increment is -8.9×10^{-16} and -7.8×10^{-16}. Both sit at the scale of solver rounding — as proposition 4.2 predicts, and as remark 4.11 declined to assume without measuring.

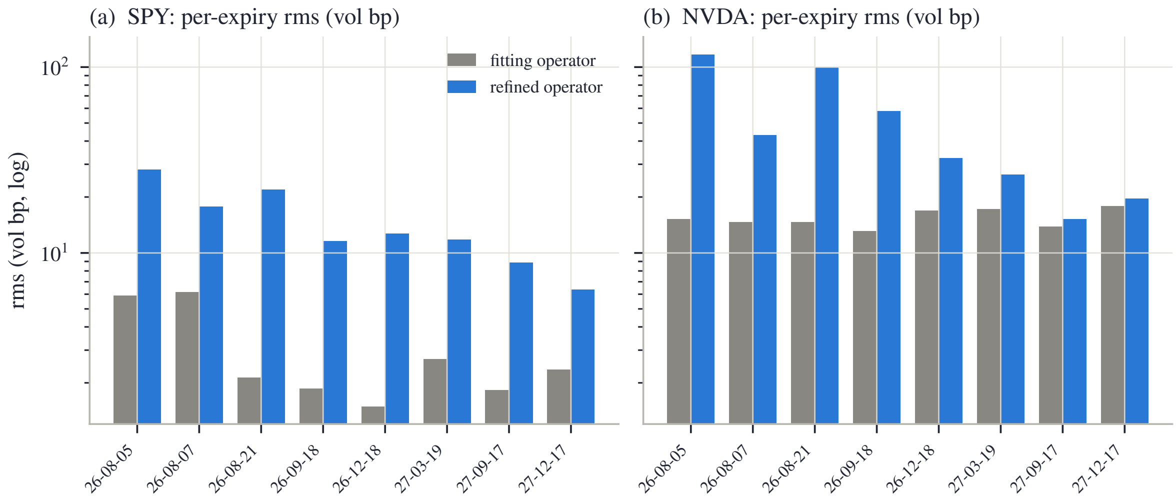

Figure 4.10: Per-expiry rms for SPY (a) and NVDA (b), on the fitting operator (grey) and on the refined operator (blue), log scale.

4.8.3 The contract

The chapter closes the way the book's model chapters do, with its contract. Read the table's three columns as decreasing strength. The first column holds by theorem, and no data can break it. The second column is not guaranteed but is *checked*, on every fit, and reported. The third column is what a user must not read into the model, with the chapter that takes each item up.

Structural	**Numerically controlled**	**Outside the claim**
Box bounds imply a positive field on the hull; a positive field implies butterfly-free, calendar-ordered marginals (propositions 4.2 and 4.5); every optimizer iterate is an admissible diffusion	Butterfly, calendar and bound cleanliness of the marched prices, measured per fit; refined-operator reprice beside the in-operator residual	Convexity preservation of the discrete scheme as a theorem; the extrapolation region beyond the hull, where basis weights are signed
Monotone, unconditionally ℓ^∞-stable march (proposition 4.9 and corollary 4.10); exact tangent Jacobian from the same factorization (proposition 4.12)	Tangent columns audited against finite differences; an independent discretization cross-checks the march	Uniqueness of the calibrated sheet: the data determines an equivalence class, the conventions of section 4.7 a representative
Bounded field $\Rightarrow$ implied variance bounded in strike, strictly inside Lee's cap on the hull (remark 4.8)	Grid placement rules resolving each expiry's $\sigma\sqrt{\tau}$; adaptive variance box keyed to ATM quotes	Jumps and scheduled-event spikes (the clock, Chapter 8); spot–volatility dynamics (Chapter 9); the deep-put tail's shape without a variance-swap quote (Chapter 5)

Table 4.1: The local-volatility model contract: theorem, numerical control, and excluded scope.

The book's separation thesis reads off the table. *Validity* is structural and cheap: a box on nodal values. *Information* is scarcer than the formula (4.3) pretends: the quotes read the field only through smoothing integrals, and only where the quotes exist. *Convention* fills the remainder — roughness, reference, front tie, wing, seed. The calibration does not eliminate those choices. It declares them, measures what the discretization leaks, and reports its error twice: once on the operator the optimizer saw, and once repriced on the refined one.

4.A Numerical facts and the chapter protocol

4.A.1 The protocol

Every fit in this chapter — synthetic and real — uses the following recipe; no constant is tuned per name.

Vertex grid. Maturity rows at $\tau = 0$ and at every quoted expiry. Strike columns delta-spaced: the ladder of put deltas $\{1, 2, 5, 10, 25, 40\}\%$, the same ladder mirrored on the call side, and the money, mapped to strikes through $y = \exp\left(\sigma_A \sqrt{\tau_{\max}}\, \Phi^{-1}(\cdot)\right)$ with σ_A the longest expiry's ATM volatility; the two end columns are extended to cover the pooled quoted range, so no quote falls outside the hull. Coverage rule: every expiry must own at least 6 columns inside its own traded range (section 4.6), enforced as follows: list the columns falling inside the expiry's range together with the range's two endpoints, and while the widest gap between consecutive entries exceeds one fifth of the range, split that gap with a new column; this is what densifies the short-dated money in fig. 4.3.

Lattice. Strike step

$$\Delta y = \min\left(0.01,\ 0.15\, \min_e \sigma_e \sqrt{\tau_e}\right),$$

capped at 800 nodes over $[0,\ \max(2.5,\ 1.05\, y_{n_y})]$ with y_{n_y} the highest vertex column, and $y = 1$ a node; every quoted expiry is a time node and each maturity interval receives $\max(8, \lceil \Delta\tau_{\text{int}}/0.01 \rceil)$ implicit steps. The refined (audit) operator halves the strike step and quadruples each interval's step count.

Box and objective. Adaptive box

$$v_{\text{lo}} = \min\left(0.0025,\ (\tfrac{1}{2} \min_e \sigma_{\text{ATM},e})^2\right), \qquad v_{\text{hi}} = \min\left(\max(0.36,\ (3\,\sigma_{\max})^2),\ 16\right)$$

— the floor a fraction of the smallest ATM implied per the averaging fact, the cap a multiple of the largest implied in view. Seed and reference $\theta^0 = \theta_{\text{ref}}$ flat at the median ATM variance. Weights $\lambda = 10^{-2}$, $\lambda_0 = 1$; residual scale η_q equal to one volatility percent of the quote's Black vega, floored at 10^{-3}. The snapshot carries no variance-swap quotes, so the wing multiple stays at its default $a = 0$ (flat clamp) throughout.

Solver. Trust-region reflective least squares [9] with the analytic Jacobian of proposition 4.12, termination tolerances 10^{-8}, at most 200 objective evaluations; the fits of section 4.8 converged in 116 (SPY), 94 (NVDA) and 20 (synthetic) evaluations from their flat seeds.

4.A.2 The synthetic market

The running example's truth is the smooth skewed field

$$\sigma_{\text{loc}}(\tau, y) = 0.21 - 0.10\, \tanh\left(\frac{y-1}{0.22}\right) + 0.03\, e^{-2\tau}, \qquad v = \sigma_{\text{loc}}^2,$$

priced by an *independently implemented* dense march (uniform $\Delta y = 0.0025$ on $[0, 3]$, $\Delta\tau = 0.002$, fully implicit) — a second discretization on different grids, so the round trip of fig. 4.8 cross-checks the chapter's march rather than validating it against itself. Quotes are read at the four expiries $\tau \in \{0.10, 0.25, 0.50, 1.00\}$, at 31 strikes per expiry uniform in the standardized moneyness $\zeta \in [-2.2, 2.2]$ of section 3.7, with $\sigma_{\text{ref}} = 0.24$, i.e. $k = 0.24\sqrt{\tau}\,\zeta$ (the coarse arm of fig. 4.2a uses 13). The “noise” of section 4.3 is a deterministic alternating ± 50 bp volatility ripple — no randomness anywhere in the chapter's figures. The synthetic sheet has rows $\{0\} \cup \{\text{expiries}\}$ and the delta-spaced columns at scale 0.24 plus one deliberate deep-put column at $y = 0.45$ for the experiment of fig. 4.6; its box is $[0.005, 0.36]$.

4.A.3 The adjoint

The tangent route of proposition 4.12 prices the Jacobian at $O(N_\tau N_y\, n_\theta)$ per evaluation, for a march of N_τ steps on N_y lattice nodes. The classical alternative makes the cost of a *gradient* independent of n_θ; deriving it costs a few lines, so the chapter records it for the grids on which it will one day pay. The derivation follows the standard optimal-control pattern — adjoin the marching constraints with multiplier vectors p^n, the *adjoint states*, require stationarity, and read the result as a backward march; a reader meeting adjoints for the first time can treat (4.11)–(4.12) as a recipe and check it numerically against the tangent Jacobian.

Proposition 4.13 (Adjoint: one backward sweep)**.** *Let $\mathcal{J}(\theta)$ be a scalar objective depending on the march through the observed values $\mathcal{O}_n U^n$, with $\mathcal{O}_n$ fixed observation rows. Define adjoint states by the backward recursion*

$$(\mathcal{M}^n)^\top p^n = \mathcal{O}_n^\top \frac{\partial \mathcal{J}}{\partial(\mathcal{O}_n U^n)} + p^{n+1}, \qquad p^{N+1} = 0. \tag{4.11}$$

Then

$$\frac{\partial \mathcal{J}}{\partial \theta_\ell} = \sum_n \Delta\tau\, (p^n)^\top\, \mathcal{L}[\mathcal{H}_\ell]\, U^n, \tag{4.12}$$

with the boundary-inclusive product of proposition 4.12, at the cost of one backward solve, $O(N_\tau N_y)$, plus the $O(n_\theta)$ accumulation of (4.12).

Proof. Form the Lagrangian (fraktur $\mathfrak{L}$, kept visually apart from the stencil operator $\mathcal{L}$) $\mathfrak{L} = \mathcal{J} - \sum_n (p^n)^\top (\mathcal{M}^n U^n - U^{n-1} - b^n)$, with b^n the boundary terms of proposition 4.12. Stationarity in each U^n — which appears in step n's system and in step $n{+}1$'s right-hand side — gives (4.11); the θ-derivative of $\mathfrak{L}$ then retains the explicit dependence of $\mathcal{M}^n$ and b^n, which combine as in proposition 4.12 into the boundary-inclusive (4.12). Structurally this is summation by parts: the transpose of a lower block-bidiagonal system is upper block-bidiagonal, i.e. a backward march with transposed tridiagonal factors. □

4.A.4 Measured cleanliness

The diagnostics quoted in section 4.8 are computed on the fitting operator's price grid: the butterfly proxy is the minimum divided second difference $(U_{i+1} - U_i)/\Delta y_i^+ - (U_i - U_{i-1})/\Delta y_i^-$ over all interior nodes and all expiries; the calendar proxy is the minimum of $U^{(e+1)} - U^{(e)}$ over adjacent expiries at fixed y (the lattice prices are already in the normalized units of section 2.8, so the fixed-y comparison is the right one); and the bounds check verifies $0 \le U \le 1$ to the same tolerance. For both calibrated sheets all three sit at solver rounding (section 4.8); they are measured per fit precisely because remark 4.11 declines to promote them to a theorem.

Chapter 5

Integrals and Wings: Variance Swaps Beyond the Last Quote

The three model chapters each ended with a fitted curve that reprices every quote. This chapter asks a harder question of those curves: what do they say where there are no quotes? The probe is a single traded number, the fair strike of a variance swap. It is an integral of the whole smile, it can be computed exactly in each of Part I's three coordinate systems, and on the book's frozen data — the one market snapshot every example reads — a fifth of it comes from strikes beyond the last quote. Models that agree to basis points at every quote therefore disagree on it. The chapter measures that disagreement, proves the two model-free facts that bound it — a ceiling on how steep any wing can be, and an envelope, fanning out from the last quote, that no arbitrage-free continuation can leave — and shows why everything inside the envelope is a choice the modeler must make and disclose. It closes with the term structure: the calendar condition of Chapter 2 forces the fair strike to grow with maturity, so the integral inherits the order the surface already keeps.

5.1 One number for the whole smile

Chapters 2–4 built three ways of fitting one expiry's smile, and each chapter ended the same way: with a curve that reprices every quote to a few basis points. This section introduces a single number, attached to the whole smile at once, that will spend the rest of the chapter testing those curves in the region the quotes never reach.

The number comes from a real instrument. A *variance swap* is a contract on how much the underlying actually moves. Over the life of the contract, each day's log-return is squared, the squares are summed, and at expiry the buyer receives that realized sum and pays a fixed amount agreed at the start. The fixed amount is called the *strike* of the swap, and the strike at which the contract is worth zero today is the *fair strike*. Quoting desks publish fair strikes the way they publish option prices. So the fair strike is not a modeling construct: it is a market number, one per (underlier, expiry), and a fitted smile implies a value for it.

To compute that implied value, work in the book's standing units. The underlying is measured as $Y = S/F$, the level divided by the forward, so Y starts at 1 and is a positive martingale — a process with no drift, whose expected future value is its current value (section 4.1). Time is variance time τ — the maturity variable the book has quoted variance against since Chapter 2, whose clock Chapter 8 will make precise — so a smile's total implied variance at log-moneyness k is $w(k) = \sigma(k)^2\tau$, with $\sigma(k)$ the implied volatility there. Let the martingale move as a diffusion,

$$\mathrm{d}Y_s = \sqrt{v_s}\, Y_s\, \mathrm{d}W_s,$$

where s runs from 0 to the expiry τ and v_s is the *instantaneous variance* at time s — the squared volatility the process is running at that instant. In the representative model of Chapter 4, v_s

is the field evaluated along the path, $v_s = v(s, Y_s)$; but nothing below needs that: v_s may be any reasonable random process. As the monitoring interval shrinks, the sum of squared returns converges to the accumulated instantaneous variance, so in these units the realized variance the swap pays is the integral $\int_0^\tau v_s \, \mathrm{d}s$, and the fair strike is its expected value under the pricing law — the law the smile encodes, the same one behind every expectation in this book. (Two conventions separate this from the listed contract, and it is worth stating them once, here. The contract sums squared *daily* returns while the integral monitors continuously, and the contract's clock is the calendar while ours is variance time; both refinements have standard corrections and Chapter 8 returns to the clock. A third gap — paths that jump — is genuine and reappears in the chapter's closing table of guarantees, section 5.7.1.)

The expected value looks like it should depend on the whole path law of Y. It does not, and the two-line computation that shows this is the heart of the chapter. Apply Itô's formula — the chain rule for diffusions, which adds a second-derivative term — to $\log Y_s$, using the multiplication rule $(\mathrm{d}W_s)^2 = \mathrm{d}s$, so that $(\mathrm{d}Y_s)^2 = v_s Y_s^2 \, \mathrm{d}s$:

$$\mathrm{d} \log Y_s = \frac{\mathrm{d}Y_s}{Y_s} - \frac{1}{2} \frac{(\mathrm{d}Y_s)^2}{Y_s^2} = \sqrt{v_s} \, \mathrm{d}W_s - \frac{1}{2} v_s \, \mathrm{d}s. \tag{5.1}$$

Integrate from 0 to τ and use $\log Y_0 = 0$:

$$\log Y_\tau = \int_0^\tau \sqrt{v_s} \, \mathrm{d}W_s - \frac{1}{2} \int_0^\tau v_s \, \mathrm{d}s.$$

Now take expectations. The first integral is a stochastic integral against a Brownian motion, and such integrals have expectation zero — they are the running gains of a bet with no edge. Writing $X = \log Y_\tau$ for the terminal log-return, the surviving terms rearrange to

$$w_{\mathrm{vs}} \;=\; \mathbb{E}\left[\int_0^\tau v_s \, \mathrm{d}s\right] \;=\; -2 \, \mathbb{E}[X]. \tag{5.2}$$

The expected realized variance equals minus twice the expected log-return. The right-hand side no longer mentions the path: it is a property of the *terminal law alone*, the same law every smile of this book encodes. A path-dependent payoff has been priced by a European one — the payoff $-2 \log y$, called the *log contract*. Chapter 2 already met this number in passing, as (2.34); this chapter is about taking it seriously. We write w_{vs} for the fair strike in total-variance units and $\sigma_{\mathrm{vs}} = \sqrt{w_{\mathrm{vs}}/\tau}$ for the same number quoted as a volatility.

Before asking what the number says about a real smile, compute it once by hand on the simplest smile there is. Suppose the smile is flat: $w(k) = w_0$ at every strike. A flat smile means the law of X is exactly the one Black–Scholes assumes, a normal with variance w_0. Its mean is pinned by the martingale condition $\mathbb{E}[e^X] = \mathbb{E}[Y_\tau] = Y_0 = 1$: for a normal with mean μ_X and variance w_0, the standard identity $\mathbb{E}[e^X] = e^{\mu_X + w_0/2}$ forces

$$\mu_X + \frac{w_0}{2} = 0 \qquad \Longrightarrow \qquad \mathbb{E}[X] = -\frac{w_0}{2},$$

and (5.2) gives

$$w_{\mathrm{vs}} = -2 \, \mathbb{E}[X] = w_0. \tag{5.3}$$

On a flat smile the fair strike *equals* the ATM total variance, exactly. This small computation earns its keep twice over. It is a sanity check — a variance swap on a constant-volatility world is worth that variance. And it is a decomposition: whatever a real fair strike shows *beyond* the ATM level is entirely a statement about the smile's departure from flatness — its curvature and, above all, its wings. Section 5.7 will measure that excess on real data and find it worth hundreds of basis points.

Why the wings, specifically? Look at the shape of the payoff. The function $-2 \log y$ is large where y is far from 1, and it grows without bound as $y \to 0$. At the same time the martingale condition $\mathbb{E}[e^X] = 1$ forces a balance: a law cannot put weight on large upside without compensating weight below, and every rearrangement of tail mass moves $\mathbb{E}[X]$. The fair strike therefore loads exactly on the part of the law the quoted strikes describe least — a suspicion the next two sections make precise and then quantify.

Figure 5.1 shows why the number deserves a chapter. It fits the three families of Part I — LQD from Chapter 2, SVI and MCS from Chapter 3 — to the same 94 mid quotes (bid–ask midpoints) of the book's running node: the (underlier, expiry) pair SPY December 2026 that Chapters 2–4 kept returning to. The fits follow Chapter 3's comparison protocol (section 3.8) — same quotes, equal weights, each family's reference settings. In panel (a), the quoted span: the three fitted curves pass through the same quotes and through each other — the worst fit error of any family is 7.5 bp, and no two of the curves are ever farther than 13.3 bp apart on the span (one vol bp is 10^{-4} of absolute volatility, the chapter's unit for volatility distances). By every measure the data offers, the three models are the same smile. In panel (b), the same three fits drawn wide, in total variance, with the quoted span shaded: outside the shading the curves separate — each family continues the smile according to its own structure — and the legend prints each family's implied fair strike. The three values of σ_{vs} span 58.2 vol bp: LQD says 19.83%, SVI says 19.33%, MCS says 19.25%. Nothing has gone wrong. Three valid laws thread the same quotes and integrate to different numbers.

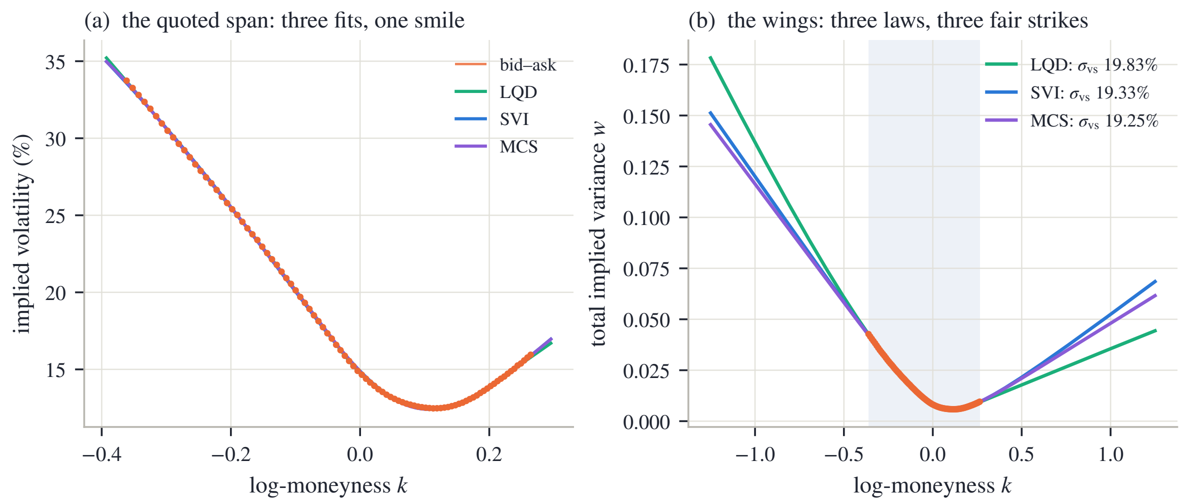

Figure 5.1: Three families, one quote set, three fair strikes (SPY December 2026, 94 quotes). (a) On the quoted span the LQD, SVI and MCS fits agree with the quotes and with each other to 13.3 bp. (b) In total variance, wide: beyond the shaded quoted span the three curves separate, and their fair var-swap strikes differ by up to 58.2 vol bp.

We now have the chapter's protagonist and its puzzle. The fair strike is one market-traded number that the whole smile determines — and that fitted smiles, in perfect agreement wherever quotes exist, do not agree on. The plan follows the thesis of the book. Section 5.2 computes the number exactly, in each model's own coordinates, and checks that all routes agree on one model at a time. Section 5.3 locates the disagreement: it lives precisely in the share of the integral that accrues beyond the last quote. Sections 5.4 and 5.5 then prove what *can* be said with no quotes at all — a ceiling and an envelope — and sections 5.6 and 5.7 treat what remains inside those bounds for what it is: a convention to be stated, policed at finite strikes, and kept consistent across the term structure.

5.2 Three integrals for one number

We have one number, $w_{\mathrm{vs}} = -2\,\mathbb{E}[X]$, defined by the terminal law. But the three model families of Part I hold that law in three different coordinate systems — three *charts*, in the word Chapter 3 introduced for a choice of coordinates: Chapter 3's families state the *smile* $w(k)$, Chapter 2's model states the *law* itself through its quantile, and Chapter 4's model states the *field* $v(\tau, y)$ that generates every law at once. This section derives one exact integral for w_{vs} in each chart. The three integrals look nothing alike, and watching them return the same number — to a tenth of a basis point — is the section's payoff.

5.2.1 Against option prices

The first route rewrites the expectation of *any* smooth payoff as a portfolio of the options themselves.

Proposition 5.1 (Spanning). *Let φ be twice continuously differentiable on $(0, \infty)$ and let $Y > 0$ have $\mathbb{E}[Y] = 1$. Write c and P for the normalized call and put at strike K, i.e. $c = \mathbb{E}[(Y - K)^+]$ and $P = \mathbb{E}[(K - Y)^+]$. Then, whenever the integrals converge,*

$$\mathbb{E}[\varphi(Y)] = \varphi(1) + \int_0^1 \varphi''(K)\, P(K)\, \mathrm{d}K + \int_1^\infty \varphi''(K)\, c(K)\, \mathrm{d}K. \tag{5.4}$$

Proof. The claim reduces to a pointwise identity: for every $y > 0$,

$$\varphi(y) = \varphi(1) + \varphi'(1)(y-1) + \int_0^1 \varphi''(K)(K-y)^+ \mathrm{d}K + \int_1^\infty \varphi''(K)(y-K)^+ \mathrm{d}K. \tag{5.5}$$

Check it for $y \ge 1$; the case $y < 1$ is the mirror image. For $y \ge 1$ the put-type kernel $(K - y)^+$ vanishes on $K \le 1$, and the call-type kernel is nonzero only for $K \in [1, y]$, so the right-hand side's integrals reduce to $\int_1^y \varphi''(K)(y-K)\,\mathrm{d}K$. Integrate by parts once, differentiating $(y - K)$:

$$\int_1^y \varphi''(K)(y-K)\,\mathrm{d}K = \Big[\varphi'(K)(y-K)\Big]_1^y + \int_1^y \varphi'(K)\,\mathrm{d}K = -\varphi'(1)(y-1) + \varphi(y) - \varphi(1).$$

Substituting back into (5.5) cancels everything but $\varphi(y)$. Now set $y = Y$ in (5.5) and take expectations. The linear term carries $\mathbb{E}[Y] - 1 = 0$ and dies; the two kernels become put and call prices. □

In words: any smooth payoff is a constant, plus a forward position, plus a strip of puts below the forward and calls above it, held in amounts given by the payoff's second derivative. The identity is completely model-free — it never asks where the prices came from.

Now specialize to the log contract, $\varphi(y) = -2\log y$. Then $\varphi(1) = 0$ and $\varphi''(y) = 2/y^2$, so (5.4) reads

$$w_{\mathrm{vs}} = 2\int_0^1 \frac{P(K)}{K^2}\,\mathrm{d}K + 2\int_1^\infty \frac{c(K)}{K^2}\,\mathrm{d}K.$$

Every strike appears, weighted by $1/K^2$: the variance swap is a strip of out-of-the-money options, densest below the forward. For computation it is tidier in log-moneyness. Substitute $K = e^k$, $\mathrm{d}K = e^k \mathrm{d}k$, and index the same prices by log-strike, writing $c(k)$ for $c(e^k)$ and likewise for the put; each $1/K^2\,\mathrm{d}K$ becomes $e^{-k}\mathrm{d}k$. Now recall put–call parity in normalized units ($P(k) = c(k) + (e^k - 1)$, from eq. (2.2)) and write $(x)^- = \min(x, 0)$ for the negative part. On the call leg ($k > 0$) the quantity $e^k - 1$ is positive, so $(e^k - 1)^- = 0$ and the bracket below is just the call; on the put leg ($k < 0$) it is negative, so $(e^k - 1)^- = e^k - 1$ and the bracket is $c(k) + (e^k - 1) = P(k)$, exactly the put parity demands. One bracket therefore covers both legs, and the two integrals merge into one integral over the smile:

$$w_{\mathrm{vs}} = 2\int_{-\infty}^{\infty} \Big[B(k, w(k)) + (e^k - 1)^-\Big]\, e^{-k}\,\mathrm{d}k, \tag{5.6}$$

where B is the Black call function of eq. (2.3), so $B(k, w(k)) = c(k)$ is the smile's call price. Equation (5.6) consumes nothing but the total-variance curve $w(k)$. Any model that can state its smile can be integrated — this is the natural route for SVI and MCS, whose $w(k)$ is a formula.

5.2.2 Against the law

The second route never leaves the law. The mean of any function of X can be taken in rank coordinates — this is eq. (2.21), the change of variables under which Chapter 2 built everything: if U is uniform on $(0, 1)$ then $Q(U)$ has the law of X, so integrating the quantile function Q over ranks *is* taking the mean. Applying this to $w_{\mathrm{vs}} = -2\,\mathbb{E}[X]$ gives Chapter 2's eq. (2.34):

$$w_{\mathrm{vs}} = -2\int_0^1 Q(u)\,\mathrm{d}u = -2\int_{-\infty}^{\infty} x(z)\,\rho(z)\,\mathrm{d}z,$$

where the second form substitutes the log-odds coordinate z, with $x(z)$ the transport and ρ the logistic density (section 2.2). For an LQD slice — one expiry's fitted law — this is one dot product against arrays the slice has already built on the fixed log-odds grid every slice carries: no Black function, no inversion, no new grid. The integrand $-2\,x(z)\rho(z)$ decays like $|z|e^{-|z|}$ — Chapter 2's exponential tails make the transport asymptotically linear in z, and the logistic weight supplies the $e^{-|z|}$ — so the slice's grid exhausts it.

5.2.3 Along paths

The third route uses neither prices nor the terminal law, but the field. Return to the left-hand side of (5.2), the expected accumulated variance, in the representative model of Chapter 4 where $v_s = v(s, Y_s)$. The expectation of a time integral is the time integral of expectations, and each instant's expectation is an integral against that instant's density $f(s, y)$ — the marginal density of Y_s that section 4.2 evolved with the forward equation:

$$w_{\mathrm{vs}} = \mathbb{E}\Big[\int_0^\tau v(s, Y_s)\,\mathrm{d}s\Big] = \int_0^\tau \mathbb{E}\big[v(s, Y_s)\big]\,\mathrm{d}s = \int_0^\tau\!\!\int_0^\infty v(s, y)\,f(s, y)\,\mathrm{d}y\,\mathrm{d}s. \tag{5.7}$$

Read the double integral as a pairing of two surfaces: the field $v(s, y)$, against the sheet of densities $f(s, y)$ that says how much time the paths spend at each level. The density sheet is called the *occupation measure* of the diffusion, and (5.7) says the fair strike is the field averaged over where the paths actually go.

Computing (5.7) does not require assembling f. Define the *expected remaining variance*

$$\mathcal{W}(s, y) = \mathbb{E}\Big[\int_s^\tau v(r, Y_r)\,\mathrm{d}r \;\Big|\; Y_s = y\Big],$$

the fair strike of a swap started at time s from level y. It satisfies a backward equation of exactly the form Chapter 4 marched, with the field itself as a source term:

$$\partial_s\mathcal{W} + \tfrac{1}{2}\,v(s, y)\,y^2\,\partial_{yy}\mathcal{W} + v(s, y) = 0, \qquad \mathcal{W}(\tau, \cdot) = 0, \qquad w_{\mathrm{vs}} = \mathcal{W}(0, 1). \tag{5.8}$$

The derivation is two applications of ideas already in hand — Itô's formula and the martingale property — and is written out in appendix 5.A. Operationally: one backward sweep of the same solver Chapter 4 marched forward, adding $v\,\Delta s$ at each step, and the number is read off at the spot. This is the natural route for the local-volatility model, which owns neither a smile formula nor a terminal quantile — but owns the field.

5.2.4 One number

Collecting the three routes with the definition they all serve:

$$\boxed{w_{\rm vs} = -2\,\mathbb{E}[X] = 2\int_{\mathbb{R}} \Big[B(k, w(k)) + (e^k - 1)^-\Big] e^{-k}\mathrm{d}k = -2\int_{\mathbb{R}} x(z)\,\rho(z)\,\mathrm{d}z = \int_0^T\!\!\int v\,f\,\mathrm{d}y\,\mathrm{d}s.} \tag{5.9}$$

One number; an integral against the smile, an integral against the law, an integral against the field. Each equality is a theorem, so any two routes evaluated on the same model must agree — and checking that they do is a free, end-to-end audit of an implementation. (Route two costs one *quadrature* — one numerical integral — and the audits below are limited only by quadrature accuracy.)

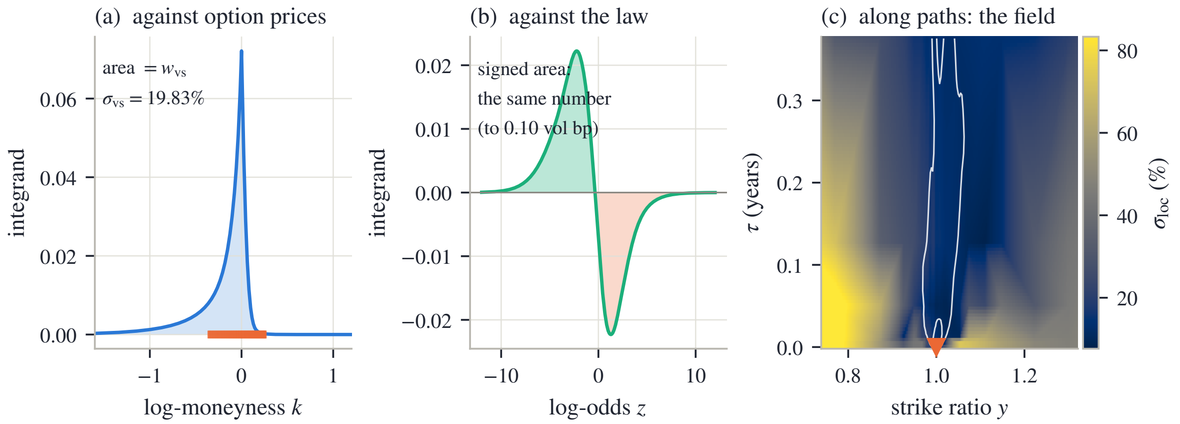

Figure 5.2: One number, three integrands. (a) The strike-side integrand of (5.6) on the running LQD slice; the orange ticks mark the quoted strikes, and the shaded area is $w_{\rm vs}$. (b) The law-side integrand $-2\,x(z)\rho(z)$ on the same slice's log-odds grid; the signed area is the same number to 0.10 vol bp. (c) The field route on Chapter 4's calibrated SPY sheet: the local volatility as a heatmap, the occupation measure as white density contours funnelling out of the spot, and the marker at $(y, \tau) = (1, 0)$ where the backward solve (5.8) reads off the number.

Figure 5.2 draws the three integrands for real fitted objects, and it repays a slow look. In panel (a), the strike side on the running LQD slice of fig. 5.1: the integrand of (5.6) peaks near the money — the e^{-k} weight decays, but the option prices it multiplies decay much faster — and the orange ticks show how little of the strike axis the quotes occupy. The area under this curve is the fair strike, $\sigma_{\rm vs} = 19.83\%$. In panel (b), the law side of the *same* slice: the integrand $-2\,x(z)\rho(z)$ against log-odds. It has two lobes, because $-2x(z)$ is positive on the downside ranks $(x < 0)$ and negative on the upside: the green lobe is downside outcomes pushing the fair strike up, the orange lobe upside outcomes pulling it down. The lobes are visibly unequal — that asymmetry is the skew — and their *signed* area agrees with panel (a)'s area to 0.10 vol bp: two utterly different integrals, one number, the agreement limited only by quadrature. In panel (c), the field route on a different model of the same market: Chapter 4's SPY sheet, the field fitted to all eight expiries at once. The heatmap is the calibrated local volatility; the white contours are the occupation measure — the densities $f(s, y)$ fanning out of the spot as time runs up the axis — and the pairing (5.7) weighs the heatmap by the contours. The marker at the origin of the fan is where (5.8) reads the answer.

The field route's audit deserves its own sentence, because it teaches the same lesson as Chapter 4's two operators. On the sheet, the strike-side route (5.6) and the field-side route (5.8) are both computed by finite differences, and both carry a first-order bias in the time step. At the march's own step the two reads differ by 23.2 vol bp; refine the step eightfold and the gap collapses to 2.9 vol bp, halving with each halving of the step. The mathematics says the integrals are equal; the computation agrees — in the limit, at the first-order rate the schemes promise, and measuring that convergence is the audit.

One structural observation organizes all of this, and it is worth stating because it predicts which route each model finds cheap. The fair strike is *linear* in the option prices and *linear* in the quantile Q — both appear under plain integral signs in (5.9) — so a model whose own coordinates are one of those objects gets the number, and by the same linearity its parameter derivatives, for the price of one quadrature: LQD holds Q, and the law route is a dot product. It is *not* linear in the smile $w(k)$ — the smile enters (5.6) through the nonlinear Black function — so a model whose coordinates are the smile must pay the Black evaluations: cheap when $w(k)$ is a formula, as for SVI and MCS, and expensive precisely when the smile must be *solved for* point by point. The field route's economy has a third source. Its pairing (5.7) is against an occupation measure the field itself generates, so the number is not linear in v. What saves it is (5.8): $\mathcal{W}$ solves a *linear* equation in which v enters only as a coefficient and a source, so the number still costs one backward sweep. This section's rule, stated once: *evaluate a shared functional in each model's own chart.* It stands behind every route above.

We can now compute the number exactly, in any model, and trust the result. The next question is what part of it the quotes actually pin down.

5.3 Where the number lives

We can compute the fair strike three ways and get one answer per model. Figure 5.1 showed that different models give different answers. This section locates the disagreement on the strike axis, and the location explains everything.

The tool is the accrual function. Write $\mathcal{S}(k)$ for the share of the strike-side integral (5.6) accrued by strikes up to k: run the integral from the far left to k, divide by the whole. Then $\mathcal{S}$ climbs from 0 to 1, and reading it at two strikes tells you what fraction of the fair strike lives between them.

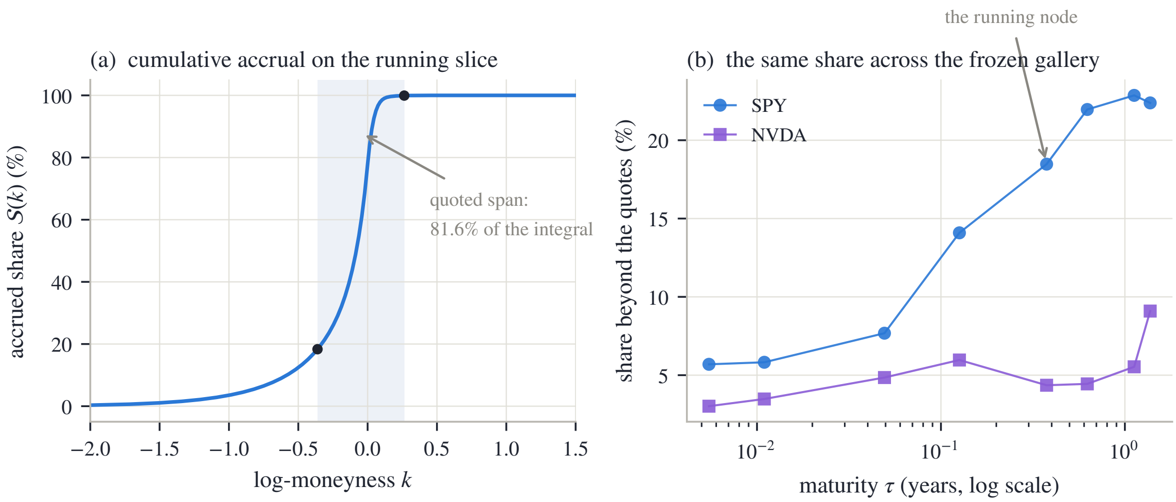

Figure 5.3: Where the fair strike accrues. (a) The accrual share $\mathcal{S}(k)$ on the running slice, with the quoted span shaded and its endpoints dotted: the span carries 81.6% of the integral, so 18.4% accrues where there are no quotes at all. (b) The beyond-quotes share for all sixteen frozen nodes (the snapshot's stored fits), against maturity: from 3.0% to 22.9%, growing with maturity for SPY.

Figure 5.3(a) draws $\mathcal{S}$ on the running slice, with the quoted span shaded. Follow the curve from the left. It is essentially flat until $k \approx -1$: the very deep puts, for all their $1/K^2$ weighting, carry prices too small to matter. It then climbs steeply through the belly — $|k| \leq 0.10$ alone accrues 52.4% of the number — and flattens again on the call side, where prices die quickly. The two dots mark the edges of the quoted span: between them the market's quotes constrain the integrand directly, and that stretch carries 81.6% of the integral. The remaining 18.4% accrues at strikes where no quote exists — where the integrand is whatever the fitted model's continuation says it is. About a fifth of this traded number is, on this node, pure extrapolation.

Panel (b) repeats the measurement across the whole frozen gallery — all sixteen (underlier, expiry) nodes of the book's snapshot, each read from the fit it was published with (the snapshot's stored fits; appendix 5.A records the provenance) and each with its own quoted span. The beyond-quotes share runs from 3.0% to 22.9%. The pattern is worth reading. What protects a node is not how many quotes it has but how many *widths of the smile* its quoted span covers. A strike matters through its standardized distance from the money, $k/(\sigma\sqrt{\tau})$ with σ that expiry's ATM implied volatility: the smile's natural width is $\sigma\sqrt{\tau}$, and a quoted span of roughly fixed width in k covers fewer of those widths as $\sqrt{\tau}$ grows. SPY's long-dated spans cover fewer, and the unquoted share climbs steadily with maturity toward a quarter of the number; the marked running node sits at 18.4%. NVDA's quoted spans are generous relative to its volatility and its share stays under ten percent.

Now put the two exhibits of the chapter together. Figure 5.1(b) showed three models disagreeing on the fair strike by up to 58.2 vol bp while agreeing on the quoted span to 13.3 bp. Figure 5.3 says the disagreement has exactly one place to live: the 18.4% of the integral that accrues beyond the quotes. The same comparison runs across model chapters: Chapter 4's whole-surface sheet, fitted to the same mid quotes, implies $\sigma_{\mathrm{vs}} = 19.70\%$ at this expiry — 13.7 vol bp from the running slice's 19.83%. None of these models is wrong. Each reprices the quotes; the quotes simply do not decide the number. A fit error of a few basis points *per quote* coexists with a fair-strike spread ten times larger, because the fair strike reads a region the fit error never visits.

This is the chapter's instance of a distinction the book keeps returning to. What the quotes pin — the belly, the well-quoted central stretch of strikes — is information. What they leave free — the wings' share — is not noise around zero; it is a quantity the model must supply, and different structural choices supply different amounts. Sections 5.4 and 5.5 will show that the freedom is bounded: no admissible continuation — none that still prices a genuine law — can push the share arbitrarily high. But within those bounds, the number is chosen, not estimated.

One consequence deserves a paragraph before we turn to the bounds, because it changes how a desk should read the instrument. A traded var-swap quote is one *observation* of precisely the combination of unquoted prices the listed strikes leave loose. Chapter 2's calibration objective (section 2.6) accommodates it directly: the model's own fair strike — computed by whichever route of section 5.2 is native — becomes one more residual against the quoted level, in volatility units, with a weight the modeler declares; the derivative of w_{vs} in the model's parameters comes in closed form by the linearity of section 5.2 (Chapter 2 already recorded it as eq. (2.46)). Weighted softly, the quote bargains with the options rather than overruling them, and it spends its influence exactly where fig. 5.3 says its information lives: in the wings. A desk that has such a quote should use it; the sections that follow are addressed to the more common case where it does not.

We now know where the number lives and why fitted models disagree about it. The next two sections establish what no model, and no market, can do in that region.

5.4 The ceiling

A fifth of the fair strike accrues where nothing is quoted. Before deciding what a model *should* draw there, establish what any model *may* draw there. This section proves the one bound that holds with no data at all: a universal ceiling on how steeply total variance can grow with strike.

Recall the objects. The wing slopes of a smile are its asymptotic growth rates in total variance,

$$\beta_R = \limsup_{k\to+\infty} \frac{w(k)}{k}, \qquad \beta_L = \limsup_{k\to-\infty} \frac{w(k)}{|k|},$$

the same β_L, β_R carried since section 2.4 (a limsup is the largest value a ratio keeps returning to; for every smile in this book the ratios simply converge). The claim is that no admissible smile — no smile that prices a positive law with mean one — can have either slope above 2.

The proof rests on one physical fact and one computation. The fact: a call at an absurdly high strike must be worth almost nothing. Formally, $c(k) = \mathbb{E}[(Y - e^k)^+]$, the payoff is dominated by Y itself (which has finite mean) and falls to zero pointwise as $k \to \infty$, so dominated convergence gives

$$c(k) \longrightarrow 0 \qquad (k \to +\infty). \tag{5.10}$$

The computation: what the Black function does when total variance grows along a ray. Fix a slope $\beta > 0$ and evaluate $B(k, \beta k)$ for large k. The Black arguments (from eq. (2.3)) are $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$; substituting $w = \beta k$,

$$d_+ = \sqrt{k}\left(\frac{\sqrt{\beta}}{2} - \frac{1}{\sqrt{\beta}}\right), \qquad d_- = -\sqrt{k}\left(\frac{1}{\sqrt{\beta}} + \frac{\sqrt{\beta}}{2}\right). \tag{5.11}$$

Everything hinges on the sign of the parenthesis in d_+: multiplying $\sqrt{\beta}/2 > 1/\sqrt{\beta}$ through by $\sqrt{\beta}$ shows it is positive exactly when $\beta > 2$. So as $k \to \infty$, d_+ runs to $+\infty$ for $\beta > 2$, to $-\infty$ for $\beta < 2$, and sits at exactly 0 for $\beta = 2$.

The second Black term dies in every case. The tool is the normal tail bound $\Phi(d) \le \varphi(d)/|d|$ for $d < 0$, with Φ, φ the standard normal distribution function and density. Its one-line proof: for $t \le d < 0$ the ratio t/d is at least 1, so

$$\Phi(d) = \int_{-\infty}^{d} \varphi(t)\,\mathrm{d}t \le \int_{-\infty}^{d} \frac{t}{d}\,\varphi(t)\,\mathrm{d}t = \frac{\varphi(d)}{|d|},$$

the last step because $t\varphi(t)$ integrates to $-\varphi(d)$. Applying the bound,

$$\log\left(e^k \Phi(d_-)\right) \le k - \frac{d_-^2}{2} - \log\left(|d_-|\sqrt{2\pi}\right), \qquad \frac{d_-^2}{2} = \frac{k}{2}\left(\frac{1}{\beta} + 1 + \frac{\beta}{4}\right),$$

and the coefficient of k collects to

$$1 - \frac{1}{2}\left(\frac{1}{\beta} + 1 + \frac{\beta}{4}\right) = \frac{4\beta - 4 - \beta^2}{8\beta} = -\frac{(\beta-2)^2}{8\beta} \ \le\ 0, \tag{5.12}$$

strictly negative except at $\beta = 2$, where the $\log|d_-|$ term still drives the bound to $-\infty$. Hence $e^k\Phi(d_-) \to 0$ along every ray, and

$$B(k, \beta k) = \Phi(d_+) - e^k \Phi(d_-) \ \longrightarrow\ \begin{cases} 0, & \beta < 2, \\ \frac{1}{2}, & \beta = 2, \\ 1, & \beta > 2. \end{cases} \tag{5.13}$$

Proposition 5.2 (The model-free ceiling)**.** *For any admissible smile, $\beta_L \le 2$ and $\beta_R \le 2$.*

Proof. Right wing. Suppose $\beta_R > 2$ and pick β strictly between 2 and β_R. By the definition of the limsup there are arbitrarily large strikes with $w(k) \ge \beta k$. The Black function increases in its variance argument, so at those strikes $c(k) = B(k, w(k)) \ge B(k, \beta k)$, and by (5.13) the right side approaches 1. A smile that steep prices calls that refuse to become worthless, contradicting (5.10).

Left wing. The mirror argument runs through the put, measured per unit of strike: $P(k)/e^k = \mathbb{E}[(1 - Ye^{-k})^+]$, which by dominated convergence tends to $\mathbb{P}(Y = 0)$ as $k \to -\infty$ — and $\mathbb{P}(Y = 0) = 1$ is impossible for a law with $\mathbb{E}[Y] = 1$. It remains to show a left ray of slope $\beta > 2$ forces that limit to be 1. Parity turns the Black call into the Black put per unit of strike,

$$\frac{P(k)}{e^k} = \frac{B(k, w) - 1 + e^k}{e^k} = \Phi(-d_-) - e^{-k}\,\Phi(-d_+),$$

and along $w = \beta|k|$ (so $k = -|k|$) the arguments (5.11) give

$$-d_- = \sqrt{|k|}\left(\frac{\sqrt{\beta}}{2} - \frac{1}{\sqrt{\beta}}\right), \qquad -d_+ = -\sqrt{|k|}\left(\frac{1}{\sqrt{\beta}} + \frac{\sqrt{\beta}}{2}\right):$$

exactly the two expressions of the right-wing computation with the roles of the arguments exchanged. So for $\beta > 2$ the first term $\Phi(-d_-) \to 1$, and the second, $e^{|k|}\Phi(-d_+)$, dies by the same exponent $-(\beta - 2)^2/(8\beta)$ as in (5.12). Hence $P(k)/e^k \to 1$, the contradiction. □

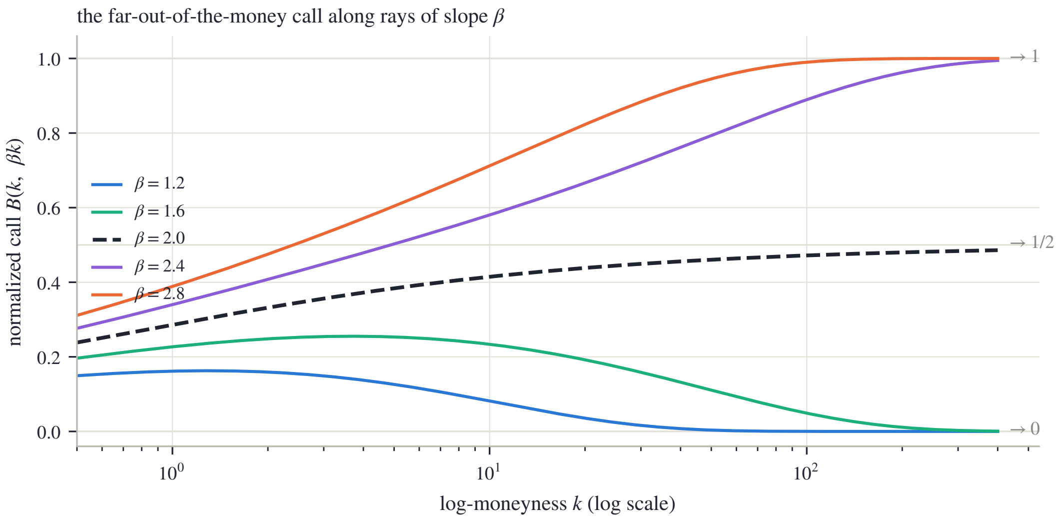

Figure 5.4: The ceiling, computed. The normalized Black call along total-variance rays $w = \beta k$, against log-strike on a log axis. Below the ceiling the call dies (at $k = 400$ the $\beta = 1.6$ ray reads 0.001); above it the call saturates at forward value (0.995 for $\beta = 2.4$); on the boundary ray it creeps up to one half (0.486 at the right edge). Only the dying curves are smiles of any law.

Figure 5.4 draws the computation. Each curve is $B(k, \beta k)$ for one slope, out to $k = 400$ on a logarithmic strike axis — far past anything tradeable, because the convergence is slow and honesty requires showing it. The three regimes of (5.13) separate cleanly: the $\beta < 2$ curves sink to zero (the $\beta = 1.6$ ray reads 0.001 at the right edge), the $\beta > 2$ curves climb toward one (0.995 for $\beta = 2.4$), and the dashed boundary ray $\beta = 2$ creeps up to one half — there $\Phi(d_+)$ equals $\frac{1}{2}$ exactly at every strike, and what remains is the second term dying at its slow boundary rate, leaving 0.486 at the right edge. Notice that the boundary ray also violates (5.10): a call permanently worth half the forward is no better than one worth the whole forward. The ceiling is therefore a *supremum, not a member*: an admissible smile can approach slope 2 from below, but no admissible smile lies on the ray itself. Chapter 3 met the same phenomenon inside the density factor: at $\beta = 2$ the leading tail term of g_{D} vanishes, and whether the far density is positive is decided by the next, smaller term — one the slope condition never examines — which is why the working cap of eq. (3.8) sits strictly inside the bound. Section 5.5 will meet the same phenomenon a third time, as the top edge of the envelope.

The ceiling says how steep a wing *cannot* be. Lee's moment formula — stated in Chapter 2 as eq. (2.23) — says how steep it *is*: $\beta = \Psi(r^*)$, with $\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r)$, where r^* is the tail's critical moment order and Ψ decreases from $\Psi(0) = 2$. The two results are faces of one mechanism, and the computation above quietly contained the link. For $\beta < 2$ the surviving Black term is $\Phi(d_+)$ with $d_+ \to -\infty$, and the same tail bound gives its decay exponent:

$$\frac{d_+^2}{2} = \frac{k}{2}\left(\frac{1}{\sqrt{\beta}} - \frac{\sqrt{\beta}}{2}\right)^2 = \frac{k\,(2-\beta)^2}{8\beta} \implies \Phi(d_+) \approx e^{-k\,(2-\beta)^2/(8\beta)} = K^{-r}, \quad r = \frac{(2-\beta)^2}{8\beta},$$

up to algebraic factors — and this r is exactly the moment budget $\Psi^{-1}(\beta)$ that Chapter 3 computed in eq. (3.9). The bridge between call decay and moments is elementary: whenever $Y \geq K$ one has $(Y-K)^+ \leq Y \leq Y\,(Y/K)^r$, so a finite moment caps the call, $\mathbb{E}[(Y-K)^+] \leq \mathbb{E}[Y^{1+r}]\,K^{-r}$ — more finite moments force faster call decay, and (this is Lee's harder half) the converse holds too. A wing of slope β prices calls that decay with power r; call decay and finite moments carry the same information; so slope pins moments and moments pin slope. Steeper wings, fatter tails, fewer moments — one dial.

What mathematics gives, then, is a cone: every admissible wing lies between slope 0 and slope 2, and its slope names its tail's moment order. What mathematics does not give is a selection. The next section sharpens the cone into a finite-strike envelope anchored at the last quote — and shows that it, too, refuses to select.

5.5 The envelope of admissible completions

The ceiling constrains slopes at infinity. A desk's question is more concrete: the quotes end at some strike — what prices may an honest model print at the strikes just past it? This section answers with an envelope: two curves, computable from the last two quotes, between which every admissible continuation must run.

Fix the setting on the call side (the put side mirrors it). A slice has been fitted through the quotes; call everything up to the last quoted strike *the belly* and regard it as settled. Bars mark quantities frozen at the last quote: $\bar{k}$ is the last quoted log-strike, $\bar{y} = e^{\bar{k}}$ the strike itself, and $\bar{c} = c(\bar{y})$ the normalized call there. Write y_1 for the second-to-last quoted strike; the *last secant slope* is the price change per unit of strike across the final quoted interval,

$$\bar{s} = \frac{c(\bar{y}) - c(y_1)}{\bar{y} - y_1} \;<\; 0.$$

A *completion* is any way of continuing the call curve to strikes beyond $\bar{y}$ so that the whole curve still prices a law. Chapter 2 established what that requires (section 2.1 and eq. (2.4)): as a function of strike, the call curve of a law is nonincreasing, and it is convex — its second derivative *is* the density, so convexity is positivity of probability. And by (5.10) it must eventually reach zero.

Those three properties already imply the envelope.

Proposition 5.3 (The admissible cone)**.** *Every admissible completion satisfies, for all $y \geq \bar{y}$,*

$$\max\bigl(0,\ \bar{c} + \bar{s}\,(y-\bar{y})\bigr) \;\leq\; c(y) \;\leq\; \bar{c}. \tag{5.14}$$

The lower edge is itself an admissible completion; the upper edge is approached by admissible completions but attained by none.

Proof. Write the completion through its slope: $c(y) = \bar{c} + \int_{\bar{y}}^{y} c'(t)\,\mathrm{d}t$. Convexity of the whole curve says c' is nondecreasing; in particular, for a convex function a chord's slope never exceeds the slope just past its right endpoint, so applying this to the chord over $[y_1, \bar{y}]$ gives $c'(\bar{y}^+) \geq \bar{s}$, hence $c'(t) \geq \bar{s}$ for all $t \geq \bar{y}$. Monotonicity says $c' \leq 0$. Integrating the two slope bounds gives $\bar{c} + \bar{s}(y-\bar{y}) \leq c(y) \leq \bar{c}$, and prices are nonnegative, which is (5.14).

For the lower edge, let the slope stay exactly at $\bar{s}$ until the line reaches zero, at the strike

$$y_0 = \bar{y} - \frac{\bar{c}}{\bar{s}}, \tag{5.15}$$

and stay at zero after. Slopes step from $\bar{s}$ up to 0, so the curve is convex and nonincreasing, and it reaches zero: admissible. Its kink at y_0 is a point mass — the law parks all remaining upside probability at the single strike y_0 and owns nothing beyond. For the upper edge, a concrete family does the work. For small $\alpha > 0$ consider the completion $c(y) = \bar{c}\,(y/\bar{y})^{-\alpha}$: it is decreasing and convex, it tends to zero, and it joins the belly with a convex kink at $\bar{y}$ (its opening slope $-\alpha\bar{c}/\bar{y}$ is closer to zero than $\bar{s}$, and slopes may jump upward at a kink). Being decreasing, convex and vanishing at infinity, it is the call curve of a genuine law (section 2.1), one that pushes its remaining mass to ever higher strikes as α shrinks. As $\alpha \to 0$ these curves approach the constant $\bar{c}$ at every strike — but the constant itself never reaches zero, so it is not admissible. $\square$

Pause on what the proposition says. At any strike beyond the last quote, the admissible call prices fill an interval: from a linear descent that hits zero — at y_0, which (5.15) computes from two quotes — up to (but excluding) the last quoted price itself. On the running node the numbers are stark: the last call-side quote sits at $\bar{k} = 0.264$, worth $\bar{c} = 1.17 \times 10^{-4}$ of the forward, and the lower edge reaches zero at $k = 0.350$. The data permits the smile to *end* — price zero, no law beyond — less than a tenth of a log-unit past the last quote. It also permits a wing that is still worth nearly the full $\bar{c}$ a log-unit out. Everything between is allowed.

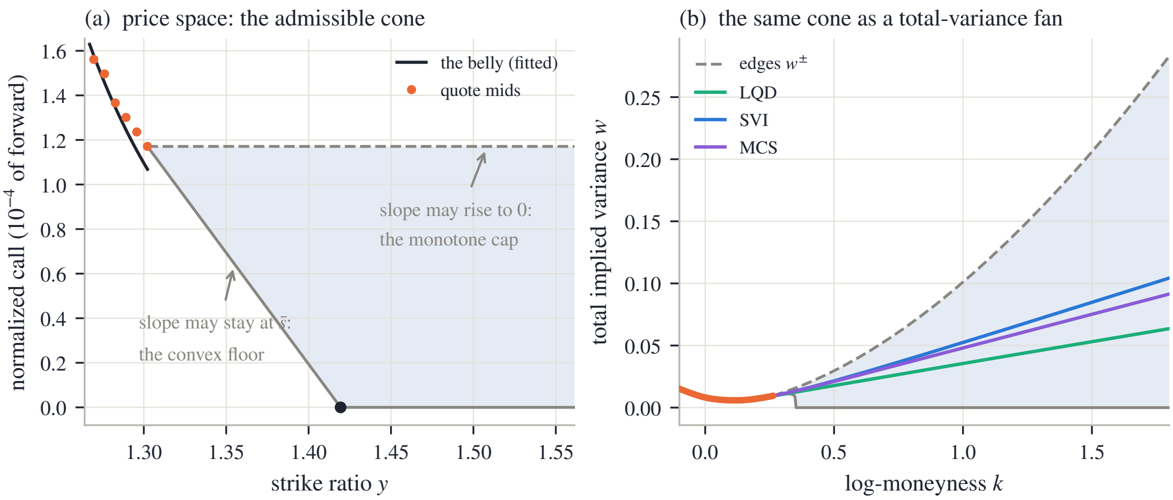

Figure 5.5: The envelope of admissible completions (running node, call side). (a) Price space: the fitted belly (black) through the last quote mids (orange), and the cone (5.14) beyond the last quote: the dashed monotone cap at $\bar{c}$, the convex floor descending at the secant slope $\bar{s}$, reaching zero at y_0 (dot). (b) The same cone mapped through the Black chart: the shaded fan $[w^-, w^+]$ of admissible total-variance completions, with the three families' fitted wings threading it.

Figure 5.5(a) draws the cone in price space on the running node. The black curve is the fitted belly, ending at the last quote; prices here are a few 10^{-4} of the forward, so the axis is in those units. From the endpoint two edges open: the dashed horizontal — the *monotone cap*, prices may not rise — and the descending line — the *convex floor*: prices may never fall faster than the last secant slope, so the straight continuation at $\bar{s}$, dying into the axis at y_0 (dot), is the steepest descent any completion is allowed. The shaded region between them is every call price any law consistent with the belly can print.

Panel (b) shows the same cone where this book draws smiles: in total variance. The Black function is increasing in w, so the price interval (5.14) maps strike by strike onto an interval

$[w^-(k),\, w^+(k)]$, the *fan*: w^- is the implied variance of the descending line (zero once the line dies), w^+ that of the constant $\bar{c}$. The fan is vast. By $k = 1$ the admissible total variance runs from 0 to several times the ATM level; the three families' wings — the same three fits of fig. 5.1 — thread it in a narrow bundle near the bottom. Numerically, each family's wing clears the floor and cap of its *own* cone — the cone rebuilt from that family's fitted call values at the same two strikes, since proposition 5.3 constrains a completion relative to its own belly (appendix 5.A) — with room to spare: worst clearances 0.0016/0.0010/0.0010 below and 0.0016/0.0012/0.0013 above, in total variance. That is proposition 5.3 confirmed on real fits. The bundle's narrowness against the fan's width restates section 5.3: the models' structural choices are far more selective than the data's constraints.

The fan's top edge has a closed form, and deriving it closes the loop with section 5.4. The upper edge holds the call at the constant $\bar{c}$, so $w^+(k)$ solves $B(k, w) = \bar{c}$. Suppose for a moment the second Black term is negligible; then $\Phi(d_+) = \bar{c}$, i.e. $d_+ = q$ with $q = \Phi^{-1}(\bar{c})$ (a negative number — on the running node $\bar{c} = 1.17 \times 10^{-4}$, far below one half). Unpack d_+:

$$\frac{\sqrt{w}}{2} - \frac{k}{\sqrt{w}} = q \qquad \Longrightarrow \qquad w - 2q\sqrt{w} - 2k = 0 \qquad \Longrightarrow \qquad \sqrt{w^+(k)} = q + \sqrt{q^2 + 2k},$$

taking the positive root of the quadratic in $\sqrt{w}$. The neglected term justifies itself with unexpected grace. The two Black arguments always differ by $\sqrt{w}$, so along this solution $d_- = d_+ - \sqrt{w} = q - \sqrt{w}$, and

$$\frac{d_-^2}{2} = \frac{q^2 - 2q\sqrt{w} + w}{2} = \frac{q^2 + 2k}{2} \qquad \Longrightarrow \qquad \log\left(e^k \Phi(d_-)\right) \le k - \frac{q^2 + 2k}{2} = -\frac{k}{2} - \frac{q^2}{2},$$

using the quadratic itself in the middle step, and the tail bound of section 5.4 (whose $-\log|d_-|$ term only strengthens the inequality) in the last: the correction decays like $e^{-k/2}$, exponentially small. (At $k = 30$ the closed form agrees with a full Black inversion by the reference implementation — the working code behind every number in this book — to 1.1×10^{-10} in total variance.) Differentiating the closed form gives the edge's slope,

$$\frac{\mathrm{d}w^+}{\mathrm{d}k} = 2\left(q + \sqrt{q^2 + 2k}\right) \cdot \frac{1}{\sqrt{q^2 + 2k}} = 2 + \frac{2q}{\sqrt{q^2 + 2k}} \longrightarrow 2 \qquad (k \to \infty), \tag{5.16}$$

rising monotonically (recall $q < 0$, and the negative term shrinks) — it reads 1.14 at $k = 30$ — and approaching 2 from below, never reaching it. The top edge of the finite-strike envelope is the ceiling of proposition 5.2, met for the third time and in its sharpest form: the steepest admissible completion climbs at slope $2 - 2|q|/\sqrt{q^2 + 2k}$, a curve that grazes the ceiling asymptotically while staying strictly under it — the supremum, not a member, made into a formula.

The section's summary is uncomfortable and important. Between the floor and the cap, mathematics is silent: every curve in the fan prices a legitimate law consistent with everything quoted. The data cannot choose. The model must — and the next section is about doing that in the open.

5.6 The wing as a stated choice

Inside the envelope the data has no further opinion, yet every fitted model draws exactly one curve there. What selects the curve is the model's structure — decisions made when the family was designed, long before this node's quotes arrived. This section reads those decisions off the three families, states them as contracts, and then demolishes a comfortable inference about them.

Recall the three families' wings; each was derived in its own chapter, and here they only need to be set side by side. The LQD slice of Chapter 2 has exponential log-return tails *by* *construction*, and its wing slopes are closed functions of its two endpoint speeds (proposition 2.9): admissibility is structural, no cap or check required. The SVI slice of Chapter 3 has exactly straight total-variance wings, $\beta_{L,R} = b_S(1 \mp \rho_S)$ — the S-subscripts mark Chapter 3's raw SVI parameters (proposition 3.1) — and its calibration fences them with a cap set strictly inside the ceiling (eq. (3.8)): strictly, because section 5.4 showed the boundary ray itself is inadmissible. The MCS slice of Chapter 3 splits the job: its base owns the wings (eq. (3.32)), and its hats are built to vanish at large strikes, so adding hats cannot move the slopes — but nothing in the family *forces* the base's slopes into the admissible cone; their admissibility is diagnosed after the fit, not enforced during it. On the running node the three contracts print the slopes of table 5.1: every wing far below the ceiling, and the put slopes several times the call slopes. That asymmetry is the equity skew — the standing pattern that downside strikes carry more implied variance than upside ones — read here in its asymptotic form.

Table 5.1: The three wing contracts on the running node. Slopes are asymptotic total-variance wing slopes; all sit far inside the ceiling of proposition 5.2.

Family	β_L (put)	β_R (call)	how the wing is governed
LQD	0.178	0.034	structural: exponential tails, slopes from proposition 2.9
SVI	0.126	0.066	capped: straight wings under eq. (3.8)
MCS	0.117	0.054	inherited: base wings, hats neutral; diagnosed, not enforced

Three families, three slopes per side, all admissible — and all different, as fig. 5.5(b) drew. It bears saying plainly: past the last quote, the divergence between the families is not an error to be averaged away. Choosing a family *is* choosing an extrapolation law. What a careful fitter owes its user is not agreement in the wings — nothing can supply that — but a stated contract: which curve will be drawn, governed by what mechanism, with what guarantee.

There is, however, a gap in those guarantees, and it is the section's second lesson. All three contracts of table 5.1 speak about *slopes* — asymptotic statements, properties of the limit $k \to \infty$. The comfortable inference is: admissible slopes, therefore admissible wing. It is false, and one deliberate construction shows why.

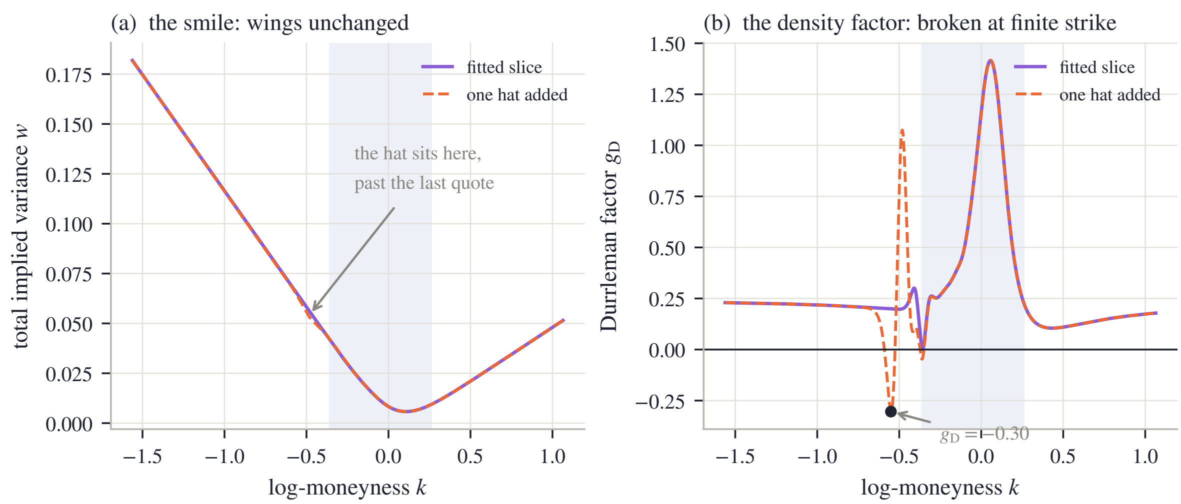

Figure 5.6: The limit is not the wing. One zero-wing hat is added to the fitted MCS slice, centered beyond the last put-side quote. (a) Total variance: the two curves are indistinguishable except for the small bump at the hat; the measured far-field wing slopes change by 0 — an exact zero in double precision. (b) The density factor g_{D}: the hat drives it to -0.30 at $k = -0.55$, negative density at a finite strike, while the fitted slice (solid) stays nonnegative to rounding (worst -0.001).

Take the fitted MCS slice of the running node and add one more of its own hat kernels (eq. (3.33)) — amplitude negative, centered just beyond the last put-side quote, where quotes no longer object. Figure 5.6(a) shows the smile before and after: the two curves coincide to the eye everywhere except a small dent near $k = -0.55$, and the wings are untouched — measuring the realized slope far out, the change is 0, an exact zero at double precision. The inheritance contract has done its job perfectly. Panel (b) shows what the contract never promised. Recall the Durrleman factor g_D from eq. (2.5): up to a positive factor it is the probability density the smile implies, so a valid smile needs $g_D \geq 0$ everywhere. The fitted slice (solid) keeps it nonnegative across the range — its worst value is -0.001, zero at rounding scale. The hatted slice (dashed) plunges to $g_D = -0.30$ at $k = -0.55$: negative probability, a butterfly arbitrage, parked at a finite strike in the extrapolated region — with both asymptotic slopes exactly as admissible as before. The dent sits in the quiet stretch just outside the quoted span, precisely where no quote penalizes it and no slope condition sees it. Asymptotic admissibility and finite-strike admissibility are independent, and the second needs its own police.

The demonstration settles how the extrapolated region must be policed, and the rule is worth stating because it cuts both ways. Conditions that are properties of a *single* curve — density positivity above, call convexity — must be checked *into* the wings: one law is being asserted there, the condition needs no second curve to compare against, and abandoning it past the last quote is exactly how the hat's dent would go unnoticed. Conditions that *compare two* curves are different. Chapter 2's calendar control (eq. (2.59)) already made the choice: it enforces the order between neighboring expiries only on the span where quotes pin both curves. Beyond that span, both curves are chosen conventions — two entries of table 5.1, possibly from different rows. A calendar inversion there — the later expiry's curve dipping below the earlier one's at some unquoted strike — is a disagreement between two fictions, not evidence of mispricing; enforcing the order there would bend the fits where the data *does* live in order to reconcile them where it does not. One curve: extend the check. Two curves: confine it to common data.

The chapter's statics are now complete for a single expiry: the number, its three computations, its unquoted share, the proved envelope, and the disclosed choice within it. One dimension remains — maturity.

5.7 The ordered term structure, and the contract

Everything so far concerned one expiry. A surface carries a fair strike per expiry, and the last question is how the family of numbers $w_{vs}(\tau)$ behaves across maturities. The answer is that the calendar condition of Chapter 2 orders it — and the proof is a three-line reuse of the spanning identity.

Recall the condition. A surface is calendar-ordered when, at every log-moneyness, the later expiry's normalized call is worth at least the earlier one's: $c_1(k) \leq c_2(k)$ for all k (eq. (2.56)). Chapter 2 showed this is the same as *convex order* of the two laws — every convex payoff is priced at least as high under the later law — and, by Kellerer's theorem, exactly the condition for some martingale to have these two laws as its distributions at the two expiries (theorem 2.16 and section 2.8).

Proposition 5.4 (Calendar order integrates)**.** *If two admissible mean-one slices satisfy $c_1(k) \leq c_2(k)$ for all k, then $w_{vs,1} \leq w_{vs,2}$.*

Proof. Puts inherit the order from calls: at a common strike K, parity gives $P = c + K - 1$ for both slices, so $c_1 \leq c_2$ implies $P_1 \leq P_2$. Now write each fair strike through proposition 5.1 with $\varphi(y) = -2 \log y$, whose second derivative $\varphi''(K) = 2/K^2$ is positive:

$$w_{vs} = 2 \int_0^1 \frac{P(K)}{K^2}\, dK + 2 \int_1^\infty \frac{c(K)}{K^2}\, dK.$$

Every integrand of slice 2 dominates the corresponding integrand of slice 1, so the integrals are ordered. □

The same argument proves more: *any* payoff with $\varphi'' \geq 0$ — any convex payoff — has its price ordered by the calendar, which is just convex order saying its own name. The variance swap is the special case a desk trades directly, and for it the order has a market reading. The difference $w_{\mathrm{vs}}(\tau_2) - w_{\mathrm{vs}}(\tau_1)$ is the *forward variance* between the two dates — the variance a client buys by holding the long swap against the short one. Proposition 5.4 says a calendar-ordered surface never prices forward variance below zero; a violation would let a desk lock in variance at a negative price, which is the calendar arbitrage of Chapter 2 restated for the integral.

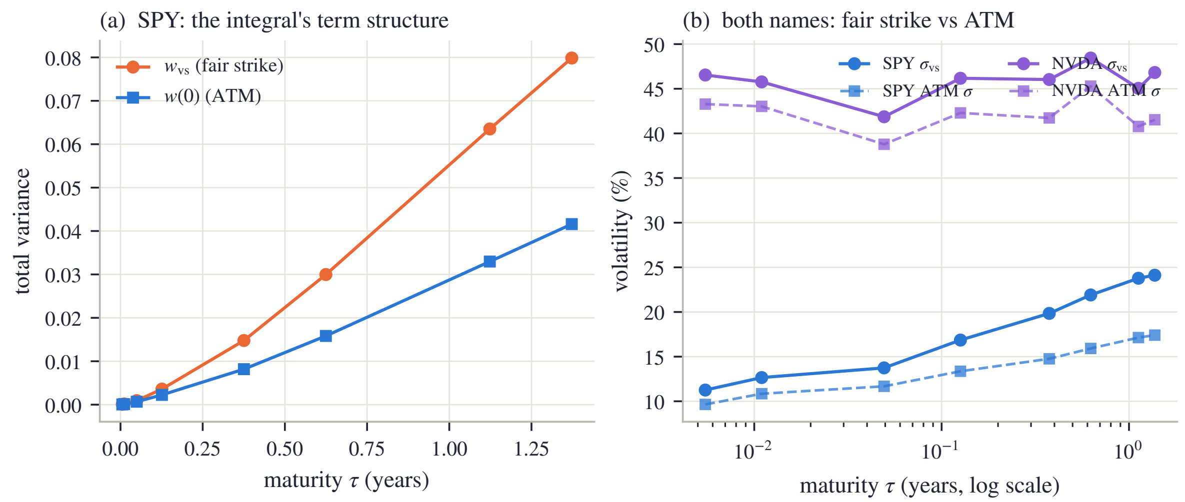

Figure 5.7: The integral's term structure on the frozen gallery (stored surface fits). (a) SPY in total variance: the fair strike w_{vs} and the ATM total variance $w(0)$ both rise with maturity, and the gap between them widens. (b) Both names in volatility units: the fair-strike vol rides above the ATM vol everywhere, by up to 671 vol bp for SPY and 529 for NVDA. NVDA's vol *levels* are not monotone in maturity — the order lives in total variance, panel (a)'s units, not in vol.

Figure 5.7 reads the term structure off the frozen gallery — the snapshot's stored surface fits, the same objects as fig. 5.3(b). In panel (a), SPY in total variance: the orange curve is $w_{\mathrm{vs}}(\tau)$, the blue is the ATM total variance $w(0)(\tau)$, both increasing. Across every adjacent pair of expiries, in both names, the fair strike's increment is positive — the smallest measured increment is +1.1 variance bp (a variance bp is 10^{-4} of total variance, the sibling of the vol bp of section 5.1), between the two front expiries a day apart, and the smallest forward variance, expressed as a forward volatility, is 13.9% — so the published surface keeps the order proposition 5.4 demands, with its thinnest margin exactly where maturities crowd together. Panel (a) shows something else: the *gap* between the curves widens with maturity. By the flat-smile computation (5.3), that gap is precisely the part of the fair strike owed to the smile's departure from flatness, and it compounds with maturity just as the unquoted share of fig. 5.3(b) did — the two figures are the same fact in two currencies.

Panel (b) restates the term structure in the units desks quote, volatility, for both names. The fair-strike vol σ_{vs} rides above the ATM vol at every expiry — by up to 671 vol bp at SPY's long end and 529 for NVDA. On a flat smile the two curves would coincide; the spread *is* the priced skew and wing mass, and a desk can read it directly as what convexity in the smile costs. One trap is visible and worth flagging: NVDA's vol curves are not monotone in maturity, and nothing is wrong — the calendar orders *total variance*, panel (a)'s units, and a falling vol term structure is perfectly consistent with rising total variance. Ordering vols is not a no-arbitrage condition; ordering their $\sigma^2\tau$ is.

5.7.1 The contract

The chapter closes the way the model chapters do, with its contract — three columns of decreasing strength. The first holds by theorem. The second is measured on the book's frozen data, and the chapter quoted every number. The third is convention: chosen, stated, and owed to the user, because no data selects it.

Table 5.2: The chapter's contract.

Proved	**Checked (frozen data)**	**Convention**
The identity chain (5.9): one fair strike, three exact integrals (smile, law, field).	Law vs strike route on one slice: 0.10 vol bp; field vs strike route on one sheet: 23.2 vol bp at the march's step, 2.9 at an eightfold refinement.	Which route a model evaluates (its native chart), and the reading of the fair strike: diffusion paths, continuous monitoring, variance time. Jumps sit outside the identity (5.2).
The ceiling (proposition 5.2): no admissible wing steeper than 2; the boundary ray itself inadmissible.	Fitted slopes (table 5.1): all wings far inside the ceiling.	The wing drawn *within* the cone: each family's stated mechanism — structural, capped, or inherited-and-diagnosed.
The envelope (proposition 5.3): every completion between the convex floor and the monotone cap; top edge climbs at the ceiling (5.16).	All three families' wings clear their cones (clearances of section 5.5); the hatted counterexample caught by the extended g_{D} check, slopes unmoved (fig. 5.6).	Where conditions are enforced: one-curve checks extended into the wings; two-curve (calendar) checks confined to common quote support (eq. (2.59)).
Calendar order integrates (proposition 5.4): the fair strike's term structure is nondecreasing; forward variance is nonnegative.	Smallest measured increment across both names: +1.1 variance bp; smallest forward vol 13.9%.	The weight given a traded varswap quote when one exists: one more residual in the objective of section 2.6, at a declared strength.

Step back, once, to the book's thesis. The fair strike is a single number in which the three ingredients the thesis separates are unusually easy to see. Validity supplied the theorems: the identity chain, the ceiling, the envelope, the order. Information is the quoted belly's 81.6%. Convention is the rest — the wing contract that completes the integral, disclosed in a table rather than hidden in an extrapolation routine. Part I is now complete: laws, smiles, fields, and the integrals that test them. Everything so far took the forward and the discount for granted at every expiry. Part II starts by earning them.

5.A Proofs and numerical protocol

5.A.1 The backward equation for the expected remaining variance

Section 5.2 stated that $\mathcal{W}(s,y) = \mathbb{E}[\int_s^T v(r,Y_r)\,\mathrm{d}r \mid Y_s = y]$ solves the backward problem (5.8). Here is the derivation, in two steps the chapter has already rehearsed.

First, a martingale. Consider the running quantity

$$M_s = \int_0^s v(r,Y_r)\,\mathrm{d}r + \mathcal{W}(s,Y_s),$$

the variance accumulated so far plus the expectation of what remains. Write $\mathcal{F}_s$ for the information available at time s. Conditioning the total $\int_0^T v\,\mathrm{d}r$ on $\mathcal{F}_s$ splits it into exactly these two pieces — the first is known at time s, and the second is its conditional expectation by the Markov property of the diffusion. So $M_s = \mathbb{E}[\int_0^T v\,\mathrm{d}r \mid \mathcal{F}_s]$: every M_s is a conditional expectation of one fixed random variable, and such a family is a martingale by the tower property — conditioning M_t on the smaller information set $\mathcal{F}_s$ ($s < t$) collapses it back to M_s, which is exactly the martingale identity $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$.

Second, Itô. Assume $\mathcal{W}$ is smooth and apply Itô's formula to $\mathcal{W}(s, Y_s)$ with $\mathrm{d}Y = \sqrt{v}\, Y \mathrm{d}W$, using $(\mathrm{d}W)^2 = \mathrm{d}s$ as in section 5.1:

$$\mathrm{d}\mathcal{W}(s, Y_s) = \Big[\partial_s \mathcal{W} + \tfrac{1}{2}\, v(s, Y_s)\, Y_s^2\, \partial_{yy}\mathcal{W}\Big]\mathrm{d}s + \partial_y \mathcal{W}\, \sqrt{v}\, Y_s\, \mathrm{d}W_s,$$

so that

$$\mathrm{d}M_s = \Big[v + \partial_s \mathcal{W} + \tfrac{1}{2}\, v\, Y_s^2\, \partial_{yy}\mathcal{W}\Big]\mathrm{d}s + \partial_y \mathcal{W}\, \sqrt{v}\, Y_s\, \mathrm{d}W_s.$$

A martingale has no drift, so the bracket must vanish at every state the process can reach — which is the PDE of (5.8). The terminal condition $\mathcal{W}(\tau, \cdot) = 0$ holds because nothing remains at expiry, and evaluating the definition at $(s, y) = (0, 1)$ gives $\mathcal{W}(0, 1) = \mathbb{E}[\int_0^\tau v\, \mathrm{d}r] = w_{\mathrm{vs}}$.

5.A.2 Numerical protocol

Every empirical number in the chapter comes from one deterministic generator reading the book's frozen snapshot; none is typed by hand. The conventions, in the order the chapter uses them.

Fits. The running node is SPY December 2026. Its three family fits are Chapter 3's comparison protocol applied verbatim (section 3.8): the same mid quotes, equal weights, each family's reference settings. The gallery and term-structure figures (fig. 5.3b and 5.7) instead read the snapshot's *stored* surface fits — the calibration the surface was published with, under the fit recipe of section 2.6 — because those are calendar-coupled across expiries, which is what a term-structure exhibit should show.

The strike-side quadrature. Equation (5.6) is evaluated by a trapezoid on $k \in [-6, 6]$ with 4001 points — a finer copy of the 801-point grid the reference implementation uses inside its optimizer loop; the total variance inside the Black call is floored at 10^{-12}. At these settings the quadrature error is far below every number quoted. The law-side integral rides each slice's own fixed log-odds grid.

The field-side audit. The sheet is Chapter 4's whole-surface SPY calibration, unchanged. Both of its var-swap routes run on an audit lattice with the calibration lattice's strike step, widened to strike ratio 12 so the static replication sees the entire wing the extrapolated field implies; the put leg is cut off below strike ratio 0.01 (moving the cutoff to 0.001 changes nothing at the quoted precision). Both routes are first-order in the time step, so the audit reports the pair of reads at the march's own step and at an eightfold refinement; the gap halves with the step, as first order promises.

The envelope. The drawn cone is built from the Black prices of the last two call-side quote *mids*. The per-family clearances are measured against each family's *own* cone — the same construction fed that family's fitted call values at the same two strikes — because proposition 5.3 constrains a completion relative to its own belly; nonnegativity of all six clearances is the proposition confirmed numerically. Clearances are measured on $\bar{k} + 0.05 \le k \le 1.8$; beyond 1.8 the flattest wing's call price falls under the double-precision floor and its Black inversion is noise.

The hat. The counterexample of fig. 5.6 adds one kernel of eq. (3.33) with half-width 0.55 and steepness 4 in standardized moneyness, centered 1.4 standardized units past the last put-side quote. Its amplitude is the smallest entry of a fixed ladder (5, 10, 20, 35, 55 percent of the base variance at the center) that drives the minimum of g_{D} below -0.05 while total variance stays positive. Realized wing slopes are measured by finite differences of w at $|k| = 8$.

Part II

The Observation

Chapter 6

Forwards, Dividends, and Carry

Part I fitted smiles in normalized coordinates: every strike was measured against a forward F, every price against a discount D, and both were treated as known. Neither is quoted anywhere. This chapter earns them. Put–call parity makes the pair the two coefficients of one straight line through the option quotes themselves, so estimating (F, D) is a regression — and the chapter reads the whole problem as a statistician would. The line determines its level superbly and its slope poorly: on a stated test board the forward is pinned to a fraction of a basis point while the implied rate wanders by an order of magnitude more, and every safeguard the chapter builds is matched to that asymmetry — a trim for outliers, a prior band for runaway rates, and a re-derivation of the forward from the part of the data that stays trustworthy. Dividends then give the carry *— the growth rate linking spot to forward — its content, and the choice among dividend conventions turns out to be a risk decision, not a cosmetic one. The stakes are measured at the end: a forward error read back through the Black formula imprints a put–call gap at the money and a fake skew — on the frozen node, tens of volatility basis points for every ten basis points of forward error — and the borrow — the fee for shorting a stock, the carry's last component — is identifiable only where a closed-form noise floor says the board can resolve it.*

6.1 Two numbers under every smile

Part I built three families of smile models, and all three lived in the same coordinates. A strike K entered as its log-moneyness $k = \log(K/F)$; a dollar price entered divided by DF (section 2.1). Two numbers sat under every computation: the *forward* F, the price agreed today for delivery of the underlying at expiry, and the *discount factor* D, the value today of one dollar paid at expiry. Part I treated both as given. This chapter asks where they come from.

The honest answer is that, for a single stock or an ETF, neither is quoted. There is no screen showing "the forward of SPY for December 2026". There is a spot price, and there is the option chain itself: for each strike, a call and a put. The forward and the discount must be *estimated* from those option prices — which makes them fitted quantities, with error bars, exactly like the smile parameters they will later support. That change of status is the subject of the chapter, and it opens Part II's theme: before a model can be fitted, the observation itself has to be built, and every step of that construction is either measured from the data or chosen and disclosed.

The estimate rests on one identity. Fix an expiry and a strike K, and write $\mathsf{C}(K)$ and $\mathsf{P}(K)$ for the dollar mid prices of the call and the put there — sans-serif letters, here and for the rest of the book, mark a raw market quote, as opposed to the normalized model prices c and P of Chapter 2. Assume for now that both options are *European* — exercisable only at expiry, in the standard vocabulary; section 6.2 meets the American reality. Consider the portfolio that is long the call and short the put, and follow it to expiry. If the underlying finishes at S_T above K, the call pays $S_T - K$ and the put pays nothing; if it finishes below, the call pays nothing and the short put costs $K - S_T$. In both cases the portfolio delivers

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K,$$

the payoff of a forward contract struck at K. What is that worth today? Price the two pieces separately. The constant $-K$ is easy: a sure payment of K at expiry is worth DK now, so the piece contributes $-DK$. For the S_T piece, use the definition of the forward price: a forward contract struck at F costs *nothing* to enter — that is what makes F the forward price — and it pays $S_T - F$. Prices of payoffs add, so the value of receiving S_T minus the value of receiving the constant F equals the value of receiving $S_T - F$, which is zero. Receiving S_T is therefore worth exactly what receiving F is worth: DF. The portfolio's price today must therefore be

$$\Pi(K) \;\equiv\; \mathsf{C}(K) - \mathsf{P}(K) \;=\; D\,(F - K). \tag{6.1}$$

This is *put–call parity* [30, 45]. Chapter 2 already carried its normalized form — the rightmost identity of eq. 2.2, $c(k) - P(k) = 1 - e^k$; multiplying it by DF and writing $K = Fe^k$ recovers (6.1) exactly. Note what the derivation did *not* use: no model of how the underlying moves, no volatility, no distribution. Parity is a validity statement in the sense of the book's thesis — any European call and put prices that violate it admit an arbitrage, whatever the model.

Now read (6.1) the other way. The right-hand side is a straight line in K, with slope $-D$ and root F. The left-hand side is observable: one number per strike, computed from quotes. So the chain itself draws a line, and the two numbers Part I assumed are that line's two coefficients. Estimating (F, D) means fitting a straight line — nothing more exotic than that.

Before any general machinery, solve the smallest possible case by hand. Suppose an expiry half a year away quotes two usable strikes, and the portfolio prices come out as

$$\Pi(90) = 11.27, \qquad \Pi(110) = -8.33$$

(constructed numbers, chosen to come out exactly; appendix 6.A). Two points determine the line. The slope is

$$\frac{\Pi(110) - \Pi(90)}{110 - 90} = \frac{-8.33 - 11.27}{20} = \frac{-19.60}{20} = -0.98,$$

so the discount is $D = 0.98$. The intercept — the line's value at $K = 0$ — is $\Pi(90) + D \cdot 90 = 11.27 + 88.20 = 99.47$, and by (6.1) the intercept equals DF, so

$$F = \frac{99.47}{0.98} = 101.50.$$

Finally, the discount converts to an interest rate. Writing t for the *calendar* year fraction to expiry ($t = 0.5$ here) and $D = e^{-rt}$,

$$r = -\frac{\log D}{t} = -\frac{\log 0.98}{0.5} \approx 4.04\%.$$

Two portfolio prices — four option quotes — have produced a forward, a discount, and an interest rate, out of prices alone. This is worth pausing on: the option market carries its own funding information, and the fitter never needs an external rate feed to normalize a smile.

One clock deserves a sentence before we go on. The book has used τ, variance time, since Chapter 2, and this chapter introduces t, the calendar year fraction. They are different objects with different jobs: variance accrues on the market's clock (Chapter 8 makes that precise), but discounting and *carry* — the growth rate that links spot to forward, given its full content in section 6.6 — accrue on the calendar: a dollar earns interest on weekends too. In everything below, $D = e^{-rt}$ and the carry exponents use t; total variance $w = \sigma^2\tau$ keeps τ. In this chapter's data the two clocks happen to agree numerically, so nothing subtle hides in the figures; the distinction starts to matter in Chapter 8.

We now know what the two numbers are and that the chain determines them. The next question is what happens with a real chain: many strikes, quoted with noise, where the line must be fitted rather than solved.

6.2 One straight line, drawn by a real chain

We know the chain draws a line. This section fits it on the book's data and meets the two facts that shape the whole chapter: the line is astonishingly straight, and its residuals are not noise.

With more than two strikes the line is overdetermined, and the natural fit is least squares: choose the slope and intercept that minimize the sum of squared residuals across strikes. Write the fitted intercept as $\hat{a}$ and the fitted slope as $\hat{b}$; then, reading (6.1) backwards as in the hand example,

$$\hat{D} = -\hat{b}, \qquad \hat{F} = \frac{\hat{a}}{\hat{D}}, \qquad \hat{r} = -\frac{\log \hat{D}}{t}. \tag{6.2}$$

Hats, here and from now on, mark estimates — numbers computed from quotes, as opposed to the true (F, D) the market is carrying. The whole estimator is four lines of arithmetic; there is nothing else inside it.

Figure 6.1 runs it on the running node — SPY December 2026, the same (underlier, expiry) pair every chapter has used — taking mid prices from the frozen snapshot's raw chain and keeping the 158 strikes where both the call and the put quote a live bid. Panel (a) shows the observable: $\Pi(K)$ against K, falling from +\$609 deep in the left wing to −\$242 at the top of the board, a range of \$850. Through that range the points hug the fitted line so tightly that the eye cannot separate them from it: the rms residual is \$2.97, which is 0.35% of the observable's range. The open circle marks the fitted root $\hat{F} = 762.71$: the strike at which a call and a put cost the same, which is by (6.1) the forward estimate. Parity, a theorem about prices that nobody enforces centrally, is visibly enforced by the market to 0.35% of the observable's own range.

Panel (b) is the same fit with the line subtracted, and it carries the section's real lesson. If the residuals were quote noise they would look like the cent-scale jitter of the hand example — a fuzz around zero with no shape. Instead they *bow*: a smooth arch, dollars deep at both wings, cresting at +\$4.8 near the money — hundreds of times the *tick*, the one-cent minimum increment a quote can move by — with every point sitting on the arch rather than scattered around it. A systematic shape in residuals means a systematic error in the model of the observation — and the model here was equation (6.1), which prices *European* options. SPY's listed options are American: each one carries the extra right to exercise early, that right has value, and the value is not the same for the call and the put at a given strike. The mids on the line are therefore each shifted by a smooth, strike-dependent premium the European identity knows nothing about. Chapter 7 is devoted to measuring and removing that premium; this chapter only needs to know it is there.

The bow has a consequence the figure also shows. Because the contamination is asymmetric — deep in-the-money puts carry much more early-exercise value than calls do — it does not average away; it tilts the fitted line and drags the root. The reference implementation — the working software whose conventions the book's figures audit — deals with this by de-Americanizing the quotes before regressing (Chapter 7's machinery); its resolved forward for this node, stored with the snapshot, is $F = 767.92$, sitting 68 basis points above the naive root $\hat{F} = 762.71$. (A *basis point*, bp, is 10^{-4} throughout: of the forward when we compare forwards, of the rate when we compare rates — that one is absolute — and later a *vol* bp, 10^{-4} of implied volatility.) Both vertical lines are drawn in panel (b); at the scale of the strike axis they almost coincide, and that is worth noticing too — 68 bp is \$5.2 on a spot of \$757.67, invisible in price space. Section 6.7 will show the same five dollars is anything but invisible in volatility space. Until then the chapter works on European boards, where (6.1) is exact, and returns to American data only when the estimation theory is in hand.

For that theory we need a controlled instrument: a board whose true (F, D) we *know*, so that estimator errors can be measured rather than argued about. The chapter's *running board* is synthetic and European: spot $S = 100$, rate $r = 4\%$, a 1% continuous dividend yield (dividends enter properly in section 6.6), half a year to expiry, and 25 strikes from 70 to 130 every 2.5, each

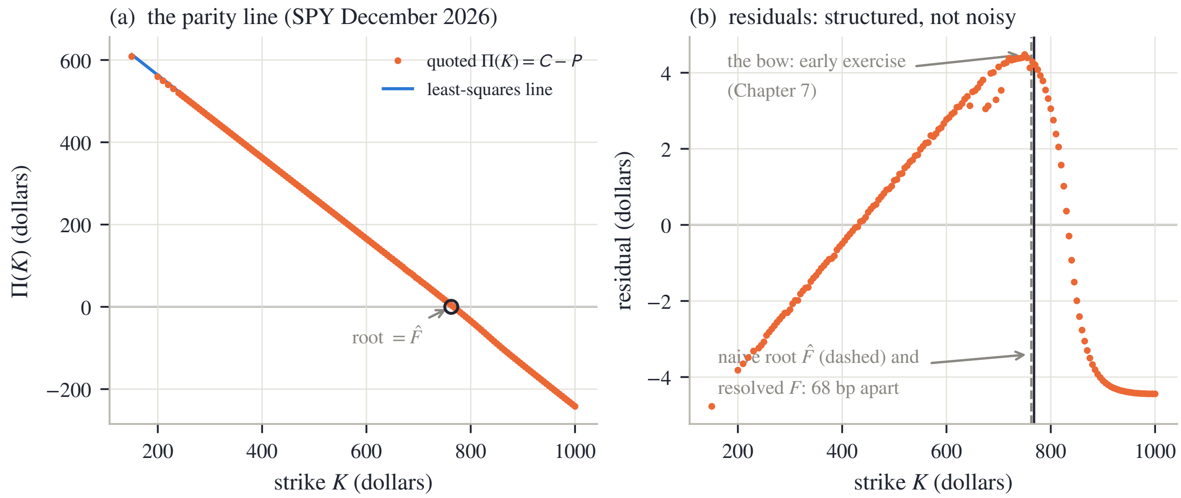

Figure 6.1: The parity line on the running node (SPY December 2026, 158 paired strikes from the frozen chain). (a) $\Pi(K) = \mathsf{C} - \mathsf{P}$ against strike: a straight line to 0.35% of its \$850 range, with the fitted root $\hat{F}$ circled. (b) The residuals bow smoothly — dollars, not cents, and all of one shape: the early-exercise premium of American quotes, the subject of Chapter 7. The naive root and the reference implementation's resolved forward differ by 68 bp.

mid priced by the Black formula at a flat 22% volatility and rounded to the cent (all constants in appendix 6.A). Its true forward is $F = Se^{(r-q_d)t} = 101.51$, where q_d is the dividend yield — take the yield-in-the-exponent form on faith for two more sections; section 6.6 derives it. Its true discount is $D = e^{-rt} = 0.9802$. On this board the least-squares fit recovers the forward to 0.15 bp and the rate to 1.0 bp — the cent rounding is the only noise. The interesting question is not whether it recovers them but *how unequally well* it recovers the two, and that question gets its own section.

We now have the estimator (6.2), a real chain where the line holds to 0.35% of its range, and a clean board with known truth. Next: the error bars.

6.3 What the line identifies well, and what poorly

The estimator (6.2) recovers a clean board's truth. This section asks the sharper question: with quote noise of a stated size, how precise is each of the two fitted numbers? The answer is lopsided — and the lopsidedness explains every safeguard built later in the chapter, so we derive it slowly and then verify it by simulation.

First the noise model. Let each quoted mid — each $\mathsf{C}(K_i)$ and each $\mathsf{P}(K_i)$ — carry an independent error of standard deviation ε, a couple of cents on a liquid board. The observable is the difference of two mids, and independent errors add in variance, so each Π_i carries error of standard deviation $\sqrt{\varepsilon^2 + \varepsilon^2} = \varepsilon\sqrt{2}$. Write n for the number of paired strikes, $\bar{K} = \frac{1}{n}\sum_i K_i$ for their mean, and

$$S_{KK} = \sum_{i=1}^{n} (K_i - \bar{K})^2$$

for the *strike dispersion* — the sum of squared distances of the strikes from their center, the number that measures how wide a lever the board gives the slope.

Proposition 6.1 (error bars of the parity line)**.** *Fit $\Pi_i = a + bK_i$ by least squares under the noise model above. Then the fitted slope and the fitted* level at the center, $\widehat{\Pi(\bar{K})}$, *are uncorrelated,* *with standard deviations*

$$\mathrm{sd}\,(\hat{D}) = \frac{\varepsilon\sqrt{2}}{\sqrt{S_{KK}}}, \qquad \mathrm{sd}\,(\widehat{\Pi(\bar{K})}) = \frac{\varepsilon\sqrt{2}}{\sqrt{n}}. \tag{6.3}$$

The induced first-order errors on the market parameters are

$$\delta\hat{r} = -\frac{\delta\hat{D}}{D\,t}, \qquad \delta\hat{F} = \underbrace{\frac{\delta\widehat{\Pi(\bar{K})}}{D}}_{\textit{level noise}} - \underbrace{(F-\bar{K})\,\frac{\delta\hat{D}}{D}}_{\textit{slope noise}\times\textit{lever}}. \tag{6.4}$$

Proof. Center the regressor: put $x_i = K_i - \bar{K}$, so that $\sum_i x_i = 0$ and the model reads $\Pi_i = \alpha + bx_i$ with $\alpha = a + b\bar{K}$ — the line's value at the center, so $\hat{\alpha}$ is exactly the $\widehat{\Pi(\bar{K})}$ of the statement. Least squares on orthogonal columns separates: minimizing $\sum_i(\Pi_i - \alpha - bx_i)^2$ over (α, b) gives

$$\hat{\alpha} = \frac{1}{n}\sum_i \Pi_i, \qquad \hat{b} = \frac{\sum_i x_i \Pi_i}{\sum_i x_i^2} = \frac{\sum_i x_i \Pi_i}{S_{KK}}.$$

Each is a fixed linear combination of the noisy Π_i, so their variances follow from the noise model directly:

$$\mathrm{var}(\hat{\alpha}) = \frac{2\varepsilon^2}{n}, \qquad \mathrm{var}(\hat{b}) = \frac{\sum_i x_i^2\,(2\varepsilon^2)}{S_{KK}^2} = \frac{2\varepsilon^2}{S_{KK}},$$

and their covariance is $\sum_i \frac{1}{n} \cdot \frac{x_i}{S_{KK}}\,(2\varepsilon^2) = 0$ because $\sum_i x_i = 0$. Since $\hat{D} = -\hat{b}$, the first half of (6.3) follows. For (6.4): the rate map is $\hat{r} = -\log\hat{D}/t$, and differentiating gives $\delta\hat{r} = -\delta\hat{D}/(Dt)$. The root satisfies $\hat{F} = \bar{K} + \hat{\alpha}/\hat{D}$ exactly — the fitted line passes through $(\bar{K}, \hat{\alpha})$ and falls with slope $-\hat{D}$, so it crosses zero $\hat{\alpha}/\hat{D}$ to the right of $\bar{K}$. Differentiate in the two independent errors, using $\alpha = D(F - \bar{K})$ for the true values:

$$\delta\hat{F} = \frac{\delta\hat{\alpha}}{D} - \frac{\alpha}{D^2}\,\delta\hat{D} = \frac{\delta\hat{\alpha}}{D} - (F-\bar{K})\,\frac{\delta\hat{D}}{D}. \qquad \square$$

Now put the running board's numbers into (6.3)–(6.4), because the asymmetry *is* the story. The board has $n = 25$ strikes and $\sqrt{S_{KK}} \approx 90$ strike-dollars, against $\sqrt{n} = 5$. With ε of 2 cents:

$$\mathrm{sd}(\hat{D}) = \frac{0.0283}{90} \approx 3.1 \times 10^{-4}, \qquad \mathrm{sd}(\hat{r}) = \frac{3.1 \times 10^{-4}}{0.9802 \times 0.5} \approx 6.4 \times 10^{-4},$$

more than six basis points of rate scatter from two cents of quote noise. The forward, by contrast, is dominated by the level term:

$$\mathrm{sd}(\hat{F}) \approx \frac{\varepsilon\sqrt{2}}{\sqrt{n}\,D} = \frac{0.0283}{5 \times 0.9802} \approx 0.6 \text{ cents},$$

about half a basis point of a \$100 forward — and the lever term adds almost nothing, because on this board $F - \bar{K} = \$1.51$: the slope's error reaches the forward only through that short lever. So the same two cents of noise leave the forward 11 times better determined than the rate — rate basis points absolute, forward basis points relative, each parameter judged against its own scale, as the section 6.2 convention set out. Why is the slope the delicate direction? Compare the two errors as *fractions of the numbers they perturb*. The slope's absolute error is tiny, but the slope itself is a discount near one, so relatively it is $3.1 \times 10^{-4}/0.98 \approx 3.2$ bp; the level's error sits on a level of $DF \approx \$100$, so relatively it is $0.006/100 \approx 0.6$ bp. The design already favors the level by a factor $F\sqrt{n}/\sqrt{S_{KK}} \approx 5.6$. Then the maps out of the line finish the job in opposite directions: the level passes to the forward almost unchanged, while the slope's error is divided by $D\,t$ on its way to the rate — doubling the gap at $t = 0.5$, and worse the shorter the expiry, since $\delta\hat{r}$ carries $1/t$.

Figure 6.2 is the audit. It runs 2000 noisy copies of the running board — fresh two-cent noise on every mid, fitted by (6.2) each time — and histograms the two errors. In panel (a), the implied rate: the measured scatter is 6.3 bp against the predicted 6.4 bp, the dashed lines marking the prediction of (6.3). In panel (b), the forward: measured 0.57 bp of F against predicted 0.57 bp. The formulas are not asymptotic decoration; they are the histograms' widths. And the $1/t$ amplification is real: rerunning the same experiment with the expiry moved from half a year to $t = 0.05$ — less than three weeks — leaves the forward essentially unchanged but blows the rate scatter up to 63 bp. A short-dated parity slope is close to meaningless as a rate estimate, and nothing about that is fixable by fitting harder: it is the design of the data, not the algorithm.

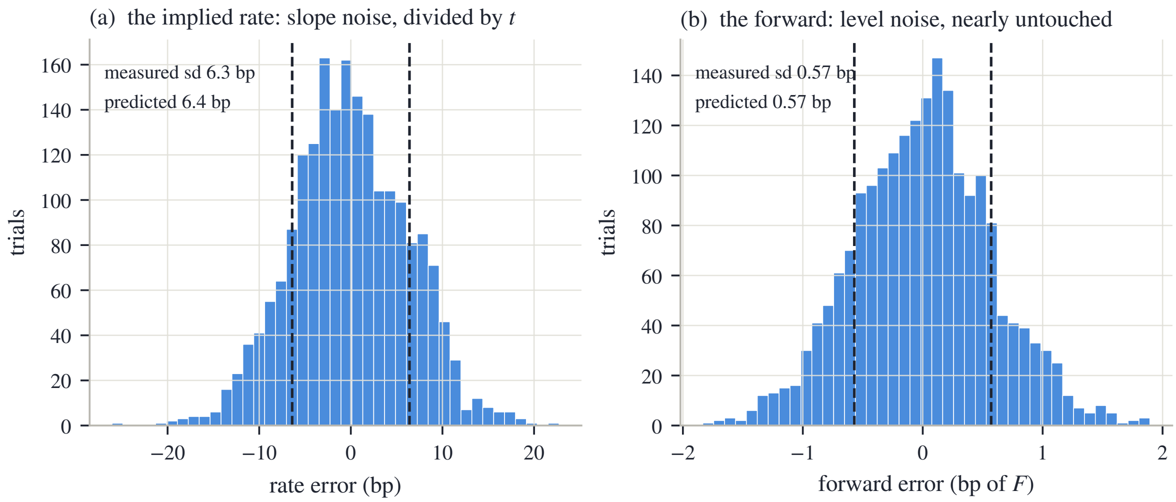

Figure 6.2: The identifiability asymmetry, measured (2000 trials of 2-cent noise on the running board, prediction of proposition 6.1 in dashed lines). (a) The implied rate scatters by 6.3 bp (predicted 6.4 bp). (b) The forward scatters by 0.57 bp (predicted 0.57 bp). Same line, same noise: the level is measured by every quote, the slope is teased out of small differences and then divided by t.

This section's fact is worth restating in one plain sentence, because the next two sections are both consequences of it: *the parity line's level — and with it the forward — is nearly noiseless, while its slope — and with it the discount and the rate — is the worst-determined object in the problem.* A defense built for real chains should therefore spend its suspicion on the slope and its trust on the level. The next section starts with the failures that hit both: quotes that are not merely noisy but wrong.

6.4 Estimating on a dirty chain

The noise model of section 6.3 was polite: every quote wrong by a little, none wrong by a lot. Real chains are not polite. A quote can be *stale* — last updated minutes ago, while the market moved — or *crossed* — its bid above its ask, a pair no live market would let stand — or simply wrong, and one such point is not two cents off the line but a dollar off. This section adds the classical defense, watches it work, and then — more important — finds its blind spot.

The defense is an *iterated trim*, and it is standard robust statistics rather than anything of ours [44]. Fit the line. Compute the residuals. Estimate their typical size robustly — take the median of the absolute residuals and multiply by 1.4826, the constant that makes this estimator agree with the standard deviation on clean Gaussian data, so one huge outlier cannot inflate the scale used to judge it; floor the estimate at one basis point of spot so a suspiciously perfect board does not set an impossibly strict bar. Call the result $\hat{\sigma}$. Drop every pair whose residual exceeds $4\hat{\sigma}$, refit, and repeat — at most three rounds, and never below three surviving pairs, since two points can always draw a perfect line. On a clean board the loop finds nothing, drops nothing, and returns the plain fit bit for bit; the defense costs nothing when it is not needed.

Figure 6.3 stages the intended failure on the running board. One deep put, at strike 75, is stale: its quoted mid is too high by \$1.20, so $\Pi(75)$ sits \$1.20 below the line. In panel (a) the corrupted point is ringed; the fitted line shown is the trimmed one, and the eye confirms there was never any doubt about where the line should go. Panel (b) shows the residuals against that trimmed fit, with the $\pm 4\hat{\sigma}$ band shaded: every clean point lives inside a band a few cents tall, and the stale put stands roughly 120 robust sigmas outside it. It is dropped in the first round (1 point trimmed), and the refit recovers the true forward to 0.1 bp.

The raw fit's damage, before the trim, is itself a lesson in section 6.3's asymmetry. The untrimmed line's forward is off by only 4 bp — one gross point among 25 barely moves the level — but its implied rate comes out at 4.8% against a true 4%. Even vandalism obeys the proposition: the damage concentrates in the slope.

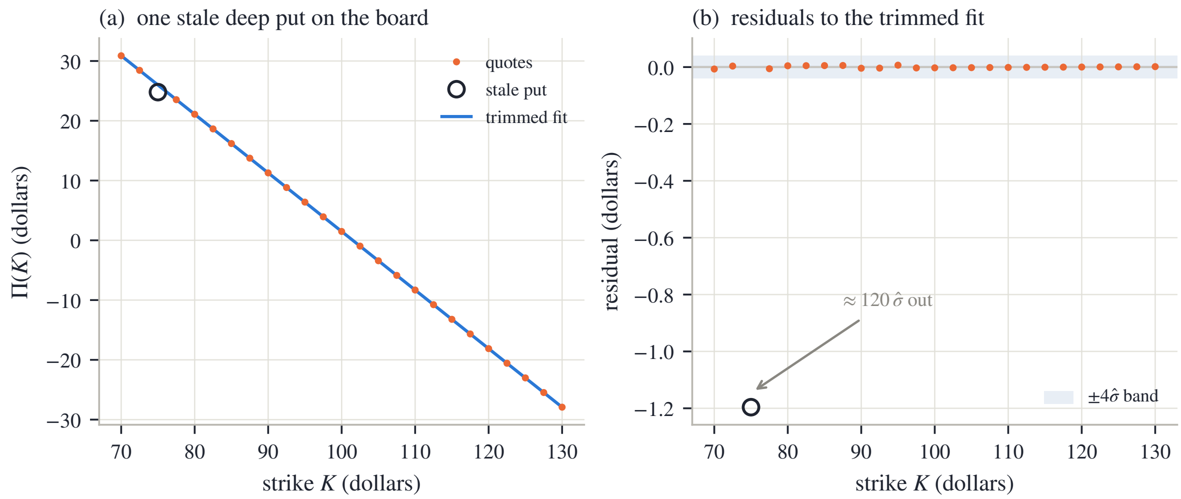

Figure 6.3: The trim doing its job (running board, one staged stale put). (a) The corrupted point (ringed) sits visibly off the line; the trimmed fit ignores it. (b) Residuals against the trimmed fit: the clean points live in the shaded $\pm 4\hat{\sigma}$ band, cents tall; the stale quote stands $\approx 120\,\hat{\sigma}$ out and is dropped in one round. The refit is exact to 0.1 bp.

Remark 6.2 (what the trim cannot see: coherent staleness). A residual trim detects points that disagree with the consensus line. Stale points that agree *with each other* are a different animal. Rerun the experiment with a coherent failure: instead of one random error, let the 5 lowest strikes — the deep put wing, exactly the quotes that update least often — all go stale *together*, their prices recomputed at a financing rate 2% away from the truth, as if that whole wing last traded under yesterday's funding. Each stale quote is now consistent with its stale neighbors, the blended least-squares line splits the difference between the two regimes, and the residuals spread out enough that *nothing* exceeds $4\hat{\sigma}$: the trim flags 0 points. This is the *masking* phenomenon of robust statistics [44] — contamination that moves the consensus toward itself hides from any test built on distance to the consensus. The fitted rate lands at 1.7% against a true 4%: badly wrong, yet entirely plausible — no test on the rate's *value* would blink at it. What survives is, once again, the level: the forward moves only 13 bp, because 5 stale quotes on one wing tilt the line without lifting its center much. Coherent staleness looks, from inside the chain, exactly like a change in the financing rate; it cannot be detected there, only bounded — and the bound is proposition 6.1's asymmetry itself.

Where we stand: the trim removes quotes that betray themselves; coherent staleness does not betray itself, corrupts the slope, and spares the level. So the slope needs protection that does not come from the chain at all: prior knowledge of where a rate can plausibly live. The next section builds that defense — and is honest about its reach. It stops slopes that leave the plausible range; the masking experiment's slope, plausible by construction, is beyond any within-chain repair, and stays bounded rather than fixed.

6.5 A prior on the rate, and the lever-arm identity

Here is where the chapter stands. The slope is the fragile parameter (section 6.3); some failures corrupt it without tripping any alarm inside the chain (remark 6.2). On a slow or stale data feed the corruption can go further than the staged examples: the fitted discount drifts above one, which means a *negative* implied rate on an ordinary US equity chain. The question this section answers is what to do when the data's least reliable parameter contradicts something we know from outside the data.

The answer is a prior — prior knowledge, imposed as a hard constraint. A financing rate for a US equity has a physical range: not below zero by percents, not above by tens. Fix a band of plausible rates (its endpoints are a disclosed convention, reviewed like any other; the reference implementation ships a wide one) and apply one rule: *if the fitted rate $\hat{r}$ falls inside the band, keep the regression's answer untouched; if it falls outside, stop trusting the slope* — set the discount to the band's edge, the nearest value prior knowledge allows. Clean chains sit strictly inside the band and pass bit for bit, like the trim before: the device only exists on demonstrably bad inputs.

But rejecting the slope creates an obligation. The forward estimate $\hat{F} = \hat{a}/\hat{D}$ has the rejected slope in its denominator; if $\hat{D}$ was untrustworthy, so is $\hat{F}$ computed from it. The forward must be re-derived from what is still trustworthy — the level — and *how* to do that well is a small piece of mathematics worth isolating, because it explains a design choice that looks arbitrary otherwise.

Solve (6.1) for F one strike at a time: on a noiseless board, $F = K_i + \Pi_i/D$ for every i. So given any discount D' — possibly the clamped one, possibly wrong — a whole family of *level estimators* is available: pick weights $\mu_i \geq 0$ summing to one (μ_i in its Chapter 3 role, a weight per quote) and average,

$$\hat{F}_\mu(D') = \sum_i \mu_i \Big(K_i + \frac{\Pi_i}{D'} \Big).$$

The question is how much a wrong D' hurts, and the answer is exact and short.

Proposition 6.3 (the lever arm of a level estimator)**.** *On a noiseless board obeying* (6.1),

$$\hat{F}_\mu(D') - F = \Big(1 - \frac{D}{D'}\Big)(\bar{K}_\mu - F), \qquad \bar{K}_\mu = \sum_i \mu_i K_i. \tag{6.5}$$

Proof. Substitute the truth $\Pi_i = D(F - K_i)$ into one term of the average:

$$K_i + \frac{\Pi_i}{D'} = K_i + \frac{D}{D'}(F - K_i) = K_i \Big(1 - \frac{D}{D'}\Big) + \frac{D}{D'} F.$$

Averaging with the weights μ_i replaces K_i by $\bar{K}_\mu$:

$$\hat{F}_\mu(D') = \bar{K}_\mu \Big(1 - \frac{D}{D'}\Big) + \frac{D}{D'} F = F + \Big(1 - \frac{D}{D'}\Big)(\bar{K}_\mu - F). \qquad \square$$

Read (6.5) in words: the transmitted error is the relative discount error times $\bar{K}_\mu - F$, the distance from the estimator's center of weight to the forward. That distance is a *lever arm.*

Where an estimator centers its strikes decides how much a wrong discount can hurt it — and the modeler chooses the center.

Figure 6.4 measures the identity on a deliberately asymmetric board (strikes from 85 to 140, so that averages are pulled away from the money; constants in appendix 6.A), imposing rate errors of up to two percent and re-deriving the forward three ways. In panel (a), the three curves. The steep one is the regression reading $\hat{a}/D'$. It belongs to the family too, in a degenerate way: the intercept $\hat{a}$ is the fitted line's value extrapolated to strike *zero* — a strike no board quotes — so reading $\hat{a}/D'$ amounts to placing all the weight at that phantom point, $\bar{K}_\mu = 0$. Its lever arm is then F itself, the worst available, and it transmits 50.0 bp of forward error per percent of rate error. The middle curve is the uniform average over the board: its lever is $\bar{K} - F$ — mean strike \$112.5 against forward \$101.5, a \$11.0 arm — transmitting 5.4 bp per percent: better by an order of magnitude, but only by the accident of where the board happens to center. The flat curve is the estimator the reference implementation uses when its band bites: weights concentrated *at the money* by a Gaussian kernel in log-moneyness (centered at spot — the one anchor that needs no estimate), whose center of weight lands at \$102.1: a lever of \$0.6, transmitting 0.31 bp per percent. The open circles confirm the middle curve against (6.5) evaluated directly — identity, not approximation. Panel (b) simply draws the three lever arms on a log scale: \$101.5 versus \$11.0 versus \$0.6. The design lesson is compact: *the clamp can afford to be crude about the rate because the level estimator it hands the problem to barely feels the rate at all.* A last-resort guard remains behind it — if even the level answer lands absurdly far from spot, the estimate falls back to spot itself and says so — and on such a chain no smile should be published at all.

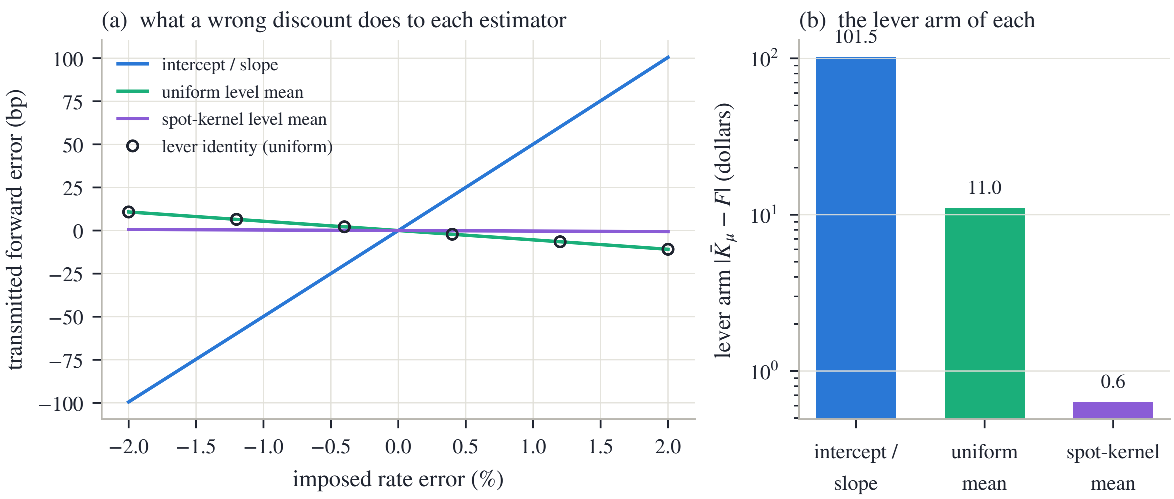

Figure 6.4: The lever-arm identity (6.5) on the asymmetric board. (a) Forward error transmitted by an imposed rate error, for the three level estimators; circles are the identity evaluated directly. (b) Their lever arms $|\bar{K}_\mu - F|$: the regression reading levers on F itself (50.0 bp per percent), the uniform mean on the board's accidental centering (5.4), the spot-centered kernel on under a dollar (0.31).

One honest caveat closes the section. The band is a prior, and a prior can be wrong: a regime of genuinely negative dollar rates would sit outside it, and the device would then clamp away information rather than noise. That is the standing trade of this chapter — stated once here, collected in the contract table of section 6.8. What we now have is a defense in two layers, each matched to a failure mode and each inert on clean data: the trim for quotes that betray themselves, the band plus the short-lever re-derivation for slopes that go absurd. Between them sits the failure neither catches — coherent, plausible staleness — bounded by the level's robustness and nothing else.

6.6 Dividends: what the carry contains

So far the pair (F, D) has been estimated without asking what makes F differ from spot in the first place. This section opens that box, because two later needs depend on it: the borrow of section 6.8 is defined as what remains of the carry after dividends are accounted for, and Chapter 7's early-exercise machinery runs on a dividend model as its fuel.

Call *carry* the growth rate the market charges between spot and forward: the number $\log(F/S)/t$. For a stock that pays nothing, carry is just the rate — $F = Se^{rt}$, because the alternative to agreeing on delivery is to buy the stock today with borrowed money, and the borrowing costs r. Dividends change the arithmetic: whoever holds the stock until expiry *collects* them, and the forward buyer does not, so their value is subtracted from what delivery is worth. How exactly it is subtracted depends on how the dividends are modeled, and there are three standard conventions.

Discrete cash. The stock will pay fixed dollar amounts d_i at *ex-dates* $t_i \le t$ — the dates from which a buyer of the stock no longer receives the next dividend, so the price drops by roughly the payment as each arrives. (Both symbols are always subscripted; the upright d of an integral is unrelated.) Replicate delivery: buy the stock today for S, borrow the money, hold to expiry. Along the way each dividend d_i arrives and can be lent out from its ex-date onward, growing to $d_i e^{r(t-t_i)}$ by expiry. The all-in cost of delivering the stock at expiry is therefore the financed purchase minus the collected, reinvested dividends:

$$F = Se^{rt} - \sum_i d_i\, e^{r(t-t_i)} = \Big(S - \sum_i d_i\, e^{-rt_i}\Big)e^{rt}. \tag{6.6}$$

The second form is the useful one: spot minus the *present value* of the dividend schedule, carried forward at the rate. This is called the escrowed-dividend forward — as if the dividends were set aside in escrow at time zero.

Discrete proportional. Each ex-date the price drops not by a fixed dollar amount but by a fixed *fraction* f_i of whatever the stock is worth that day. The replication changes in one place: reinvest each dividend into the stock itself rather than lending it. Follow one ex-date to see the bookkeeping. Just before it, a share is worth some price S^-; the holder receives $f_i S^-$ in cash, and the price drops to $(1 - f_i)S^-$. Spending the cash on shares at the dropped price buys

$$\frac{f_i S^-}{(1-f_i)S^-} = \frac{f_i}{1-f_i} \quad \text{new shares per share held,}$$

multiplying the holding by $1 + \frac{f_i}{1-f_i} = \frac{1}{1-f_i}$. Each ex-date therefore cancels one factor $(1 - f_i)$: start with $\prod_i (1 - f_i)$ of a share today and exactly one share exists at expiry. The cost today is $S \prod_i (1 - f_i)$, and

$$F = S \prod_i (1 - f_i)\, e^{rt}. \tag{6.7}$$

Continuous yield. Smear the proportional drops into a constant leak q_d (subscripted — bare q is Chapter 2's quantile density): the limit of (6.7) with many small events is

$$F = S\, e^{(r-q_d)\,t}. \tag{6.8}$$

This is the cheapest convention, and the one the running board has been using since section 6.2.

Which convention should a modeler pick? At first sight it barely matters. Figure 6.5(a) shows the chapter's running schedule — \$0.65 per quarter on a \$100 stock, first ex-date 30 days out — through the lens of the *implied dividend yield*: the constant q_d that would produce the same forward at each maturity, $q_d(T) = r - \log(F(T)/S)/T$. The cash schedule draws a sawtooth. Each vertical jump is one ex-date entering the horizon: just after the first one enters, a single \$0.65 payment thirty days away annualizes to 7.9% — violent at the short end, where dividing by a small T amplifies everything (the same $1/t$ the rate suffered in section 6.3). By a year the teeth have shrunk toward the flat equivalent yield of 2.59% drawn dashed, and the two conventions price long-dated forwards almost alike. If the term structure of F were the whole story, the choice would be cosmetic.

It is not, and panel (b) shows the honest discriminator: what happens to the forward when *spot moves*. Differentiate each convention in S at a fixed schedule. For the yield and proportional forms, (6.8) and (6.7), F is proportional to S, so

$$\frac{\partial \log F}{\partial \log S} = 1 :$$

the forward rides spot one for one. For the cash form, write $\mathrm{PV} = \sum_i d_i e^{-rt_i}$ for the schedule's present value; then $F = (S - \mathrm{PV})\, e^{rt}$ with PV *fixed in dollars*, and

$$\frac{\partial \log F}{\partial \log S} = S\,\frac{\partial}{\partial S} \log\left(S - \mathrm{PV}\right) = \frac{S}{S - \mathrm{PV}} \;>\; 1. \tag{6.9}$$

A cash schedule does not scale with the stock, so on a rally the forward outruns spot: the dividends claim a fixed slice, and the slice matters relatively less as the pie grows. Panel (b) plots this *spot elasticity* against maturity. The cash staircase climbs — each step one more ex-date inside the horizon — reaching $S/(S - \mathrm{PV}) = 1.053$ at two years, where the escrowed value is $\mathrm{PV} = \$5.01$; the dotted line is the closed form (6.9) and the solid one a direct bump measurement, in agreement. The proportional and yield conventions sit at exactly one, at every maturity.

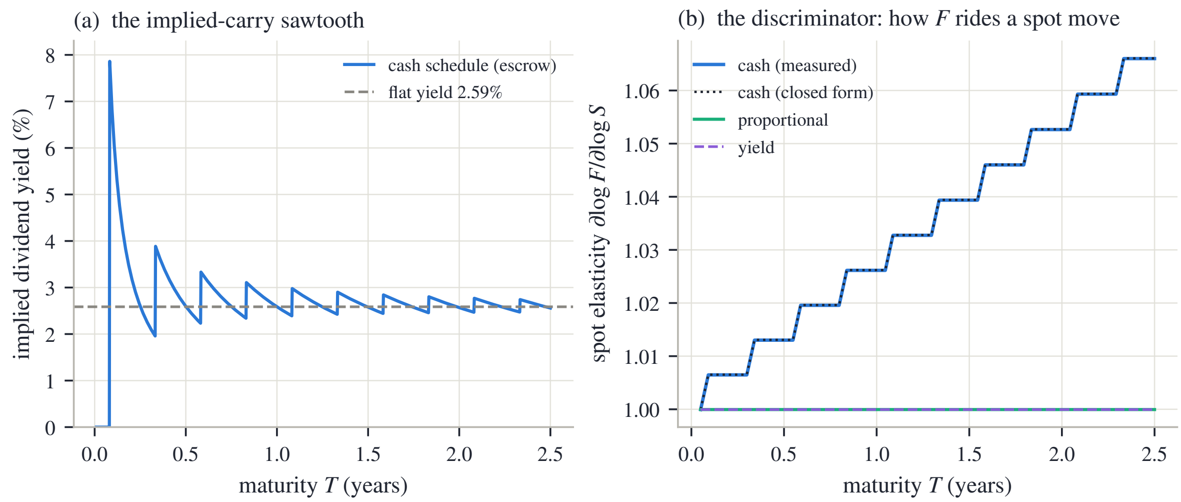

Figure 6.5: Dividend conventions on the running schedule ($0.65 quarterly, first ex-date 30 days out). (a) Implied dividend yield of the cash schedule against maturity: a sawtooth, each tooth one ex-date entering the horizon, spiking to 7.9% at the short end; dashed, the flat equivalent yield 2.59%. (b) The spot elasticity (6.9): cash climbs to 1.053 by two years (dotted: closed form; solid: measured by bumping spot); proportional and yield sit at exactly one.

Why does an elasticity deserve this much attention in a chapter about statics? Because the smile is drawn against F. When spot moves and the surface is transported to the new spot — the scenario logic Chapter 9 develops — every strike's log-moneyness $k = \log(K/F)$ moves by *minus* however much $\log F$ moves, and (6.9) says the dividend convention decides that amount. Two modelers with identical fits today, one carrying cash dividends and one a yield, will disagree about which strike is at the money *after* a two-percent rally — a risk disagreement, not a pricing one. The dividend model is thus a convention in the full sense this book gives the word: not inferable from today's chain (panel (a) showed the term structures nearly agree), consequential for what the desk does next (panel (b)), and therefore to be stated, not assumed silently.

The carry now has two named parts — rate and dividends — and the chapter can ask its last estimation question: whether what is left over, the borrow, can be measured at all. One tool is still missing: how a carry error shows up *in the smile.* That tool is the next section, and it doubles as the payoff of the whole chapter.

6.7 What a wrong forward does to the smile

The chapter opened by claiming that a small forward error is a large smile error. Every tool needed to prove that claim is now on the table, and the proof is a short computation with a memorable shape.

Set the scene precisely. The market's quotes are what they are; the error lives in the modeler's head. The true forward is F, but the smile is built at a misread forward $F' = Fe^{\delta}$ — δ is the relative error, a few basis points. Every dollar quote is then pushed through the Black formula at the wrong normalization: the strike's log-moneyness is computed as $k' = \log(K/F') = k - \delta$, and the price is divided by DF' instead of DF. The discount is read correctly throughout — the forward error is the subject. The question: how far does the re-implied volatility move, per unit of δ?

Take a call first. Its dollar price, held fixed, is $DF\, B(k, w)$ with the normalized Black call of Chapter 2 (eq. 2.3),

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2},$$

where Φ and φ are, as throughout the book, the standard normal distribution function and density. The call's two partial derivatives were computed once in Chapter 2's appendix, and this chapter reuses them:

$$\partial_k B = -e^k\, \Phi(d_-), \qquad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

Differentiate the fixed dollar price along the misreading. Three things move at once: the prefactor F (its log by $\mathrm{d}\delta$), the moneyness k (by $\mathrm{d}k = -\mathrm{d}\delta$), and the implied total variance w (by whatever it must, to keep the price fixed). To first order,

$$0 = \frac{\mathrm{d}[F\, B(k, w)]}{F} = B\, \mathrm{d}\delta + \partial_k B\, \mathrm{d}k + \partial_w B\, \mathrm{d}w = (B - \partial_k B)\, \mathrm{d}\delta + \partial_w B\, \mathrm{d}w,$$

so

$$\mathrm{d}w = -\frac{B - \partial_k B}{\partial_w B}\, \mathrm{d}\delta.$$

The numerator collapses — this is the pleasant step:

$$B - \partial_k B = \Phi(d_+) - e^k \Phi(d_-) + e^k \Phi(d_-) = \Phi(d_+),$$

and inserting $\partial_w B$ — the price's sensitivity to total variance, the object desks call the (Black) *vega* when quoted per unit of volatility —

$$\mathrm{d}w = -\frac{2\sqrt{w}\, \Phi(d_+)}{\varphi(d_+)}\, \mathrm{d}\delta.$$

Convert to volatility: $\sigma = \sqrt{w/\tau}$, so $\mathrm{d}\sigma = \mathrm{d}w/(2\sqrt{w\tau})$, and the $\sqrt{w}$ cancels:

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\delta}\right|_{\text{call}} = -\frac{\Phi(d_+)}{\varphi(d_+)\sqrt{\tau}}, \qquad \left.\frac{\mathrm{d}\sigma}{\mathrm{d}\delta}\right|_{\text{put}} = +\frac{\Phi(-d_+)}{\varphi(d_+)\sqrt{\tau}}. \tag{6.10}$$

The put case is the same computation run on the normalized put $P = B + e^k - 1$ (parity again): its numerator is $P - \partial_k P = B - \partial_k B - 1 = \Phi(d_+) - 1 = -\Phi(-d_+)$, which flips the sign and swaps $\Phi(d_+)$ for $\Phi(-d_+)$.

Before the figure, read (6.10) at the money, where $k = 0$ and $d_+ = \sqrt{w}/2$ is small: both $\Phi(d_+)$ and $\Phi(-d_+)$ are close to $\frac{1}{2}$, and $\varphi(d_+) \approx \varphi(0) = 0.399$. A forward misread *up* by δ therefore moves the two sides in opposite directions — call vols down, put vols up, each by about $\frac{1}{2}\delta/(0.4\sqrt{\tau})$. Now recall how a smile is assembled from quotes: at each strike the market's liquid side is the *out-of-the-money* (OTM) option — the put below the money, the call above — so the drawn curve switches sides at $k = 0$, and the opposite moves become a *jump* there, of size

$$\text{ATM gap} \;\approx\; \frac{\delta}{\varphi(0)\sqrt{\tau}} \;\approx\; \frac{2.5\,\delta}{\sqrt{\tau}}. \tag{6.11}$$

The intuition for the two directions is worth one sentence each. Thinking the forward is higher makes the call look closer to the money than it is, and a closer-to-the-money option carrying the same price must carry *less* volatility; the same misreading pushes the put deeper out of the money, and a deeper option at an unchanged price must carry *more*.

Figure 6.6 draws both halves of the story on the running node's fitted smile — the frozen SPY December 2026 fit of the earlier chapters, δ set to 10 bp. Panel (a) is the closed form (6.10) across the quoted span, scaled to vol basis points per 10 bp of forward error: the put side (positive curve) and call side (negative curve) approach the money from opposite signs, and both fade toward the wings — far from the money an option's price responds weakly to the forward relative to how strongly it responds to volatility, so the damage concentrates exactly where the book is most liquid. Panel (b) is the measurement: every out-of-the-money quote on the node re-implied at the bumped forward, minus its true volatility. The orange curve is the measurement; it lands on the dotted linearization so exactly that the two read as one curve — and it shows the two symptoms a desk would actually see. At the money, the advertised jump: 41 vol basis points of put–call gap for 10 basis points of forward error — (6.11) with the node's numbers. Away from the money, a tilt: the put wing lifted by 11 bp at $k = -0.10$, the call wing dropped by 9 bp at $k = +0.10$. Lifted left wing, lowered right wing — that is precisely what a steepening of the skew looks like. A desk that trusts its forward would read this picture as the market repricing crash risk; the correct reading is a data error \$5.2 wide.

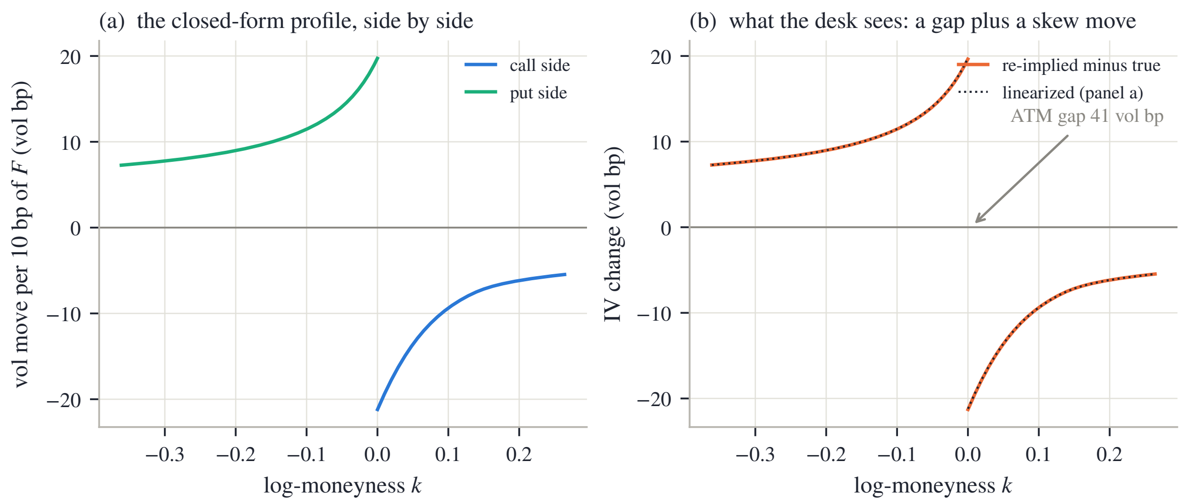

Figure 6.6: A forward error of 10 bp on the running node's smile. (a) The closed form (6.10): put side up, call side down, both largest near the money. (b) Every OTM quote re-implied at the bumped forward (orange; dotted: the linearization): an ATM put–call gap of 41 vol bp plus a wing tilt that reads exactly as a skew move.

Now close the loop the chapter opened in section 6.2. There, the naive whole-chain regression on the real (American) node missed the resolved forward by 68 bp — five dollars, invisible in price space. Push that error through (6.11): it imprints an ATM put–call gap of about 278 vol basis points — nearly three volatility *points* of artificial structure at the most liquid point of the surface, dwarfing the few-basis-point fit errors Part I worked to achieve. This is why the forward cannot be an afterthought: every error in it is repaid, amplified, at the money. And it is why the early-exercise bow of fig. 6.1(b) must be removed rather than tolerated — the task Chapter 7 takes up.

One tool from this section travels forward. Equation (6.10) converts *any* carry-side error into smile units: a rate error, a dividend error, or — next section — a borrow error, each reaching the smile through the forward it misplaces. The chapter's last question is whether that final component of carry can be measured well enough to be worth carrying at all.

6.8 The residual rate: borrow, at the edge of identifiability

The carry now has two named parts, rate and dividends. On most names, that is the whole story. On some it is not, and the leftover has a name and a market of its own.

To sell a stock short, one must first *borrow* the shares, and the lender charges a fee. For most large names the fee is a few basis points and nobody prices it. But when a stock is crowded on the short side — *hard to borrow*, in desk language — the fee can run to hundreds or thousands of basis points a year, and it enters the forward through the same replication logic as everything else in this chapter — from both sides. The arbitrage that keeps the forward from sitting too *low* requires selling the stock and buying it forward, and that arbitrageur pays the fee: the forward can sink below its frictionless level by the fee before pushing it back turns a profit. And the fee pulls from above too: whoever *holds* the stock can lend it out and collect the same fee, so holding shares unlent while the forward trades higher leaves money on the table, and competition among lenders presses the forward down to the fee-adjusted level. (In practice lendable supply and fee splits leave a band rather than a line; the equation below reads as its middle.) Writing b for the borrow rate, the carried forward becomes

$$F = S\,e^{(r-q_d-b)\,t} \qquad \Longleftrightarrow \qquad b = r - q_d - \frac{1}{t}\log\frac{F}{S}. \tag{6.12}$$

Read the second form as a measurement recipe: given the rate, the dividend model, and the parity-fitted forward, the borrow is the carry *residual* — the rate left over when the named parts are subtracted. Nothing new has to be fitted; the chapter's line already contains the number. The question, in this chapter's spirit, is not whether b can be computed — it always can — but whether the computation *means* anything. Two closed forms answer that, one for the stakes and one for the precision, and the chapter has already built both.

Materiality. A borrow error is a carry error, and a carry error reaches the smile through the forward it misplaces: by (6.12), a shift δb moves $\log F$ by $-t\,\delta b$ — the sign says a costlier borrow lowers the forward; the size is what matters here — and section 6.7's conversion (6.10) turns that into implied volatility. At the money,

$$\left|\frac{\mathrm{d}\sigma}{\mathrm{d}b}\right| = t\cdot\frac{\Phi(d_+)}{\varphi(d_+)\sqrt{\tau}} \;\approx\; 1.25\,\sqrt{t} \qquad \text{(clocks agreeing, } d_+ \text{ small)}, \tag{6.13}$$

about $125\sqrt{t}$ volatility basis points per hundred basis points of borrow. Figure 6.7(a) draws the exact curve at a flat 20% volatility: 65 vol bp at three months, 136 at a year. On a genuinely hard-to-borrow name, the borrow is not a financing footnote; it is a first-order input to the smile's *level*, on par with the volatility quotes themselves.

Precision. How well does the board pin the residual? By proposition 6.1, the parity forward is known to about $\varepsilon\sqrt{2}/\sqrt{n}$ in dollars — the level line — hence to $\varepsilon\sqrt{2}/(\sqrt{n}\,S)$ as a relative error, taking $DF \approx S$ for this back-of-envelope. Inverting (6.12), a borrow perturbs the log-forward through the factor t, so the forward's noise divides by t on its way to b: the smallest borrow the board can distinguish from zero is

$$b_{\min} \approx \frac{\varepsilon\sqrt{2}}{\sqrt{n}\, S\, t}. \tag{6.14}$$

This is the borrow's *identifiability floor* — one standard error of the residual, so "resolve" here means the one-sigma convention — and its shape is the now-familiar $1/t$: precisely the amplification that made short-dated parity rates meaningless in section 6.3 makes short-dated borrow reads meaningless too. Figure 6.7(b) draws (6.14) for a board sized like a typical single name rather than the running board — 20 pairs, quote noise stated in basis points of spot; constants in appendix 6.A — at two noise levels, against the shaded range of borrow fees typical of genuinely hard names. Read it maturity by maturity. At one week, even a clean board resolves nothing finer than 164 bp — larger than most hard-to-borrow fees themselves, so a week-dated "borrow read" is noise wearing a suit. At three months and ten basis points of quote noise the floor is 13 bp; at a year it is 3.2 bp on a clean board and 16 bp on a noisy one — comfortably below the shaded band. A borrow read is a measurement exactly where its floor sits under the fee it claims to measure; elsewhere it is a guess dressed in the estimator's output format.

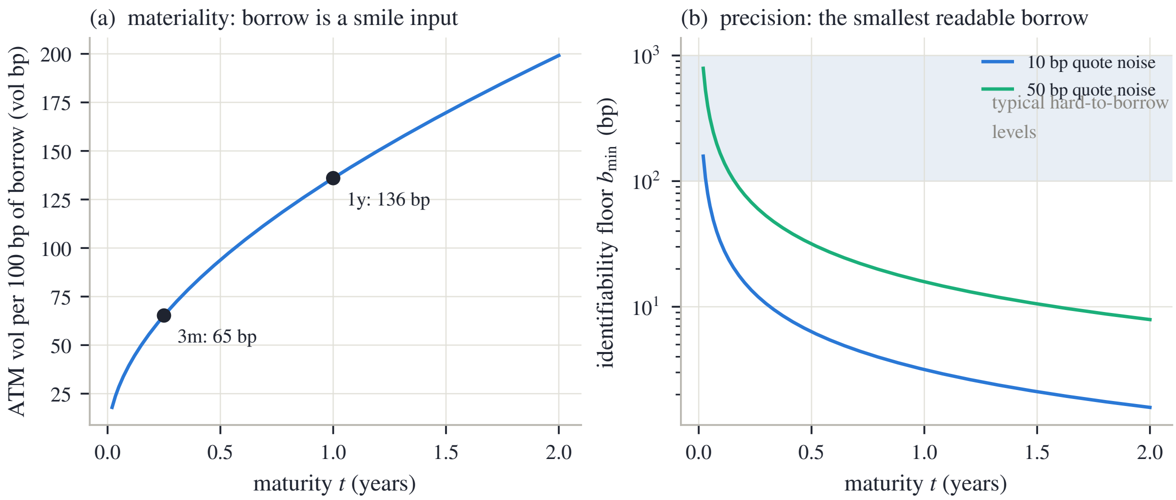

Figure 6.7: The borrow's two closed forms (flat 20% volatility, 20 paired strikes). (a) Materiality (6.13): ATM implied vol moved per 100 bp of borrow — 65 vol bp at three months, 136 at a year. (b) Precision (6.14): the identifiability floor at two quote-noise levels, against typical hard-to-borrow fee levels (shaded). A read is publishable where the floor clears the band.

The tension between the two panels is the section's conclusion, and it is a design principle rather than a formula. Borrow matters most at the short end of a stressed name — exactly where (6.14) says it cannot be measured from one chain. The honest output of an estimator in that position is not a number but a declared state: *unidentified*, reported as such, with the floor that justifies it — never a silent zero, which is also a borrow estimate, just an undeclared one. The reference implementation carries its carry as a per-expiry decomposition in which every leg is tagged with its source — measured, supplied, or unidentified — and lets a fit consume a borrow only where the read clears both its floor and a materiality gate. Two caveats bound the whole enterprise, each one sentence. A flat rate error is indistinguishable from a borrow error at a single expiry — both are carry — so splitting them honestly needs an external rate curve, and (6.12) inherits any rate bias one for one. And on American chains the early-exercise premium moves with the very carry being solved for, so the borrow and the de-Americanization must close jointly — one more reason this chapter hands its output to Chapter 7 rather than declaring victory.

6.8.1 The contract

The chapter closes the way the earlier ones do, with its contract — three columns of decreasing strength. The first holds by theorem. The second is measured, on the frozen chain or on the stated boards, and the chapter quoted every number. The third is convention: chosen, stated, and owed to the user, because no data selects it.

Table 6.1: The chapter's contract.

Proved	**Checked**	**Convention**
Parity (6.1): European call and put prices determine (F, D) as one line's coefficients, model-free.	The frozen node's line is straight to 0.35% of its range over 158 strikes (fig. 6.1a).	Mids as the regression target, and the both-sides-live pairing rule; fit-to-bid/ask variants change the observable, not the identity.
Error bars (6.3)–(6.4): level $\sim \varepsilon/\sqrt{n}$, slope $\sim \varepsilon/\sqrt{S_{KK}}$ amplified by $1/t$; the two uncorrelated.	Predicted vs measured on 2000 trials: 6.4 bp vs 6.3 bp (rate), 0.57 bp vs 0.57 bp (forward) (fig. 6.2).	The noise model itself (independent, homoskedastic mids) — adequate for design decisions, not a law.
The lever identity (6.5): a level estimator transmits a discount error through $\bar{K}_\mu - F$ only.	50.0 / 5.4 / 0.31 bp per percent for the three estimators (fig. 6.4).	The trim's thresholds, the rate band's endpoints, the kernel's width and spot centering; all inert on clean chains. Coherent staleness (remark 6.2) is bounded, not detected.
Escrow (6.6)–(6.8) and the elasticity split (6.9): cash > 1, proportional and yield $= 1$.	Sawtooth vs flat yield 2.59%; elasticity 1.053 at two years, closed form on top of the bump measurement (fig. 6.5).	The dividend convention per name — a risk decision (section 6.6), disclosed, never inferred from one chain.
The conversion (6.10) and its corollaries: ATM gap (6.11), borrow materiality (6.13), floor (6.14).	41 vol bp of gap per 10 bp of forward error on the frozen node; the naive American-mid forward would imprint 278 vol bp (fig. 6.6).	Below its floor a borrow is reported *unidentified*, never zero; a read enters fits only past a declared materiality gate.

Step back to the book's thesis, one last time for this chapter. Validity was parity itself — an arbitrage statement no model can opt out of. Information was distributed exactly as proposition 6.1 said it would be: abundant in the line's level, scarce in its slope, absent past the borrow's floor. Convention was everything the data could not select — the rate band, the dividend model, the declared treatment of what cannot be measured. One contamination ran through the chapter unresolved: the American bow of fig. 6.1(b), tolerated here, priced in section 6.7, and entangled with the borrow just above. The next chapter removes it.

6.A Numerical protocol

Every measured number in the chapter — every fit result, scatter, gap, share, or error — comes from one deterministic generator and enters the text as a generated macro. Three kinds of numbers are typed instead, deliberately: the constants of the constructions themselves (the boards, schedules, and noise levels below, which *define* the experiments); arithmetic the reader is meant to reproduce by hand (the two-strike solve of section 6.1, whose portfolio prices were placed exactly on the line with $D = 0.98$ and $F = 101.50$, and the plug-in evaluations of sections 6.3, 6.7 and 6.8); and stated approximation constants such as $\varphi(0) = 0.399$. Everything else derives from the frozen snapshot or from the boards, in the order the chapter uses them.

The real chain (figs. 6.1 and 6.6). The raw SPY chain embedded in the book's frozen snapshot, December 2026 expiry. A strike enters the parity fit when both its call and its put quote a live (positive) bid; the observable is the difference of the two mids, $\Pi = \mathsf{C} - \mathsf{P}$ with each mid the bid–ask midpoint. The naive fit is plain least squares over all surviving pairs. The *resolved* forward quoted beside it is the one stored with the snapshot — the reference implementation's resolution for that node, which de-Americanizes quotes before regressing (Chapter 7) — and the smile used in fig. 6.6 is the node's stored fit from the same snapshot. The re-implied volatilities of fig. 6.6(b) solve the one-dimensional Black inversion by bracketing to machine tolerance, put side for $k < 0$ and call side for $k > 0$; in this data the calendar and variance clocks coincide, as section 6.1 noted.

The running board (figs. 6.2 and 6.3). European and synthetic: spot $S = 100$, rate $r = 4\%$, continuous dividend yield $q_d = 1\%$, flat volatility 22%, half a year to expiry; strikes from 70 to 130 every 2.5 (25 pairs); every mid priced by the Black formula and rounded to the cent. The identifiability audit adds independent Gaussian noise of standard deviation two cents to every call and put mid separately, refits (6.2), and repeats for 2,000 trials from a fixed-seed generator — the chapter's only use of randomness; the short-dated variant repeats the experiment at $t = 0.05$. (The noise rides on top of the board's cent rounding, which adds ≈ 0.3 cents in quadrature; the predictions ignore it, and the measured–predicted agreement shows how little it matters.) Predictions come from (6.3)–(6.4) with the board's n, S_{KK}, and truth.

Trim and masking (fig. 6.3, remark 6.2). The trim is the loop of section 6.4 verbatim: robust scale 1.4826× the median absolute residual, floored at one basis point of spot; threshold four robust sigmas; at most three rounds; never below three surviving pairs. The staged failure marks the strike-75 put mid \$1.20 high. The masking variant re-prices the five lowest strikes at a financing rate two percent above the truth: their portfolio differences Π are recomputed exactly at the stale (F', D') and rounded to the cent directly (the difference, not the two mids — immaterial at this size, noted for exactness), and the identical trim runs.

The lever board (fig. 6.4). Same construction, strikes from 85 to 140, prices kept exact (no rounding) so the figure isolates the identity (6.5). Imposed rate errors span ±2%. The kernel estimator weights each strike by a Gaussian in $\log(K/S)$ of width 0.10, centered at spot.

Dividends (fig. 6.5). The schedule pays \$0.65 each quarter, first ex-date 30 days out, horizon 2.5 years, at $r = 4\%$; the proportional variant converts each payment to the fraction 0.65/100 of the prevailing level; the flat yield is the schedule's one-year equivalent. Elasticities are measured by recomputing each convention's forward under a one-basis-point spot bump, against the closed form (6.9).

Borrow (fig. 6.7). Closed forms only: (6.13) at a flat 20% volatility, and (6.14) with $n = 20$ paired strikes at quote noise of 10 and 50 basis points of spot. The shaded hard-to-borrow range, 100–1,000 bp per year, is an illustrative convention of the figure, not an estimate.

Chapter 7

Removing Early Exercise

Chapter 6 fitted the forward and the discount from the option chain and left one contamination standing: the residuals of the parity line bowed, dollars deep, because listed equity options carry a right the European pricing identity knows nothing about — the right to exercise early. This chapter measures that right and subtracts it. A little optimal-stopping theory maps where the right has value before anything is computed: never for a call on a name that pays no dividends (a theorem of Merton's), always for a deep enough put under a positive rate, and nowhere at all in the dead zone where a price equals its exercise floor and carries no volatility information. A two-step binomial tree, solved by hand, prices the right; the removal is then one root find per quote — find the volatility at which the tree's American price equals the quoted mid, and report that volatility as the European equivalent. The subtraction is a controlled cancellation: the market's premium cancels against the model's, and most of the tree's own discretization error cancels between its American and European legs. Dividends enter as dates, not rates — the premium is exactly zero before an ex-date and jumps across it, which no smeared yield can reproduce. The chapter closes Chapter 6's loop on the frozen SPY node: the parity residual bow is reproduced by the tree to the noise level, the de-Americanized line comes out straight, and its root lands within a few basis points of the stored resolved forward. On the quotes a fit actually consumes, the whole effect is a few volatility basis points — on the discarded deep in-the-money side it is a hundred or more, and knowing which number to quote is part of the lesson.

7.1 A price with an extra right

Chapter 6 ended with a debt. The parity line on the running node — SPY December 2026, the (underlier, expiry) pair every chapter has worked on — was almost perfectly straight: its residuals ran to 0.35% of the full range the observable spans. But those residuals were not noise. They bowed, cresting at +\$4.8 near the money, hundreds of times the one-cent tick (fig. 6.1b). Chapter 6 identified the culprit in one sentence — SPY's listed options are American — and promised that this chapter would measure and remove it. We now pay the debt.

Start with the vocabulary, because everything follows from it. A *European* option can be exercised only at expiry; every model in Part I priced European options, as terminal expectations under one probability law (Chapter 2). An *American* option can be exercised at any moment its holder chooses, up to and including expiry. US-listed single-stock and ETF options — the raw quotes $\mathsf{C}(K)$, $\mathsf{P}(K)$ this book's frozen chains carry — are American. Index options, and the synthetic test boards the book constructs for itself, are European; for them, every step in this chapter leaves the quotes untouched.

The extra right cannot make the option cheaper. Whatever plan a European holder has — namely, wait to expiry — an American holder may copy; the American holder simply has more plans to choose from, and the best of a larger set of plans is worth at least the best of a smaller one. Write $A(\sigma)$ for the model price of one American option as a function of the volatility σ, everything else (its strike, expiry, and carry) held fixed, and $E(\sigma)$ for its *European twin*: the same contract with the early-exercise right removed. Then

$$A(\sigma) \;\geq\; E(\sigma),$$

and the difference $A - E$ — the *early-exercise premium*, the dollar value of the right itself — is never negative. It is often exactly zero, and section 7.2 proves where. By the same larger-set logic, an American price can never sit below the option's *intrinsic value* — what exercising this instant would pay, $(S-K)^+$ for a call and $(K-S)^+$ for a put — because "exercise this instant" is itself one of the holder's plans. Section 7.2 makes both statements precise. One notation note, once: italic E is always a *price* in this book; the expectation operator keeps its blackboard $\mathbb{E}$. The premium itself gets no symbol — we write $A - E$, or say "the premium".

Here is why the premium matters to a volatility fitter. The market prints one number per quote: the American price. It does not print E, and it does not print the premium; the decomposition is real but unobservable, quote by quote. Part I's machinery inverts a *European* price into an implied volatility — that is what the Black formula's argument is — so feeding it an American mid hands it a price that is too high for what it thinks it is pricing. The inversion has no slot for the premium, so it books the whole excess as volatility, and the smile comes out biased upward.

How badly? Recall the dictionary from Chapter 6: a price error ΔP moves an inverted volatility by roughly $\Delta P/\text{vega}$, where *vega* — the price sensitivity to volatility, $\partial C/\partial\sigma = DF\,\varphi(d_+)\sqrt{\tau}$ for the dollar call, with $d_\pm$ the Black formula's usual arguments (first courses write d_1, d_2) — peaks at the money and collapses both deep in the money and far out of it. A premium of fixed dollar size therefore reads as a *large* phantom volatility exactly where vega is small: deep in the money. The damage is measured in vol basis points (one vol bp is 10^{-4} of absolute volatility, as in Chapter 6) not because the dollars are large — they are cents to a few dollars — but because the wings divide by a small vega.

Figure 7.1 runs the whole experiment on a synthetic board where the truth is known: American calls on a stock paying a 6% dividend yield against a 2% rate, half a year to expiry, priced by this chapter's tree at a true volatility of exactly 25% (all constants in appendix 7.A; why dividends make call premiums live is section 7.2's subject). In panel (a), the orange curve is the naive reading: each American price handed to the analytic European inversion. At the forward the phantom is already 92 vol bp; moving into the money it grows without mercy, reaching 3872 vol bp at the deepest strike. The blue curve is the subtraction this chapter builds (section 7.4): it returns a flat 25% across the board, to an rms of 0.7 vol bp. Notice where the blue curve *stops*: below \$78 there are no blue points at all, because there the price has collapsed onto its exercise floor and contains no volatility to recover — the dead zone section 7.2 maps and section 7.4 handles honestly. Panel (b) shows the premium itself in dollars: at most \$2.35, largest deep in the money and fading through the forward (dashed line) toward zero out of the money. Hold the two panels side by side: a two-dollar premium and a four-thousand-basis-point lie. The quote is not wrong; the model reading it is.

The chapter now walks the repair from first principles. Section 7.2 proves where the premium lives, so we know where work is needed before doing any. Section 7.3 builds the pricing machine — a binomial tree small enough to solve by hand. Section 7.4 performs the subtraction and accounts for its errors. Section 7.5 handles dividends, which enter as dates, not rates. Section 7.6 closes the loop with Chapter 6 — the forward needed the de-Americanization that needed the forward — and reproduces the residual bow. Section 7.7 measures what all of it is worth on the frozen gallery, on the population a fit actually consumes and on the one it discards, and writes the chapter's contract.

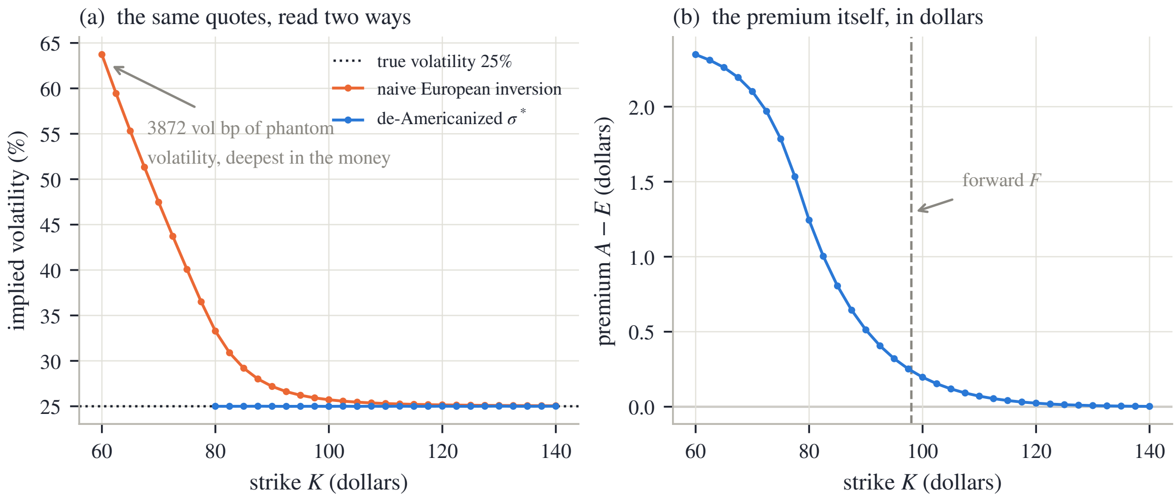

Figure 7.1: The naive reading of American prices (synthetic calls, 6% dividend yield against a 2% rate, true volatility 25%, half a year). (a) The same converged tree prices inverted two ways: the naive European inversion books the premium as volatility — up to 3872 vol bp deep in the money — while the de-Americanization of section 7.4 recovers the flat truth (rms 0.7 vol bp); below \$78 the price equals its exercise floor and no volatility exists to recover. (b) The premium $A - E$ in dollars: at most \$2.35. Cents to dollars on the left panel's axis become thousands of vol bp on the right of panel (a) because the wings divide by a small vega.

7.2 Where the premium lives

We know the premium biases the naive reading. Before removing it, it pays to know where it is — because over much of the strike–maturity plane it is *exactly zero*, and the theory that says so is short enough to carry in full.

First the formal object. A *stopping time* is a rule for choosing when to act that may use only what has happened so far — "exercise the first time the stock drops below 80" is a stopping time; "exercise at the low of the path" is not, because recognizing the low requires seeing the future. Write $\mathcal{T}$ for the set of all stopping times ϑ taking values in $[0, t]$. The American value is the value of the *best plan*:

$$A = \sup_{\vartheta \in \mathcal{T}} \mathbb{E}\left[e^{-r\vartheta}\, \mathrm{payoff}(S_\vartheta)\right], \tag{7.1}$$

with the payoff $(S_\vartheta - K)^+$ for a call and $(K - S_\vartheta)^+$ for a put. The expectation is under the risk-neutral law — Chapter 2's pricing measure, and the only fact about it this section uses is that it grows the spot at the carry, the rate the forward implies. Discounting runs at the rate r on the calendar clock t, as all financing has since Chapter 6 [40].

Proposition 7.1 (Dominance, both ways)**.** *$A \geq E$, and $A \geq$ intrinsic; hence the premium $A - E$ is never negative, and an American quote never sits below its exercise floor.*

Proof. The plans "always wait to expiry" ($\vartheta \equiv t$) and "exercise immediately" ($\vartheta \equiv 0$) are both stopping times. The first values the European twin; the second values the intrinsic. A supremum is at least every member of its feasible set. □

One line, but it already fixes two working conventions. Because the true premium cannot be negative, a negative *estimate* of it can be clamped to zero with a clear conscience (section 7.4). And an American quote *below* intrinsic is not a puzzle about premiums — it is a broken price, to be discarded, not interpreted. The substantive question is when the inequality $A \geq E$ is an equality: when is the extra right worth nothing?

Theorem 7.2 (Merton: no dividends, no call premium)**.** *For a call on an underlying that pays no dividends, with $r \geq 0$, early exercise is never strictly better than waiting: $A = E$ [37].*

Proof. Consider the option at any interior moment, with spot s and time $u > 0$ still to run, and compare exercising now against one specific alternative plan: waiting to the end. Waiting is worth the European call value, and three displayed steps bound it from below. The hinge function $x \mapsto (x - K)^+$ is convex, so by Jensen's inequality — the average of a convex function is at least the function of the average —

$$e^{-ru}\,\mathbb{E}\big[(S_u - K)^+ \,|\, S_0 = s\big] \;\geq\; e^{-ru}\big(\mathbb{E}[S_u \,|\, S_0 = s] - K\big)^+.$$

With no dividends, the risk-neutral spot grows at the full rate r (nothing leaks out of the stock): $\mathbb{E}[S_u \,|\, S_0 = s] = s\,e^{ru}$. Substituting,

$$e^{-ru}\big(s\,e^{ru} - K\big)^+ = \big(s - Ke^{-ru}\big)^+ \;\geq\; s - K,$$

where the last inequality holds because $Ke^{-ru} \leq K$, and it is *strict* whenever $r > 0$ and $u > 0$. So at every interior moment, the value of waiting strictly exceeds $s - K$, which is everything exercising now can pay. Exercising early forfeits value at every opportunity; the best plan in (7.1) is $\vartheta \equiv t$, and that plan is the European contract. □

In words a colleague could repeat: exercising a call early hands over the strike K today when the contract only ever requires handing it over at expiry — giving up the interest on the strike — and collects a stock that, paying no dividends, was growing at the full rate anyway. There is nothing to gain and interest to lose. The right is legally present and economically worthless.

Puts are different, and the same interest-on-the-strike logic now works *for* exercising.

Proposition 7.3 (Deep puts must carry a premium)**.** *For a put with $r > 0$, the premium is strictly positive deep enough in the money: whenever $S < K\,(1 - e^{-rt})$, exercising now strictly beats waiting to expiry, so $A > E$.*

Proof. A put can never pay more than K, and it pays at expiry, so its European value is capped by the discounted strike: $E \leq Ke^{-rt}$. Deep enough in the money, the intrinsic value exceeds that cap:

$$K - S \;>\; Ke^{-rt} \qquad\Longleftrightarrow\qquad S \;<\; K\big(1 - e^{-rt}\big).$$

There, $A \geq K - S > Ke^{-rt} \geq E$: the exercise-now plan strictly beats the European plan, and the premium $A - E$ is positive. □

Read the trade-off plainly. The put holder deep in the money is essentially sure to receive K; receiving it *now* rather than at expiry earns the interest. Waiting to the end pays only the option's remaining *time value* — its price in excess of intrinsic — which deep in the money shrinks toward zero while the interest does not. The threshold in the proposition is crude — it only proves the exercise region is nonempty — but the mechanism is exact, and it has a mirror image: a *dividend* is value the *call* holder collects only by exercising before the stock goes ex-dividend, so dividends (or any carry q_d exceeding r) open an exercise region on deep in-the-money calls. Section 7.5 prices that mechanism properly.

Figure 7.2 draws the whole geography, computed by the tree of section 7.3 at a fixed 25% volatility. Panel (a) maps the put premium $A - E$ in dollars over strike (as a percentage of spot) and maturity, at $r = 4\%$ with no dividends. Two features to notice. The premium concentrates deep in the money and grows with maturity — more calendar for the interest to accrue over — reaching \$8.35 per hundred dollars of spot at the top right. And the orange edge marks something stronger than a large premium: to its right the price *equals* intrinsic, to the plateau tolerance stated in appendix 7.A — at half a year, every strike beyond 129% of spot. In that dead zone — the intrinsic *plateau*, and the chapter will use both names: the price sits flat on its floor there —

the quote is the exercise floor and nothing else: no volatility enters it, so no volatility can be read out of it. Section 7.4 shows how the inversion recognizes the zone and refuses it honestly. Panel (b) runs the mirror image — calls under the running example's 6% dividend yield against a 2% rate — and the picture flips left to right: the premium sits on deep in-the-money calls, and the dead zone opens below 78% of spot. One number is deliberately absent from the figure: on the same grid with the dividend yield set to zero, the largest call premium anywhere is 0 — not small, *zero*: at every node of every tree the waiting value exceeded intrinsic, the exercise comparison never once engaged, and the American and European recursions returned identical arrays. That is theorem 7.2, measured.

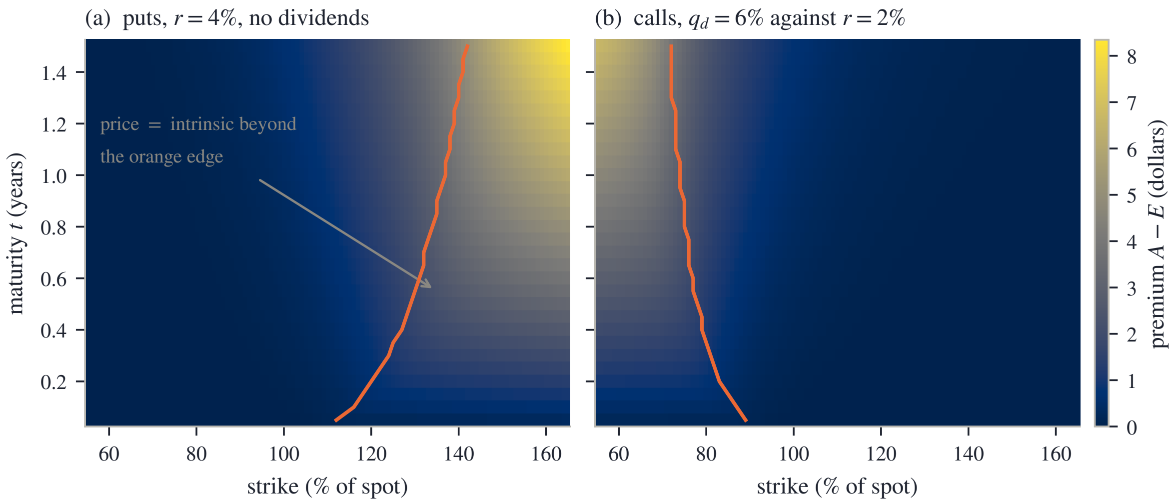

Figure 7.2: The premium $A - E$ (dollars per 100 spot) over the quote plane, at a fixed 25% volatility. (a) Puts at $r = 4\%$, no dividends: the premium loads on deep in-the-money strikes and long maturities; right of the orange edge the price equals intrinsic — at $t = 0.5$, beyond 129% of spot — and carries no volatility information. (b) Calls under a 6% dividend yield against a 2% rate: the mirror image, dead zone below 78% of spot. With the dividend yield set to zero the call premium is exactly zero everywhere on the grid: Merton's theorem, measured.

We now know the premium's address: deep in-the-money puts under positive rates, deep in-the-money calls around dividends, growing with maturity, zero for no-dividend calls, and — past the plateau — replaced by a dead zone where the quote has nothing to say about volatility. What we lack is a machine that prices the best-plan value (7.1) at will. Building one small enough to run by hand is the next section's job.

7.3 The smallest possible market: a two-step tree

We need a machine that evaluates the best-plan value (7.1). What makes the problem interesting is that at every moment the holder compares "stop now" — whose value, intrinsic, is trivial — against "continue", whose value is a conditional expectation of everything that may happen later. The simplest honest machine for carrying such expectations is the binomial tree of Cox, Ross and Rubinstein [13], and it is small enough that this section builds one from scratch and solves it by hand.

Chop the calendar time to expiry into N_t steps of length $\Delta t = t/N_t$. Over one step, let the spot do one of exactly two things: multiply by $e^{\sigma\sqrt{\Delta t}}$ (an "up" move) or by its reciprocal $e^{-\sigma\sqrt{\Delta t}}$ (a "down" move). Why this particular size? Because each step then moves the *log* of the spot by $\pm\sigma\sqrt{\Delta t}$, contributing $\sigma^2\Delta t$ of squared log-move; over N_t independent steps the contributions add to $\sigma^2 t$ — the total variance a volatility σ is supposed to generate over t years of calendar time. One matching condition down, one to go: the up-move probability p is fixed by requiring the tree to grow the spot at the carry — the rate $r - q_d$ that links spot to forward, with q_d the dividend yield exactly as in Chapter 6:

$$p\, e^{\sigma\sqrt{\Delta t}} + (1-p)\, e^{-\sigma\sqrt{\Delta t}} = e^{(r-q_d)\Delta t} \qquad \Longrightarrow \qquad p = \frac{e^{(r-q_d)\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}\,. \tag{7.2}$$

For p to be a probability we need $0 < p < 1$, which by (7.2) says the carry factor must land strictly between the down and up factors:

$$e^{-\sigma\sqrt{\Delta t}} < e^{(r-q_d)\Delta t} < e^{\sigma\sqrt{\Delta t}} \quad \Longleftrightarrow \quad |r-q_d|\,\Delta t < \sigma\sqrt{\Delta t} \quad \Longleftrightarrow \quad \sigma > |r-q_d|\sqrt{t/N_t}\,.$$

The one-step move must be able to outrun the drift. Plug in working numbers — a carry gap of 4%, half a year, and $N_t = 256$, the depth the chapter uses when it inverts whole chains (appendix 7.A) — and the floor is $0.04\sqrt{0.5/256} \approx 0.18\%$ of volatility: the condition excludes nothing any equity board quotes, but the inversion of section 7.4 keeps its brackets above it so that every tree it prices is a valid one.

Two structural facts make the lattice cheap. Multiplication commutes, so up-then-down and down-then-up land on the same spot: after n steps of which j were up, the spot is $S\, e^{(2j-n)\sigma\sqrt{\Delta t}}$, and layer n holds only $n+1$ distinct values (the tree *recombines*). And the value can be computed backward. At expiry the option is worth its payoff at each terminal node. One layer earlier, the holder at node j compares stopping against continuing, and continuing is worth the discounted p-average of the two successor values. Writing $A_j^{(n)}$ for the American value at layer n, node j:

$$A_j^{(N_t)} = \text{payoff}_j, \qquad A_j^{(n)} = \max\Big\{\text{intrinsic}_j^{(n)},\ e^{-r\Delta t}\big(p\, A_{j+1}^{(n+1)} + (1-p)\, A_j^{(n+1)}\big)\Big\}. \tag{7.3}$$

This is dynamic programming in its plainest form: start where the answer is known and walk backward, letting each node pick the better of its two options. Delete the max against intrinsic and the same recursion prices the European twin $E_j^{(n)}$ — the two legs of section 7.1's decomposition come from one lattice, differing by a single comparison. Readers of Chapter 4 will recognize the family: each value is a nonnegative combination of the next layer's, and max preserves order, so the rollback is a monotone scheme — raise any input and no output falls — which is the discrete reason tree prices are stable and free of invented oscillation.

Now solve one completely. Take an American put with $S = 100$, $K = 105$, $\sigma = 30\%$, half a year to expiry, $r = 6\%$, no dividends, and just $N_t = 2$ steps — the smallest tree in which early exercise can happen strictly inside the lattice. The step is $\Delta t = 0.25$. The up factor is $e^{0.30\sqrt{0.25}} = e^{0.15} = 1.1618$; the down factor is its reciprocal 0.8607. The one-step carry factor is $e^{0.06\times 0.25} = 1.0151$ and the one-step discount is $e^{-0.015} = 0.9851$. The up-probability (7.2) comes out

$$p = \frac{1.0151 - 0.8607}{1.1618 - 0.8607} = 0.5128.$$

The spot lattice is: after one step, 116.18 or 86.07; after two, 134.99, back to 100.00, or 74.08. The terminal payoffs of the $K = 105$ put are 0, 5.00 and 30.92. Follow fig. 7.3(a) node by node. At the up node (spot 116.18), continuing is worth

$$0.9851 \times \big(0.5128 \times 0 + 0.4872 \times 5.00\big) = 2.40,$$

and the put is out of the money there (intrinsic 0), so the node's value is 2.40. At the down node (spot 86.07) — the orange node, and the whole lesson — continuing is worth

$$0.9851 \times \big(0.5128 \times 5.00 + 0.4872 \times 30.92\big) = 17.37,$$

but exercising pays $105 - 86.07 = 18.93$ immediately. The holder exercises: the node is worth 18.93, and the best plan has stopped strictly before expiry. Rolling back once more, the root is worth

$$0.9851 \times \big(0.5128 \times 2.40 + 0.4872 \times 18.93\big) = 10.30,$$

comfortably above the root's own intrinsic of 5. Now delete the max and reprice: the down node keeps its continuation value 17.37, and the root comes out 9.55. The difference,

$$A - E = 10.30 - 9.55 = 0.75,$$

is the first early-exercise premium this book has computed by hand — earned entirely at one node, where interest on the strike beat the remaining time value. Every number above is reproducible with a pocket calculator, and checking one or two is the best way to make the rollback (7.3) feel inevitable.

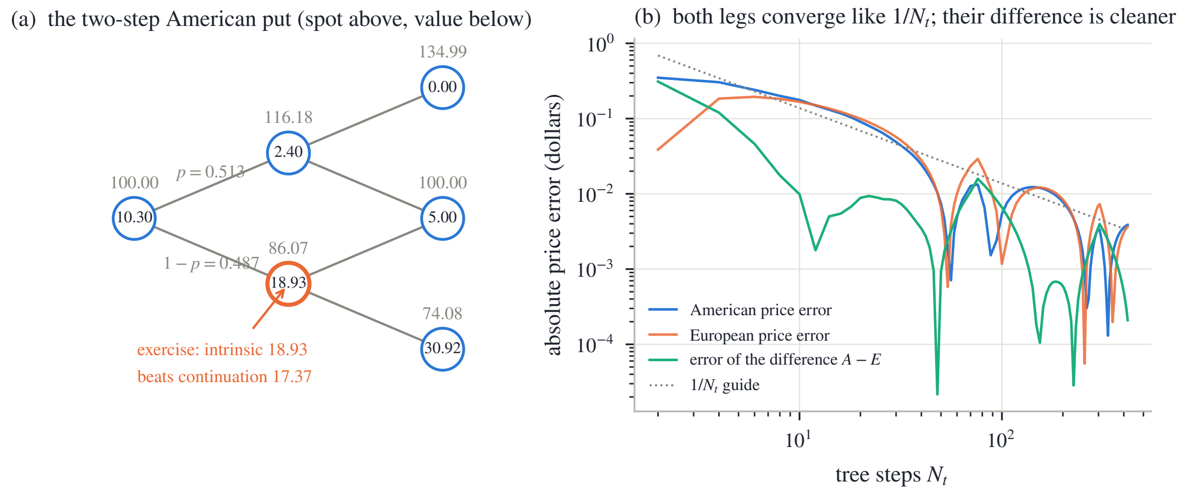

Figure 7.3: (a) The hand-solved two-step American put ($S = 100$, $K = 105$, $\sigma = 30\%$, $t = 0.5$, $r = 6\%$): spot above each node, option value inside it. At the orange node intrinsic 18.93 beats continuation 17.37 and the best plan stops early; the European twin keeps 17.37 there, and the root difference is the premium 0.75. (b) Convergence of the same put's price in the step count (even N_t, against an $N_t = 4001$ reference): both legs converge like $1/N_t$ (the dotted guide) with an oscillatory fine structure, and the *difference* $A - E$ (green) runs below either leg — in the median across depths beyond 64, 3× smaller.

Two steps prove the mechanism; accuracy needs more. Panel (b) of fig. 7.3 reprices the same put at even step counts from 2 to 400 and plots the absolute error against a deep reference tree. Read three things off it. Both the American and the European leg converge like $1/N_t$ — halve the step, halve the error — with an oscillatory fine structure that is a known cosmetic feature of binomial lattices (the strike weaves between node levels as N_t changes). At the whole-chain working depth $N_t = 256$ the American leg is within 0.10 cents of converged. And the green curve — the error of the *difference* $A - E$ — runs below either leg's, in the median 3× smaller: the two legs share one lattice, so most of their discretization error is common and cancels in the subtraction. Keep that asymmetry in mind; it is the quiet reason the next section's method reports cleaner numbers than a $1/N_t$ scheme has any right to.

We can now price the stopping right to known accuracy. The next question is how to remove it from a quote we did not price ourselves.

7.4 The subtraction

We can price the stopping right at any volatility we choose. The market, meanwhile, hands us one American mid per quote and tells us nothing else. This section removes the right from the quote, and the whole method fits in one line:

$$\boxed{\text{find } \sigma^* \text{ such that } A(\sigma^*) = \mathsf{C}(K) \text{ (or } \mathsf{P}(K)\text{), then report } \sigma^*.} \tag{7.4}$$

In words: turn the tree's volatility dial until its *American* price reproduces the quoted mid, and hand the dial's position to Part I as if it had come from a European quote. This is the standard desk repair, called *de-Americanization*, and its properties — pragmatic, fast, exact inside the model and only there — were studied critically by Burkovska et al. [11]. The rest of the section explains why the line works, when it fails, and what each error costs.

7.4.1 Why the root is the right volatility

At first sight (7.4) looks like it answers the wrong question: we wanted the premium subtracted, and instead we matched an American price. The resolution is an identity. At the root, add and subtract the tree's European leg:

$$\underbrace{\mathsf{C}(K)}_{\text{the quote}} - \underbrace{[A(\sigma^*) - E(\sigma^*)]}_{\text{the model's premium at } \sigma^*} = E(\sigma^*), \tag{7.5}$$

which holds *exactly*, because $A(\sigma^*)$ equals the quote by construction. Read it right to left: the volatility σ^* that reprices the American quote is, at the same time, the volatility at which the *European* tree price equals the quote minus the model's premium. Solving (7.4) therefore *is* subtracting a premium — the model's — and the subtraction is honest to the extent the model's premium matches the market's. Inside the model it matches perfectly; outside, the difference between the true premium and the modelled one (a different diffusion, a different dividend path) passes straight into σ^*. That is the method's one irreducible approximation, and the chapter's contract (section 7.7) states it rather than hiding it.

There is a second, quieter cancellation, and it is why a $1/N_t$ scheme can serve error budgets below a volatility basis point — the standard Part I's fits are held to. The tree is wrong about *both* of its legs by discretization; but the two legs share one lattice, so most of that error is common — fig. 7.3(b) measured the difference $A - E$ converging 3× cleaner than either leg in the median. This pairing has a name worth knowing: the tree acts as a *control variate* — the statistician's term for computing an error-prone quantity twice, once on each side of a subtraction, so that the shared error cancels in the difference. What the pairing cannot cancel is the European leg's own residual gap against the analytic Black price, and that gap is what survives into the reported volatility. The budget below prices it.

7.4.2 When the root exists, and the honest failure when it does not

Rooting a scalar equation is legitimate when the function is monotone, and here it is a theorem: in the Black–Scholes model, the value of an American option with convex payoff is nondecreasing in the volatility — convexity of the payoff propagates to the value function of the stopping problem, and more volatility then means more value [19]. Away from the intrinsic floor the monotonicity is strict, so when a root of (7.4) exists it is unique. The theorem is about the continuous problem, not about any finite tree; the working method leans on something weaker and verifiable — *bisection*, which needs only a bracket $[\sigma_{\text{lo}}, \sigma_{\text{hi}}]$ with the quote between $A(\sigma_{\text{lo}})$ and $A(\sigma_{\text{hi}})$, and halves it until the root is pinned (the bracket floor sits above section 7.3's drift condition; all constants in appendix 7.A).

"When a root exists" is not decoration. Figure 7.4 draws the map $\sigma \mapsto A(\sigma)$ for three American puts under a 4% rate at half a year. For the out-of-the-money strike ($K = 80$) and the moderately in-the-money one ($K = 110$), the curve rises strictly and crosses its quote — the dots, priced at a true 25% — exactly once: the roots are the open circles, and both recover 25%. The deep strike ($K = 160$) is another animal. Its curve *starts* on the intrinsic floor of \$60 — at the true volatility, the converged price carries 0.00 cents of time value; the quote *is* the floor, to the cent — and only lifts off the floor at volatilities roughly double the market's. A quote sitting *at* that floor is never strictly crossed from above: there is no root, because there is no volatility in the price to find. This is section 7.2's dead zone met in the flesh, and the honest output is a refusal — "this quote determines no volatility" — not a number. The refusal costs nothing downstream: a quote with no volatility content had nothing to contribute to a volatility fit.

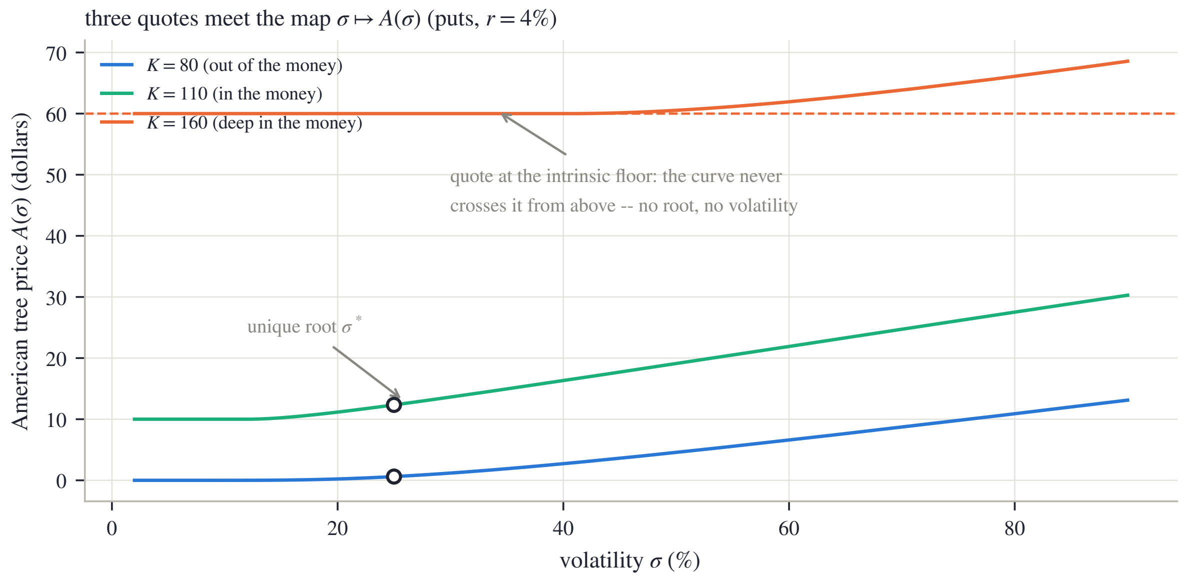

Figure 7.4: The map $\sigma \mapsto A(\sigma)$ for three American puts ($S = 100$, $r = 4\%$, $t = 0.5$). The $K = 80$ and $K = 110$ curves cross their quotes (dots, priced at a true 25%) exactly once — the open-circled roots. The $K = 160$ curve starts *on* the intrinsic floor (dashed): at the true volatility the price carries 0.00 cents of time value, the quote equals the floor, and no root exists — the quote determines no volatility, and the method says so rather than inventing one.

7.4.3 From the root to the prepared quote

One root per strike is also one *premium estimate* per strike: by (7.5), the model premium at the root is

$$\max\left(\mathsf{C}(K) - E(\sigma^*),\, 0\right), \tag{7.6}$$

where the clamp at zero is proposition 7.1 enforced against noise — the true premium cannot be negative, so a negative estimate is quote or tree noise, not information. The same single estimate is subtracted from the bid, the mid, and the ask alike, so the quoted dollar spread — the market's own statement of uncertainty, which Chapter 6 was careful to preserve — survives the surgery untouched. The shifted band then flows on exactly as Part I assumed: parity conversion to a normalized call (Chapter 2's price in units of DF), inversion to total variance, fitting. A quote whose root find failed keeps its raw band and an estimate of zero; the ordinary European sanity filters decide its fate.

One remark about clocks, before the clocks chapter makes it a theme. The root σ^* is a *calendar-clock* Black volatility: the tree marched on t, where discounting and carry live. What Part I actually consumes is the total variance w — Chapter 2's $\sigma^2\tau$ — inverted from the de-Americanized *price*, and a price is clock-free: only the reported quotient $\sigma = \sqrt{w/\tau}$ depends on which time τ one divides by. In this book's frozen data the two clocks agree to the day, so nothing subtle hides in the figures; Chapter 8 is about the cases where they do not.

7.4.4 One strike, by the numbers

Return to the worked board of fig. 7.1 and pull out the $K = 90$ call ($S = 100$, $r = 2\%$, $q_d = 6\%$, true $\sigma = 25\%$, half a year; single-quote tree depth in appendix 7.A). The tree prices

$$A = 11.7856, \qquad E = 11.2725, \qquad A - E = 0.5131 :$$

with a 6% yield to capture against a 2% rate, the stopping right is genuinely valuable on an in-the-money call — 4.4% of this option's worth sits in the premium. Hand A to the analytic European inversion and it returns $\sigma_{\text{naive}} = 27.19\%$ — a +219 vol bp phantom, the premium booked as volatility. Solve (7.4) instead and the root comes back $\sigma^* = 25.00\%$: the truth, to bisection tolerance.

7.4.5 The error budget

The reported σ^* carries four error sources; they scale differently and are worth separating once.

- *Bisection tolerance.* The working inversion runs 48 halvings of a bracket at most 4.0 wide: $4 \cdot 2^{-48} \approx 1.4 \times 10^{-14}$ of volatility. Nothing.
- *The finite tree.* The control-variate pairing protects the root, but the European-leg gap against analytic Black survives. Its honest, measured size on real data: halving the whole-chain depth from $N_t = 256$ to 128 moves the de-Americanized volatilities of the frozen running node's fitted quotes by 2.0 vol bp in the median and 8 at worst — so depth is a numerical-target dial with measurable stakes, not a free speed knob. Its small systematic footprint reappears, labelled, in section 7.7.
- *The carry.* The tree runs on Chapter 6's (F, D) and dividend model, and σ^* inherits them at full strength: the method removes the early-exercise bias and cannot repair a wrong forward. (Circularly, the forward was fitted from American quotes — section 7.6 untangles that loop.)
- *The model.* The lattice diffusion between dividend dates, the deterministic dividend path of section 7.5, and the one-sided clamp (7.6) are modelling choices. Exact in the model, approximate outside it; disclosed in the contract.

We now own the subtraction end to end: a root, its meaning as a premium removal, its unique-existence theorem, its honest refusal, and its priced errors. What we have not yet earned is the tree's *carry input* when real cash dividends are on the calendar.

7.5 A dividend is a date, not a rate

So far the tree has run on a smooth carry: the pair (r, q_d) implied by Chapter 6's fitted forward, with the dividend yield q_d draining value out of the spot continuously. For the *European* leg that is often good enough — a European price cares about the forward at expiry, and Chapter 6 showed how to make any dividend convention reproduce the right forward. The *stopping right* is pickier. Section 7.2 located the call-side premium in capturing a dividend by exercising *before its ex-date* — the last moment the stock still carries the payment. A date, not a rate. This section teaches the tree about dates, and measures what smearing them into a rate gets wrong.

The obvious construction fails, instructively. Take the lattice of section 7.3 and simply subtract the cash amount d_1 — Chapter 6's subscripted dividend symbol, not the Black argument $d_\pm$ — from every node at the ex-date. Recombination dies on the spot: follow the two two-step paths that should meet, with an ex-date between the steps, writing $\mathrm{u} = e^{\sigma\sqrt{\Delta t}}$ just for this display:

up, drop, then down: $(S\mathrm{u} - d_1)/\mathrm{u} = S - d_1/\mathrm{u}$, $\qquad$ down, drop, then up: $(S/\mathrm{u} - d_1)\,\mathrm{u} = S - d_1\mathrm{u}$.

The multiplicative moves scale the subtracted cash differently on each path, the two nodes no longer coincide, and the lattice explodes from $n+1$ nodes per layer into 2^{N_t} distinct paths — from a few hundred values to more than there are atoms in a spreadsheet.

The classical repair is the *escrowed spot* [30]. Split the stock into the part that will be paid out and the part that will not: set aside ("escrow") the present value of the scheduled dividends, and let only the remainder diffuse. The diffusing base starts at

$$X_0 \;=\; S - \mathrm{PV}_0, \qquad \mathrm{PV}_0 = \textstyle\sum_i d_i\, e^{-rt_i},$$

with cash amounts d_i at ex-date year fractions $t_i \le t$, exactly Chapter 6's schedule objects. The base lattice is purely multiplicative — it recombines, cash never touches it — and the *true* spot at any node is reconstructed when needed: base value plus the present value of the dividends *not yet paid* at that layer's date. The exercise comparison in (7.3) tests intrinsic against that true cum-dividend spot, so the holder's decision sees every upcoming payment; at expiry every scheduled dividend has been paid and the payoff reads the bare base. One honest label: under escrow the thing that diffuses lognormally is the dividend-stripped spot, so the model's spot distribution is a *displaced* lognormal — a slightly different law than the smooth-carry tree at the same forward. That is a modelling choice inside the premium, and it goes in the contract with the others.

One consistency condition ties the schedule to Chapter 6, and it must hold *exactly*: the escrowed tree's forward, $(S - \mathrm{PV}_0)\, e^{rt}$, has to reproduce the fitted forward F — otherwise the premium machine and the parity line disagree about the same chain. When the forecast schedule misses by a little, the repair follows Chapter 6's escrow lesson — the *timing* is the forecast's, the *amounts* bend to the market — by rescaling every cash amount by the common factor

$$\frac{S - Fe^{-rt}}{\sum_i d_i\, e^{-rt_i}}\,, \tag{7.7}$$

which is exactly the ratio that makes the escrowed forward land on F. (A nonsensical factor — negative, or far from one — is a sign the schedule itself is wrong, and rejects it.)

Now the measurement: what does the date-aware tree know that the smeared one cannot? Figure 7.5 prices a deliberately heavy payer — one \$6 dividend on a \$100 stock, ex-date at 0.35 years, against a 2% rate — both ways. In panel (a), fix the expiry at $t = 0.5$ and hand both models the *same* forward (\$94.99; the matching smooth yield is $q_d = 12.29\%$). The premiums still disagree — everywhere, and by up to 3.1× where both are meaningfully positive. Matching the forward is not enough, because the premium prices the *timing* of the payout: a lump of value leaving at one date is worth more to an exerciser — who can position just ahead of it — than the same value bleeding out smoothly.

Panel (b) is the sharper exhibit. Fix the in-the-money $K = 85$ call and sweep the *expiry* across the ex-date. The date-aware premium (blue) is *exactly zero* — machine zero, 0 — for every expiry before 0.35: an option that expires before the ex-date has no dividend inside its life to capture, so theorem 7.2 applies verbatim. At the ex-date the premium does not fade in; it *jumps*. The reason is worth one plain sentence: for an expiry just past the date, the holder can exercise the instant before the stock goes ex and collect the full cum-dividend intrinsic, while the European twin must ride through the drop — so crossing the date hands the best plan, at once, roughly the dividend times the probability of finishing in the money (the panel's jump is a little under \$5 of the \$6, this call being deep but not certain). The smeared comparator (green, the schedule's one-year-equivalent flat yield $q_d = 6.14\%$) misses both regimes: before the date it *invents* \$0.65 of premium where the truth is exactly zero, and after the date it underprices the lump it spread out. A dividend forecast, for the stopping right, is a calendar — the amounts matter, and the dates matter separately.

For the frozen SPY chains that thread this book, the chapter's own real-data figures run the smooth-carry tree with the forward-consistent q_d: an ETF's quarterly dividends are small,

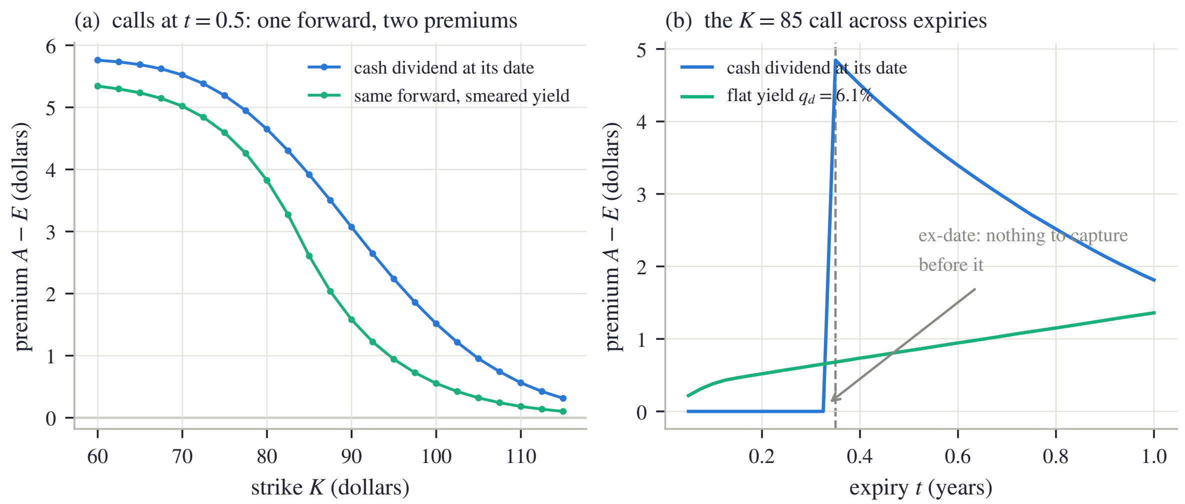

Figure 7.5: One \$6 cash dividend, ex-date 0.35 years, $r = 2\%$, $\sigma = 25\%$. (a) Calls at $t = 0.5$ under the escrowed schedule versus the smooth yield matched to the *same* forward (\$94.99): equal forwards, unequal premiums — up to 3.1× — because the premium prices the payout's timing. (b) The $K = 85$ call across expiries: the date-aware premium is exactly zero before the ex-date (nothing inside the option's life to capture: theorem 7.2 verbatim) and jumps across it; the flat smear ($q_d = 6.14\%$) invents \$0.65 of premium where the truth is zero and underprices the lump after.

and the fitted forward already carries their information. The residual cost of that convention is measured, not assumed, in the next section — a few basis points of forward. For a single name with heavy dates, the escrowed tree is the honest default, and either choice rides into σ^* as part of the model.

We now have the machine, the subtraction, and the carry. One debt to Chapter 6 remains — the forward that all of this runs on was itself fitted from American quotes.

7.6 The loop with Chapter 6, closed

Here is the circle we have been walking around. To run the tree, we need the carry — Chapter 6's pair (F, D), the forward and the discount factor. Chapter 6 fitted them from put–call parity. But parity, as derived in section 6.1, is a statement about *European* portfolios, and the quotes on the line were American: each mid shifted by its own premium, more on the put side than the call side, which is exactly why the residuals bowed and the naive fitted forward sat 68 bp low. (Basis points of *forward* in this section, not of volatility — we are back on Chapter 6's ground.) So the subtraction needs the forward, and the forward needs the subtraction. Neither comes first.

The resolution is the standard one for a fixed point: start with the biased answer and iterate, because each half-step improves the input to the other. Concretely, on the running node's board of 158 paired strikes:

1. *Pass one.* Take the naive fitted forward $\hat{F} = 762.71$ from fig. 6.1 as the starting carry. De-Americanize every quote on the board — both legs of every pair — through (7.4), and subtract each leg's premium estimate (7.6) from its mid. Refit the parity line *on the de-Americanized mids.* The fitted forward moves.

2. *Pass two.* Rebuild the carry from the new forward and do it again. This time the fitted forward moves by only 2.6 bp — the loop has converged for any purpose in this book, and a third pass is not worth its arithmetic.

One input is deliberately *not* taken from the board: the rate. Chapter 6 proved the line's slope is the poorly identified coefficient, and this board makes the point personally — its fitted slope implies $r = -0.24\%$, a negative rate in mid-2026, which is noise wearing a suit. Following section 6.5's discipline, the rate is pinned by a disclosed convention (a money-market 4.3%; appendix 7.A) and the *level* — the well-measured number — carries the information, through $q_d = r - \log(F/S)/t$.

Not every pair survives the loop, and the dropouts are old friends: on 35 of the 158 pairs, the deep in-the-money put sits on its intrinsic floor — section 7.4's no-root lane, live on a real board — so no premium can be measured there and the pair is left out of the refit. The 123 both-leg pairs that remain are the population of fig. 7.6.

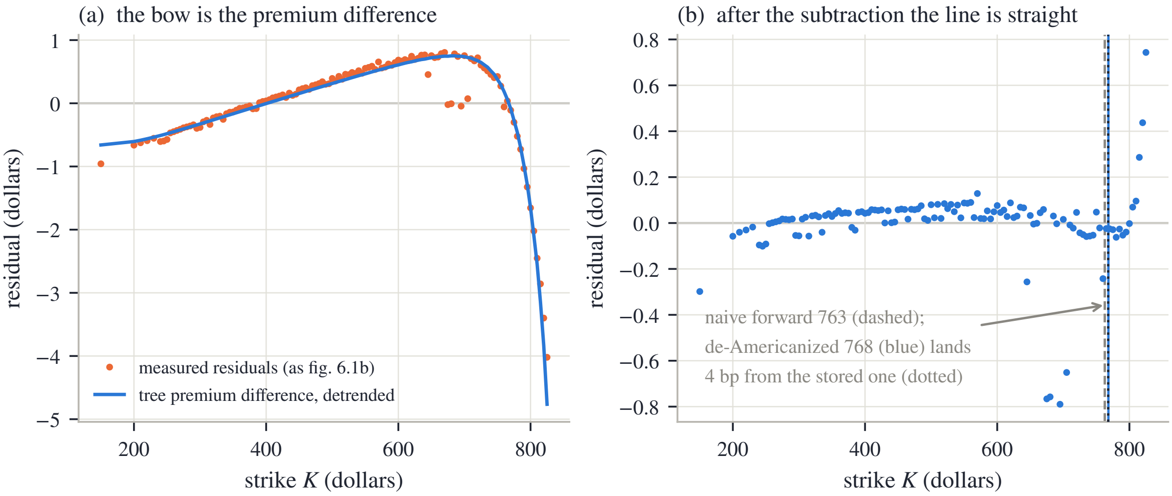

Figure 7.6: Chapter 6's debt, paid (SPY December 2026, 123 of 158 pairs; the rest sit on the intrinsic floor). (a) The measured parity residuals against the tree's premium difference, both detrended by their own line: the dots sit on the curve to \$0.17 rms — the bow *is* the premium difference. (b) Residuals after subtracting each leg's premium: rms \$2.97 down to \$0.17, and the refitted forward (blue) lands 4 bp from the snapshot's stored resolved forward (dotted) — against 68 bp for the naive one (dashed).

Panel (a) settles an account left open since fig. 6.1(b). The orange dots are the measured residuals of the parity line on the kept pairs. The blue curve is *the model's prediction of them*: the tree's premium difference, call leg minus put leg, at each pair's own fitted volatilities. It is detrended by its own least-squares line, exactly as the measured residuals are, because a parity regression absorbs anything affine — only the arch can survive it. (For the same reason the arch here is shallower than fig. 6.1(b)'s: the line is refitted on the kept pairs alone, which reshuffles how much of the bow the affine part absorbs; the shape, not the split, carries the information.) The dots sit on the curve, to an rms of \$0.17 — the same size as the quote noise left over in panel (b), which is to say: to within what this board can measure at all, the bow is *entirely* the premium difference. The shape now reads itself. At low strikes the calls are deep in the money, but SPY's dividend carry is mild against the rate, so the call premium stays small — Merton's logic with a small leak. At high strikes the put premium grows dollar-deep and drags $\mathsf{C} - \mathsf{P}$ *down*; after the fitted line absorbs the overall tilt, what remains is precisely the arch Chapter 6 could describe but not explain.

Panel (b) completes the repair. With each leg's premium subtracted, the residuals collapse from \$2.97 rms to \$0.17 — from a structured bow, dollars deep, to unstructured cent-scale noise — and the refitted forward moves from 762.71 to 768.20. That lands 4 bp from 767.92, the forward stored with the book's frozen snapshot — the resolution the reference implementation (Chapter 6's working software, whose conventions the book's figures audit) computed with its own discrete dividend schedule. Read that residual gap for what it is: the distance between two disclosed dividend conventions (section 7.5), not an unexplained error — and compare it with the 68 bp the naive forward was carrying. Chapter 6 measured, on this very node, what such basis points cost if ignored: 41 vol bp of put–call gap at the money for every 10 bp of forward error, so the naive forward would imprint 278 vol bp of artificial smile structure (section 6.7's own number). The loop is not bookkeeping; it is where the smile's raw material gets clean.

One extension is beyond this chapter, and Chapter 6 already flagged it: on a genuinely hard-to-borrow name, the borrow b enters the carry, the carry moves the premium, and the premium moves the fitted forward that identifies the borrow — section 6.8's closing caveat. The resolution is this section's loop with one more unknown riding inside it, and it inherits section 6.8's identifiability floors — the smallest borrow a board's noise lets it distinguish from zero — unchanged: below the floor, the jointly closed answer is still *unidentified*, reported as such. On ordinary names — the frozen board included — the borrow leg is immaterial and the two passes above are the whole story.

7.7 How much it matters, honestly

The single most misquotable number in this subject is "the effect of de-Americanization" with the population left unsaid. On one and the same expiry it is a few basis points, or hundreds, depending on which quotes you average over — and both numbers are true. This section measures both on the frozen gallery — the snapshot's eight SPY expiries — so the reader owns the distinction, then writes the chapter's contract.

First, name the populations. A fitting desk reads the *out-of-the-money side* of the board: puts below the forward, calls at or above it. The reason is worth one careful sentence, because it is *not* that the out-of-the-money option has more vega — at any one strike the call and the put have exactly the same vega, since parity ties their prices by a volatility-free line. It is that the out-of-the-money leg packs that same vega into a small price with a tight spread, while its in-the-money twin dilutes it through a wider spread on a bigger price and, by parity, says nothing the other leg has not already said. The book's frozen chains follow that convention. So the number that describes a fit's exposure is the wedge — naive minus de-Americanized implied volatility, quote by quote — *on the out-of-the-money side*; the wedge on the discarded in-the-money side is a stress exhibit about quotes the fit never touches.

Figure 7.7(a) shows the fitted side of the running node, quote by quote (93 quotes; construction in appendix 7.A). Walk the two halves separately. The calls (blue, $k \geq 0$) sit within a few basis points of zero, median -1.7 vol bp — and the sign deserves a sentence: a *negative* wedge cannot be a negative premium (proposition 7.1 forbids one); it is the finite tree's European-leg gap from section 7.4's budget, the numerical floor of the method showing through where the true premium is essentially zero. The puts (orange, $k < 0$) are the real payload: a few basis points out in the wing, rising steadily toward the money and reaching 61 vol bp just below $k = 0$. That rise has a clean reason worth owning. The fitting coordinate k measures moneyness against the *forward*, but the exercise right compares strike against the *spot* — and between the spot and the forward (757.67 to 767.92 here) sits a band of strikes whose puts are out of the money to the fit and already in the money to the holder: real intrinsic today, real premium, at nearly full vega. The two coordinate systems disagree exactly there, and the wedge marks the spot. Median over the whole fitted node: 3.6 vol bp — small, but not small against the sub-basis-point error budgets Part I's fits run on.

Panel (b) sweeps the whole SPY gallery, one point per expiry. The fitted-population median (filled circles) grows with maturity, from 1.32 vol bp on the one-day board to 6.8 at the long end — exactly the maturity dependence the premium map of fig. 7.2 promised, more calendar for the rate to work on — and the worst single fitted quote (open circles) reaches 221 vol bp, always in the put band just described. The orange square is the other population: the running node's *discarded* in-the-money puts. Of those, 35 sit on the intrinsic floor with no volatility to read at all — the same 35 floor-bound puts whose pairs dropped out of section 7.6's refit — and the 12

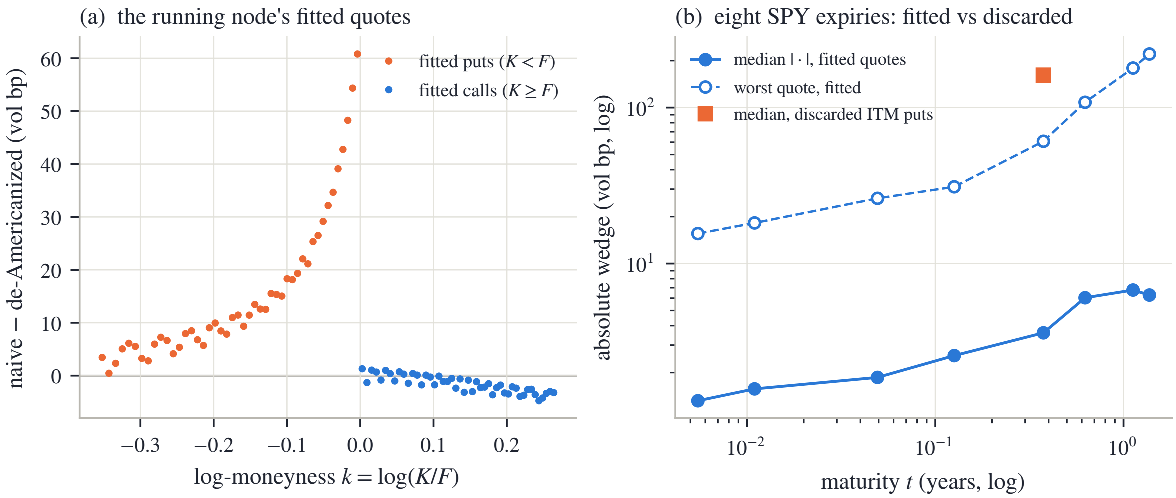

Figure 7.7: What the subtraction is worth, on the population that counts. (a) The running node's fitted (out-of-the-money) quotes: calls hug zero (median -1.7 vol bp — the tree's European-leg gap, not a negative premium); puts rise toward the money, to 61 vol bp in the band between spot and forward where the fit's coordinate and the exercise right disagree. (b) Eight SPY expiries: the fitted-population median grows from 1.32 vol bp at one day to 6.8 at the long end (filled), worst single fitted quote 221 (open); the discarded in-the-money puts of the running node (orange square) run a median of 161 vol bp — a different population, an order of magnitude away.

that do invert carry a median wedge of 161 vol bp, up to 448. An order of magnitude separates the two populations. When someone quotes "the de-Americanization effect", ask which book they averaged over; when this book quotes one, the population is in the sentence.

One last disclosure before the contract, about what the subtraction leaves behind. Each root (7.4) is solved strike by strike — no coupling across strikes — and the clamp (7.6) nudges each strike from one side only. Nothing in that construction *enforces* coherence across strikes, and Chapter 2 taught what coherence means here: the call curve convex in strike, because its second derivative is the density. On a liquid board the incoherence injected by noise and clamping hides below the quoted spread; on a sparse wing it can surface as a small *butterfly violation* — a locally non-convex stretch of the call curve, hence a patch of negative implied density — handed to the model layer as input. The reference implementation, when it repairs such a wing, does so under two self-imposed rules that are really one rule — *a repair aimed where the data is weak must be structurally unable to move where the data is strong*: wings only, never the near-money core, and never outside the quoted band. This book's fitting chapters assumed arbitrage-aware inputs; this is the sentence that says who cleans them.

7.7.1 The contract

The chapter closes as the others do: three columns of decreasing strength — proved, measured, chosen.

Step back to the book's thesis. *Validity* was the stopping theory: dominance and Merton's theorem hold for every model, and no inversion may return a negative premium or read volatility off an exercise floor. *Information* was the geography — the premium identified off the plateau, unidentifiable on it — and the honest bookkeeping of populations: a few basis points where the fit lives, hundreds where it does not. *Convention* was everything inside the subtraction: the lattice, the dividend calendar, the clamp, the depth, the disclosed rate. Chapter 6 fitted the observation's frame; this chapter removed the one right the frame could not see. Every tree in it marched on calendar time t, while the smile's variance kept its own clock τ — two clocks this book has so far been allowed to conflate, because its frozen data happens to agree on both. The

Table 7.1: The chapter's contract.

Proved	**Checked**	**Convention**
Dominance (proposition 7.1): $A \geq E$, $A \geq$ intrinsic. Merton (theorem 7.2): no dividends, no call premium. Deep puts must pay (proposition 7.3).	No-dividend call premium exactly zero over the whole mapped grid; plateau onsets at $t = 0.5$: puts beyond 129%, calls below 78% of spot (fig. 7.2).	The map's exhibit constants (a 25% volatility, the plateau tolerance) — illustrations, not estimates.
Monotonicity of $A(\sigma)$ (cited): off the plateau, the root σ^* of (7.4) is unique when it exists.	The synthetic board: truth recovered flat to 0.7 vol bp rms; the $K = 90$ case file, +219 vol bp naive against an exact root (fig. 7.1, section 7.4).	Depth N_t, bracket, and bisection count (appendix 7.A); halving the depth measurably moves σ^* (2.0/8 vol bp median/worst) — a numerical-target dial, disclosed.
The control-variate identity (7.5): at the root, quote minus model premium *is* the European leg, exactly.	The bow: measured residuals equal the predicted premium difference to \$0.17 rms; the line straightens (\$2.97 to \$0.17); the root lands 4 bp from the stored forward (fig. 7.6).	Exact in the model only [11]: the lattice diffusion, the deterministic dividend path, the one-sided clamp (7.6) all ride into σ^*.
The premium prices dividend *dates*: zero before an ex-date (theorem 7.2 verbatim), a jump across it (section 7.5).	Cash schedule vs matched-forward smear: 3.1× at one forward; \$0.65 of premium invented pre-ex-date by the flat smear (fig. 7.5).	The dividend convention per name — Chapter 6's risk decision — and the escrow rescale (7.7) tying it to the fitted forward.
The plateau is an information boundary: a quote at its exercise floor determines no volatility (section 7.4).	35 of the running node's discarded in-the-money puts sit at the floor; fitted vs discarded medians 3.6 vs 161 vol bp (fig. 7.7).	A failed root reports *no volatility*, never a number; any quoted "de-Am effect" names its population.

next chapter is about the days when they disagree.

7.A Numerical protocol

Every measured number in the chapter is computed by one deterministic program and inserted into the text mechanically — nothing empirical is typed by hand — and the chapter uses no randomness anywhere. Three kinds of numbers are typed instead, deliberately: the constants of the constructions themselves (the boards and schedules below, which *define* the experiments); arithmetic the reader is meant to reproduce by hand (the two-step tree of section 7.3, whose inputs are stated there, and plug-in evaluations such as the drift floor $0.04\sqrt{0.5/256} \approx 0.18\%$); and universal constants of the method (the bisection count and bracket below). Everything else derives from the frozen snapshot or from the stated constructions.

Trees. All trees are Cox–Ross–Rubinstein with the moves and probability of section 7.3. Depths: $N_t = 501$ for single-quote exhibits (the worked strike, the root-map curves, the escrow figure), $N_t = 256$ for whole-chain inversions (the loop and the gallery), $N_t = 4001$ for converged reference prices (the synthetic "quotes" of figs. 7.1 and 7.4 and the convergence reference of fig. 7.3b). The bisection bracket is $[\max(10^{-4}, 1.5\,|r - q_d|\sqrt{t/N_t}), 4.0]$ — the floor sits $1.5\times$ above the drift condition — with 48 halvings; a quote below intrinsic, above its static cap (S for a call, K for a put), or not bracketed — the intrinsic plateau — returns no root, and the quote is reported as carrying no premium estimate and no volatility.

The wedge board (fig. 7.1, section 7.4's case file). American calls: $S = 100$, $r = 2\%$, dividend yield $q_d = 6\%$, true volatility 25%, $t = 0.5$, strikes 60 to 140 every 2.5. "Quotes" are converged-tree American prices at the true volatility; the naive curve inverts them through the analytic European Black formula at the board's forward and discount; the de-Americanized curve solves (7.4) at depth 501. The case file's strike is $K = 90$.

The premium map (fig. 7.2). Fixed $\sigma = 25\%$; strikes 55% to 165% of spot, maturities 0.05 to 1.5 years; puts at $r = 4\%$, $q_d = 0$; calls at $r = 2\%$, $q_d = 6\%$; whole-chain depth. A node is on the intrinsic plateau when the American price exceeds a *positive* intrinsic value by less than \$0.001 per 100 of spot. The Merton check reprices the call panel at $q_d = 0$ over three maturities and records the largest $|A - E|$ anywhere.

The hand tree (fig. 7.3). American put, $S = 100$, $K = 105$, $\sigma = 30\%$, $t = 0.5$, $r = 6\%$, $q_d = 0$, $N_t = 2$; every node value is a generated number. The convergence panel reprices the same put at even depths 2–62 and depths 64–420 in steps of six, against the $N_t = 4001$ reference (the reference is odd, but at that depth the odd/even alternation sits far below the panel's scale).

The root map (fig. 7.4). American puts, $S = 100$, $r = 4\%$, $q_d = 0$, $t = 0.5$, strikes 80/110/160, volatilities 2% to 90% at depth 501; quotes for the two rootable strikes are converged prices at 25%; the deep strike's quote is its intrinsic floor, which the converged price matches to 0.00 cents.

The escrow comparison (fig. 7.5). One cash dividend $d_1 = \$6$ at $t_1 = 0.35$ years, $S = 100$, $r = 2\%$, $\sigma = 25\%$. Panel (a): calls at $t = 0.5$, strikes 60 to 115; the smooth-yield comparator uses the q_d that reproduces the escrowed forward at $t = 0.5$ exactly. Panel (b): the $K = 85$ call at expiries 0.05 to 1.0 in steps of 0.025; the comparator holds the schedule's one-year-equivalent flat yield at every expiry. The quoted ratio maximum is taken where the comparator's premium exceeds one cent.

The real chain (figs. 7.6 and 7.7). The raw SPY chains embedded in the frozen snapshot; a strike pairs into the parity board when both legs quote a live (positive) bid, mids are bid–ask midpoints — Chapter 6's rules verbatim. Carry convention for every real-chain tree: the rate is the stated money-market constant $r = 4.3\%$ per year, continuously compounded, for the snapshot date (the board's own fitted slope, -0.24%, is discarded per section 6.5); the dividend carry is $q_d = r - \log(F/S)/t$ from the working forward, so the tree's forward matches it exactly; the discount used by the naive European comparator is e^{-rt} under the same convention. The loop of section 7.6 starts from the naive fitted forward, and each pass de-Americanizes at the whole-chain depth and refits the line on the pairs whose two legs both rooted. The gallery's fitted population per expiry: the out-of-the-money side (puts strictly below the resolved forward, calls at or above), live bid, inside the wing cut $|\log(K/F)| \leq 4\,\sigma_{\mathrm{atm}}\sqrt{t}$, with σ_{atm} the node's prepared mid volatility nearest the money; the naive leg inverts the same mid through the analytic Black formula at the node's resolved forward. The stress population is the running node's puts with K above the resolved forward. The depth-sensitivity row of the contract repeats the running node's fitted inversion at $N_t = 128$ and compares roots lane by lane.

Chapter 8

The Market's Clock

Since Chapter 2 every smile in this book has carried a maturity variable τ called variance time, and every chapter so far has quietly set it equal to the calendar. This chapter earns the distinction. A scheduled event — an earnings report, a macro print — packs several ordinary days' variance into one calendar day, and a volatility read against the calendar then misbehaves in a recognizable way: the frozen NVDA board opens the chapter with an 18-day expiry quoted below both of its neighbours — more than four volatility points below the shorter one. Nothing is wrong with those prices. The lesson is that implied volatility is a fraction — a price-determined total variance divided by a clock — and the clock is a modeling choice. A short computation with independent normal days shows that the law at expiry knows only the variance budget, not its schedule; a classical theorem extends this to every continuous martingale. The chapter then builds the day-weighted event clock, proves that relabeling time preserves prices, butterfly validity and calendar order while volatility readings scale by exactly $\sqrt{t/\tau}$, and works through an earnings week on paper — the famous overnight volatility crush appears as a scheduled correction of a reading, with no price moving at all. The clock then earns its keep twice: dense term-structure curves must be interpolated on it, or a known event smears into phantom volatility at unquoted maturities; and the clock itself can be read off the board's own prices, by asking which small, sparse set of events makes forward variance per clock unit as flat as possible. That inverse problem has measured limits — shrinkage, a materiality wall, dates that no data can pin inside an interval — and the chapter closes by running it on the frozen boards: the solver hands NVDA's earnings interval about seven extra days, hands the index board almost nothing, and, if allowed candidates everywhere, flattens genuine term structure too. The calendar it installs is an estimate to review, not an oracle.

8.1 Three expiries and a broken ruler

Part II has been rebuilding the observation, one input at a time. Chapter 6 fitted the forward and the discount; Chapter 7 removed the early-exercise right and handed every quote back as a European price. One input has ridden along unexamined since Chapter 2: the maturity variable τ that sits under every total variance $w = \sigma^2\tau$. The book has been calling it *variance time* from the start — while quietly setting it equal to the calendar, since no chapter yet had a reason to distinguish them. Chapter 7 closed by naming the debt: "two clocks this book has so far been allowed to conflate, because its frozen data happens to agree on both." This chapter is about the days when they disagree.

Start with real prices, because the disagreement is sitting in the book's own frozen data. The snapshot behind every chapter — the Monday evening of 2026-08-03 — carries two names, and so far the chapters have worked mostly on SPY. Turn instead to NVDA, the snapshot's single-stock name, and read its at-the-money implied volatility across the eight quoted expiries — the readings computed as $\sqrt{w/t}$, total variance over calendar time, which this chapter will call the *calendar reading* and write σ_{cal}. (At the money means at $k = 0$, the forward; the precise read-off rule is one line in appendix 8.A, which also lists the eight expiry dates. As before, one *vol bp* is 10^{-4} of absolute volatility; one *variance bp* is 10^{-4} of total variance.) Three of the eight readings make the puzzle by themselves:

$$\underbrace{43.13\%}_{\text{4 days}} \qquad \underbrace{38.84\%}_{\text{18 days}} \qquad \underbrace{42.29\%}_{\text{46 days}}.$$

The same underlier, the same afternoon, three expiries two and four weeks apart — and the middle one is quoted 429 vol bp below the shortest. Is NVDA's volatility falling or rising with maturity? The honest answer is that the question is broken, and the goal of this section is to see why.

Ask the question in the right units. A volatility is a rate; before trusting a rate, look at the quantity it is the rate *of.* The quantity is total implied variance $w = \sigma^2 t$ — for now we write t, the calendar year fraction of Chapter 6, because the calendar is the only clock we have so far. The three expiries carry

$$w(\text{4d}) = 20.4, \qquad w(\text{18d}) = 74.4, \qquad w(\text{46d}) = 225.4 \quad \text{variance bp.}$$

Now do the division a first course would do: variance gained per calendar day, interval by interval. Between the 4-day and the 18-day expiries, 14 calendar days pass, so

$$\frac{74.4 - 20.4}{14} = 3.86 \text{ variance bp per day;}$$

between the 18-day and the 46-day expiries, 28 days pass, and

$$\frac{225.4 - 74.4}{28} = 5.39 \text{ variance bp per day.}$$

The days are not alike. A day in the second window carries 1.40 times the variance of a day in the first. And the second window is not anonymous. The 18-day expiry is August 21; the 46-day expiry is September 18; and NVDA reports earnings once a quarter, on a schedule published well in advance, with the next report due in late August — after the first of those dates, before the second. The market is quoting exactly what it believes: the report's day will move the stock like several ordinary days, every expiry that *contains* that day inherits its variance, and the expiry that *dodges* it — August 21 — gets to be cheap.

Figure 8.1 shows the whole board. In panel (a), the orange curve is the calendar reading σ_{cal} of all eight expiries against calendar days to expiry, on a logarithmic axis. Notice the shape around the annotated dip: the two shortest expiries (2 and 4 days, both before August 21) read above 43%, the 18-day expiry reads 38.84%, and the 46-day expiry recovers most of the drop, at 42.29%. Further out, the curve wanders rather than trends — the highest reading on the whole board belongs to the 228-day expiry. Read as "the market's forecast of NVDA's volatility as a function of horizon," the curve is erratic. Panel (b) draws the same eight quotes as what they actually pin down: accrual rates $\Delta w_i/\Delta t_i$, as horizontal steps — one step per interval, the first interval running from the valuation date (where accrued variance is zero) to the first expiry, each later one between consecutive expiries. (This per-interval rate is called *forward variance*, and section 8.6 gives it a symbol and a starring role.) The two computations above are the third and fourth steps: the low step is the pre-report lull, and the shaded interval — the one containing the scheduled report — runs hot. Panel (b) is panel (a) with the division done honestly: nothing here is erratic, it is one hot window sitting among cooler ones.

Why call this a *broken ruler* rather than an interesting market fact? Because of what it does to every tool the book has built. Each single-expiry fit of Part I is untouched — a smile model fits its own expiry's quotes and never asks how fast variance accrued on the way — but everything that couples expiries inherits the kink. A term-structure view drawn through these readings shows volatility falling four points in two weeks and then climbing back, which no

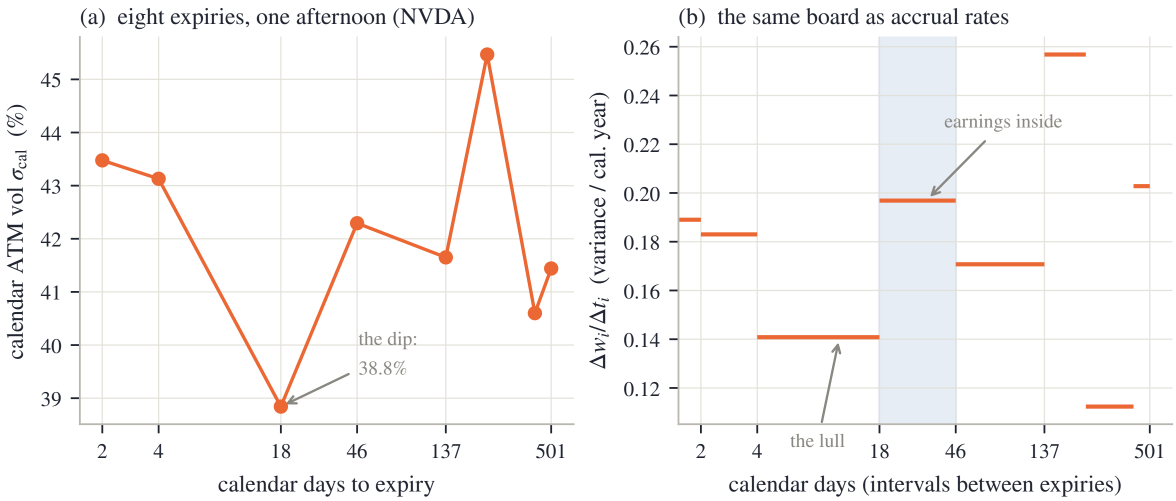

Figure 8.1: The opening puzzle (NVDA, frozen snapshot of 2026-08-03). (a) At-the-money implied volatility against calendar days to expiry: the 18-day expiry — the one that expires *before* the scheduled late-August earnings report — is quoted 429 vol bp below the 4-day reading. (b) The same board as accrual rates: total variance gained per calendar year in each interval between consecutive expiries. The interval that contains the report (shaded) accrues 1.40× the variance per day of the lull interval before it.

trader believes is a forecast. An interpolation between the quoted expiries must invent readings for unlisted maturities, and section 8.5 shows the calendar-clock rule inventing them badly. The local volatility field of Chapter 4 consumes $\partial_\tau w$ directly — its numerator *is* the accrual rate of panel (b). Even the calendar-order condition of Chapter 2 (total variance non-decreasing in maturity at fixed k) offers no comfort: it holds on this board, and the board is still a mess to read, because order says nothing about the accrual being bunched into one window.

There are two ways to respond. One is to enrich the model: add a jump of random size at the report date, price options on the mixture, and carry the machinery of jump models from here on. That road is real but expensive, and this book does not take it (section 8.7 says where it lives in the literature). The other response is older and much cheaper: keep the model, change the *clock*. If an earnings day is worth several ordinary days of variance, then measure time in units in which it *is* several days. On that clock, variance accrues at a steady rate again, and the constant-rate diffusion picture — the one behind every formula in this book — fits without a fight. The next section shows that this is not a trick: it is how the mathematics of martingales says time should be measured in the first place.

8.2 The walk keeps its own clock

We know what the board says: some days carry more variance than others. This section shows that the mathematics has a natural home for that fact. The claim to be earned, first in a discrete computation a first course can check and then as a named theorem, is this: *the law of the underlying at expiry depends on the days only through their total variance — the budget — and not on how that budget is scheduled.* Once that is true, "which clock to use" becomes a genuine choice, because the prices cannot object.

Do the discrete case honestly. Let the log return of the underlying over trading day i be a random shock ε_i, normally distributed with mean zero and variance Δw_i — day i's contribution to the total. (Mean zero is a deliberate simplification: it drops the $-\frac{1}{2}$ drift correction Chapter 5 derived, which shifts the mean and not the variance; the correction reappears, intact, in the continuous computation below.) Let the days be independent: today's surprise does not read yesterday's. After n days the cumulative log return is the sum $\varepsilon_1 + \cdots + \varepsilon_n$, and two first-course facts finish the argument. A sum of independent normals is normal. And its variance is the sum of the variances:

$$\mathrm{Var}(\varepsilon_1 + \cdots + \varepsilon_n) = \Delta w_1 + \Delta w_2 + \cdots + \Delta w_n.$$

The right-hand side is a *sum*, and sums do not care about order or arrangement. Put the big day first, last, or in the middle; split it into two medium days; the terminal law is the same normal with the same variance. The law at expiry knows the budget, not the schedule.

Make it concrete with the numbers this section's figure uses: 30 trading days, each ordinary day's shock worth 2% of standard deviation — that is, $(2\%)^2$ of variance — and one scheduled event on day 20 carrying 5 ordinary days' variance in one calendar day. Take one ordinary day's variance as the unit. The 30 days are 29 ordinary ones plus the event day; the ordinary days contribute one unit each and the event day contributes 5, so the terminal standard deviation is

$$2\% \times \sqrt{\underbrace{29 \times 1}_{\text{ordinary days}} + \underbrace{1 \times 5}_{\text{the event day}}} = 2\% \times \sqrt{34},$$

and it would be $2\% \times \sqrt{34}$ with the event on day 3, day 20, or day 30. An option struck on the terminal level — any option, since Chapter 2 taught that a full strike ladder pins the terminal law and nothing more — cannot tell those calendars apart.

But something *can* tell them apart: any reading that divides by elapsed calendar time. Figure 8.2 shows the same simulated walk twice. In panel (a), the horizontal axis is the trading day. The blue band with grey edges is the ± 1-standard-deviation envelope, $\pm 2\%\sqrt{\text{variance so far}}$: the statistical corridor the walk lives in. Follow the band's upper edge with your eye: it grows like a square root, then takes a visible step up at the dotted line — day 20, where one day adds five days' variance — and resumes its square-root growth from a higher base. The kink is the calendar clock failing to keep pace with the walk. In panel (b), the *same* paths are drawn against a different horizontal axis: accumulated variance, measured in *variance days* — ordinary days count one, the event day counts 5. The event day is now simply a wider slot on the axis (the orange strip), the envelope is one smooth square root from end to end, and nothing about the walk marks the event at all. Same randomness, same terminal law; the kink lived in the ruler.

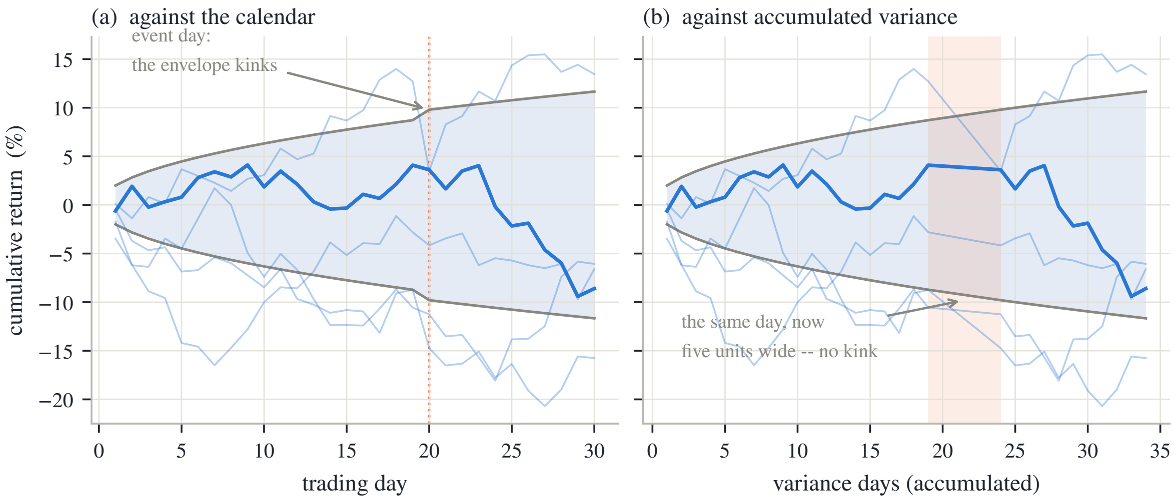

Figure 8.2: One walk, two rulers (5 simulated paths, one highlighted; construction and seed in appendix 8.A). (a) Against the trading-day index, the ± 1-sd envelope (grey edges, blue band) kinks at the event day: day 20 carries 5 days' variance. (b) Against accumulated variance the event day is a wider slot on the axis (the orange strip), and the envelope is one smooth square root. The walk itself never noticed the event's position; only the ruler did.

The continuous version of this bookkeeping is a classical theorem. Chapter 5 wrote the underlying's motion as a diffusion, $\mathrm{d}Y_s = \sqrt{v_s}\, Y_s\, \mathrm{d}W_s$ — Y the forward-normalized level, W

a Brownian motion (the continuous limit of an unbiased random walk), v_s the instantaneous variance, the squared volatility the process runs at instant s (section 5.1). There the time variable was variance time itself; run it now on the *calendar*, where v_s is free to schedule itself unevenly — quiet Tuesdays, a violent earnings evening. Chapter 5's Itô computation (5.1) splits the log return into a drift and a martingale part — a *martingale* being, as before, a process with no drift, whose expected future value is its current value:

$$\log Y_t = \underbrace{\int_0^t \sqrt{v_s}\,\mathrm{d}W_s}_{M_t,\ \text{the martingale part}} - \frac{1}{2}\underbrace{\int_0^t v_s\,\mathrm{d}s}_{\mathcal{C}(t),\ \text{accumulated variance}}.$$

The second integral, written $\mathcal{C}(t)$, is the continuous analogue of "variance days": the running total of instantaneous variance, growing fast through violent stretches and slowly through quiet ones. It is the clock the market itself keeps. For a general continuous martingale — one not handed to us as an integral against a Brownian motion — the accumulated variance is defined directly from the path, as the small-step limit of the sum of squared increments; for our $M_t = \int_0^t \sqrt{v_s}\,\mathrm{d}W_s$ that limit is exactly the $\mathcal{C}(t) = \int_0^t v_s\,\mathrm{d}s$ above. With the term defined, the theorem says the martingale part is nothing but a Brownian motion read on that clock:

Theorem 8.1 (Dambis; Dubins–Schwarz [14, 17])**.** *Let M be a continuous martingale with $M_0 = 0$ whose accumulated variance $\mathcal{C}$ grows without bound. Then there is a standard Brownian motion Z such that*

$$M_t \;=\; Z_{\mathcal{C}(t)} \qquad \textit{for all } t.$$

Two remarks keep the statement honest at this book's level. The growth condition just ensures the relabeled clock runs on forever, so that Z is a full Brownian motion; it is harmless here, where the market keeps moving past any one expiry. And the delicate step — reading a Brownian motion at the *random* time $\mathcal{C}(t)$ and getting the original martingale back — is precisely what the proof establishes, by a careful change-of-time argument in the standard reference [42, Ch. V]. We use the statement, not the proof. What it buys us is the discrete computation again, now for every continuous martingale at once: running the motion to calendar date t, the log return is $\log Y_t = Z_{\mathcal{C}(t)} - \frac{1}{2}\mathcal{C}(t)$, so when the schedule v_s is deterministic the terminal law is normal with variance $\mathcal{C}(t)$ — the budget — and mean $-\frac{1}{2}\mathcal{C}(t)$, no matter how the accrual was arranged along the way. The market keeps one clock, $\mathcal{C}$; the calendar is a second clock that we brought.

Now say what this means for implied volatility, in one careful paragraph. A quoted option price, pushed through the Black formula of Chapter 2, delivers a *total variance* $w(k)$ — and that inversion never asks about time at all. This is worth pausing on, because a first course writes Black–Scholes with σ and T as separate inputs. Chapter 2's normalized form (2.3) shows they never act separately: the formula is $B(k, w)$ with $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$, and maturity enters only through the product $w = \sigma^2\tau$. Everything after the inversion is division: quoting the same w as a volatility requires choosing a clock and computing $\sqrt{w/\text{clock}}$. Divide by the calendar t and we write

$$\sigma_{\text{cal}} = \sqrt{w/t}\,;$$

divide by a clock τ built to follow the market's own accrual schedule and we get $\sigma = \sqrt{w/\tau}$ — which is precisely the σ this book has used since Chapter 2, where $w = \sigma^2\tau$ was the founding convention. Until today the two clocks agreed, because no event calendar was in force; the NVDA board is where they part. The dip of section 8.1 is now diagnosed: the 18-day expiry dodges a scheduled burst of $\mathcal{C}$, its w is accordingly modest, and dividing that modest w by an *undiscounted* 18 days of calendar produces a number that looks like a volatility collapse. The prices are consistent; the reading σ_{cal} is the artifact.

One honest boundary before building the clock. The theorem licenses a *relabeling* of time; it does not predict $\mathcal{C}$. The market's clock is random — nobody knows Tuesday's variance in advance — and what a fitting desk can carry is a deterministic *forecast of its schedule*: ordinary days advance the clock at rate one, scheduled event days advance it by more. Building that forecast, as a small object a desk can quote and edit, is the next section's job.

8.3 The day-weighted clock

We now know the market keeps its own clock, and that a desk can only carry a forecast of its schedule. This section builds that forecast in the simplest currency a trading desk speaks: *days*.

Each scheduled event — an earnings report, a central-bank decision, a court ruling — is recorded as a pair (t_e, N_e): its date t_e, a calendar year fraction like any expiry's, and its size N_e, the number of *extra* equivalent days the event is worth. Mind the word extra: an earnings day with $N_e = 4$ counts as five ordinary days of variance — its own one, plus four extra. The walk's event day in section 8.2, five variance units in one calendar day, is in this currency an event of size $N_e = 4$; the five-unit-wide strip of fig. 8.2(b) and the four-day jump about to appear in fig. 8.3(a) describe the same event in the two bookkeepings. Given a calendar of such events, the variance time of maturity t is defined by counting days:

$$\boxed{\tau_{\text{days}}(t) \;=\; 365\,t \;+\sum_{t_e \le t} N_e, \qquad \tau(t) \;=\; \frac{\tau_{\text{days}}(t)}{365}.} \tag{8.1}$$

The sum runs over events at or before t: an expiry pays for exactly the events it contains. With an empty calendar, $\tau(t) = t$ — the convention every previous chapter ran under, now visible as the special case.

Compute one by hand. Take a 14-day expiry and a single event worth $N_e = 4$ extra days, scheduled 10 days out. The event is inside the expiry, so

$$\tau_{\text{days}} = 14 + 4 = 18, \qquad \tau = \frac{18}{365} = 0.0493,$$

against a calendar $t = 14/365 = 0.0384$. Eighteen variance days will elapse in fourteen calendar days. For an expiry of 9 days — one that *dodges* the event — the sum is empty and $\tau_{\text{days}} = 9$: nothing changes. That is the whole mechanism; everything else in the chapter is consequences.

Figure 8.3(a) draws (8.1) for this calendar. The blue line is τ_{days} against calendar days: it tracks the dashed calendar diagonal exactly until day 10, jumps by 4 days there, and runs *parallel* to the diagonal afterwards — an event shifts the clock once and is never paid for again. (The dotted variant is the normalized clock, explained at the end of this section; at this zoom it hugs the plain one.) Panel (b) is read after the proposition below.

What may a change of clock do, and what may it never do? The answer deserves to be a proposition, because it is the safety case for everything that follows. Recall two terms from Chapter 2, one line each: a smile is *butterfly-valid* when its implied density is nonnegative — the Durrleman check $g_D \ge 0$ of (2.5) — and two expiries are in *calendar order* when total variance does not decrease with maturity at fixed log-moneyness (section 2.8).

Proposition 8.2 (Relabeling time: invariance and covariance)**.** *Let $\tau(\cdot)$ be any strictly increasing relabeling of maturity with $\tau(0) = 0$ — in particular the clock (8.1). Then:*

(i) option prices, and with them every total variance $w(k)$ and the terminal law of each expiry, are unchanged;

(ii) butterfly validity of each smile and calendar order across smiles are unchanged;

(iii) the volatility reading is covariant *— it does not stay fixed but transforms with the clock, by a known factor: $\sigma = \sqrt{w/\tau} = \sigma_{\text{cal}}\sqrt{t/\tau}$;*

(iv) the accrual rate between consecutive expiries is covariant in the same sense, by the interval's clock ratio: $\Delta w/\Delta\tau = (\Delta w/\Delta t)\cdot(\Delta t/\Delta\tau)$.

Proof. A relabeling attaches new time coordinates to the same expiries; no random variable, no expectation, and no price changes, which is (i). Butterfly validity is a property of one expiry's law, unchanged by (i); calendar order compares the same pairs $w(k,T_1)\le w(k,T_2)$ in the same order because the relabeling is strictly increasing, which is (ii). For (iii) and (iv), divide the same invariant quantities (w, Δw) by the relabeled denominators and cancel: $\sqrt{w/\tau}=\sqrt{w/t}\cdot\sqrt{t/\tau}$, and likewise for the ratios of increments. $\square$

Everything a desk can trade sits in the invariant column; everything the clock touches is a *reading*. Part (iv) looks like bare cancellation, and that is exactly its content: the increments Δw carry all the information, and a clock change re-divides them without adding any — a fact section 8.6 will lean on. In particular, installing an event can never move a price and never create an arbitrage — not because an implementation is careful, but because the clock enters the mathematics in exactly one place: as a denominator.

Part (iii) is worth turning around, because it is the formula desks feel. Fix an expiry t and insert one event of N_e extra days before it. Total variance w is pinned by the price, so the clock reading drops by exactly the ratio of the two clocks:

$$\sigma=\sigma_{\text{cal}}\sqrt{\frac{t}{\tau}}=\sigma_{\text{cal}}\sqrt{\frac{365\,t}{365\,t+N_e}}. \tag{8.2}$$

Keep the direction straight, once, on the hand example. The clock reading is the *smaller* number: (8.2) multiplies the calendar reading by $\sqrt{14/18}$, taking 34.0% down to 30.0%. Turned over, the calendar reading is the clock reading *lifted* by $\sqrt{18/14}=1.134$: the same pair of numbers, the same price. How big is the effect in general? For an event small against the maturity ($N_e\ll 365\,t$), expand the square root with the binomial approximation $\sqrt{1+x}\approx 1+\frac{x}{2}$, whose error is of order x^2:

$$\frac{\sigma_{\text{cal}}}{\sigma}=\sqrt{1+\frac{N_e}{365\,t}}\;\approx\;1+\frac{N_e}{2\cdot 365\,t}.$$

The lift of the calendar reading scales like N_e over *twice the calendar days to expiry*: four extra days lift a one-year reading by about half a percent of itself (4/730), and a two-week reading by roughly a seventh (4/28; the exact lift is 1.134 − 1, a touch less). Events are a short-maturity phenomenon — exactly where desks watch implied volatility most closely.

Now read fig. 8.3(b), which plots the lift direction: the blue curve is $\sigma_{\text{cal}}/\sigma=\sqrt{\tau/t}$ across expiries, for the same one-event calendar. Expiries before day 10 dodge the event: the ratio is exactly one. The expiry *on* the event date pays the full toll, $\sqrt{14/10}=1.183$ — the annotated peak. Past it the ratio decays toward one as the event becomes a smaller and smaller fraction of the maturity. The dashed curve is the small-event approximation just derived: generous at the peak, where N_e is anything but small against $365\,t$, still visibly above the exact curve out to about a month, and indistinguishable beyond.

One bookkeeping choice completes the clock. As defined, (8.1) makes a year with events *longer* than 365 variance days: the events add variance on top of an unchanged baseline. Some desks prefer the opposite convention: the year's total budget is what it is, and events say *when* it accrues, not that there is more of it. The normalized clock implements that by rescaling all days — ordinary and event alike — by one factor that restores the annual budget:

$$\tau_{\text{days}}(t)\;\longmapsto\;\tau_{\text{days}}(t)\cdot\frac{365}{365+\sum_{t_e\le 1}N_e}, \tag{8.3}$$

where the sum in the denominator collects the events of the first year ($t_e\le 1$). For the worked calendar the factor is $365/369=0.9892$ — the dotted line of panel (a). Under (8.3) the one-year

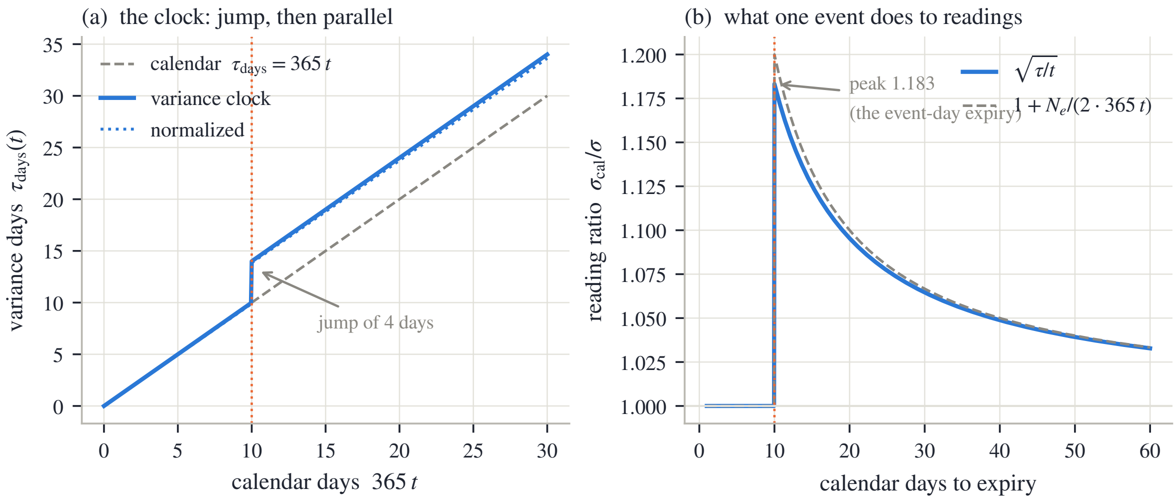

Figure 8.3: The day-weighted clock, one event of 4 extra days at day 10. (a) $\tau_{\text{days}}(t)$: the calendar diagonal, a jump of 4 days at the event, parallel growth after; the normalized variant (dotted) rescales the whole line by 0.9892. (b) The reading ratio $\sigma_{\text{cal}}/\sigma = \sqrt{\tau/t}$ across expiries: one before the event, a peak of 1.183 at the event-day expiry, decay like $1 + N_e/(2 \cdot 365\,t)$ (dashed) beyond it.

variance time, and with it the one-year volatility reading, exactly match a no-event year. Sub-year readings shift slightly: the rescaling makes every day worth a little less than before, so a maturity that dodges all events now counts marginally fewer variance days, while the events' *relative* weight within the year is exactly what it was. Neither convention is more correct: un-normalized treats events as extra variance against a one-per-day baseline, normalized treats the annual budget as known and the events as its scheduled lumps. What matters — the book's recurring refrain — is that the choice is stated. The chapter's contract table records it, and the whole rest of the chapter runs un-normalized.

We now own the clock: a desk-sized object — a list of dated day counts — that turns the erratic readings of section 8.1 into steady ones without touching a single price. The next section watches it work through the week every option trader knows best.

8.4 An earnings week on paper

The clock is built; watch it explain the most famous pattern in listed options. Every trader knows the shape: implied volatility on a name climbs into its earnings date, day after day, and collapses the morning after — the *vol crush*. The usual telling makes it sound like a repricing, as if the market changed its mind overnight. This section constructs an earnings week where the market never changes its mind about anything, and the whole pattern appears anyway — in the readings.

The construction could not be plainer. A synthetic name runs at a flat clock volatility of 30% — on the variance clock, every expiry, every day, $\sigma = 30\%$ exactly — with one scheduled event worth 4 extra days at day 10. All prices follow from Chapter 2's formulas with $w(t) = (0.30)^2\,\tau(t)$ and the clock (8.1); there is no other input. Whatever patterns appear below are therefore not market opinions: they are arithmetic.

Figure 8.4(a) is the view across expiries, priced at the start of the week. The blue line is the clock reading: flat at 30% by construction. The orange curve is the calendar reading $\sigma_{\text{cal}}(t) = \sigma\sqrt{\tau(t)/t}$ of the *same prices*, with dots at the weekly listed expiries. Expiries before day 10 dodge the event and read exactly 30% — the annotated 7-day dot. The expiry on the event day pays the whole toll at its shortest possible maturity and reads 35.5% — the hump's peak, the ratio 1.183 of fig. 8.3(b) applied to 30%. (The peak falls between listed expiries; the weekly dots at days 7 and 14 straddle it.) Beyond it the hump decays as the event thins out over more calendar: by eight weeks the reading is within about a point of the clock's. This is the shape the NVDA board of section 8.1 showed in the wild, reproduced here from a two-line construction — with the dip now explained as the dodging expiries' side of the hump.

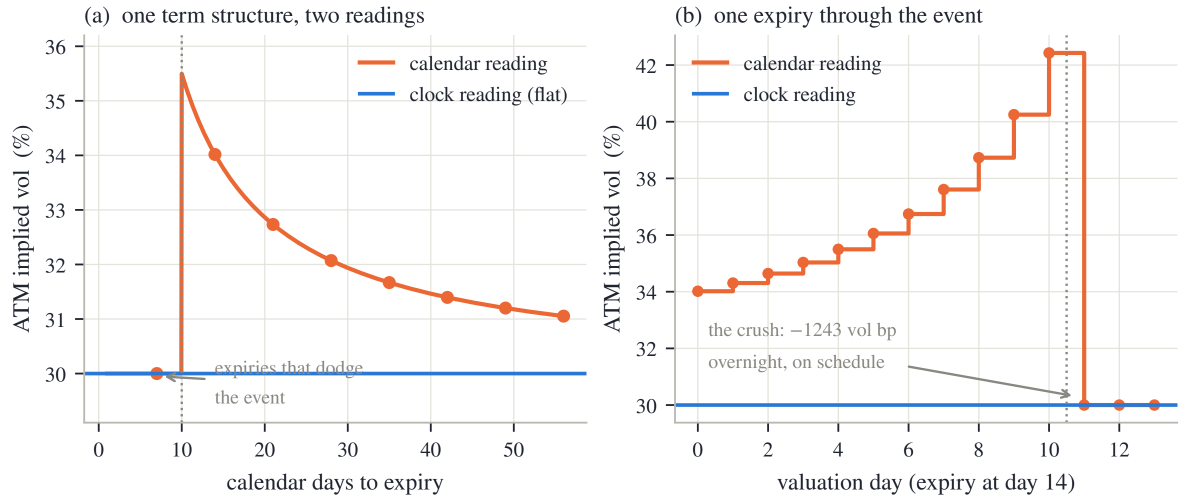

Figure 8.4: An earnings week on paper: flat 30% clock volatility, one event of 4 extra days at day 10. (a) Across expiries: the calendar reading (orange, weekly dots) is flat at 30% where expiries dodge the event, peaks at 35.5% for the expiry on the event date, and decays; the clock reading (blue) is flat by construction. (b) One expiry (day 14) re-read each morning: the calendar reading ramps from 34.0% to 42.4% as the event's four extra days weigh on fewer and fewer remaining calendar days, then gives them up overnight — −1243 vol bp — when the event resolves. No price surprised anyone at any point.

Panel (b) is the view a desk actually watches: one fixed expiry — day 14, chosen to contain the event — re-read every morning as the week passes. On valuation day s the remaining calendar is $14 - s$ days while the event is still ahead, so the remaining variance days are $(14 - s) + 4$, and the calendar reading of the unchanged 30% clock volatility is

$$\sigma_{\text{cal}}(s) = 30\% \times \sqrt{\frac{18 - s}{14 - s}}\,.$$

Both the numerator and the denominator lose one day per morning, so the *fixed* four-day premium weighs on a shrinking base and the ratio climbs: $30\%\sqrt{18/14} = 34.0\%$ on day zero, then day by day up the staircase of panel (b), reaching $30\%\sqrt{8/4} = 30\%\sqrt{2} = 42.4\%$ on the last morning before the report — four calendar days left, eight variance days. That evening the event resolves. The next morning one more night has passed, so three calendar days remain, and the sum in (8.1) no longer contains the event: three variance days over three calendar days, and the reading is exactly 30%. The crush: −1243 vol bp overnight (computed unrounded: the peak is $30\sqrt{2} = 42.43\%$), the entire climb surrendered at once, and at no point did any price do anything unscheduled. The hump is a property of the ruler, not of the prices.

Be precise about "unscheduled," because the word carries the whole claim. Prices *do* change across the event night: the option sheds the event's variance as it is realized, exactly as it sheds an ordinary day's time value. That is scheduled decay — known to the dollar in advance, priced from the start — not news. What the construction lacks, and what the clock deliberately does not model, is the *surprise*: the realized move itself. On a real board the report's outcome repositions the whole surface; the clock only ever reprices the schedule. And the clock is a forecast — if the market decides mid-week that this quarter's report is worth six extra days rather than four, the calendar's N_e must be re-estimated, and the reading of *that* change is a genuine repricing, not an artifact. What the ruler explains is the part every calendar-clock chart gets wrong on schedule: the climb and the crush.

8.5 The curve between the quotes

We can now read quoted expiries on the right clock. But a board quotes a handful of maturities, and half the daily uses of a term structure need values *between* them: plotting a dense curve, marking an unlisted maturity, feeding a model that wants variance at arbitrary dates. Between the quotes something must be assumed, and this section shows that the assumption is the clock choice again — in the sharpest form the chapter has to offer.

The rule the clock recommends is simple. Interpolate *total variance*, not volatility — variance is the additive quantity, the thing that accumulates day by day, while a volatility is a ratio and ratios do not add. And interpolate it *linearly in* τ: between two quoted expiries, let w grow at a constant rate per variance day. Constant accrual per unit of the market's clock is exactly the steady-diffusion picture of section 8.2, restated between quotes; the alternative — linear in t, constant accrual per calendar day — is the same picture on the wrong clock, and we are about to measure the difference.

Build the test so that the truth is known. A synthetic name runs at a flat 30% clock volatility — the previous section's name — with one event worth 4 extra days at day 45. Three expiries are quoted: 7, 30 and 90 days. The event falls *between* the second and third quotes: no quoted expiry sits near it on either side. Generate the three quotes from the construction ($w = (0.30)^2\tau$ at each), then hand them — plus the event calendar — to the two interpolation rules and ask each for the whole dense curve.

Figure 8.5(a) shows both answers as volatility readings against calendar days. Both curves pass through the three quoted dots — at a quote there is nothing to interpolate, so the rules cannot disagree there. Between 7 and 30 days they also agree: no event inside, so a variance day is a calendar day and the two rules coincide. Everything happens between 30 and 90. The blue curve — linear in τ — runs flat at 30% to day 45, jumps as the event enters the maturity, and decays along the hump shape of section 8.4: it reproduces the generating truth exactly (to 0.00 vol bp — exactness is by construction here, since the quotes sit on a curve that is itself linear in τ; the honest general statement closes the section). The orange dashed curve — linear in t — takes the interval's total variance budget, which is correct at both endpoints, and spreads it *evenly across all sixty calendar days.* The event's four days of variance are smeared over the whole bracket.

Where does the smear hurt, and by how much? Work the two sides of the event date; the reading at maturity T (in days) is $30\% \times \sqrt{x(T)/T}$ with $x(T)$ the interpolated variance-day count. First the bracket's budget. By the clock (8.1), the 30-day quote carries $x(30) = 30$ variance days and the 90-day quote carries $x(90) = 90 + 4 = 94$ — the event is inside it — so the bracket holds $94 - 30 = 64$ variance days over 60 calendar days, and the calendar rule pays them out evenly: $\frac{64}{60}$ variance days per calendar day. Just *before* day 45, the truth carries no event: $x_{\text{true}} = 45$. The calendar rule has been paying the event out steadily since day 30, so it carries $x_{\text{cal}} = 30 + 15 \cdot \frac{64}{60} = 46$ — one phantom variance day — and reads $30\%\sqrt{46/45} = 30.33\%$: 33 vol bp of phantom volatility at a maturity that contains no event at all. Just *after* day 45, the truth jumps to $x_{\text{true}} = 49$ while the calendar rule still shows $x_{\text{cal}} = 46$, three days short, and reads $30\%\sqrt{46/45}$ against a truth of $30\%\sqrt{49/45}$: 97 vol bp *too low* on the first maturities that do contain the event. Panel (b) plots this error across the whole axis: zero wherever a quote or an eventless bracket pins the rules together, a sawtooth around the event — climbing to +33, snapping to −97, decaying back to zero at the 90-day quote.

Both signs are money. A desk asked to price an option expiring the day before the event — an expiry that dodges it — off the calendar-interpolated curve quotes essentially the full 33 bp of volatility that does not exist; asked for one expiring the day after, the one instrument

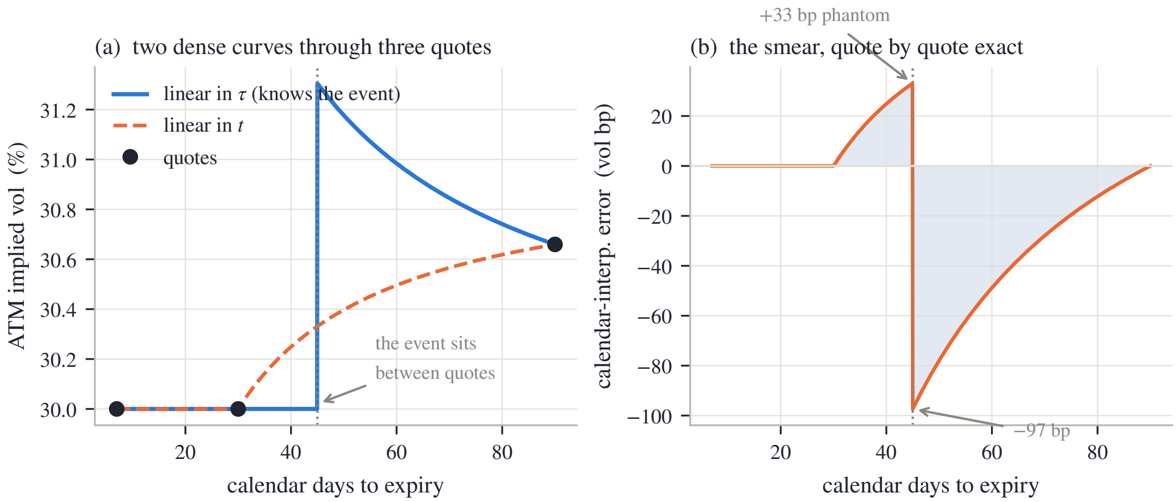

Figure 8.5: Interpolate in the clock, not the calendar (three quotes, flat 30% clock truth, event of 4 extra days at day 45). (a) Linear-in-τ (blue) reproduces the truth exactly; linear-in-t (orange, dashed) smears the event across the whole 30–90-day bracket. Both pass through every quote. (b) The calendar rule's reading error: +33 vol bp of phantom volatility just before the event date, −97 vol bp just past it.

whose whole point is carrying the event, it quotes nearly the full 97 bp too little. The errors sit precisely at the maturities an event trader cares about, they point in opposite directions across one calendar day, and no quoted expiry ever reveals them — the rules agree wherever a price exists. Interpolation error between quotes is invisible until it is traded against.

The honest general statement: on a real board the truth between quotes is not known, and linear-in-τ is itself a convention — the cleanest available, being the constant-accrual statement on the market's own clock, but a convention nonetheless. What it guarantees, and what the calendar rule cannot, is that *known* events stop leaking into maturities that do not contain them. The guarantee is two lines of algebra: between two quotes, w is linear in τ, and by (8.1) τ grows linearly in t except at an event date, where it jumps by $N_e/365$ — so, composed, the dense variance curve is linear in calendar time within each stretch, pays each event's variance as a jump exactly at its date, and changes slope only at the quotes. The remaining question — where the calendar's dates and sizes come from — is the chapter's last construction, and its most delicate.

8.6 Reading the clock off prices

Everything so far consumed a calendar: someone typed in "four extra days on August 26," and the clock did the rest. For scheduled earnings that is the right workflow — the dates are published, and a desk should use them. But the sizes N_e are nobody's published number, and the board's own prices carry an opinion about them: we *saw* that opinion in fig. 8.1(b), where one interval accrued 1.40× the variance per day of its neighbour. This section turns that observation into an estimator — read the calendar *off* the prices — and, just as importantly, measures what such an estimator can and cannot promise.

Set up the objects. The data is the quoted ladder: expiries $t_1 < \cdots < t_n$ with at-the-money total variances $w_1, \ldots, w_n$ (the same per-interval increments $\Delta w_i = w_i - w_{i-1}$, $\Delta t_i = t_i - t_{i-1}$ as before, with $w_0 = t_0 = 0$). The unknowns are candidate event sizes: one candidate per interval, $N_i \geq 0$ extra days somewhere inside interval i — where exactly inside turns out not to matter, and a proposition below says why. Given a candidate vector $N = (N_1, \ldots, N_n)$, each interval's width on the variance clock is

$$\Delta\tau_i = \Delta t_i + \frac{N_i}{365},$$

and the chapter's central diagnostic gets its symbol: the *forward variance* of interval i,

$$\mathcal{F}_i = \frac{\Delta w_i}{\Delta\tau_i}$$

— variance accrued per variance-clock year between two consecutive expiries. (Calligraphic $\mathcal{F}$; the roman F remains the forward price of Chapter 6.) Two structural facts, each one line of algebra, shape everything.

First: the data half of $\mathcal{F}_i$ is untouchable. The increments Δw_i are fixed by prices (proposition 8.2(i)), and the calendar widths Δt_i by the listing. The candidate N enters only through the denominators — an estimator of the clock adjusts how the accrual is *divided*, never how much accrual there *is*.

Second: an event can only pull its own interval down. Interval j's width $\Delta\tau_j$ contains only interval j's own candidate — an event in interval i shifts the running clock τ of every later expiry equally at both endpoints, so every other width is unchanged. And N_i can only *increase* $\Delta\tau_i$, hence only *decrease* $\mathcal{F}_i$. The estimator can lower a spike toward its neighbours; it cannot raise a valley. Its entire vocabulary is "this interval's days were worth more than ordinary days."

Now the estimator itself. On the market's own clock, accrual should be steady (section 8.2); so ask for the candidate vector that makes the $\mathcal{F}_i$ ladder *as flat as possible* — while insisting, because many candidate vectors flatten almost equally well, that events be few and small. Formally, minimize over $N \geq 0$:

$$J(N) = \sum_{i<n} \left(\mathcal{F}_{i+1} - \mathcal{F}_i\right)^2 + \lambda_{\text{mono}} \sum_{i<n} \left(\min(\mathcal{F}_{i+1} - \mathcal{F}_i,\, 0)\right)^2 + \lambda_{\text{sparse}} \sum_i \frac{N_i}{365} + \lambda_{\text{ridge}} \sum_i \left(\frac{N_i}{365}\right)^2. \tag{8.4}$$

Read the four terms one at a time. The first is the flatness ideal: squared steps of the forward-variance ladder. The second charges *decreases* again, on top. Why single out decreases? Because a genuine pre-event board leaves a specific signature: the event interval spikes above both neighbours, so the ladder rises into it and falls out of it. Lowering that one interval repairs both edges at once. An ordinary *upward-sloping* term structure, by contrast, has no falling edges at all — markets show that shape constantly, no event caused it, and the extra charge makes sure the solver is not tempted to "repair" it. The third term is the cost of speech: a charge on each installed day, linear in N_i. Linear, not squared, and the difference matters. Chapter 2's calibration carried a *ridge* penalty — a squared-size charge — and a squared charge shrinks everything a little while zeroing nothing; a linear charge (with $N_i \geq 0$ it is exactly the absolute-value penalty statisticians call ℓ_1, or *sparsity*) prefers exact zeros, paying for an event only where the flatness gain clears a threshold. The last term is a tiny squared charge that keeps the optimization well conditioned; it never decides anything. The minimization is a small bounded solve — n variables, reference weights and the routine in appendix 8.A — and solved events below half a day are dropped as noise.

Before trusting any output, be precise about what the data can identify at all.

Proposition 8.3 (Dates inside an interval are not identified)**.** *Every quantity the pipeline consumes — fits, readings, interpolation, the objective (8.4) — depends on the event calendar only through the clock values $\tau(t_1), \ldots, \tau(t_n)$ at the quoted expiries, hence only through the per-interval totals $N_i = \sum_{t_e \in (t_{i-1}, t_i]} N_e$. Consequently two events inside the same interval are indistinguishable from one event of their combined size placed anywhere in that interval, and an event's date becomes identified only when a listed expiry separates it from its neighbours.*

Proof. The clock (8.1) at a quoted expiry t_i is $365\, t_i + \sum_{t_e \leq t_i} N_e$: it reads the calendar only through cumulative sums up to quoted dates. Two calendars with the same per-interval totals give the same sums at every t_i, hence the same $\tau(t_i)$, the same $\Delta\tau_i$, and the same value of every quantity listed. The date within an interval never enters. $\square$

So the solver's placement of its candidate — mid-interval, by convention — is exactly that: a convention, recorded in the contract table, not an estimate. When the report date is public, type it in; the *size* is what the board can be asked for.

What can the size estimator promise? Figure 8.6 is the audit, run on boards where the truth is planted: a flat clock volatility, one event of known size, solve (8.4), compare. Panel (a) plots recovered against planted size on three boards. On a dense board — expiries from a week to a year, listed like a liquid single name's — at 40% volatility, the curve is the diagonal to the eye: recovery within 0.03 days at every planted size of two days and up. On the same board at index-like 20% volatility, the curve runs parallel to the diagonal but below it, short by up to 0.40 days. And on a coarse quarterly board at 20%, the curve is *zero* until the planted event exceeds 2.5 days, then rises roughly parallel, several days short. (On every board the toe of the curve — planted events under a day or so — reports as exactly zero; the last paragraph of the derivation below says why.) Three behaviours, one mechanism, and it is worth deriving because it is the honest price tag of the sparsity charge.

Work on the audit's configuration: one live candidate N (the others parked at zero, where the audit's solve indeed leaves them) against a planted truth N^*. Two facts start the derivation. On this board the flatness part of (8.4) is *minimized exactly at the truth*: the generator's ladder is flat by construction, so installing $N = N^*$ restores a perfectly flat $\mathcal{F}$ and no other value does better. And near its minimum a smooth function is a parabola: value flat at N^*, some curvature J'', giving $\frac{1}{2}J''\,(N - N^*)^2$. Adding the per-day charge $(\lambda_{\text{sparse}}/365)\,N$ tilts the parabola, and setting the derivative to zero moves its minimum:

$$J''(\widehat{N} - N^*) + \frac{\lambda_{\text{sparse}}}{365} = 0 \qquad \Longrightarrow \qquad \widehat{N} = N^* - \frac{\lambda_{\text{sparse}}}{365\,J''}.$$

Every sparsity-charged estimator shrinks; the shortfall is the charge over the curvature. And the curvature is where the market enters. The candidate moves the objective only through its own interval's forward variance, at rate

$$\frac{\partial \mathcal{F}_i}{\partial N_i} = -\frac{\Delta w_i}{\Delta\tau_i^2} \cdot \frac{1}{365} = -\frac{\mathcal{F}_i}{365\,\Delta\tau_i},$$

so the flatness curvature is of the order of the square of that rate, $J'' \sim \big(\mathcal{F}_i/(365\,\Delta\tau_i)\big)^2$, up to the count of squared terms touched. Since $\mathcal{F}_i$ is of order σ^2, the shrinkage scales like

$$N^* - \widehat{N} \;\sim\; \lambda_{\text{sparse}}\,\frac{\Delta\tau_i^2}{\sigma^4}$$

— inversely with the *fourth* power of the board's volatility, and directly with the *square* of the interval's width, with the day-count conversions and term counts absorbed into the proportionality (the audit's measured pair, 0.03 against 0.40 days, carries rounding at the small end). Halve the volatility and the shrinkage grows roughly sixteenfold; stretch the bracketing interval from a month to a quarter and it grows another order of magnitude.

One mechanical fact completes the picture. The estimate is constrained to $N \geq 0$ and solved events below half a day are dropped, so whenever the shifted optimum $\widehat{N}$ lands at or below that floor, the solver reports *exactly zero* — the flat toes at the left of panel (a), and the walls of panel (b), are this clipping in action, not a failure to compute. Panel (b) shows the consequence directly: fix the planted event at 2 days and sweep the board's volatility. The dense board starts seeing the event at about 12% volatility; the quarterly board not until about 22%. Below the wall the flatness gain a small event offers cannot clear the per-day charge, and the solver — correctly, by its own lights — says nothing. One more audit line completes the promise: handed

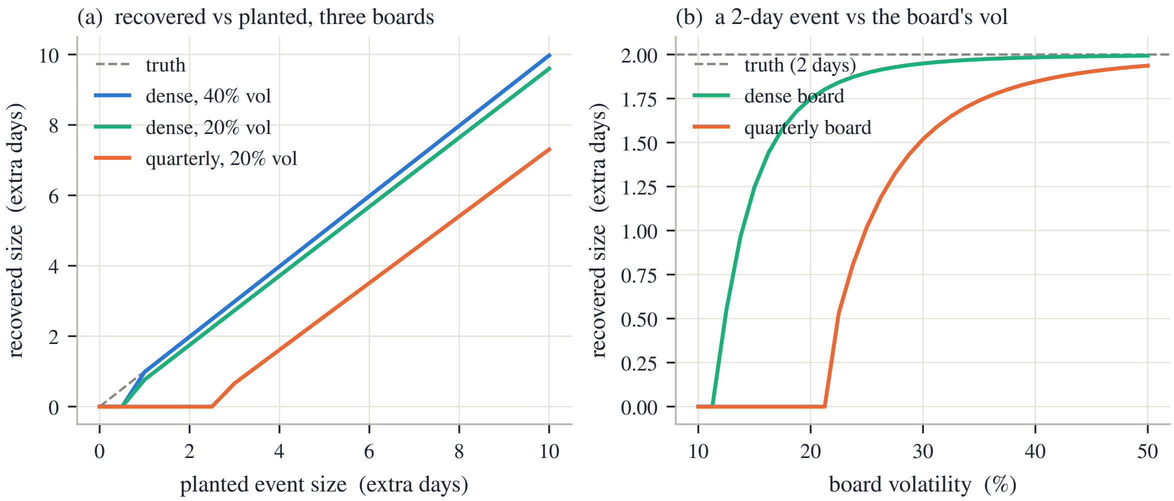

Figure 8.6: The planted-event audit. (a) Recovered vs planted size: near-exact on the dense 40% board (shrinkage $\leq 0.03\,\mathrm{d}$ for planted sizes of $2\,\mathrm{d}$ and up), shrunk by up to $0.40\,\mathrm{d}$ at 20%, and blind below $2.5\,\mathrm{d}$ on a quarterly 20% board — the $\lambda_{\text{sparse}}\,\Delta\tau^2/\sigma^4$ law at work, with the $N \geq 0$ clipping flattening every curve's toe. (b) A fixed 2-day event against board volatility: the dense board's materiality wall sits near 12%, the quarterly board's near 22%.

a perfectly flat ladder, the solver installs exactly 0.000 days. A sparsity prior's virtue is precisely that silence is its default.

We now know the estimator's character. It reads exactly the kinks the clock was built to explain; it can only pull hot intervals down; it shrinks what it reports and stays silent below a wall it computes from the board's own volatility and spacing; and it cannot date an event more finely than the listing brackets it. What it has never been told is *why* an interval runs hot. The last section runs it on the real frozen boards and confronts that limit directly.

8.7 The clock of the frozen board

The estimator is audited; point it at the real data. This section runs (8.4) on the two frozen ladders — NVDA, where the chapter began, and SPY — and reads the results with the discipline section 8.6 paid for.

Figure 8.7(a) is the disciplined run. Candidates are allowed at the first five expiries — through December 2026, a *horizon* choice disclosed in the contract — while the later intervals carry no candidates and hold the ladder's far end in place (they sit beyond the panel's right edge; panel (b) shows them). The orange steps are the calendar ladder of fig. 8.1(b); the blue steps are the same increments divided by the solved clock. The solver spends its budget exactly where the chapter has been pointing: 7.3 extra days on the earnings-bearing interval (shaded), a combined 1.3 days on the two short-dated intervals at the left edge — the board's hot Monday-to-Friday opening, a real if modest acceleration — and nothing anywhere else: 8.6 days in total. Look at what the blue ladder does. (Units, once: both ladders are variance per year of their own clock — calendar year for orange, solved variance year for blue.) The three intervals before the report meet at 0.1409 — the solver has equalized the pre-earnings stretch — and the earnings interval comes down from 0.1969 to 0.1559, most of the way to its neighbours but not all of it: the per-day charge keeps the last few days unbought, the live shrinkage of fig. 8.6(a) on a real board. The December interval is untouched — orange peeking out behind blue — and the in-horizon spread of the ladder tightens from 560 to 299 variance bp.

Now read panel (a) honestly, in two directions. What the solver has genuinely done is recover, *from prices alone*, that the window between August 21 and September 18 carries about seven ordinary days more variance than its calendar length — with no access to a news feed. The public schedule says why: NVDA's quarterly report sits in that window. But the solver does not know that, and cannot. It reads kinks; earnings, a product launch, a macro date, or a genuinely lumpy term structure all look identical through (8.4). The installed calendar is an *estimate of sizes at conventional dates*, to be reviewed against the public schedule — not an oracle that discovered earnings.

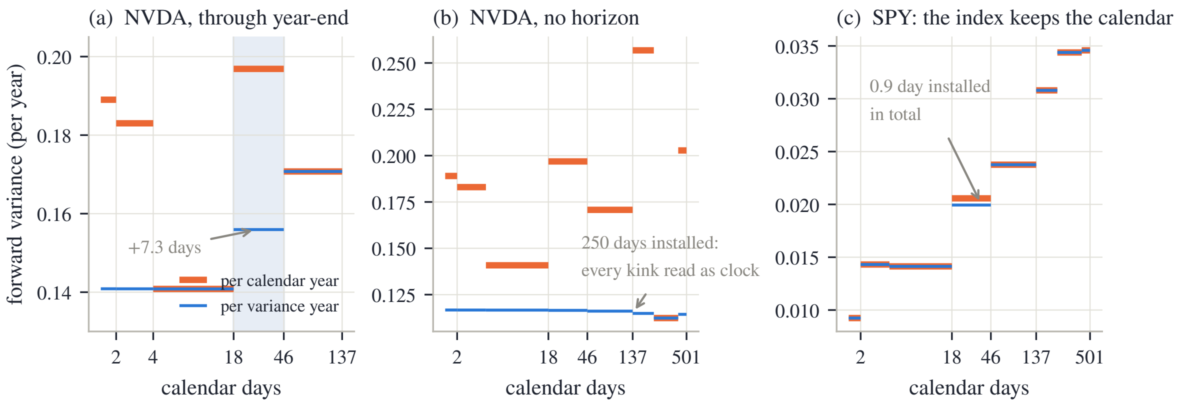

Figure 8.7: Reading the frozen boards' clocks (forward variance per calendar year, orange, vs per solved variance year, blue). (a) NVDA with candidates through year-end: 7.3 extra days land on the earnings interval (shaded), 1.3 on the short end, none elsewhere; the pre-report intervals equalize at 0.1409 and the in-horizon spread tightens from 560 to 299 variance bp. (b) NVDA with candidates everywhere: 250 days installed, the ladder forced flat (1444 to 44 variance bp) — including 112 days against a single ordinary interval. (c) SPY: 0.9 day installed in total; the rising ladder is left standing.

Panel (b) shows what happens when that review is skipped. Same board, same objective, but candidates at every expiry and no horizon. The solver installs 250 extra days and flattens the entire ladder, from a spread of 1444 variance bp to 44 — including 112 days against the December-to-March interval alone, an ordinary three-month stretch whose elevated accrual is some mixture of two more earnings reports and genuine term-structure shape. Nothing malfunctioned. Handed enough candidates, the flatness ideal is achievable, and the objective achieves it — by telling the clock story about *everything*. A perfectly flat blue ladder is the optimizer satisfied, not the truth found; the solver has no way to say "this interval is hot for a reason the clock should not absorb." The horizon of panel (a), and the human review of what it installs, are not decorations on the method — they are the method.

Panel (c) is the control experiment the chapter owes the reader: the index board. SPY's calendar ladder rises steadily — the familiar upward term structure; its largest decrease anywhere is 2 variance bp — and the solver, run with no horizon at all, installs 0.9 day in total, leaving the ladder's 253-variance-bp spread standing. Both halves of the design show up in that restraint. The monotonicity charge in (8.4) fires only on *decreases* of forward variance, and this ladder has essentially none; and at the index's forward volatilities — the board's median is 15% — the σ^4 materiality wall of section 8.6 prices small events out entirely. A diversified index dilutes any single name's report to a fraction of a day; the solver's silence on SPY is the audit's flat-ladder promise kept on real data. The single name keeps an event clock; the index keeps the calendar.

Three paragraphs of scope complete the chapter's subject. First, below one day. This chapter's smallest unit of time is the day, and that too is a convention: variance does not accrue evenly *within* a day either. The classic evidence is the weekend — measured across decades, an exchange holiday adds far less variance than a trading day, because most variance arrives while markets trade [21] — and the same asymmetry separates a session hour from an overnight gap. The day-weight idea continues below the day boundary without new machinery: a day becomes a profile (so much of its weight during the session, so much overnight), and a scheduled morning print — a CPI release at 8:30 — is simply an event whose date t_e is a fraction of a day, entering (8.1) unchanged. The reference implementation carries such a sub-day clock; this book stops at day granularity, stated here once.

Second, what the clock does not model. The clock is a deterministic forecast of the schedule of variance; it prices the *anticipated* part of an event and is silent on the realized surprise, on random event sizes, and on volatility that is itself dynamic — the stochastic-volatility world, where an event is a random burst rather than a known dilation. That is model territory, developed in Bergomi [6], and it is the road this book declined at the end of section 8.1: richer, costlier, and unnecessary for the readings this chapter set out to repair.

Third, where the idea comes from. Reading asset prices on a non-calendar clock is one of the oldest ideas in the subject: Clark [12] explained heavy-tailed daily returns by subordinating a Brownian motion to a transaction-volume clock, and Ané and Geman [3] showed returns recover normality when read on such an activity clock — the empirical cousins of theorem 8.1. This chapter's contribution to the reader is the desk-sized version: a clock parameterized in days, priced into readings by proposition 8.2, and estimable from the board with stated limits.

8.7.1 The contract

The chapter closes as the others do: three columns of decreasing strength — proved, measured, chosen.

Table 8.1: The chapter's contract.

Proved	**Checked**	**Convention**
The law at expiry knows the variance budget, not its schedule — exactly, for independent normal days; for every continuous martingale by theorem 8.1 (cited).	The walk of fig. 8.2: envelope kink 1.08% on the calendar axis, one smooth square root on the variance axis.	The clock τ is a deterministic forecast of the schedule; realized surprises and random event sizes are out of scope (6).
Relabeling time (proposition 8.2): prices, butterfly validity and calendar order invariant; readings covariant by $\sqrt{t/\tau}$, interval rates by $\Delta t/\Delta\tau$.	The worked week (fig. 8.4): ramp 34.0% to 42.4%, overnight crush −1243 vol bp, no unscheduled price move; the NVDA dip (429 vol bp) diagnosed as the dodging expiry.	Event currency: extra equivalent days N_e per event, 365 days per year; normalization (8.3) off throughout (a stated budget choice).
Linear-in-τ interpolation bends the dense curve at the calendar's dates and nowhere else (section 8.5).	The smear (fig. 8.5): +33 vol bp phantom before the event, −97 past it, zero at every quote.	Total variance linear in τ between quotes — the constant-accrual convention on the market's clock.
Dates inside an interval are not identified (proposition 8.3); an event can only lower its own interval's $\mathcal{F}_i$.	The audit (fig. 8.6): shrinkage 0.03/0.40 d at 40%/20%, quarterly wall at 2.5 d, flat ladder → 0.000 days.	Candidate placement mid-interval; solver weights, half-day drop threshold and horizon (appendix 8.A); solved sizes are shrunk estimates.
— (this row is empirical by design)	The frozen boards (fig. 8.7): 7.3 days onto NVDA's earnings interval (spread 560→299 var bp); the no-horizon run's 250-day overreach; SPY left at 0.9 day.	The installed calendar is reviewed against the public schedule before use; the solver measures how much, never why.

Step back to the book's thesis, one last time for Part II. *Validity* was proposition 8.2: a clock is a relabeling, and relabelings can never touch prices, densities, or order — the entire construction is arbitrage-proof by arithmetic. *Information* was what the board genuinely pins down: total variances at quoted expiries, hence per-interval accrual rates — and, through the inverse problem, event *sizes* above a computable wall, but never dates inside an interval and never causes. *Convention* was everything else, each piece now on the table: the day currency, the normalization choice, the interpolation rule, the candidate placement, the solver's weights and horizon. With this chapter, Part II's work is complete: the raw board has been turned into forwards and discounts (Chapter 6), European prices (Chapter 7), and now readings on an honest clock — the observation is fully prepared. What Part III asks is what none of this statics can answer: when the spot moves, what happens to the smile? The surface gives every derivative in the strike direction and none in the spot direction — a missing derivative, and the next chapter is about the conventions that supply it.

8.A Protocol: data, constructions, and solver constants

Every number and figure in this chapter regenerates from one deterministic script reading the frozen snapshot (`scripts/ch08/gen_figures.py`); nothing is refetched, and no smile is refitted. This appendix records the constructions.

The ATM ladders (figs. 8.1 and 8.7). For each (ticker, expiry) node of the frozen snapshot, the at-the-money total variance w_i is read from the node's prepared mid quotes by linear interpolation of $w = \sigma^2 t$ in log-moneyness k at $k = 0$, using the node's stored resolved forward (Chapter 6's object) to define k. No smoothing, no fitting: two bracketing quotes decide each ladder point. The maturities t_i are the snapshot's calendar year fractions. Both names quote the same eight expiries — 2026-08-05, 08-07, 08-21, 09-18, 12-18, then 2027-03-19, 09-17 and 12-17: from the 2026-08-03 valuation date, 2, 4, 18, 46, 137, 228, 410 and 501 calendar days. The NVDA 2-day node is the snapshot's shortest and noisiest (Chapter 2's gallery); its ATM read is used as stored, and none of the chapter's conclusions rests on it alone.

The walk (fig. 8.2). 30 trading days; ordinary daily return standard deviation 2% (Δw_i constant); the event day (index 20) carries 5 ordinary days' variance. 5 paths from a fixed-seed generator (seed 8) — the chapter's only randomness; the envelopes are computed, not simulated.

The worked calendar (figs. 8.3 to 8.5). One event, $N_e = 4$ extra days at day 10 (figs. 8.3 and 8.4) or day 45 (fig. 8.5); flat clock volatility 30% where prices are needed; quoted expiries weekly to eight weeks (fig. 8.4) or at $\{7, 30, 90\}$ days (fig. 8.5). The interpolation exhibit's quotes sit exactly on $w = \sigma^2\tau$, so the linear-in-τ rule reproduces the generator to 0.00 vol bp by construction; the figure's content is the calendar rule's deviation from it.

The solver (eq. (8.4); figs. 8.6 and 8.7). Reference weights $\lambda_{\text{mono}} = 1$, $\lambda_{\text{sparse}} = 10^{-3}$, $\lambda_{\text{ridge}} = 10^{-4}$; one candidate per interval, dated mid-interval (proposition 8.3 makes the date a convention); box bounds $0 \leq N_i \leq 1825$ days (a wide safety box, never active in any run of this chapter); minimization by a standard bounded quasi-Newton routine from a zero start; solved events below 0.5 day dropped. The audit boards of fig. 8.6: the dense board's expiries are $\{7, 14, 28, 56, 91, 182, 365\}$ days with the planted event at day 40, the quarterly board's $\{91, 182, 273, 365\}$ days with the event at day 120. The scaling law of section 8.6 deliberately drops its constants (term counts, day conversions), so these weights reproduce the measured shrinkage scale — 0.40 day at 20% on the dense board — through the audit, not through the formula; the audit is the reference. The NVDA hero run of fig. 8.7(a) allows candidates at expiries through 2026-12-18 (the horizon); the panel (b) and SPY runs allow candidates at every expiry. Spreads are max-minus-min of the forward-variance ladder, quoted in variance bp; the in-horizon spread uses the intervals up to the horizon.

Units. Throughout the chapter: one variance bp is 10^{-4} of total variance (or of a forward-variance rate per year); one vol bp is 10^{-4} of absolute volatility; days convert to years at 365 calendar days per year, the same convention as the clock (8.1).

Part III

Dynamics and Inference

Chapter 9

Spot–Vol Dynamics: The Missing Derivative

Part II closed with the observation fully prepared: forwards, European prices, an honest clock. Part I turned such observations into surfaces. Everything so far is a photograph taken at one spot. This chapter is about the first question the photograph cannot answer: when the underlying moves, what happens to the smile? The strike direction is fully known — differentiating the surface across strikes gives digitals and densities, and Chapter 2 built whole laws from it. The spot direction is empty: through today's surface pass many admissible tomorrows, and no quoted price separates them. Yet a desk must pick one, because the true sensitivity of an option to spot is the Black delta plus its volatility sensitivity — vega — times exactly this missing derivative: on the book's frozen SPY node the standard candidate rules sit fourteen delta points apart at an ordinary put strike. The chapter builds the candidates slowly: two sign rules that prevent the classic transport errors, then a one-line transport whose single dial $\mathcal{R}$ — the skew-stickiness ratio — measures how much of the skew is realized as ATM movement, with the named regimes — the smile pinned to moneyness, pinned to strikes, or generated by a frozen local-volatility field — at $\mathcal{R} = 0, 1, 2$. The second half develops the one regime that derives its answer instead of assuming it: freeze Chapter 4's local-volatility field in absolute strike and reprice. A short computation — the half-rule — shows the implied skew is half the field's slope, so the frozen field moves the ATM vol by twice the skew: $\mathcal{R} = 2$ is a consequence, not a convention. Beyond the money the consequence is subtler and more honest: two fields that agree wherever today's quotes look produce visibly different tomorrows in the wings, so the wings of the answer belong to the field — which Chapter 4 showed the quotes do not pin down. Repricing measures the ratio as an output: exactly 2 at every maturity for a straight field, and lifted by a computable curvature term for a bent one. The chapter closes by running a -5% scenario across the frozen SPY board, where the regime choice spreads the two-day ATM reading across twenty-one volatility points and is worth ten to twenty-odd delta points on the benchmark put of every expiry — and by stating plainly what remains a choice.

9.1 The derivative the surface does not carry

Take stock of what the book can now do. Part II turned a raw board into a clean observation: Chapter 6 fitted the forward and the discount, Chapter 7 removed the early-exercise right, Chapter 8 put the readings on an honest clock. Part I turned such observations into surfaces: at every quoted expiry, a fitted smile, a full probability law. All of it is a photograph. Every object was built from the prices of one afternoon, at one level of the underlying.

Look at how much the photograph contains. Differentiate a fitted smile's call prices once in strike and you get the digital price — the probability of finishing below the strike (Chapter 2 built its whole model on this). Differentiate twice and you get the density. Differentiate total variance in maturity and you get the forward variance of Chapter 8. In the surface's own coordinates — strike and expiry — every derivative is available, exact, and priced. Chapter 8 closed on precisely this point, and it is the one aphorism this chapter allows itself: the surface gives every derivative in the strike direction and none in the spot direction.

Here is the question that needs the missing direction. The frozen SPY December 2026 smile — the book's running node, 137 days to expiry, ATM volatility 14.8% — was fitted at the snapshot's forward. Suppose the next tick takes the forward down 4%. What is the new smile, *now*? No new option quotes have printed yet, and a desk cannot wait for a recalibration to know its risk. It needs a rule that turns the old smile plus the move into a new smile.

The uncomfortable fact is that today's prices do not contain that rule.

Remark 9.1 (A snapshot does not identify dynamics). Every price quoted today is an integral against today's laws: Chapter 2 wrote each vanilla as an expectation under its expiry's distribution, and the fit determined those distributions. A *dynamics* is a different kind of object: an assignment of a whole surface to each future level of the underlying. Consistency with today's quotes constrains that assignment at exactly one point — today's level, where it must reproduce today's surface — and nowhere else. To see this concretely, take any family of arbitrage-free surfaces that passes through today's surface. Declaring "after a move of size H the surface is the family member at H" *is* choosing a dynamics. Every instrument quoted today is priced identically under every such declaration, so today's quotes cannot rank them. The derivative of the smile along spot is extra structure. The market data of one afternoon cannot supply it; the desk must choose it, or derive it from a model — which is where this chapter ends up in section 9.5.

The choice would not matter much if it only changed a chart between ticks. It matters because it is a *hedge ratio*. The spot sensitivity of an option has two parts: the Black delta — the sensitivity at frozen volatility — plus vega times the very derivative the surface does not carry (vega, recall, is the price's sensitivity to volatility). Two desks holding the same option against the same fitted surface, but assuming different smile responses, will compute different deltas and trade different hedges. Section 9.4 prices the disagreement on the frozen node: about 14.4 delta points at an ordinary put strike. Nothing about the choice is cosmetic.

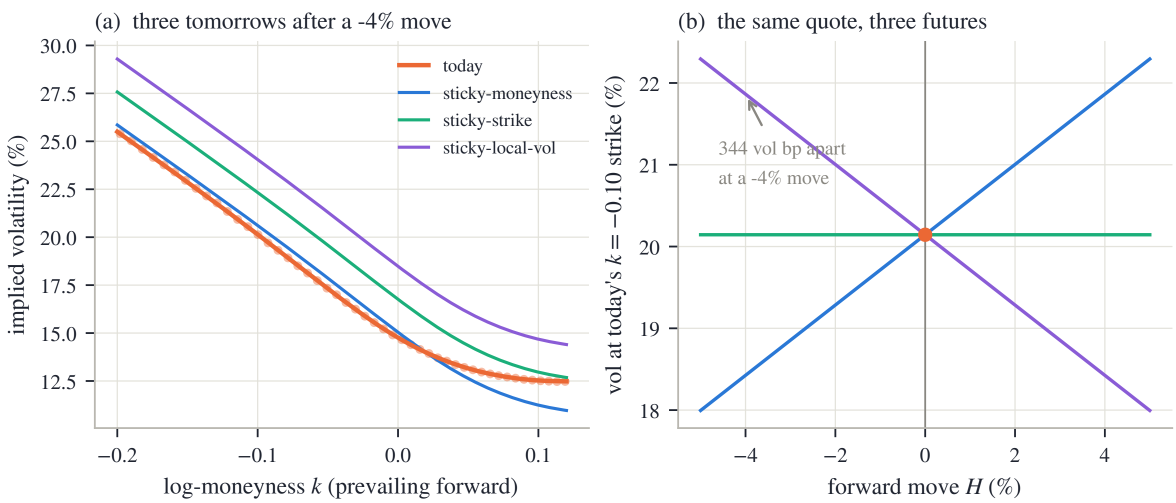

Figure 9.1: The missing derivative (frozen SPY December 2026 node). (a) Today's fitted smile (orange, with the prepared mid quotes as dots) and three candidate smiles after a -4% forward move, drawn with the transport built in section 9.3 — all three consistent with every price in panel (a)'s dots. (b) The volatility of one fixed strike (today's $k = -0.10$ put, fitted today at 20.1%) against the move size, under the three rules: at a -4% move the answers sit 344 vol bp apart.

Figure 9.1 draws the predicament before any machinery. In panel (a), the orange curve is today's fitted smile with its prepared mid quotes ("prepared" in Part II's sense: the cleaned, forwarded, Europeanized observation); the three thinner curves are three candidate tomorrows after a -4% forward move, each produced by one of the standard rules this chapter will build, and each labeled with the name a desk would use. They stack in the order of the eventual dial — after this *down* move the sticky-local-vol tomorrow sits highest, sticky-moneyness lowest, sticky-strike between, at every strike drawn — for reasons section 9.3 derives. Nothing in the orange dots prefers one of the three: they are different completions of the same photograph (remark 9.1). Panel (b) reads the same disagreement at a single fixed strike — the put ten percent below the money, whose fitted volatility today is 20.1%. Each rule makes that strike's tomorrow volatility a straight line in the move size; the lines share only the point at zero move. At -4% they span 344 vol bp (a vol bp, as throughout the book, is 10^{-4} of absolute volatility) — three and a half volatility points for the same strike, same surface, same tick.

We now know what is missing and why it must be chosen. The rest of the chapter builds the choices properly: first the bookkeeping that keeps signs honest (section 9.2), then the one-parameter family of rules and its dial (section 9.3), then the price tag in delta (section 9.4), and then the one construction in this book that *answers* the question instead of asking it (sections 9.5 and 9.6).

9.2 Two coordinate systems, one discipline

Before any rule can be written down, one piece of bookkeeping must be fixed, because it is where nearly every practical error in smile transport lives. The market trades *strikes*: dollar levels, printed on contracts, unchanged when the underlying moves. The book draws *smiles*: curves in log-moneyness $k = \log(K/F)$, measured against the forward. When the forward moves, every strike stays put and every label changes. Two coordinate systems, one of which just slid under the other.

Fix the conventions once, and hold them for the whole chapter.

Definition 9.2 (Sign conventions)**.** (i) The move is $H = \log(F_{\text{new}}/F_{\text{old}})$: positive when the forward rises. (ii) Log-moneyness $k = \log(K/F)$ is always measured against the *prevailing* forward: today's curve uses today's forward, tomorrow's curve uses tomorrow's. (iii) A transported *curve* is a function of new moneyness; a transported *quote* — a fixed strike — is a point whose moneyness label changes.

One wording convention travels with (i), stated once for the whole chapter: a "four percent move" always means the *log* move $H = \pm 0.04$ — so a 4% rise lands the forward at $100\,e^{0.04} = 104.08$, not 104 — and, likewise, a strike "ten percent below the money" means $k = -0.10$.

Now do the two computations these conventions force, by hand. Let today's forward be $F = 100$ and let it rise 4%, so $H = +0.04$ and $F_{\text{new}} = 100\,e^{0.04} = 104.08$.

First, follow one *quote*. The strike $K = 100$ has today's label $k = \log(100/100) = 0$: it is at the money. Tomorrow its label is

$$k_{\text{new}} = \log\frac{100}{104.08} = \log\frac{K}{F_{\text{old}}e^{H}} = k - H = -0.04.$$

A fixed strike's label slides *down* by the move: the **quote rule**, $k \mapsto k - H$. This is just relabeling; no price and no volatility has been touched.

Second, build the new *curve*, under the simplest hypothesis a desk uses: every fixed strike keeps yesterday's volatility ("sticky-strike"). To evaluate the new curve at new label k, ask which strike now sits there. It is $K = F_{\text{new}}e^{k}$, and its *old* label was

$$\log\frac{K}{F_{\text{old}}} = \log\frac{F_{\text{new}}e^{k}}{F_{\text{old}}} = k + H.$$

That strike's volatility is unchanged, so the new curve reads the old curve there:

$$\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k + H) \qquad \text{(sticky-strike)}. \tag{9.1}$$

The **curve rule** reads *forward*, at $k + H$ — a plus — while the quote rule slides labels to $k - H$ — a minus. Both signs are correct, and they point in opposite directions in k. This is the trap. Swap either sign and the transported smile's skew response flips; the error is silent, because the resulting picture still looks like a plausible smile. The discipline is to derive each sign the way we just did — "which strike sits here now?" for curves, "where did my strike's label go?" for quotes — and never to guess.

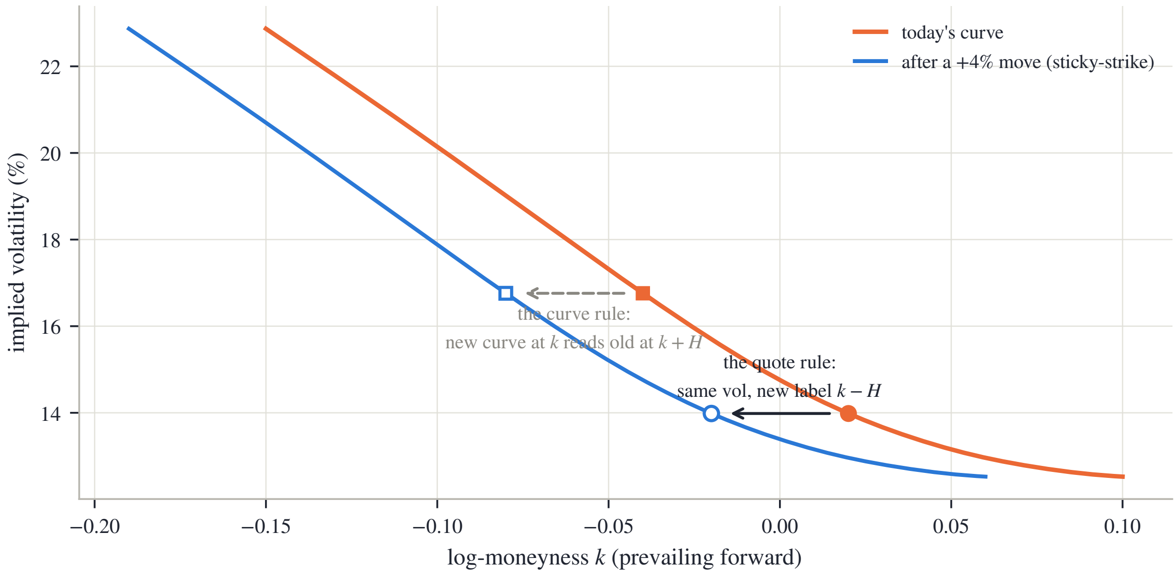

Figure 9.2: The two signs on one +4% move (frozen hero smile, sticky-strike). The fixed strike whose fitted volatility is 14.0% keeps that volatility while its label slides left to $k - H$ (filled to open circle: the quote rule). The new curve (blue) at any label k reads the old curve (orange) at $k + H$ (filled to open square: the curve rule). Opposite directions, both correct.

Figure 9.2 draws both rules on the frozen node's smile under the same +4% move. The orange curve is today's smile; the blue curve is the sticky-strike tomorrow, the whole shape carried left by the re-indexing (9.1). Follow the circles first: the strike carrying 14.0% sits at label +0.02 today (filled circle) and at −0.02 tomorrow (open circle), at exactly the same height — same volatility, new label, the quote rule. Now follow the squares: to know tomorrow's curve at label −0.08, walk *right* by H to the old curve at −0.04 (filled square) and carry that value across (open square) — the curve rule. The circles move one way, the squares' reading goes the other, and the figure is worth returning to whenever a sign feels doubtful.

One more reading of (9.1) is worth a sentence, because it explains the name of the whole subject. Under sticky-strike the curve slides along the strike axis; under the opposite pure hypothesis — the smile is a fixed function of moneyness, "sticky-moneyness" — the curve would not move at all in k. Real desks argue precisely about how sticky the smile is in which coordinate. The conventions are now fixed; the next section shows that, once shapes are preserved, the entire argument collapses to a single number.

9.3 One free dial

We know the bookkeeping; now build the rule. The guiding idea is modest: for small moves, a transport should not invent new smile *shape* — no quotes have printed that would justify a new shape. A curve can be changed in only two ways without changing its shape: slide its argument, or add a constant. A shape-preserving transport is therefore

$$\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k + aH) + bH$$

for two numbers a and b per unit move. The slide is fixed by anchoring the family to the one hypothesis whose meaning is operational: if the level term is zero, the transport should be exactly sticky-strike — volatilities pinned to strikes, the pure re-indexing (9.1) of the last section. That sets $a = 1$. What remains free is one number: the level.

Write s_0 for the *ATM skew*: the slope of the smile at the money, $s_0 = \partial_k \sigma|_{k=0}$, in volatility per unit of log-moneyness. On the frozen hero node $s_0 = -0.430$: the fitted curve loses about four volatility points per ten percent of moneyness as k rises through zero. The natural yardstick for the level is then $s_0 H$ — the skew times the move, the amount of volatility the smile's own tilt attaches to a move of size H. Writing the level as $(\mathcal{R} - 1)\, s_0 H$, a multiple of that yardstick — the reason for the -1 appears in one moment — gives the chapter's transport, its one boxed display:

$$\boxed{\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k + H) + (\mathcal{R} - 1)\, s_0\, H.} \tag{9.2}$$

The dial $\mathcal{R}$ earns its name at the money.

Proposition 9.3 (ATM response)**.** *Under* (9.2)*, the at-the-money volatility responds to the move by*

$$\sigma_{\text{new}}(0) - \sigma_{\text{old}}(0) = \mathcal{R}\, s_0\, H + O(H^2), \qquad \textit{hence} \qquad \mathcal{R} = \frac{\mathrm{d}\sigma_{\text{atm}}/\mathrm{d}\log F}{s_0}.$$

Proof. Evaluate (9.2) at $k = 0$ and expand the re-indexed term to first order in H:

$$\sigma_{\text{new}}(0) = \sigma_{\text{old}}(H) + (\mathcal{R}-1)s_0 H = \sigma_{\text{old}}(0) + s_0 H + (\mathcal{R}-1)s_0 H + O(H^2) = \sigma_{\text{old}}(0) + \mathcal{R}s_0 H + O(H^2).$$

The second equality is the point: the re-indexing alone already moves the ATM reading by $s_0 H$, because sliding along a tilted curve changes the value at the money. The level term adds the rest. ☐

So $\mathcal{R}$ is the fraction of the skew realized as ATM movement per unit log-forward move. The literature calls this the *skew-stickiness ratio* [6], and the classical taxonomy of Derman [16] names three of its values. Read them through a falling market, since $s_0 < 0$: the forward drops ($H < 0$), and the question is what the ATM volatility does.

- $\mathcal{R} = 0$, *sticky-moneyness*: to first order the ATM volatility does not move; the whole smile rides with the forward as a fixed function of moneyness. ("To first order" is doing real work in this bullet — section 9.7 shows a case where the leftover curvature term is worth volatility points.)
- $\mathcal{R} = 1$, *sticky-strike*: the ATM volatility rises to the value the old smile assigned to the new ATM strike — each strike kept its volatility, and the money walked down the skew to a higher reading.
- $\mathcal{R} = 2$, *sticky-local-vol*: the ATM volatility rises *twice* as far — it overshoots the old curve. The name, and why two is the natural value for it, is the business of section 9.5.

A plain reading of the dial: $\mathcal{R}$ asks how much of the skew is a forecast. The skew says lower strikes carry higher volatility. When spot actually goes lower, $\mathcal{R}$ is the fraction of that promised extra volatility that the at-the-money reading actually collects.

Compute the whole transport once by hand, on a toy smile where every number is two-digit arithmetic. Let today's smile be the straight line $\sigma_{\text{old}}(k) = 0.20 - 0.15\, k$ (so $s_0 = -0.15$, and the ATM volatility is 20%), let $F = 100$, and let the forward rise 4% ($H = +0.04$), as in section 9.2.

(The taxonomy above read the market *downward*; the toy runs it *upward*, so every intuition from the bullets flips sign — what rose there falls here.) Then $\sigma_{\text{old}}(H) = 0.20 - 0.15 \times 0.04 = 0.194 = 19.4\%$ and $s_0 H = -0.15 \times 0.04 = -0.006$, i.e. -60 vol bp, and (9.2) gives, at the new money ($k = 0$) and at the old strike $K = 100$ (new label $k = -H$, so the re-indexed term is $\sigma_{\text{old}}(0)$):

	ATM (strike 104.08)	strike 100
$\mathcal{R} = 0$ (sticky-moneyness)	20.0%	20.6%
$\mathcal{R} = 1$ (sticky-strike)	19.4%	20.0%
$\mathcal{R} = 2$ (sticky-local-vol)	18.8%	19.4%

Check each against the two rules. The ATM column is $20\% + \mathcal{R}s_0H$: unchanged, down 60 bp, down 120 bp — proposition 9.3 exactly, since the toy smile has no curvature. The strike-100 column is $20\% + (\mathcal{R} - 1)s_0H$: under sticky-moneyness the strike is now below the money and picks up the skew's 60 bp; under sticky-strike it keeps 20% by construction; under sticky-local-vol it *loses* 60 bp, because the level term pushes the whole curve past it. The reader should reproduce all six numbers on paper before going on; everything later in the chapter is this computation with real curves.

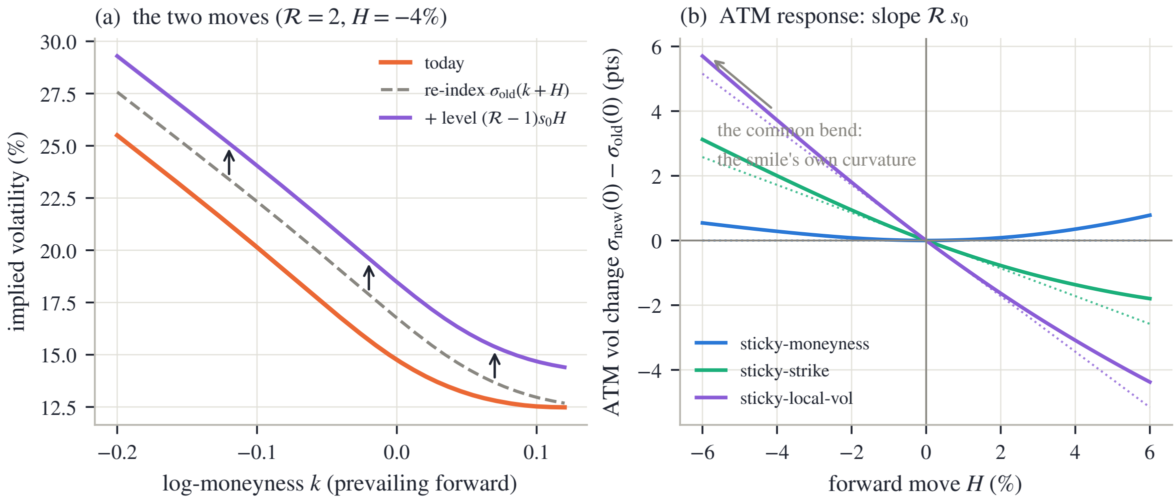

Figure 9.3: The dial (frozen hero smile). (a) The transport (9.2) built in its two moves for $\mathcal{R} = 2$, $H = -4\%$: the dashed curve is the pure re-index $\sigma_{\text{old}}(k + H)$; the arrows add the uniform level $(\mathcal{R} - 1)s_0H = +172$ vol bp. (b) The exact ATM response of the transport against the move for the three regimes (solid), with the linear law $\mathcal{R}s_0H$ dotted: the slopes are $\mathcal{R}s_0$, and all three curves share one bend — at $H = -6\%$ it is worth 54 vol bp — which is the smile's own curvature entering through the re-index term.

Figure 9.3 shows the same construction on the real node. In panel (a), the orange curve is today's smile and the dashed grey curve is the pure re-index $\sigma_{\text{old}}(k + H)$ for a -4% move: the shape has slid bodily to the left, and at the money it already reads higher — that is the s_0H from the proof above. The three arrows — drawn at three different strikes to show the lift is the same everywhere — then add the level $(\mathcal{R} - 1)s_0H$, here $+172$ vol bp for $\mathcal{R} = 2$, lifting the dashed curve to the violet one. Every transported smile in this chapter is those two moves and nothing else. Panel (b) plots the exact ATM change $\sigma_{\text{new}}(0) - \sigma_{\text{old}}(0)$ against the move for the three regimes, with the straight lines $\mathcal{R}s_0H$ dotted on top. Near zero move the three solid curves are the three lines: slope $\mathcal{R}s_0$, proposition 9.3. Away from zero they all bend by the same amount — 54 vol bp at $H = -6\%$ — and the bend has nothing to do with $\mathcal{R}$: it is $\frac{1}{2}\sigma''_{\text{old}}(0)H^2$, the smile's own curvature picked up by the re-index term — the 54 vol bp is that term measured on the fitted curve, and it is the first $O(H^2)$ the proposition swept aside. The same term explains a detail the sharp-eyed reader may have caught in fig. 9.1(a): the sticky-moneyness tomorrow does not lie exactly on today's curve, though the words "the smile is a fixed function of moneyness" say it should. The transport (9.2) realizes each regime to first order in the move, and its second-order honesty is exactly this one curvature term.

Two remarks close the section. First, the level did not have to be vertical: one could instead slide the argument farther, $\sigma_{\text{old}}(k + \mathcal{R}H)$, and match the same ATM response. The two versions differ by curvature times H^2 — the same second order the transport already declines to model — so the choice between them is a convention, and this book's is (9.2). Second, the dial is not unconstrained folklore: within one-factor stochastic-volatility models — models in which a single random factor drives the volatility — short-dated smiles force $\mathcal{R}$ between the sticky-strike and sticky-local-vol values [6], so the named regimes bracket the modeling mainstream. What no model choice can escape is remark 9.1: today's surface prices every $\mathcal{R}$ identically. The next section computes what the choice costs.

9.4 The price of the choice: delta

We have a one-dial family of transports, and a claim from section 9.1 to make good: the dial is a hedge ratio. This section derives the delta correction each regime implies and evaluates it on the frozen node. Two small reminders before the computation. *Vega* is an option price's sensitivity to its implied volatility. And on this board variance time equals calendar time — Chapter 8 ran its clock solver on SPY and found nothing to install — so $\tau = t$ throughout, and we write τ.

Start with the object the transport controls directly: the volatility of a *fixed strike*. Compose the two rules of definition 9.2. A strike with today's label k_0 has tomorrow's label $k_0 - H$ (quote rule), and the transported curve evaluated there (curve rule, i.e. (9.2)) gives

$$\sigma_K^{\text{new}} = \sigma_{\text{new}}(k_0 - H) = \sigma_{\text{old}}\big((k_0 - H) + H\big) + (\mathcal{R} - 1)s_0 H = \sigma_K^{\text{old}} + (\mathcal{R} - 1)\, s_0\, H. \tag{9.3}$$

The re-index and the label slide cancel *exactly* — no first-order expansion was needed — and what is left is the level. A fixed strike's volatility responds at the rate $(\mathcal{R} - 1)s_0$ per unit log-forward move, at every strike: zero under sticky-strike (by construction), $-s_0$ under sticky-moneyness, $+s_0$ under sticky-local-vol. This is the missing derivative of the chapter's title, made explicit: choose $\mathcal{R}$ and you have chosen $\mathrm{d}\sigma_K/\mathrm{d}\log F$.

Now push it into the delta. A dollar call is $C = D\, F\, B(k, w)$, with B the normalized Black call of Chapter 2 (eq. 2.3) and $w = \sigma_K^2 \tau$ the strike's total implied variance; the discount D rides along untouched and we set it aside, quoting deltas in the forward (Black) convention. Differentiate along the move. Three things change at once: the prefactor F, the label $k = \log(K/F)$ (down by $\mathrm{d}\log F$), and — this is the new part — the volatility, by (9.3). Per unit of $\mathrm{d}\log F$, and dividing by F to convert to a delta (since $\mathrm{d}F = F\,\mathrm{d}\log F$),

$$\Delta_{\text{tot}} = B - \partial_k B + \partial_w B\, \frac{\mathrm{d}w}{\mathrm{d}\log F}.$$

Both Black partial derivatives were computed in Chapter 2's appendix and recalled in Chapter 6, which also performed the collapse the first two terms need: $B - \partial_k B = \Phi(d_+)$, the Black forward delta. For the third term, differentiate $w = \sigma_K^2 \tau$ and insert (9.3) together with the vega formula $\partial_w B = \varphi(d_+)/(2\sqrt{w})$:

$$\frac{\mathrm{d}w}{\mathrm{d}\log F} = 2\sigma_K \tau\, (\mathcal{R} - 1)s_0, \qquad \partial_w B\, \frac{\mathrm{d}w}{\mathrm{d}\log F} = \frac{\varphi(d_+)}{2\sqrt{w}}\, 2\sigma_K \tau\, (\mathcal{R} - 1)s_0 = \varphi(d_+)\sqrt{\tau}\, (\mathcal{R} - 1)s_0,$$

where the last step used $\sqrt{w} = \sigma_K \sqrt{\tau}$: the volatility cancels. Together,

$$\Delta_{\text{tot}} = \Phi(d_+) + \varphi(d_+)\, \sqrt{\tau}\, (\mathcal{R} - 1)\, s_0. \tag{9.4}$$

Read it in words: the true delta is the Black delta plus a smile correction — the normal density at d_+, times the square root of maturity, times the fixed-strike volatility response the regime implies.

Between the extreme named regimes the correction swings from $-\varphi(d_+)\sqrt{\tau}\,s_0$ to $+\varphi(d_+)\sqrt{\tau}\,s_0$, so the $\mathcal{R}=0$ versus $\mathcal{R}=2$ gap is

$$\text{delta gap} = 2\,\varphi(d_+)\,\sqrt{\tau}\,|s_0|, \tag{9.5}$$

a pure product of three quantities the desk already knows.

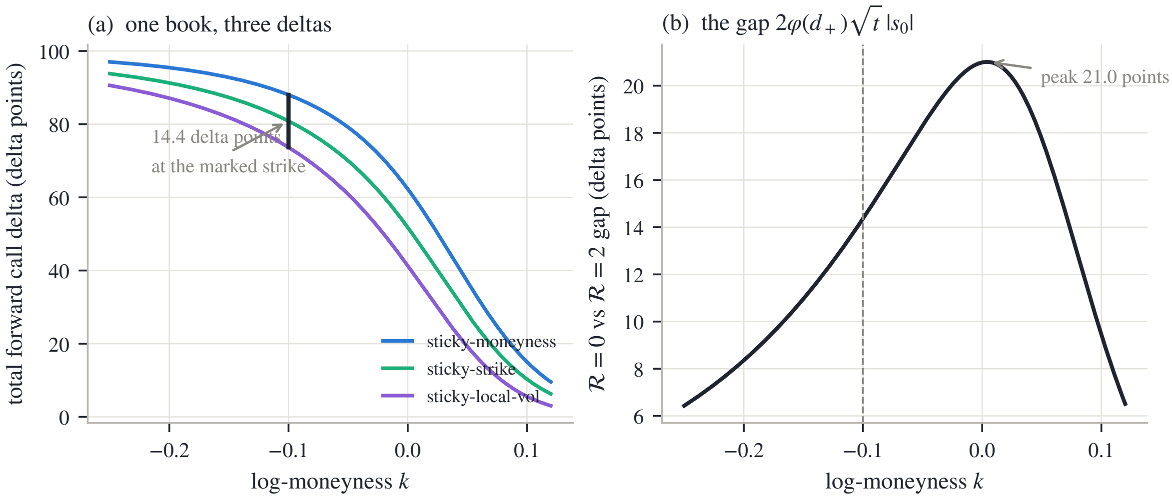

Figure 9.4: One book, three deltas (frozen hero node, eq. (9.4)). (a) Total forward call delta across strikes under the three regimes; at the marked $k=-0.10$ strike — a 19-delta put — the $\mathcal{R}=0$ and $\mathcal{R}=2$ answers are 14.4 delta points apart. (b) The gap (9.5) across strikes: a scaled vega profile, peaking at 21.0 delta points at the money.

Figure 9.4 evaluates both formulas on the frozen node, whose skew is $s_0=-0.430$. In panel (a), each curve is Δ_{tot} across strikes for one regime, in delta points (hundredths). The three curves tell one story with three endings: the sticky-moneyness delta is the largest and the sticky-local-vol delta the smallest — at every strike, since the correction $\varphi(d_+)\sqrt{\tau}(\mathcal{R}-1)s_0$ carries one sign across the axis. The ordering is worth one plain sentence. Under sticky-local-vol a falling market raises this put's volatility (eq. (9.3) with $\mathcal{R}=2$, $s_0H>0$ when $H<0$); a long call at that strike therefore loses less on the way down than Black alone predicts, which is the same as saying its effective delta is smaller. At the marked strike — ten percent below the money, a 19-delta put in Black's own units — the vertical bracket measures the disagreement: 14.4 delta points between the two outer regimes, on the same book, against the same fitted surface. A desk running a hundred such contracts hedges a difference of 14.4 underlying units, purely on the choice of dial.

Panel (b) plots the gap (9.5) across strikes, and its shape should look familiar: $\varphi(d_+)$ is (up to normalization) the vega profile, so the gap is vega rescaled by the skew. It peaks at the money — 21.0 delta points on this node — and fades in the far wings, where prices respond too weakly to volatility for the regime to matter. Notice what the peak location means: the regime choice is *most* consequential exactly where the book is most liquid and the desk feels most confident. The dashed vertical marks the panel (a) strike: at 14.4 points it is nowhere near the worst case.

We now know what the dial costs: double-digit delta points through the liquid belly of an ordinary index node. Remark 9.1 says the surface cannot set the dial. The natural next question is whether anything else in this book can. One object can try: the one model of Part I that carries every expiry at once, Chapter 4's local-volatility field. Holding *it* fixed while spot moves produces a transport with no dial at all — that is the next section.

9.5 The regime that answers: freezing the field

Every value of the dial so far is an assumption. This section builds the one transport in the book that is *derived*. The ingredient is Chapter 4's local-volatility field: one function $v(y)$ of the normalized strike $y = K/F$ — a local variance at every level the underlying might visit — chosen so that the model reprices every expiry of the surface at once. (Chapter 4 wrote $v(\tau, y)$; the fields of this section and the next are deliberately time-homogeneous, so v depends on the level alone.) Now make one hypothesis: *the field is a property of the underlying at absolute price levels*. If the local volatility at \$650 is high, it is high because of what the market believes about the world at \$650 — not because of where spot happens to stand today. Under that hypothesis, when spot moves, the field *stays where it is* in absolute strike, and the new smile is whatever the frozen field reprices to. (Mechanically: after the forward moves to $F_{\text{old}}e^{H}$, the same absolute strikes carry the same local variances under new labels, so the normalized coefficient is read as $v(y\,e^{H})$ — the quote rule of definition 9.2, applied to the field.) No dial is left: the transport is a computation.

The computation has a famous shortcut, and the shortcut explains the number 2 that has been hanging over the chapter since section 9.3. For short maturities, an option's implied volatility is approximately an *average of the local volatility along the way from the money to the strike*: the price is built from paths that drift from spot toward the strike, sampling the field as they go. (Made precise, the right average is a *harmonic* one — the reciprocal of the averaged reciprocal — along the connecting chord, a result of Berestycki et al. [5]; the midpoint reading used below goes back to Hagan et al. [29]. The distinctions do not matter here: to first order in the field's slope every such average agrees, and first order is all the half-rule needs.) Write the field in volatility form and in log-moneyness, $\sqrt{v}$ as a function of $x = \log y$, and take it to be a straight line with slope b near the money. Then the chord average is elementary:

$$\sigma(k) \;\approx\; \frac{1}{k}\int_0^k (\sigma_0 + b\,x)\,\mathrm{d}x = \sigma_0 + \frac{b}{2}\,k \qquad\Longrightarrow\qquad s_0 = \frac{b}{2}.$$

This is the **half-rule**: *the implied skew is half the field's slope.* The market's smile shows half of what the generator carries, because each strike only averages the field part of the way out.

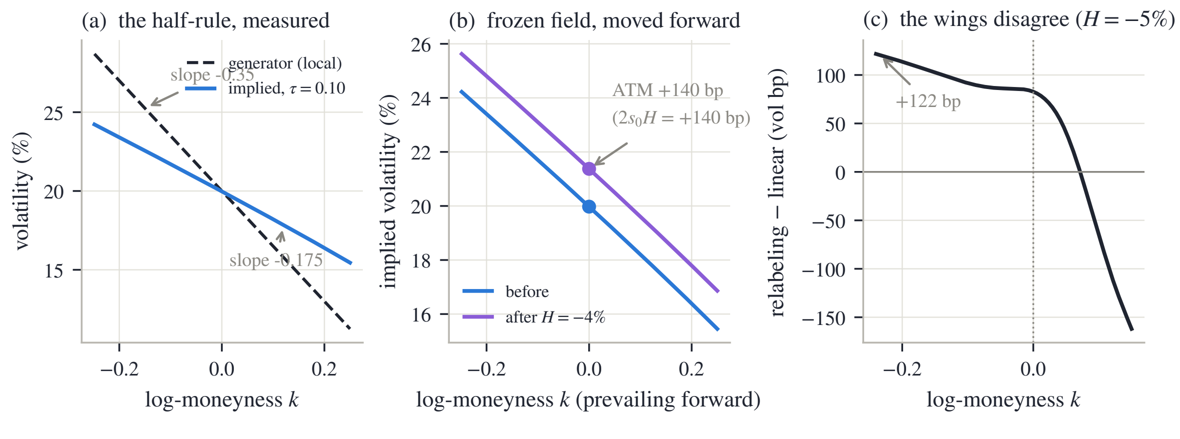

Figure 9.5: The frozen field's answer (synthetic straight-line field; panel (c) the frozen hero smile). (a) The generator (dashed, slope -0.35) and the implied smile it produces at $\tau = 0.10$ (blue, measured slope -0.175): the half-rule, measured ratio 0.499. (b) The field frozen in absolute strike, the forward moved -4%: the implied ATM rises +140 vol bp against the half-rule's prediction $2s_0H = +140$ vol bp. (c) The two closed-form transports on the real node at $H = -5\%$: midpoint relabeling minus linear $\mathcal{R} = 2$ transport, zero at the money, $+122/-162$ vol bp in the wings.

The rule is not folklore; it is measurable. Figure 9.5(a) shows a synthetic world built for exactly this test: a time-homogeneous field whose volatility is a straight line in log-strike with slope -0.35 (dashed), and the implied smile obtained by actually pricing through the forward equation with Chapter 4's marching scheme and inverting Black (blue), at maturity $\tau = 0.10$. The measured implied slope is -0.175 — ratio 0.499 to the generator. Half, to a tenth of a percent, with no fitting anywhere.

Now freeze and move. Let the forward move by H — take $H < 0$, a drop — and hold the field. At the *new* money, the implied volatility averages the field around the new spot, so it reads the straight generator there:

$$\sigma_{\text{atm}}^{\text{new}} \approx \sigma_0 + b\,H = \sigma_0 + 2\,s_0\,H,$$

an *increase* for a drop, since b and H are both negative. The ATM reading slides along the *generator's* slope b — and the smile only ever showed half of b. The frozen field therefore moves the ATM volatility by *twice the skew* per unit move: $\mathcal{R} = 2$, derived. The overshoot of section 9.3's taxonomy is the market paying out the half of the slope the smile had been holding back. Panel (b) runs the experiment in the synthetic world: forward down 4%, field untouched, both smiles repriced through the forward equation. The implied curve slides up its generator; the measured ATM change is +140 vol bp against the half-rule's $2s_0H = +140$ vol bp. Exact to the basis point.

At the money, then, the frozen field's answer is settled. Away from the money one can push the same logic further and ask for a closed form: which *old* smile point should tomorrow's smile at label k read? The tool is the *midpoint rule*, and it is one line from what was just derived: for a straight-line field, the average along the chord equals the field's value at the chord's *midpoint*, so the implied volatility at strike K reflects the field near the dollar midpoint $(F + K)/2$. Two (forward, strike) pairs that share a midpoint therefore share a reading. Tomorrow's pair is $(F_{\text{old}}e^H,\ K)$ with $K = F_{\text{old}}e^He^k$; the old strike K'' sharing its midpoint solves

$$\frac{F_{\text{old}} + K''}{2} = \frac{F_{\text{old}}e^H + K}{2} \qquad\Longrightarrow\qquad K'' = F_{\text{old}}\big(e^H(1 + e^k) - 1\big),$$

and its old label is the log of that ratio:

$$\ell(k, H) = \log\big(e^H(1 + e^k) - 1\big), \qquad w_{\text{new}}(k) = w_{\text{old}}\big(\ell(k, H)\big) \tag{9.6}$$

— the *midpoint relabeling*, written in total variance (the chart Chapter 2 fits in; at a single maturity the volatility form says the same thing, since $\sigma = \sqrt{w/\tau}$). Check it at zero move: $\ell(k, 0) = \log(1 + e^k - 1) = k$, the identity. Expand it in the move: differentiating,

$$\frac{\partial \ell}{\partial H} = \frac{e^H(1 + e^k)}{e^H(1 + e^k) - 1}, \qquad \frac{\partial \ell}{\partial H}\bigg|_{H=0} = \frac{1 + e^k}{e^k} = 1 + e^{-k},$$

so

$$\ell(k, H) = k + (1 + e^{-k})H + O(H^2). \tag{9.7}$$

At the money, $\ell(0, H) \approx 2H$: the new smile reads the old curve *two* units of moneyness per unit of move — the half-rule's factor again, now recovered from pure geometry, which is a reassuring consistency check. Away from the money the displacement $(1 + e^{-k})H$ is strike-dependent: at $k = -0.3$ it is $2.35\,H$, at $k = +0.3$ only $1.74\,H$. The put wing reads farther along the old curve than the call wing does.

That strike dependence is exactly what the linear transport (9.2) at $\mathcal{R} = 2$ flattens into one uniform level. So the frozen-field idea now has *two* closed-form candidates: the linear transport, and the relabeling (9.6). Panel (c) of fig. 9.5 measures their disagreement on the real frozen node, for a -5% move: the difference is zero at the money — both say $2s_0H$ there — and grows into the wings, to +122 vol bp at the span's put edge ($k = -0.24$) and -162 at its call edge ($k = +0.15$). (The asymmetric span is not a whim: it is the widest window on which the relabeling's read-back point $\ell(k, H)$ stays inside the node's own quoted strikes; appendix 9.A.)

Which one is the frozen field's true answer? Here the chapter owes the reader a confession about the derivation just given. The half-rule was derived by averaging along a chord in *log*-strike; the relabeling by matching midpoints in *dollar* strike. At the money, at first order, the choice of chart cannot matter — both gave $2s_0H$. In the wings the two charts stop agreeing, and neither rule of thumb outranks the other. The honest arbiter is the computation itself: reprice the frozen field and see. That is the next section — and the answer turns out to depend on something today's smile does not show.

9.6 The wings belong to the field, and the ratio measured

The last section left a tie to break: two closed forms, one frozen field, and only a computation can adjudicate. This section runs the computation — and finds that the question was subtler than the tie. The experiment needs fields whose shape we control exactly, so it runs in the synthetic world, with every construction stated (grid and solver details in appendix 9.A).

Take *two* time-homogeneous fields, built to agree at the money: both have volatility 20% at the forward, and both have log-slope $b = -0.35$ there. They differ only in what "a straight skew" means:

$$\sqrt{v}(y) = 0.20 + b\,\log y \qquad \text{(straight in } \textit{log}\text{-strike)},$$
$$\sqrt{v}(y) = 0.20 + b\,(y-1) \qquad \text{(straight in } \textit{dollar} \text{ strike)}.$$

Figure 9.6(a) draws both (dashed) with the smiles each produces today at $\tau = 0.25$ (solid), priced through the forward equation as before. Walk the panel from the money outward. At $k = 0$ the four curves agree by construction. The dashed fields separate quickly — straight in one chart is curved in the other — but the two *solid* curves stay nearly on top of each other across the whole span: the smiles differ only through second-order averaging effects, a few tens of basis points at the edges. Squint, and today's market looks the same under either field. Chapter 4 proved this is no accident: many sheets reprice the same quotes. The two fields here are a minimal, fully explicit instance of that ambiguity.

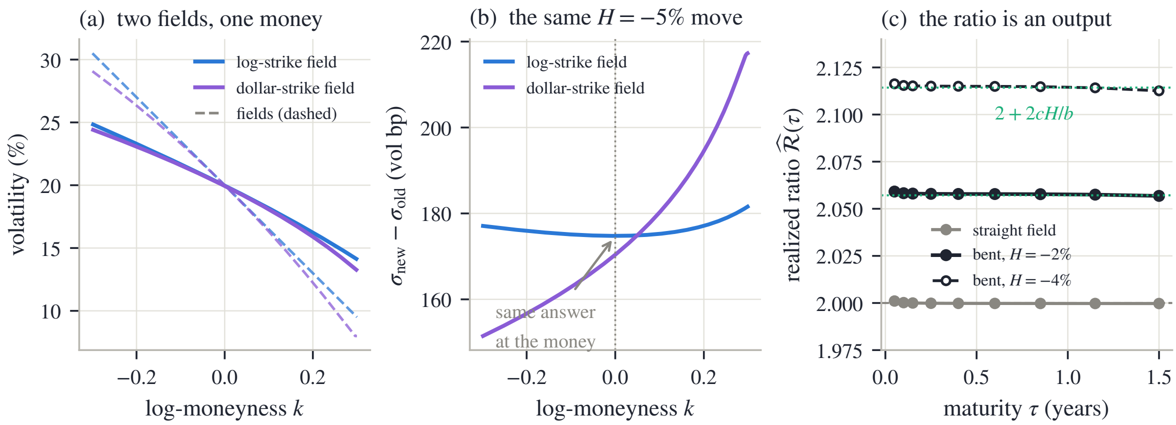

Figure 9.6: The reprice adjudicates (synthetic fields, forward-equation pricing throughout). (a) Two fields (dashed) sharing ATM value and ATM slope, one straight in log-strike, one in dollar strike, and their nearly indistinguishable smiles today at $\tau = 0.25$ (solid). (b) Each field frozen, the forward moved -5%, both surfaces repriced: the smile changes agree at the money (+175 and +170 vol bp — the half-rule) and part in the wings, by 26 vol bp at $k = -0.30$ and 36 at $k = +0.30$: flat response against tilted response. (c) The realized ratio $\widehat{\mathcal{R}}(\tau)$ of the reprice: the straight field pins 2.00 at every maturity (spread 0.001); bending the field lifts the ratio to the computable level $2 + 2cH/b$ (dotted guides): 2.06 measured against 2.06 predicted at $H = -2\%$, 2.11 against 2.11 at $H = -4\%$.

Now freeze each field and run the same scenario: forward down 5%, reprice, and record how the smile *changed* — $\sigma_{\text{new}}(k) - \sigma_{\text{old}}(k)$, each curve against its own before, both in prevailing moneyness. Panel (b) shows the two answers. At the money they coincide to a few basis points: +175 vol bp for the log-strike field, +170 for the dollar-strike one, both the half-rule's $2s_0H$ in action. (Even the small difference is explainable rather than numerical: the dollar field's exact ATM answer is $b(e^H - 1)$ rather than bH, a distinction of second order in the move.) Away from the money the two answers have visibly different *shapes*. The log-strike field answers with an almost uniform lift — the flat blue line: its smile slides up as a block, which is precisely what the linear $\mathcal{R} = 2$ transport (9.2) builds. The dollar-strike field answers with a tilt — the violet curve, rising from the put side to the call side — and the tilt is worth deriving, because a naive read of the expansion (9.7) points the wrong way: the displacement $(1 + e^{-k})H$ is *largest* on the put wing, yet the response there is *smallest*. Two lines resolve it. By the midpoint rule, the dollar-straight field's own smile today is

$$\sigma_{\text{old}}(k) \approx 0.20 + \tfrac{b}{2}(e^k - 1), \qquad \text{with local slope} \qquad \sigma'_{\text{old}}(k) \approx \tfrac{b}{2}\, e^k :$$

straight in dollars is curved in log-moneyness, and the slope fades on the put side. The response at a fixed label is displacement times slope,

$$\sigma'_{\text{old}}(k)\,(1 + e^{-k})\,H \;\approx\; \tfrac{b}{2}\, e^k\,(1 + e^{-k})\,H \;=\; \tfrac{b}{2}\,(1 + e^k)\,H,$$

so the large displacement lands on the flat part of the curve and the product tilts the *other* way: proportional to $1 + e^k$, smallest on the put side — the violet shape, and exactly what the relabeling (9.6) builds when applied to this smile. (At $k = 0$ the product is $bH = 2s_0H$ again.) By $k = \pm 0.30$ the two fields' answers are 26 and 36 vol bp apart. One note of honesty on those two numbers: the first-order logic just traced puts the separation near 26 vol bp on *both* wings; the call side's excess is each field's own second-order response to a move this large — the reprice keeps score exactly where the expansions stop.

So the tie from section 9.5 dissolves rather than resolves: *each* closed form is the exact first-order answer — for a different field. Two fields that today's quotes can barely tell apart give measurably different tomorrows in the wings. The conclusion deserves plain words, because it is the chapter's deepest point. **At the money, the frozen-field answer is universal**: twice the skew, no matter the field's shape — that is the half-rule, and every construction in this chapter reproduces it. **In the wings, the answer belongs to the field**, and Chapter 4 showed the quotes do not pin the field down there (section 4.6). The missing derivative, chased into the one regime that promised to derive it, reappears one level deeper: the model answers, but only as precisely as the data determined the model.

The frozen field also completes the dial in a satisfying way. All three named regimes — not just $\mathcal{R} = 2$ — can be written as one rule about *how the field follows the spot.* Let the field's strike axis be relabeled, per unit move, by

$$x \;\longmapsto\; x - \tfrac{\mathcal{R}}{2}\,H \qquad \text{in prevailing log-moneyness } x = \log y. \tag{9.8}$$

For $\mathcal{R} = 2$ this is $x \mapsto x - H$: exactly the label change of a field frozen in absolute strike (the quote rule of definition 9.2, applied to the field's nodes). For $\mathcal{R} = 0$ the labels do not move: the field rides with the forward. For $\mathcal{R} = 1$ it follows halfway. The consistency check takes two lines through the half-rule. Relabeling by (9.8) means the field seen in prevailing coordinates is $\sqrt{v}_{\text{new}}(x) = \sqrt{v}_{\text{old}}(x + \tfrac{\mathcal{R}}{2}H)$, so the new ATM implied volatility — the field's value at the new money, by the chord average — is

$$\sigma^{\text{new}}_{\text{atm}} \approx \sigma_0 + b\,\tfrac{\mathcal{R}}{2}H = \sigma_0 + \mathcal{R}\,s_0\,H,$$

using $b = 2s_0$. One relabeling rule for the field generates the entire dial of section 9.3; the stickiness debate is a debate about how many halves of the move the generator follows.

Finally, make the section's title quantitative: measure the ratio as an *output*. Define the realized ratio of a reprice,

$$\widehat{\mathcal{R}}(\tau) = \frac{\sigma^{\text{new}}_{\text{atm}}(\tau) - \sigma^{\text{old}}_{\text{atm}}(\tau)}{s_0(\tau)\, H},$$

the hat marking, as in Chapter 6, a measured estimate: everything on the right comes out of the pricer. Panel (c) of fig. 9.6 plots it across maturities from 0.05 to 1.5 years. For the straight log-strike field at $H = -2\%$, the curve is a ruler: $\widehat{\mathcal{R}}$ from 2.00 at the shortest maturity to 2.00 at the longest, with a total spread of 0.001 across the whole range. For this construction, 2 is not merely a short-maturity limit — at first order in the skew the frozen field pays out double at *every* maturity, and the measurement confirms it to the third digit. The maturity-independence has a one-sentence reason: at first order the numerator (the ATM response) is an average of the field's slope over the neighbourhood the paths visit around the money, the denominator (the skew) is *half the same average*, and the quotient cancels whatever maturity dependence the average carries. Then bend the field: add curvature $c\,(\log y)^2$ with $c = 0.5$ to the straight generator. The ratio lifts off 2 — to 2.06 at $H = -2\%$ and 2.11 at $H = -4\%$ — and stays flat in maturity. The lift is not mysterious; it is one more chord average. At the new money the bent generator reads $\sigma_0 + bH + cH^2$, while the bent field's implied skew is still $b/2$ — the curvature term's chord average is $ck^2/3$, whose slope vanishes at the money — so

$$\widehat{\mathcal{R}} = \frac{bH + cH^2}{(b/2)\,H} = 2 + \frac{2c}{b}\,H,$$

and the dotted guides in the panel sit at exactly these levels: predicted 2.06 and 2.11, measured 2.06 and 2.11. The ratio is a genuine output: it reports the field's shape (through c/b) and the size of the move (through H), neither of which the constant "2" of the regime's name can see.

One honest boundary closes the section. The frozen field is itself a hypothesis about the world, and empirically it is not quite right: measured markets move their smiles by *less* than the frozen field predicts — short-dated skew-stickiness ratios sit between the sticky-strike and sticky-local-vol answers [6]. The reprice is the exact consequence of an approximate premise. That is why this regime is one setting of the dial and not the dial's replacement: it answers precisely, and the desk must still decide whether to believe the premise.

9.7 Scenarios on the frozen board

The machinery is complete; point it at the real data. This section runs one scenario — the forward down 5% — across the entire frozen SPY board, all eight expiries, under the three named regimes, and reads the results with everything the chapter has taught. Each expiry is transported with its *own* skew $s_0(T)$, read from its own stored fit, and the ATM value after the move is the exact transport: the node's fitted curve read at $k = H$, plus the level $(\mathcal{R}-1)s_0(T)H$.

Walk fig. 9.7(a) from right to left. At the long end the board barely notices the regime: the December 2027 skew is gentle ($s_0(T) = -0.29$), and even the sticky-local-vol response is about 2.9 volatility points. Moving left, the three transported curves fan apart, slowly and then violently. At the two-day node, today's ATM reading of 9.6% becomes 15.2% under sticky-moneyness, 25.7% under sticky-strike, and 36.2% under sticky-local-vol — a spread of twenty-one volatility points on one node, from one tick. The whole spread is one number's doing: the two-day skew is $s_0 = -2.10$ — a two-day smile prices a 5% move as a catastrophe, and its put wing is priced accordingly — and every formula in this chapter multiplies that skew by the move.

Two honest observations about the same panel. First, look closely at the short end of the sticky-moneyness curve: it does *not* sit on today's orange curve, though "the smile rides the forward" suggests it should. The exact transport reads each node's own curve at $k = H$, and on a two-day smile the point five percent below the money is far up the put wing — the curvature

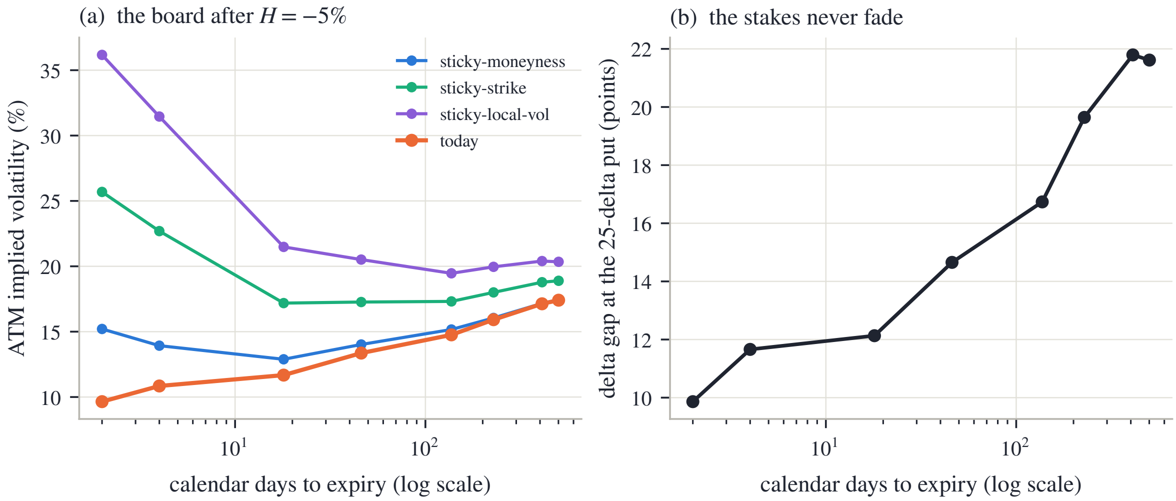

Figure 9.7: The frozen SPY board under a -5% scenario. (a) ATM implied volatility by expiry (log time axis): today (orange) and the three transported boards. The two-day node moves from 9.6% to 15.2/25.7/36.2% under $\mathcal{R} = 0/1/2$ — its skew is $s_0 = -2.10$. (b) The $\mathcal{R} = 0$ vs $\mathcal{R} = 2$ delta gap (9.5) at each expiry's 25-delta put: from 9.9 delta points at two days to 21.6 at the far end.

term of fig. 9.3(b), grown large because short-dated smiles are sharply curved. Even the mildest regime moves short-dated ATM readings on a move this size. Second, on the two shortest expiries that $k = H$ read lies *beyond the last quoted strike* — the two-day node's quotes stop less than three percent from the money. What the transport consumes there is the fitted model's published wing: Chapter 5's disclosed completion, not a quoted value. A scenario engine cannot avoid this — a five-percent question to a two-day smile is a question about its wing — but a desk should know which of its scenario numbers stand on quotes and which on convention.

Panel (b) prices the same fan in the units of section 9.4: the $\mathcal{R} = 0$ versus $\mathcal{R} = 2$ delta gap (9.5), evaluated at each expiry's 25-delta put — the standard liquid put benchmark; its strike solves $\Phi(d_+) = 0.75$ on the node's own fitted smile. The gap runs from 9.9 delta points at two days, through 16.7 at the running December 2026 node, to 21.6 at the end of the board. It *grows* with maturity on this board: the gap is $2\varphi(d_+)\sqrt{\tau}\,|s_0|$, and SPY's skew decays more slowly than $1/\sqrt{\tau}$ across these expiries (scaled by $1/\sqrt{\tau}$ from the two-day node's -2.10, the skew would have fallen to -0.25 by the December 2026 expiry; the fitted value there is -0.430), so the maturity factor wins. There is no safe corner: the choice of dial is worth roughly ten to twenty delta points on the benchmark put at *every* expiry of an ordinary index board.

One paragraph on validity, where it bites. A transported smile is a new smile, and nothing in (9.2) re-checks the butterfly or calendar conditions of Chapter 2: for small moves the transported surface inherits validity to first order, but at scenario sizes the outputs should be re-audited before anything downstream consumes them — and there is a matching theorem: an arbitrage-free dynamics cannot move the smile by level shifts alone indefinitely [43], so every rule in this chapter is a local scenario device, not a global model. The one output that is arbitrage-free by construction is the frozen-field *reprice* of section 9.6, because it prices an actual model rather than transporting a curve.

9.7.1 The contract

The chapter closes as the others do: three columns of decreasing strength — proved, measured, chosen.

Step back to the book's thesis one more time. *Validity* in this chapter is modest and structural: transports preserve arbitrage-freeness only locally, and the one globally safe path is to move the

Table 9.1: The chapter's contract.

Proved	**Measured**	**Convention**
A snapshot does not identify dynamics: every dial value prices today's quotes identically (remark 9.1).	The fan of tomorrows on the frozen node: 344 vol bp at one strike for a 4% move (fig. 9.1).	The dial $\mathcal{R}$ itself: assumed, not estimated — nothing in this chapter regresses realized smile moves on skew.
Shape preservation leaves one free level; ATM response $\mathcal{R}s_0H$ (proposition 9.3); fixed-strike response $(\mathcal{R}-1)s_0H$, exactly (9.3).	The common second-order bend of every regime: 54 vol bp at $H = -6\%$ (fig. 9.3).	The transport is first-order and shape-frozen: no convexity response; vertical level chosen over horizontal slide (equivalent at first order).
Total delta $\Phi(d_+) + \varphi(d_+)\sqrt{\tau}(\mathcal{R} - 1)s_0$; regime gap $2\varphi(d_+)\sqrt{\tau}\lvert s_0\rvert$ (9.4)–(9.5).	14.4 delta points at the 19-delta put, 21.0 at the money (fig. 9.4); 9.9–21.6 points across the board (fig. 9.7).	Deltas quoted in the forward (Black) convention, discount set aside; $\tau = t$ on this board (Chapter 8's SPY finding).
The half-rule $s_0 = b/2$; frozen-field ATM response $2s_0H$; the relabeling (9.6) and its expansion (9.7); the grid rule (9.8) generating the dial.	Half-rule ratio 0.499; ATM move +140 vs predicted +140 vol bp (fig. 9.5); $\widehat{\mathcal{R}}$ flat at 2.00 for the straight field, $2 + 2cH/b$ (2.06, 2.11) for the bent one (fig. 9.6).	The frozen field is an approximate premise: measured markets sit below $\mathcal{R} = 2$ [6]; the wings of the answer depend on field shape the quotes do not pin down.
— (this row is empirical by design)	The board scenario: two-day ATM spread 15.2–36.2%; short-expiry reads beyond the last quote (fig. 9.7).	Scenario reads on the two shortest expiries consume the published wing (Chapter 5's disclosed completion); transported surfaces are to be re-audited before downstream use [43].

model, not the curve. *Information* is the chapter's sharpest lesson, because here the information runs out: today's surface determines every strike derivative and no spot derivative (remark 9.1) — and even the regime that derives its answer inherits the field's under-determination in the wings. *Convention* is therefore load-bearing as nowhere else in this book: the dial is a disclosed choice, worth double-digit delta points, that no snapshot can audit. What *could* audit it is time — watching how smiles actually move as spot moves, tick after tick, and weighing each new observation against what the transported surface expected. Weighing new evidence against a moving prior is a filtering problem, and it is the subject of the next chapter.

9.A Protocol: data, constructions, and solver constants

Every number and figure in this chapter regenerates from one deterministic script reading the frozen snapshot (`scripts/ch09/gen_figures.py`); nothing is refetched, no smile is refitted, and nothing anywhere in the chapter is random.

Frozen smiles (figs. 9.1 to 9.4; panel (c) of fig. 9.5; fig. 9.7). All real-data curves are the snapshot's *stored* haircut LQD fits — Chapter 2's model, in the bid–ask-shrinking ("haircut") fit mode of the recipe stamped in the snapshot manifest — rebuilt from the stored parameters: the same objects every earlier chapter's gallery used. The hero node is SPY December 2026 (137 days). The ATM skew s_0 of a fitted curve is a central difference of the fitted volatility at $k = \pm 10^{-3}$. The marked strike of figs. 9.1 and 9.4 is $k = -0.10$ in today's label. Figure spans stay inside each node's prepared-quote span, with two disclosed exceptions on the two shortest expiries of fig. 9.7, discussed in section 9.7.

Delta conventions (figs. 9.4 and 9.7). Deltas are forward (Black) deltas, the discount factor set aside; one delta point is 10^{-2} of delta. d_+ is evaluated on the node's own fitted smile, $d_+ = -k/\sqrt{w} + \sqrt{w}/2$. The 25-delta put strike of fig. 9.7(b) solves $\Phi(d_+) = 0.75$ on the fitted smile (put forward delta -0.25); the gap plotted there is eq. (9.5) with each node's own (τ, s_0).

The synthetic fields (fig. 9.5(a,b); fig. 9.6). Time-homogeneous local-variance fields on the normalized strike axis $y = K/F_{\text{old}}$, zero carry, variance time equal to calendar time. Volatility form $\sqrt{v}$, with $x = \log y$: straight-in-log $0.20 + bx$; straight-in-dollar $0.20 + b(y - 1)$; bent $0.20 + bx + cx^2$, with $b = -0.35$ and $c = 0.5$. All are clamped to $[0.08, 0.60]$; over every span drawn or read in the chapter the clamps are inactive, except the dollar-strike field's lower clamp in the last few points of fig. 9.6(a)'s dashed curve.

Pricing and inversion. Chapter 4's fully implicit forward-equation marcher (appendix 4.A's scheme, the same code the Chapter 4–5 figures used) on a uniform strike grid $y \in [0, 4]$ with $\Delta y = 0.002$ (the grid contains $y = 1$ exactly) and time step $\Delta\tau = 5 \times 10^{-4}$, refined to 1.25×10^{-4} for the basis-point-scale panels (fig. 9.6(a,b)). Implied variances are recovered from the marched call prices by bisection on the normalized Black formula, at grid-node strikes only — no interpolation touches any reported number, and the at-the-money read sits on the $y = 1$ node. A moved scenario reprices with the coefficient $v(y\,e^H)$: the frozen field read at the same absolute strikes under the new normalization.

Reading the realized ratio (fig. 9.6(c)). σ_{atm} and $s_0(\tau)$ are the value and slope at $k = 0$ of a quadratic fitted to the smile's grid nodes with $|k| \le 0.03$; $\widehat{\mathcal{R}}(\tau)$ is then the displayed quotient at maturities $\{0.05, 0.10, 0.15, 0.25, 0.40, 0.60, 0.85, 1.15, 1.50\}$. Marching-step audit: halving to quartering the time step moves the bent field's ratio at $\tau = 0.25$ by 0.00% — the reported digits are converged.

The board scenario (fig. 9.7). Move $H = -0.05$ applied uniformly to all eight SPY expiries. The transported ATM value of a node is the exact transport: the node's stored fitted curve read at $k = H$, plus $(\mathcal{R} - 1)s_0(T)H$. On the 2-day and 4-day nodes the $k = H$ read lies beyond the prepared-quote span (their quotes end at -2.8% and -4.5% log-moneyness respectively), so those reads consume the fitted model's published wing, as disclosed in section 9.7.

Units. As throughout the book: one vol bp is 10^{-4} of absolute volatility; volatility points are 10^{-2}; delta points are 10^{-2} of delta. Days convert to years at 365.

Chapter 10

Inference Under Weak Information: Filtering and Priors

Chapter 9 ended on a promise: watching how new evidence lands on a transported surface is a filtering problem. This chapter keeps that promise, and it starts one step earlier, with a static question that every fitter faces before any dynamics: what does a thin morning of quotes actually determine? On the book's frozen SPY node, thinned to its at-the-money band, the answer is measured directly — the fitting error does not move at all while the unquoted wing is dragged across twelve volatility points. The likelihood is flat along whole directions, and something other than the data must choose along them. The chapter builds the two instruments that make that choice honestly. In parameter space*: yesterday's surface enters today's fit as extra least-squares rows, admitted through a gate that is exactly zero wherever today's quotes identify a coordinate — a dead zone, proved, not a small bias. Each persisted quantity is priced for identifiability before it is admitted, and the persisted coordinates are chosen — risk reversals and butterflies, not absolute wing prices — so that a market-wide level jump cannot be damped. In* time*: the smile's ATM handles become a filtered state; yesterday's estimate, transported by Chapter 9's rule, meets today's noisy reading, and the update is a precision-weighted average whose gain nobody sets by hand — it is computed from two variances, each built honestly: the observation's from the fit's own information in stated bid–ask units, inflated when the chain contradicts itself, and the prediction's from a budget accrued on Chapter 8's clock. The two instruments answer different questions — absence and noise — and the chapter keeps them separate all the way to the accounting identity that prevents a quote from being counted twice. The closing audit is the chapter's standard of truth: the standardized error $\mathcal{Z}$ should scatter with standard deviation one if the stated error bars mean what they say, and on a seeded synthetic history a starved prediction budget fails that audit in both directions at once — overconfident about a surface it updates too slowly.*

10.1 What a sparse morning does not determine

Chapter 9 closed with a moving prior: a surface transported to the new spot, waiting to be compared with fresh evidence. Before building the machinery that weighs the two, it pays to ask a static question — one that needs no dynamics at all. Take one morning's quotes, alone. How much of the smile do they actually determine?

The book's running node answers concretely. The frozen SPY December 2026 chain carries 94 prepared quotes (137 days to expiry). Keep only the 31 quotes inside the at-the-money band $|k| \leq 0.10$ and set the rest aside. This staged morning is not artificial meanness: on an illiquid name, or in the first minutes of a session, a band like this is all the market offers, and the desk still needs a whole smile — the wings included — to price, hedge, and feed every integral of Chapter 5.

What can the band say about the wing? Here is a direct experiment, run with a deliberately flexible curve family — a *spline*, which is just a smooth curve pieced together from cubic polynomials and controlled by its values at a handful of chosen strikes (the *knots*; the exact knots and the small smoothness penalty are in Appendix 10.A) — so that no parametric habit answers in the data's place. Fit the 31 band quotes, but *impose* the wing: pin the fitted volatility at $k = -0.30$ to a chosen value, and refit. Sweep the pinned value from 24% to 36% — a span of 12 volatility points — and record, for each imposed wing, how well the curve fits the quotes it was actually given. Each quote's misfit is measured against that quote's own stated noise, the bid–ask half-spread; "fits the band" always means "fits within the noise the market itself states."

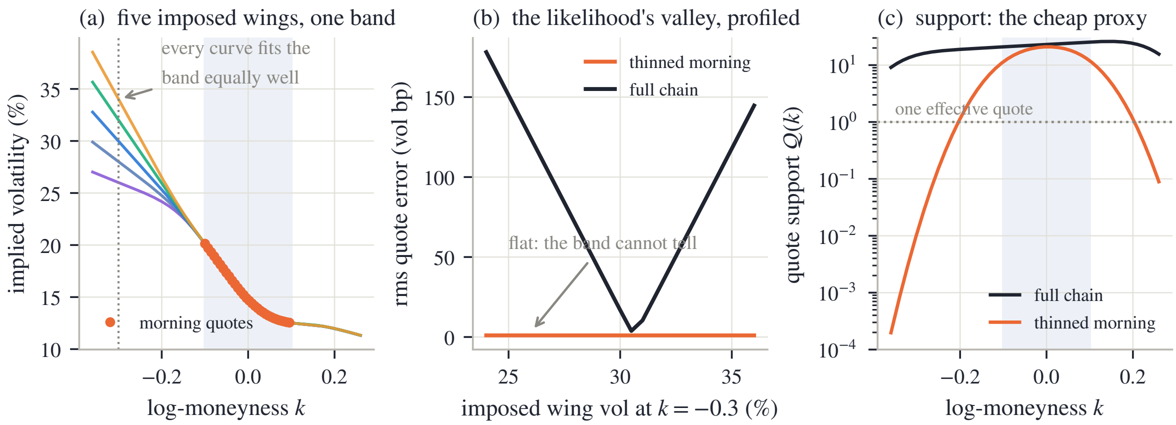

Figure 10.1: What a sparse morning does not determine (frozen SPY December 2026 node, 31 quotes kept in the shaded band). (a) Five fits of the same band, each with the wing volatility at $k = -0.30$ (dotted vertical) pinned to a different imposed value; through the band they are one curve. (b) The root-mean-square quote error against the imposed wing value: on the thinned morning (orange) it varies by 0.02 vol bp across the whole twelve-point sweep — flat; on the full 94-quote chain (ink) the same sweep meets a sharp V. (c) The quote-support profile $\mathcal{Q}(k)$ of eq. (10.1) for both quote sets: the thinned morning's support collapses outside the band, exactly where panel (b)'s valley is flat.

Figure 10.1 shows the result, and it is worth walking slowly. In panel (a), the orange dots are the morning's quotes and the five curves are five fits, each pinned to a different wing value — 26% up to 34% at $k = -0.30$. Through the shaded band the five curves lie on top of one another; to the left they fan apart by eight volatility points. Panel (b) makes the visual precise. The orange line is the fitting error of the band quotes, in vol bp, as the imposed wing sweeps across its twelve-point range: it moves by 0.02 vol bp in total — the band cannot tell the imposed wings apart. The ink line repeats the identical sweep on the *full* chain, where real quotes sit at $k = -0.30$: now the error climbs steeply away from one preferred value, and an imposed wing only half a volatility point off already costs five bp of fitting error (0.5 points is the half-width at that cost). Same family, same sweep, same machinery — the only difference is which quotes exist.

This deserves a name. Think of the fit's objective — the sum of squared, noise-scaled quote errors — as a landscape over all candidate curves. (Statisticians call the corresponding probability of the data the *likelihood*; for Gaussian noise the objective is its negative logarithm, so "the likelihood is flat" and "the objective is flat" say the same thing, and the chapter uses both.) A *flat direction* is a way of changing the curve that does not change the objective at all: the data assigns the same score to every curve along it. Panel (b)'s orange line exhibits one flat direction explicitly (move the wing, hold the band), and there are many more — every reshaping of the smile outside the band, and finer wiggles between quotes inside it. Statisticians say the wing is *unidentified*: no amount of reweighting or clever optimization extracts from these quotes a number the quotes do not contain.

Two remarks before anything is built on this.

First, agreement between models is not evidence. Refit the same thinned band with the reference implementation's three parametric families at several resolutions and smoothing strengths — eight fits in all — and their wing readings at $k = -0.30$ land within 0.3 volatility points of one another. Eight rules agreeing looks like knowledge; panel (b) says the quotes would have accepted any answer across a twelve-point range. The agreement reflects a shared habit of smooth extrapolation, a convention baked into every one of those families — not information. A desk that reads consensus wings as measured wings is mistaking its tools for its data.

Second, flatness is measurable in advance, cheaply. The profile experiment is exact but costs a refit per candidate value. A proxy does almost as well: count how many quotes sit *near* a given strike. Define the *quote support* at log-moneyness k as the kernel-weighted count — a *kernel* being nothing more than a smooth bump, here a Gaussian, that decides how much a quote at some distance still counts —

$$\mathcal{Q}(k) = \sum_i \exp\left(-\frac{(k_i - k)^2}{2 \cdot 0.06^2}\right), \tag{10.1}$$

where the sum runs over the morning's quote locations k_i (when quotes carry unequal fit weights, each term is multiplied by its weight, normalized to average one). A quote sitting exactly at k contributes one; a quote a few bandwidths of 0.06 away contributes essentially nothing. Panel (c) plots $\mathcal{Q}$ for both quote sets on a logarithmic axis: the full chain holds double-digit support everywhere the book has ever read that smile, while the thinned morning's support falls off a cliff at the band edge — $\mathcal{Q}(0) = 21.0$ at the money, $\mathcal{Q}(-0.30) = 0.01$ in the wing. Where support is large the profile is a V; where it vanishes the profile is flat. Support is a proxy, not a theorem — section 10.3 returns to exactly how far it can be trusted — but it prices the geometry of panel (b) without a single refit.

We now know precisely what is missing: along the flat directions the data has no opinion, and yet a published smile must take a value there. *Something* always chooses — a family's tail habit, a smoothness penalty, a default. The thesis of this book says the choice should be disclosed and defensible, and the most defensible candidate is obvious: yesterday's surface, which was fitted to a full chain and has merely aged a day. The rest of the chapter builds the two instruments that let yesterday speak *only* where today is silent or noisy. Section 10.2 through section 10.4 work in parameter space: where may a prior pull at all? Section 10.5 through section 10.7 work in time: when today's data is present but noisy, how hard may yesterday's prediction resist it?

10.2 Yesterday as evidence: rows and the gate

We know where the data is silent. This section builds the mechanism that lets yesterday fill the silence — and proves that the mechanism cannot touch anything the data has measured.

10.2.1 Two estimates of one number

Start with the smallest possible case, because the whole chapter is this case repeated. Suppose two independent, unbiased estimates of the same unknown quantity are on the table — say yesterday's (transported) reading of some smile feature, and today's data-only reading — with known variances $\mathcal{V}_1$ and $\mathcal{V}_2$. ("Unbiased" means each is right on average; "variance" is the familiar mean squared error of an estimate around the truth.) How should they be combined? Try a weighted average with weight a on the first,

$$\text{combined} = a \cdot \text{first} + (1 - a) \cdot \text{second}.$$

Because the two are independent and unbiased, the combination is unbiased for every a, and its variance is

$$a^2\, \mathcal{V}_1 + (1 - a)^2\, \mathcal{V}_2.$$

Minimize over a: differentiate and set to zero,

$$2a\,\mathcal{V}_1 - 2(1-a)\,\mathcal{V}_2 = 0 \qquad \Longrightarrow \qquad a = \frac{\mathcal{V}_2}{\mathcal{V}_1 + \mathcal{V}_2}.$$

Each estimate is weighted by the *other's* variance — equivalently, by its own *precision*, the reciprocal $1/\mathcal{V}$ of its variance: dividing numerator and denominator by $\mathcal{V}_1\mathcal{V}_2$,

$$a = \frac{\mathcal{V}_2}{\mathcal{V}_1 + \mathcal{V}_2} = \frac{1/\mathcal{V}_1}{1/\mathcal{V}_1 + 1/\mathcal{V}_2}.$$

Substituting the optimal a, the combined variance is

$$\mathcal{V}_{\text{comb}} = \frac{\mathcal{V}_1\,\mathcal{V}_2}{\mathcal{V}_1 + \mathcal{V}_2}, \qquad \text{i.e.} \qquad \frac{1}{\mathcal{V}_{\text{comb}}} = \frac{1}{\mathcal{V}_1} + \frac{1}{\mathcal{V}_2} : \tag{10.2}$$

precisions add. A worked number to keep: yesterday says 20.0% with standard deviation 0.30%; today's fit says 20.4% with standard deviation 0.15%. Variances 0.09 and 0.0225 (in $\%^2$); weight on yesterday $0.0225/0.1125 = 0.20$; combination $0.20 \times 20.0 + 0.80 \times 20.4 = 20.32\%$. Today, four times as precise in variance, gets 80% of the say. These particular numbers return in section 10.5.

This is ordinary Bayesian updating in its simplest clothing: the first estimate plays the *prior* (what was believed before today's data), the second the data, and the combination is the *posterior* (the belief after). The share of the posterior owned by the prior is

$$\text{prior share} = \frac{1/\mathcal{V}_1}{1/\mathcal{V}_1 + 1/\mathcal{V}_2}, \tag{10.3}$$

its precision divided by the total. Keep eq. (10.3) in sight; the whole design question of this section is what that share should be when the "data" side is a whole fit rather than one number.

10.2.2 The prior as extra rows

How does yesterday enter a real calibration? Chapters 2–4 fitted a parameter vector θ (the model's coordinates, in whatever chart) by least squares over quote equations. The prior joins the same objective as *pseudo-observations*: pick a set of smile functionals O_j — a functional here is just a number read off a smile, such as the fitted volatility at one strike, or a difference of two such volatilities — evaluate each on yesterday's transported surface to get a target $\hat{O}_j^-$, and append one row per functional:

$$\text{row } j: \qquad \sqrt{\lambda_j}\,\bigl(O_j(\theta) - \hat{O}_j^-\bigr), \tag{10.4}$$

carried at weight $\lambda_j \geq 0$ alongside the quote rows. (The minus superscript marks a value believed *before* today's data; it will do heavy duty from section 10.5 on.) Squared and summed with the data misfit, such a row is exactly the prior side of eq. (10.3) for the coordinate O_j, with prior precision λ_j. Everything therefore reduces to one question: what should λ_j be?

10.2.3 The design fork: a ridge, or a gate

The familiar answer is a *ridge* — Tikhonov regularization, a fixed small λ_j for every row, the standard cure for ill-posed fitting (a problem is ill-posed when its data do not determine a unique, stable answer; Chapter 4's extraction was one, and Chapter 4 used exactly this cure on the local-volatility sheet) [47]. For the present purpose the ridge has a disqualifying property. By eq. (10.3), a fixed $\lambda > 0$ owns the share $\lambda/(\lambda + \mathcal{I})$ of *every* coordinate, where $\mathcal{I}$ is the precision with which today's data pins that coordinate. The share shrinks as the data improves — but it never reaches zero. A ridge biases the superbly-quoted at-the-money level toward yesterday exactly as surely as it stabilizes the dark wing, only less. If yesterday's level was 20% and today's true level is 24%, a ridge pulls today's published level below 24% — it *damps* a genuine move the market has plainly made. For a fitter whose first duty is to report what the market says, that is not a small flaw; it rules the ridge out.

What is wanted instead is a weight that depends on the data: full strength where the data is silent, and *exactly zero* — not merely small — once the data identifies the coordinate. The reference implementation uses one function for this everywhere:

Definition 10.1 (Activation gate)**.** For an identification precision $\mathcal{I} \geq 0$, a required precision $\mathcal{I}_{\mathrm{req}} > 0$, and a sharpness exponent $\gamma > 0$,

$$\mathrm{gate}(\mathcal{I}) = \Big(\min\big(\max(1 - \mathcal{I}/\mathcal{I}_{\mathrm{req}},\, 0),\, 1\big)\Big)^{\gamma}. \tag{10.5}$$

Read it from the inside out: $1 - \mathcal{I}/\mathcal{I}_{\mathrm{req}}$ is the shortfall of precision below the requirement; the $\max(\cdot, 0)$ clips it at zero once the requirement is met; the $\min(\cdot, 1)$ caps it at one when there is no data at all; and the exponent γ shapes how fast the gate closes as data accumulates. Each persisted row then receives $\lambda_j \propto \mathrm{gate}(\mathcal{I}_j)$: a fixed budget of prior weight is split across the rows in proportion to their gate values, and a row whose gate is zero is simply dropped.

Proposition 10.2 (The no-damp guarantee)**.** *If coordinate j satisfies $\mathcal{I}_j \geq \mathcal{I}_{\mathrm{req}}$, then* $\mathrm{gate}(\mathcal{I}_j) = 0$*, hence $\lambda_j = 0$, and row j of eq.* (10.4) *is the zero function of θ. The objective, its gradient, and every downstream quantity are then independent of the prior value $\hat{O}_j^-$. In particular, when every gate is closed the fit is the prior-free fit.*

Proof. With $\lambda_j = 0$ the row is $\sqrt{0} \cdot (O_j(\theta) - \hat{O}_j^-) = 0$ identically: it contributes zero to the objective and a zero row to the derivative stack, for every θ. Nothing that is computed from the objective can depend on $\hat{O}_j^-$ through a row that is identically zero. □

The proof is three lines because the content is structural, not analytic: the gate does not make the prior's influence small where data speaks — it makes the prior *absent* there. That is the difference between gating and shrinkage, and it turns "do not damp the signal" from a preference into a property of the objective.

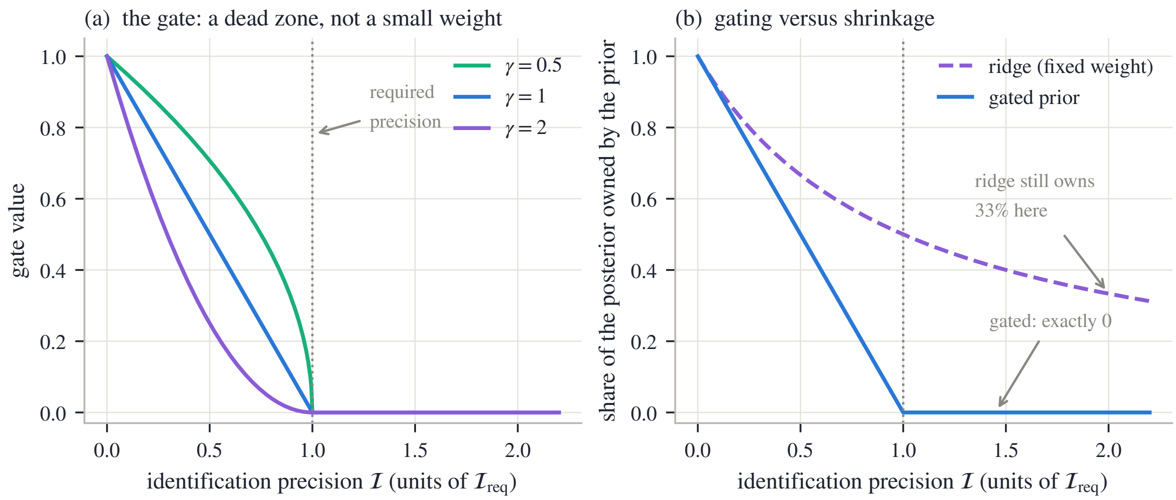

Figure 10.2: The gate, and why a ridge cannot replace it. (a) The gate eq. (10.5) for three sharpness exponents, against precision in units of the requirement: all three reach exactly zero at the dotted requirement line and stay there — the dead zone. (b) The share of the posterior owned by the prior, eq. (10.3), with prior weight λ fixed for the ridge (dashed) and gated (solid), both at unit budget: at twice the required precision the ridge still owns 33% of the coordinate; the gated prior owns exactly nothing.

Figure 10.2 draws both halves of the argument. In panel (a), the three curves are the gate at $\gamma = 0.5$, 1, and 2; all three start at one where the data says nothing, descend as precision builds, and hit zero exactly at the required precision — after which they stay at zero, which is the whole point. The exponent only shapes the descent: $\gamma = 2$ releases the prior early, $\gamma = 0.5$ holds on longer. Panel (b) translates gates into posterior shares via eq. (10.3). The dashed curve is the ridge: at a precision of twice the requirement — a coordinate the data pins twice as well as demanded — the ridge still owns 33% of the answer. The solid curve is the gated prior: identical in the dark, and exactly zero past the requirement. Same stabilization where it is wanted, none where it is not.

We now have the mechanism and its guarantee. What the gate consumes is a precision $\mathcal{I}_j$ per persisted functional — a number that must be produced cheaply, before fitting, for functionals more interesting than a single strike. Producing it is the next section's job.

10.3 The information price of a functional

The gate needs a precision per persisted coordinate. This section prices it — first for a single strike, then for the composite objects a desk actually persists — and meets, honestly, the one place where the price is misquoted.

10.3.1 Legs and baskets

For the fitted volatility at one location k_a — call it a *leg* — the natural currency is the quote support $\mathcal{Q}_a = \mathcal{Q}(k_a)$ of eq. (10.1): the effective number of quotes sitting on that leg. More effective quotes, proportionally more precision; this is the usual $1/n$ law of averaging n measurements, taken as the working model.

But a desk does not think in isolated strikes. Its vocabulary is composite: the at-the-money level; the *risk reversal*, the volatility of a call-side strike minus that of a put-side strike — the trader's one-number summary of skew; the *butterfly*, the average of the two wing volatilities minus the level — the one-number summary of curvature. Each is a *signed basket* of legs,

$$O = \sum_a \omega_a \, \sigma(k_a), \tag{10.6}$$

with fixed coefficients ω_a: the level has one leg with $\omega = 1$; the risk reversal has $\omega = (+1, -1)$ on its two wing legs; the butterfly has $\omega = (\frac{1}{2}, \frac{1}{2}, -1)$ on wing, wing, and money. What is the precision of a basket, given the support of its legs?

Proposition 10.3 (The harmonic law)**.** *Suppose each leg volatility is estimable from today's quotes with variance proportional to $1/\mathcal{Q}_a$, independently across legs. Then the implied estimate of the basket eq. (10.6) has precision (in the same proportionality)*

$$\mathcal{I}(O) = \Big(\sum_a \frac{\omega_a^2}{\mathcal{Q}_a} \Big)^{-1}. \tag{10.7}$$

In particular, one unquoted leg ($\mathcal{Q}_a \to 0$) drives $\mathcal{I}(O) \to 0$ regardless of how well the other legs are quoted.

The reciprocal-of-summed-reciprocals shape is a (weighted) harmonic mean of the legs' supports — hence the name the chapter uses, *the harmonic law* — and harmonic means are dominated by their smallest entry, which is the dead-leg rule in one phrase.

Proof. For independent estimates, the variance of a linear combination is the coefficient-squared-weighted sum of variances:

$$\mathrm{Var}\Big(\sum_a \omega_a \hat{\sigma}_a \Big) = \sum_a \omega_a^2 \, \mathrm{Var}(\hat{\sigma}_a) \propto \sum_a \frac{\omega_a^2}{\mathcal{Q}_a}.$$

Precision is the reciprocal. As any one $\mathcal{Q}_a \to 0$, the corresponding term $\omega_a^2/\mathcal{Q}_a \to \infty$, so the sum diverges and $\mathcal{I}(O) \to 0$ [the standard sampling theory of a linear functional; 35]. $\square$

The law has two consequences worth computing by hand.

The dead-leg rule. Take a risk reversal whose call leg is well quoted, $\mathcal{Q}_{\text{call}} = 6$, but whose put leg has almost nothing, $\mathcal{Q}_{\text{put}} = 0.1$. Then

$$\mathcal{I}(\mathrm{RR}) = \frac{1}{\frac{1}{6} + \frac{1}{0.1}} = \frac{1}{0.167 + 10} = 0.10.$$

Six quotes' worth of call-side evidence buys almost nothing: the basket is a *difference*, and a difference is only as known as its least-known part. At any requirement of order one, this basket's gate stands essentially fully open — correctly, because a risk reversal with a missing put leg is a number the morning does not know.

The factor of four. Now suppose both wing legs have the same support $\mathcal{Q}$, while the money leg is quoted so densely that its term in eq. (10.7) contributes essentially nothing. The risk reversal's coefficients are ± 1: $\mathcal{I}(\mathrm{RR}) = 1/(2/\mathcal{Q}) = \mathcal{Q}/2$. The butterfly's wing coefficients are $\frac{1}{2}$, and they enter eq. (10.7) *squared*: $\mathcal{I}(\mathrm{BF}) \approx 1/(2 \cdot \frac{1}{4}/\mathcal{Q}) = 2\mathcal{Q}$. The same legs make the butterfly 4 times easier to identify than the risk reversal. Identifiability belongs to the functional, not to the strikes: two baskets over identical legs can sit on opposite sides of the requirement, and the gate — which acts per functional — will treat them differently, exactly as it should.

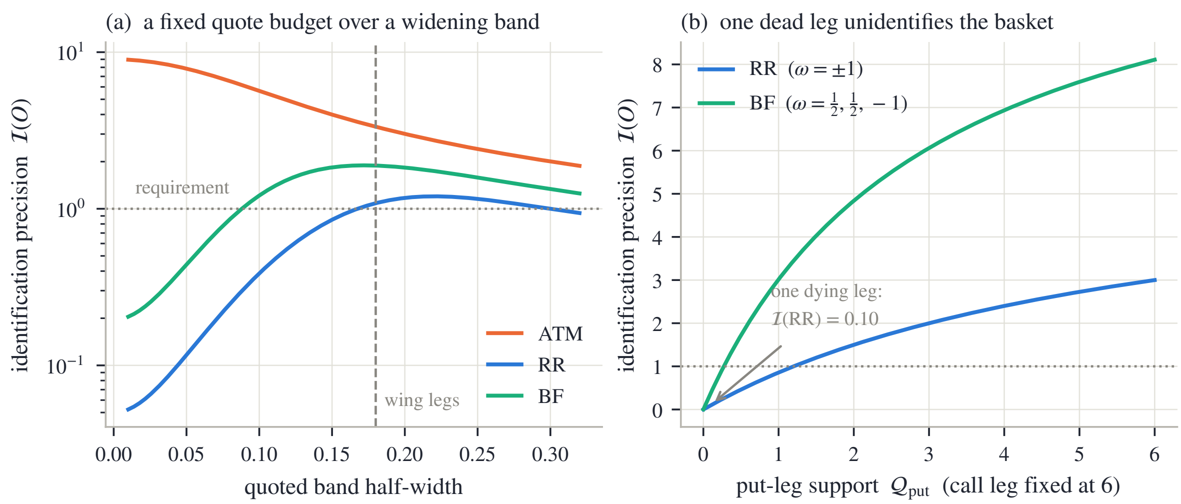

Figure 10.3: The information price of a basket (eq. (10.7) on the support measure eq. (10.1)). (a) A fixed budget of nine quotes spreads over a widening band; the wing legs sit at ± 0.18 (dashed vertical). The at-the-money leg is identified from the first moment and slowly *loses* precision as the fixed budget thins; the butterfly crosses the dotted requirement at half-width 0.09, the risk reversal only at 0.17 — and both begin to rise before the band actually reaches the legs, the kernel bleed discussed in the text. (b) The dead-leg rule: call-leg support fixed at six, put-leg support swept to zero — the basket's precision collapses with its weakest leg, the butterfly running $4\times$ above the risk reversal throughout (its half-coefficients enter the harmonic law squared, and the abundant money leg adds almost nothing).

Figure 10.3 computes both effects. In panel (a), nine quotes — the size of the sparse morning cluster that returns in section 10.5 — are spread uniformly over a band whose half-width grows along the horizontal axis, and the three baskets' precisions are drawn against it. The orange curve (the level) starts high and *declines*: a fixed budget spread ever wider leaves fewer effective quotes at the money. The green curve (butterfly) and blue curve (risk reversal) start dark and rise as the band approaches their ± 0.18 legs, the butterfly crossing the requirement first — at half-width 0.09 against the risk reversal's 0.17 — and both roll over and decline again once further widening only thins the legs. Panel (b) is the dead-leg rule as a curve: with the call leg held at support six, the basket precisions fall to zero with the put leg's support, the butterfly running 4× above the risk reversal throughout.

10.3.2 Where the price is misquoted

One honest caveat belongs here, because this is where it bites. Support is a kernel sum, and a kernel radiates: quotes up to a bandwidth away from a leg still credit it. In panel (a) both wing baskets begin to rise — and the butterfly actually crosses its requirement — *before* the quoted band reaches the legs. The proxy declares a leg identified when the nearest quotes are still a bandwidth short of it, which is precisely the region just beyond the last quote that a wing prior exists to protect. So the gate, fed by support, can close on functionals the morning has not truly measured. The consequence is a design rule rather than a repair: gated shape baskets cannot be the only mechanism holding a dark wing, and section 10.4 pairs them with anchors placed by a coverage test that does not radiate (the test is defined there). The exact alternative — measure each functional's realized precision from a data-only fit, then gate on that — exists and costs one extra fit; the support proxy is the price of gating before fitting.

We can now price any persisted functional and gate it. What remains open is the most consequential choice: *which* functionals to persist. The next section shows that this choice — mere coordinates, seemingly — decides whether the prior can damp a market move.

10.4 Choosing coordinates: shape, not level

The gate acts coordinate by coordinate. That makes the choice of coordinates load-bearing: gating can only keep yesterday out of what today has measured if "measured" and "unmeasured" land in *different* coordinates. This section proves that the trader's baskets achieve exactly that separation, and then watches the proof work in a controlled overnight experiment.

10.4.1 What a penalty can damp

Consider one basket $O = \sum_a \omega_a \sigma(k_a)$ over L legs, and compare two ways of persisting the same yesterday values: one *basket row*, $\sqrt{\lambda}\,(\sum_a \omega_a\sigma_a - \sum_a \omega_a\sigma_a^{\text{prior}})$, against L *per-leg rows*, $\sqrt{\lambda_a}\,(\sigma_a - \sigma_a^{\text{prior}})$ — here σ_a stands for the fitted volatility at leg a, the quantities the rows actually constrain.

Proposition 10.4 (Baskets annihilate the level direction)**.** *The basket row is unchanged by every joint move $\delta \in \mathbb{R}^L$ of the leg volatilities with $\sum_a \omega_a\delta_a = 0$ — an $(L-1)$-dimensional family of moves. For the risk reversal ($\omega = (+1, -1)$) and the butterfly ($\omega = (\frac{1}{2}, \frac{1}{2}, -1)$) the coefficients sum to zero, so this family contains the* uniform lift *$\delta = (1, \dots, 1)$: adding the same amount to every leg leaves both rows exactly unchanged. The per-leg rows, by contrast, are unchanged only by $\delta = 0$; under a uniform lift of size t their summed squared cost grows as $t^2 \sum_a \lambda_a$.*

Proof. The basket row depends on the leg volatilities only through the single number $\sum_a \omega_a\sigma_a$; any move δ with $\sum_a \omega_a\delta_a = 0$ leaves that number, hence the row, fixed. The condition is one linear equation on $\delta \in \mathbb{R}^L$, so its solutions form an $(L-1)$-dimensional space. For RR, $\sum_a \omega_a = 1 - 1 = 0$; for BF, $\frac{1}{2} + \frac{1}{2} - 1 = 0$; in both cases $\delta = (1, \dots, 1)$ satisfies the condition. For the per-leg rows, the uniform lift changes row a by $\sqrt{\lambda_a}\,t$, so the summed squared cost is $t^2 \sum_a \lambda_a$, zero only at $t = 0$. □

Read as market language: a risk reversal and a butterfly are statements about *shape* — they are blind, by construction, to the overall volatility level. Persisting yesterday's shape through baskets therefore exerts *no* force against a market-wide level move, while persisting yesterday's wing *prices* strike by strike fights such a move with weight equal to the full budget. The same information, expressed in different coordinates, either respects a level jump or damps it.

10.4.2 The overnight experiment

Figure 10.4 runs the proposition as an experiment (constants in Appendix 10.A). Yesterday's smile is a smooth curve with a genuine wing — skew and curvature both nonzero. Overnight the market jumps: today's true smile is yesterday's *lifted bodily by 4 volatility points*, shape unchanged. Today's quotes cover only the at-the-money band, fifteen mids at stated noise of 30 bp. The same flexible spline family of section 10.1 is fitted three ways: with no prior at all; with gated *shape baskets*; and with *absolute anchors* — yesterday's wing volatilities imposed directly at four wing strikes, the seemingly natural "hold the wings at yesterday" rule. The basket prior is built exactly by the rules of the last two sections, with no hand in it: five candidate rows (the level, and a risk reversal and a butterfly at each of the leg pairs ± 0.20 and ± 0.30), each priced by the harmonic law, each gated, and a fixed budget split over the open gates in proportion to their values.

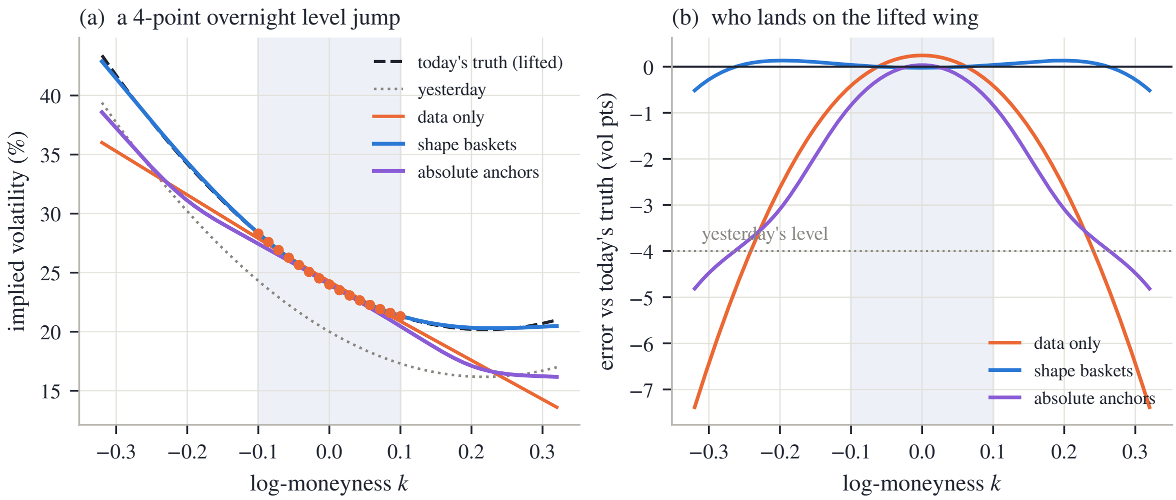

Figure 10.4: Coordinates decide what gets damped (controlled 4-point overnight level jump; quotes only in the shaded band, truth dashed, yesterday dotted). (a) Three fits of the same morning: data only (orange), gated shape baskets (blue), absolute anchors (violet). (b) The same three curves as errors against today's lifted truth: the basket fit tracks zero across the whole wing (0.28 points at $k = -0.30$); the anchor fit converges to the dotted "yesterday's level" line — clinging, 4.5 points low; the data-only fit misses by 6.4 points.

Walk panel (a). The orange dots are the morning; the dashed ink curve is today's truth, the dotted grey curve yesterday's smile, four points below it everywhere. The data-only fit (orange curve) matches the band and then extrapolates by its smoothness rule alone: straight, missing the true wing's own curvature by 6.4 volatility points at $k = -0.30$ — the flapping of section 10.1, now with a truth to be wrong against. The violet curve is the anchor fit: it matches the band too (the anchors sit outside it), but its wing bends down to yesterday's *absolute* values. The blue curve is the basket fit: it rides today's level — which it learned from the band — and rebuilds the wing as "today's level plus yesterday's shape," landing on the dashed truth.

Panel (b) replots the three fits as errors against today's truth, which makes the mechanism quantitative. The blue basket curve hugs zero across the entire drawn range, 0.28 points off at the deep readout. The violet anchor curve descends, in the wings, toward the dotted line at -4 points — yesterday's level exactly: the anchors have reconstructed *yesterday's* wing under *today's* market, 4.5 points wrong. That is proposition 10.4 in the flesh: absolute price rows penalize the uniform lift, so they spend their budget pulling the smile back down; basket rows are invariant under the lift, so their entire budget goes into shape.

Two computed details close the case. First, no damping: the basket fit's at-the-money level agrees with the data-only fit's to 26.9 vol bp — a fifteenth of the 4-hundred-bp jump. Even that residue is indirect: by proposition 10.4 the basket rows exert exactly zero force along the lift, but they do repaint the wings, and the smoothness rule leaks a little of the repainting back into the band. Second, no hand-tuning: all five gates were computed, not set. The at-the-money support on this morning is $\mathcal{Q}(0) = 9.8$, far past the requirement, so the level row's gate is gate $= 0.00$ — the prior contains *no* row that could contest the level even in principle (proposition 10.2). The moderate-wing risk reversal's gate stands open at 0.68 — while the moderate-wing *butterfly's* is 0.00: on the same legs, its factor-of-four cheaper information price (proposition 10.3) already clears the requirement, so the gate drops it. The deep-wing pair, dark to every measure, enters essentially fully open (1.00 and 0.99). The budget concentrated itself on exactly the three coordinates this morning left dark.

10.4.3 The division of labour

The experiment also shows why absolute anchors survive at all. The basket fit reconstructed the wing because the wing's *shape relative to the band* was yesterday's to give and the legs were reachable. But a basket needs its money-side reference: beyond the last leg pair, in the truly deep tail, shape baskets run out — and section 10.3.2's kernel bleed makes their gates unreliable just past the band edge besides. There, and only there, absolute anchors are the tool, and they are placed by a different test than the gate's. Instead of radiated support, each candidate anchor strike is scored by its *coverage deficit*: compare the quote mass actually present in that strike's own neighbourhood against the mass a desired even coverage of the smile would put there, and give the anchor weight only in proportion to the shortfall. A dense band a few strikes away contributes nothing to a wing anchor's neighbourhood, so this test — unlike the kernel sum of eq. (10.1) — cannot be talked out of a deficit by nearby liquidity; where quotes already meet the desired coverage the deficit, and the anchor, is zero. The anchors then hold the deep tail at yesterday's transported prices. Their known cost is exactly what panel (b) displays — across a level jump the deep tail is pulled toward yesterday's absolute prices — and the division of labour confines that cost to strikes where no alternative information exists at all. Shape on baskets as far as legs reach; absolute anchors only beyond; the level always to the market.

Half of the chapter's program is now complete, and it is worth saying in one sentence what has been built: *along each coordinate, the published smile is today's data where today's data speaks, and yesterday's transported value where it does not, with "speaks" priced per functional and the handover proved not to touch the identified coordinates.* What this half cannot handle is a morning whose quotes are present but sick — dense, well inside every gate's requirement, and internally contradictory. For that, the prior is the wrong instrument entirely, and time is the right one.

10.5 The same question in time: the filter

Here is the morning the gate cannot help with. A one-month smile has nine near-the-money quotes — dense, liquid, well past every requirement, so every gate of section 10.2 is closed, and rightly so. But one of the nine is stale: its market moved and the quote did not. Chase the mids literally and the fit grows a curvature kink that nobody traded. This is not a gap — the region is superbly observed by count — and using the prior to pull the curvature back would violate the very guarantee proposition 10.2 establishes. The problem is not absence of data; it is *noise* in data that is present. The right instrument treats today's reading as one more noisy estimate of a moving quantity, to be weighed — not vetoed — against what yesterday leads us to expect. That instrument is the filter [1, 31], and after section 10.2.1 it can be assembled in a page.

10.5.1 The state, and the cycle

What should be filtered? Not raw option mids: contracts appear and disappear, the out-of-the-money side flips when the forward moves, and a smoothed vector of raw volatilities has no arbitrage guarantee. The reference implementation filters the smile's three *ATM handles* — the level $\sigma(0)$, the skew s_0 (Chapter 9's ATM slope), and the curvature (the second derivative of the fitted volatility at the money) — read off a fitted, arbitrage-checked smile. Three numbers, model-agnostic, stable in meaning from day to day: a small state that summarizes what the belly of the smile is doing.

Each handle then lives the same three-step cycle at every snapshot. Write O for the handle's true (unknown) value today.

Predict. Yesterday's combined estimate is pushed to today's market. Chapter 9 built exactly this transport: for a log-forward move H under the regime dial $\mathcal{R}$, the level moves by $\mathcal{R}\, s_0 H$, the skew by curvature times H, the curvature is carried unchanged. The transported value is the *prediction* $\hat{O}^-$, with a variance $\mathcal{V}^-$ that must account for everything that can have happened since yesterday (section 10.6 builds it).

Observe. Today's prepared quotes are fitted — data only, no temporal prior — and the handle is read off the fit: the *observation* $\hat{O}_{\text{obs}}$, with a variance $\mathcal{V}_{\text{obs}}$ that must reflect how precisely, and how consistently, today's quotes actually pin this handle (section 10.6 again).

Update. Two independent, unbiased estimates of the same number, with variances: this is section 10.2.1 verbatim. The precision-weighted average, rearranged into its standard form, is the update:

$$\boxed{\begin{aligned} &\mathcal{K} = \frac{\mathcal{V}^-}{\mathcal{V}^- + \mathcal{V}_{\text{obs}}} \in (0,1), \\ &\hat{O}^+ = \hat{O}^- + \mathcal{K}\,(\hat{O}_{\text{obs}} - \hat{O}^-), \\ &\mathcal{V}^+ = (1-\mathcal{K})\,\mathcal{V}^-. \end{aligned}} \tag{10.8}$$

The first line defines the *gain* $\mathcal{K}$: the fraction of the surprise the update accepts. The second applies it to the *innovation* $\hat{O}_{\text{obs}} - \hat{O}^-$ — the gap between what arrived and what was expected. The third is the posterior variance; it deserves its two lines of algebra. Precisions add, eq. (10.2), so

$$\frac{1}{\mathcal{V}^+} = \frac{1}{\mathcal{V}^-} + \frac{1}{\mathcal{V}_{\text{obs}}} = \frac{\mathcal{V}^- + \mathcal{V}_{\text{obs}}}{\mathcal{V}^-\mathcal{V}_{\text{obs}}} \quad \Longrightarrow \quad \mathcal{V}^+ = \frac{\mathcal{V}_{\text{obs}}}{\mathcal{V}^- + \mathcal{V}_{\text{obs}}}\,\mathcal{V}^- = (1-\mathcal{K})\,\mathcal{V}^-,$$

and the posterior mean, checked the same way, is $\mathcal{V}^+(\hat{O}^-/\mathcal{V}^- + \hat{O}_{\text{obs}}/\mathcal{V}_{\text{obs}}) = \hat{O}^- + \mathcal{K}(\hat{O}_{\text{obs}} - \hat{O}^-)$. Tomorrow, $\hat{O}^+$ and $\mathcal{V}^+$ become the input to the next predict step, and the cycle repeats.

10.5.2 The dial nobody may own

Pause on what eq. (10.8) does *not* contain. Every fitter that carries state through time owns, somewhere, a dial marked *trust*: how far should this morning's quotes move yesterday's surface? The tempting implementation is a slider — tuned by hand, per desk, per mood. In eq. (10.8) that dial is the gain, and the gain is not a setting: it is an *output*, the ratio of two variances. Every design question about trust becomes a question about building $\mathcal{V}^-$ and $\mathcal{V}_{\text{obs}}$ honestly — which is auditable — rather than about where to set a slider — which is not. When both variances tell the truth, the dial turns itself: a noisy morning arrives with a large $\mathcal{V}_{\text{obs}}$ and is barely admitted; a clean morning after a quiet night arrives against a small $\mathcal{V}^-$... and the update follows the market. The gate of section 10.2 and the gain of eq. (10.8) are the same philosophy on the two sides of the chapter: the first answers "*where* did the market not speak?" with a dead zone; the second answers "the market spoke here — *how noisy was its voice?*" with a ratio that is never zero and never one.

10.5.3 The stale strike, filtered

Now run the morning that opened this section through eq. (10.8), handle by handle. The prediction — yesterday's state, transported — says level 20.0%, skew −0.35, curvature 0.10, with standard deviations (0.30%, 0.08, 0.05). Today's data-only fit reads level 20.4%, skew −0.37, curvature 0.55: a plausible level move, a plausible skew — and a curvature five times the prediction, manufactured by the one stale strike. The observation variances, built by the machinery of the next section, come out as standard deviations (0.15%, 0.05, 0.30): level and skew tight, curvature wide, because the cluster that produced it contradicts itself. Take these six variances as given for one page; *building* them is section 10.6's whole subject.

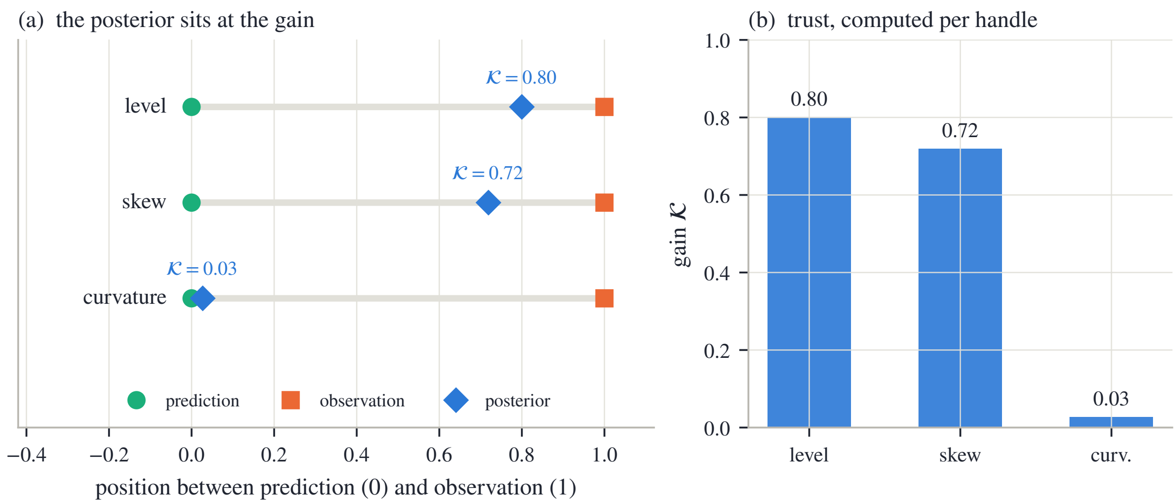

Figure 10.5: The stale-strike morning through the scalar update eq. (10.8), one handle per row. (a) Each handle drawn on its own normalized segment from prediction (circle, at 0) to observation (square, at 1); the posterior (diamond) sits at exactly its gain. The level moves 80% of the way to the market; the curvature kink is admitted at less than 3%. (b) The three computed gains. Nobody set anything: the gains are ratios of stated variances.

Figure 10.5(a) draws each handle on a normalized segment: prediction at zero, observation at one, so that — by the second line of eq. (10.8) — the posterior's position *is* its gain, and eq. (10.8) can be checked by eye. The level, predicted to 0.30% and observed to 0.15%, carries gain $\mathcal{K} = 0.09/(0.09 + 0.0225) = 0.80$ (the worked number of section 10.2.1): the posterior, 20.32%, sits four-fifths of the way to the market, with posterior standard deviation 13 bp. The skew behaves likewise at gain 0.72 (posterior -0.364). The curvature is the story: prediction ten times tighter than the observation, gain $\mathcal{K} = 0.05^2/(0.05^2 + 0.30^2) = 0.027$, posterior 0.112 — the five-fold kink is not vetoed by any rule; it is *priced*, and at that price the update accepts under three percent of it. Panel (b) collects the three gains as bars: trust, computed per handle, unequal because the evidence is unequal.

One remark on dimensions before moving on. The three handles were updated *separately* — three scalar filters side by side. A full three-dimensional update also exists, in which the handles' estimated correlations let one handle's innovation move another's posterior; it is the optimal version when the stated correlations are true. The danger is that on a coarse chain they are not: with few strikes, level and curvature are estimated *jointly* rather than separately, and the correlation entries of the observation covariance are then precisely its least reliable numbers. Spent at face value, they let a junk curvature innovation drag the level along with it. A per-handle update makes that failure structurally impossible — each scalar posterior lies between its own prediction and its own observation, full stop. The reference implementation therefore combines per handle; the discarded cross-handle information is real, and this is the disclosed price of not spending unreliable correlations.

Everything now rests on the two variances that fed eq. (10.8). Where does $\mathcal{V}_{\text{obs}}$ come from, such that a self-contradictory cluster shows up as a wide one? And what makes $\mathcal{V}^-$ grow honestly between snapshots?

10.6 Two honest budgets

The update was one boxed display; the work is the two variances it consumes. This section builds each from first principles — the observation's from the fit itself, the prediction's from the clock — because any dishonesty in either budget becomes, through eq. (10.8), a wrong gain with nobody's hand on it.

10.6.1 The observation budget: what did the market pin down?

The observation is not a quote; it is a number read off a *fit*. Its variance must answer: given where today's quotes sit and how noisy each is, how precisely is this handle determined?

Warm up with one strike. A single quote at the money, with stated noise ε (its bid–ask half-spread, read in volatility), measures the level with variance ε^2; n such quotes, independently noisy, measure it with variance ε^2/n. Precision is quote count over noise squared — support again, now in proper units.

The general case keeps exactly this structure. Divide each quote equation by its own stated noise ε_i, so that every row of the least-squares stack is measured in units of its own noise ("whitened," in the standard jargon). Near the optimum the fit is locally linear in its parameters θ, and the whitened stack carries an *information matrix* — the natural generalization of "count over noise squared":

$$\mathcal{I}_\theta = \sum_i \frac{1}{\varepsilon_i^2} \frac{\partial\sigma_{\text{fit}}(k_i;\theta)}{\partial\theta} \frac{\partial\sigma_{\text{fit}}(k_i;\theta)}{\partial\theta}^{\top}, \tag{10.9}$$

one outer product per quote, each weighted by that quote's precision. ($\sigma_{\text{fit}}(k_i;\theta)$ is the model volatility at quote i's strike; the derivative is the row vector of its sensitivities to the parameters.) Why is the *inverse* of $\mathcal{I}_\theta$ the covariance of the fitted parameters? The one-parameter case shows the whole mechanism with first-course algebra. Write the whitened quote equations as $a_i\theta \approx b_i$, where a_i is quote i's sensitivity divided by its noise and b_i the whitened quote; least squares solves them as $\hat\theta = \sum_i a_i b_i / \sum_i a_i^2$. Each b_i is the true value plus unit-variance noise, $b_i = a_i\theta_{\text{true}} + \text{noise}_i$, so

$$\hat\theta - \theta_{\text{true}} = \frac{\sum_i a_i\,\text{noise}_i}{\sum_i a_i^2}, \qquad \operatorname{Var}(\hat\theta) = \frac{\sum_i a_i^2 \cdot 1}{\left(\sum_i a_i^2\right)^2} = \frac{1}{\sum_i a_i^2} = \frac{1}{\mathcal{I}_\theta}:$$

the estimation error is a noise-weighted average whose variance is exactly one over the information. With several parameters the same computation, carried through with matrices, gives $\mathcal{I}_\theta^{-1}$ as the covariance [?]. Any handle $O(\theta)$ then inherits its variance by first-order error propagation — the delta method:

$$\mathcal{V}_{\text{obs}}(O) = \frac{\partial O}{\partial\theta}^{\top} \mathcal{I}_\theta^{-1} \frac{\partial O}{\partial\theta}. \tag{10.10}$$

In the one-strike warm-up, eq. (10.9) is the scalar n/ε^2 and eq. (10.10) returns ε^2/n: the general formula is the simple one with the geometry of the strike grid kept. (One guard on the inverse: a parameter direction the quotes do not touch makes $\mathcal{I}_\theta$ nearly singular, and a naive inverse would assign that direction either an infinite variance — harmless — or, after aggressive rounding, *zero*, which is the one lie a filter must never inherit; the reference implementation floors the small eigenvalues so unidentified directions come out large-but-finite.)

Two disciplines make eq. (10.10) honest in practice, and fig. 10.6 measures both.

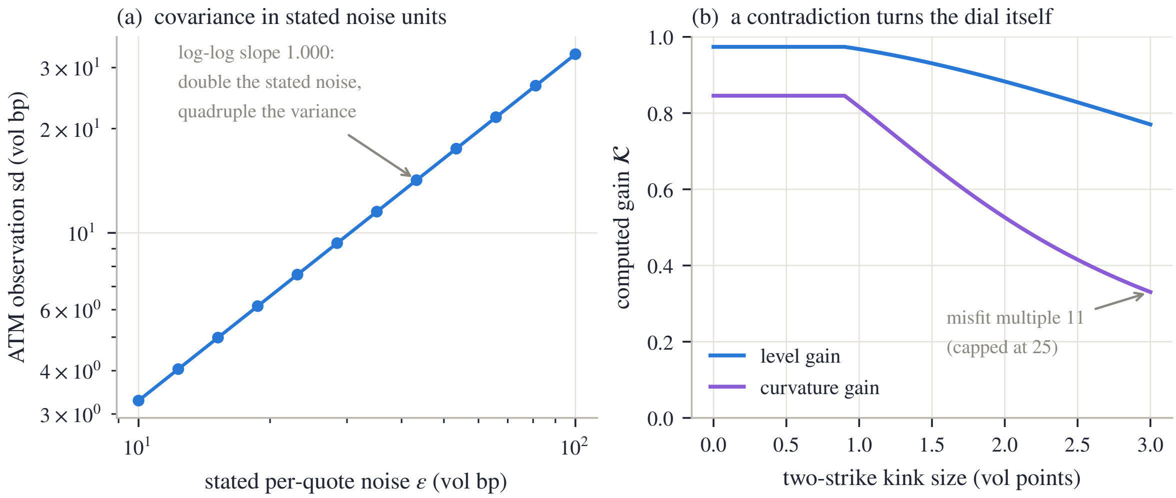

Figure 10.6: The observation budget is built, not asserted (synthetic 21-quote chain; the fitted family's parameters are the three handles, so eq. (10.10) is exact). (a) The ATM observation standard deviation against the stated per-quote noise, log–log: slope 1.000 — double the stated noise, double the sd, quadruple the variance. (b) Two adjacent strikes are kinked apart by a growing amount and the fit's gains are recomputed at each step: the realized-misfit multiple leaves the budget alone while the chain fits within its stated noise (the flat start), then inflates it, and the curvature gain falls from 0.85 to 0.33 while the level gain — read almost noiselessly relative to its prediction — falls far less, from 0.97 to 0.77.

Units. Information is only information *in stated noise.* If the fit's weights are merely relative — equal weights, say — then eq. (10.9) silently prices every quote as if its noise were one full volatility point, and the resulting "covariance" is fiction. The whitening by the actual ε_i is what ties the budget to the market's own stated uncertainty, and it implies a checkable contract: variance is noise *squared*, propagated. Panel (a) of fig. 10.6 checks it on a synthetic 21-quote chain, sweeping the stated noise and recomputing eq. (10.10): the ATM observation standard deviation follows with log–log slope 1.000. Double what the market says its quotes are worth and the computed observation variance rises exactly fourfold — by construction, not by tuning.

Consistency. Geometry is not agreement. Equations eq. (10.9)–eq. (10.10) see *where* the quotes sit and how noisy each claims to be — but not whether they agree with one another. The stale-strike morning of section 10.5 is exactly that gap: plenty of quotes, fine stated noise, and a cluster that cannot all be right at once. The fix reads the fit's own residuals. After fitting, compute the average of the squared whitened residuals — each quote's misfit divided by its stated noise, squared, averaged over the quotes (less a small correction for the parameters fitted; the exact expression is in Appendix 10.A). If the quotes scatter as their stated noise claims, this average is close to one. If it comes out at, say, eleven, the chain is misbehaving eleven times worse than it admits — and the observation variance is multiplied by that realized-misfit multiple (floored at one, capped so a single broken chain cannot poison the state). Panel (b) shows the mechanism end to end: two adjacent strikes are kinked apart by a growing amount, and the computed gains respond. While the kink hides inside the stated noise, the multiple sits at its floor and nothing changes — the flat start of both curves. Beyond it, the multiple climbs (it reaches 11 at a three-point kink) and the curvature gain collapses from 0.85 to 0.33, while the level gain — a handle this chain reads to 9.8 bp against a much wider prediction — falls far less, from 0.97 to 0.77. The one multiple distrusts the whole chain, and it bites hardest where the observation was weakest relative to its prediction. No threshold fired and no rule named "stale strike": the contradiction raised the stated variance, and the gain, a ratio of variances, turned itself.

10.6.2 The prediction budget: how far can the smile have walked?

The other variance belongs to the prediction, and it answers: how much can this handle genuinely have changed since it was last combined? The reference implementation grows it additively,

$$\mathcal{V}^- = \mathcal{V}^+_{\text{prev}} + q^2_{\text{walk}}\, \Delta t_{\text{days}} + (\text{widenings}), \tag{10.11}$$

starting from yesterday's posterior variance. The second term is the *clock term*: surfaces diffuse, and a handle drifts about q_{walk} per square-root day — the square-root scaling of any diffusion, Chapter 8's accrual logic applied to the handle itself. Two honesty points live inside it. The elapsed time Δt_{days} should be counted on the market's clock, not the wall's: Chapter 8 measured that a closed weekend moves the smile about as much as one overnight, so a calendar count pays three days of variance for one overnight's worth of movement and no single q_{walk} can then be right at every cadence. And the scale q_{walk} itself is an empirical commitment — the audit of section 10.7 is precisely the instrument that convicts a wrong one. The remaining widenings are event-shaped and listed in words: a spot move of size H adds transport uncertainty in proportion to $|H|$ (the transported prediction leans on Chapter 9's dial, which is itself a convention); a scheduled event inside the interval widens the budget while it resolves; a change of data source resets the state outright rather than pretending continuity.

One last mechanism guards the tails of real life. A genuine five-point overnight jump is, under a 30-bp-per-$\sqrt{\text{day}}$ budget, a surprise of tens of standard deviations; a filter that believes its budget will crawl toward it for days. The honest repair is not a bigger everyday budget — that would fail the audit in the other direction — but a *surprise widening*: when the standardized innovation $|\hat{O}_{\text{obs}} - \hat{O}^-|/\sqrt{\mathcal{V}^- + \mathcal{V}_{\text{obs}}}$ exceeds a few standard deviations, multiply $\mathcal{V}^-$ up until the surprise reads as merely unusual, and let the gain rise toward the data. (A widening, not another gate: nothing here switches off — a variance is enlarged.) Below its threshold the update is untouched; and a *contradictory* chain does not trip it, because its own inflated $\mathcal{V}_{\text{obs}}$ already shrinks the standardized innovation — the two honesty mechanisms compose rather than compete. Section 10.7 measures all of this.

10.7 Counting once, and checking the bars

Two matters remain, and they are the chapter's bookkeeping and its conscience: how to let the filter steer a real calibration without counting any quote twice, and how to test — against data — that the whole construction's stated uncertainty is true.

10.7.1 One objective, one count

Section 10.5 computed the update *after* a data-only fit: fit the quotes, read the handle, combine with the prediction. When the filter is allowed to steer the fit itself, a tempting implementation is fatal: build the posterior from today's quotes, then feed that posterior back into the same fit as one more prior row *alongside those same quotes*. The quotes then speak twice — once through the likelihood, once smuggled inside the "prior" — and the published uncertainty is too small by exactly the double count. The correct implementation is a single objective: today's quote rows, plus *one* extra row per handle anchoring the model's handle to the prediction $\hat{O}^-$ at weight $1/\mathcal{V}^-$ — the row form eq. (10.4) with the prediction as target. That this is the *same* answer as the two-step update is a one-line calculation worth displaying:

Proposition 10.5 (The joint fit is the filter)**.** *For a handle observed directly, the minimizer over O of the joint objective*

$$\frac{(O - \hat{O}_{\text{obs}})^2}{\mathcal{V}_{\text{obs}}} + \frac{(O - \hat{O}^-)^2}{\mathcal{V}^-}$$

is the posterior $\hat{O}^+$ *of eq.* (10.8)*, and the objective's second derivative is twice the posterior precision* $1/\mathcal{V}^+$*.*

Proof. Set the derivative in O to zero:

$$\frac{2(O-\hat{O}_{\text{obs}})}{\mathcal{V}_{\text{obs}}}+\frac{2(O-\hat{O}^-)}{\mathcal{V}^-}=0 \implies O\Big(\tfrac{1}{\mathcal{V}_{\text{obs}}}+\tfrac{1}{\mathcal{V}^-}\Big)=\frac{\hat{O}_{\text{obs}}}{\mathcal{V}_{\text{obs}}}+\frac{\hat{O}^-}{\mathcal{V}^-},$$

the precision-weighted average — which section 10.5 already showed equals $\hat{O}^- + \mathcal{K}(\hat{O}_{\text{obs}} - \hat{O}^-)$. The second derivative is $2(1/\mathcal{V}_{\text{obs}} + 1/\mathcal{V}^-) = 2/\mathcal{V}^+$ by eq. (10.2). □

So the filter, done inside the objective, is one ungated row per handle — and the accounting rule follows: once the fit is committed, its handles *are* the posterior, and no second update may be applied against the same quotes. Note the word *ungated*: the prediction row carries its covariance weight always, closed gates or no. This is the chapter's two instruments meeting in one objective, each keyed to its own symptom — the prior rows of section 10.2 act where quotes are *absent* (their gate reads support), the filter row acts because quotes are *noisy* (its weight reads a variance) — and they must not overlap, because both pull toward the same transported yesterday. Yesterday enters once. When the filter runs, its prediction rows *are* yesterday's belly — level, skew, curvature, transported and priced — so the shape baskets of sections 10.3 and 10.4, which carry that same yesterday in other clothes, stand down; what persistence keeps is only what the filter state does not transport: the deep-tail anchors beyond the last basket leg, and — for a node with no quotes at all — the baseline the next chapter builds. Anchoring the same handle to yesterday twice — once through $\hat{O}^-$, once through a persistence row — would count yesterday twice, the mirror image of the quote double count.

10.7.2 The audit: were the error bars true?

A filter can always make charts smoother. Whether it is *right* is a different question, and it has a precise form: the posterior comes with a stated variance, so the construction claims that its errors, divided by their claimed scale, are of size one. Define the *standardized error*

$$\mathcal{Z}=\frac{\text{truth}-\hat{O}^+}{\sqrt{\mathcal{V}^+}}, \tag{10.12}$$

whose standard deviation should be 1 if the error bars mean what they say. The test statistic is that standard deviation, taken over a run of days: $\text{std}(\mathcal{Z})$ well above one convicts the filter of overconfidence (bars too narrow); well below one, of timidity (bars too wide). The audit needs a truth to compare against, so it runs wherever one exists — held-out quotes on real history, or, as here, a synthetic world where the truth is known exactly.

Figure 10.7 runs the audit end to end on a seeded synthetic history — 500 days in which the true level diffuses at exactly 30 bp per square-root day, jumps +5 volatility points on two days, and is observed daily through honestly-stated noise (Appendix 10.A lists the generator). The same scalar filter eq. (10.8) is run with different assumed walk scales q_{walk}, and every claim of the last two sections becomes measurable.

Panel (a) is the texture. Around the first jump, the honest-budget filter (blue, its one-standard-deviation band shaded) tracks the truth within about two days, its band wide enough to contain its own errors. The starved filter — walk scale a third of the truth — is the violet curve: it enters the jump with a narrow band, trusts its prediction too much, and crawls toward the new level for a week.

Panel (b) is the chapter's verdict in one curve: the standard deviation of $\mathcal{Z}$, eq. (10.12), against the assumed walk scale. At the true scale the curve sits near the dotted target — $\text{std}(\mathcal{Z}) = 1.14$, the small excess being the two jumps, which the plain clock term does not model; switch the surprise widening on and it drops to 1.01. To the left the curve climbs steeply:

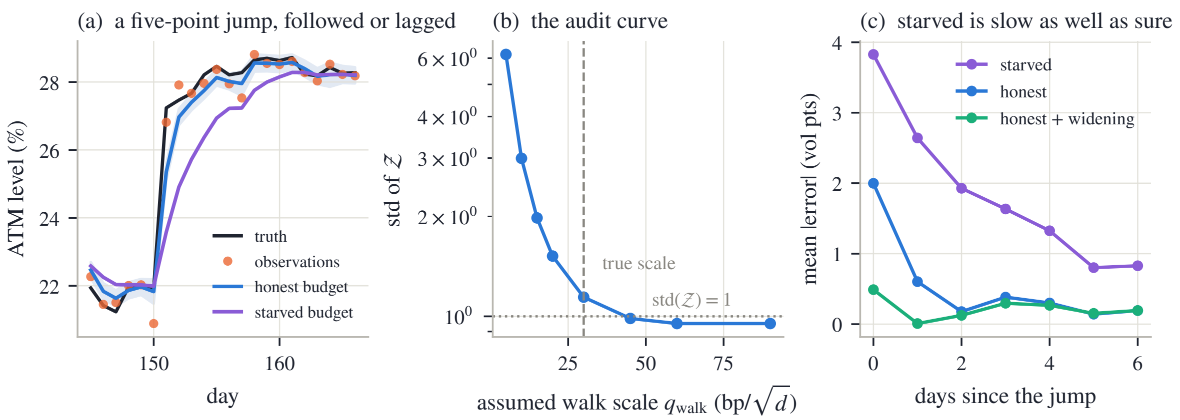

Figure 10.7: The audit on a seeded synthetic history (500 days; the true level diffuses at 30 bp/$\sqrt{\text{day}}$ and jumps +5 points twice; observations carry honestly stated noise, wide on every seventh day). (a) A window around a jump: the honest-budget filter (blue, with its ±1-sd band) follows in two days; the starved budget (violet) crawls. (b) The audit curve: std of $\mathcal{Z}$ against the assumed walk scale — starved budgets sit far above the dotted target of 1 (3.0 at 10 bp), the true scale reads 1.14. (c) Mean error against days since the jump: starved 3.8 points on the day and still 1.6 three days later; honest 2.0 then pennies; the surprise widening cuts the jump-day error to 0.49.

at a third of the true scale the standardized errors have standard deviation 3.0 — the filter's stated bars are three times too narrow. And panel (c) shows the same starved filter's *speed*: 3.8 points of error on the jump day, still 1.6 points three days later, against the honest budget's 2.0-then-pennies, and 0.49 on the day with the surprise widening. This is the general lesson, and it is worth stating plainly because intuition suggests the opposite: shrinking the prediction budget does not make a filter cautious. It makes it *overconfident and slow at once* — sure of a surface it updates too slowly — the worst available combination, and the audit catches both symptoms in one statistic.

10.7.3 The contract

The chapter closes as the others do: three columns of decreasing strength — proved, measured, chosen.

Step back to the book's thesis. *Validity* in this chapter is inherited, not new: every fit underneath carries the arbitrage machinery of Part I, and the filtered state is read off arbitrage-checked smiles. *Information* is the chapter's subject made explicit: for the first time the book has measured not just what the data says but where it says *nothing* — the flat valley — and where it speaks with a noisy voice — the inflated budget. *Convention* is what fills the remainder, and the chapter's discipline is that every convention entered through a stated, auditable quantity: a gate with a dead zone, a variance with a units contract, a walk scale that an audit can convict. Posterior equals data where the data speaks, and yesterday — transported, priced, and counted once — where it does not: the thesis of this book, made dynamic.

One boundary remains, and it is the book's last. Everything here is *per node*: one smile, its own yesterday, its own morning. But a desk's universe is not a list of unrelated smiles — it is a web of underliers and expiries whose laws move together, and a morning that is sparse on one node is often rich on its neighbours. What a lit node learns should reach the dark ones, at a price set by how related they are — the same precision language, now flowing across a graph. Building that graph, and completing a whole universe of smiles from a sparse set of observed ones, is the final chapter.

Table 10.1: The chapter's contract.

Proved	**Measured**	**Convention**
The no-damp guarantee: a closed gate contributes nothing, for every parameter vector (proposition 10.2).	The flat valley: 0.02 bp of rms movement across a 12-point wing sweep on the thinned morning; the full chain's V (fig. 10.1).	The required precision, the gate exponent γ, and the prior budget: set by judgment, disclosed, not derived.
The harmonic law and the dead-leg rule; the RR/BF factor of 4 (proposition 10.3).	Gates computed on the staged mornings: ATM 0.00, moderate RR 0.68 (fig. 10.4).	Support eq. (10.1) is a kernel proxy for identification; it over-credits one bandwidth past the last quote (section 10.3.2).
Baskets annihilate the level direction; per-leg anchors penalize it (proposition 10.4).	The overnight experiment: basket wing error 0.28 points, anchor 4.5, data-only 6.4 (fig. 10.4).	Deep-tail anchors persist absolute prices beyond the last basket leg: right when the level is stable, wrong across a jump — confined there by design.
The update eq. (10.8) is the precision-weighted average; the joint fit reproduces it exactly (proposition 10.5).	The stale-strike morning: gains 0.80/0.72/0.027 computed from stated variances (fig. 10.5).	The state is three ATM handles, combined per handle: estimated cross-handle correlations are deliberately not spent (section 10.5).
The quadratic contract: observation variance is stated noise squared, propagated (eq. (10.9)–eq. (10.10)).	Slope 1.000 in log–log; the contradiction sweep: curvature gain 0.85→0.33 with the level barely moved (fig. 10.6).	The prediction budget eq. (10.11): walk scale, market-clock Δt_{days}, transport and event widenings, the surprise widening's threshold — each an empirical commitment.
— (this row is empirical by design)	The audit: std($\mathcal{Z}$) 3.0 starved vs 1.14 honest (1.01 gated); jump-day errors 3.8/2.0/0.49 points (fig. 10.7).	std($\mathcal{Z}$) = 1 is the acceptance test this book adopts for any machinery that publishes uncertainty.

10.A Protocol: data and constants

Every number and figure in this chapter regenerates from one deterministic script reading the frozen snapshot (`scripts/ch10/gen_figures.py`); nothing is refetched and no stored fit is altered. The audit figure's seeded walk is the chapter's only randomness.

The staged morning (figs. 10.1 and 10.3). The real-data exhibit is the book's canonical SPY December 2026 node (137 days, 94 prepared quotes), thinned to the 31 quotes with $|k| \leq 0.10$. Each quote's stated noise is its bid–ask half-spread in volatility, floored at one vol bp.

The spline family (figs. 10.1 and 10.4). The deliberately flexible family is a natural cubic spline in k — linear in its knot values, so every least-squares statement in the chapter is exact rather than linearized. Its knots sit at

$$k \in \{-0.36, -0.30, -0.22, -0.15, -0.09, -0.04, 0, 0.04, 0.09, 0.15, 0.26\}$$

for the profile experiment and at

$$k \in \{-0.30, -0.20, -0.12, -0.05, 0, 0.05, 0.12, 0.20, 0.30\}$$

for the overnight experiment. A mild second-difference penalty on the knot values (weights 0.5 and 3.0 respectively, in whitened units) keeps the unquoted knots finite; it is the "smoothness rule" named in the text.

The pinned-wing profile (fig. 10.1). For each imposed value the fitted volatility at $k = -0.30$ is constrained by one row at effectively infinite weight (10^{12}) and the family is refitted; the plotted statistic is the plain rms of (fit $-$ mid) over the fitted quote set, in vol bp. The sweep runs 24–36% in 25 steps. The ensemble sentence in section 10.1 refits the thinned band with the reference implementation under Chapter 3's protocol: one family at three resolutions and two ridge strengths, plus the two other parametric families — eight fits, wing readings compared at $k = -0.30$.

Support and gates (eqs. (10.1) and (10.5); figs. 10.2 and 10.3). Kernel bandwidth 0.06 in log-moneyness throughout; fit weights normalized to average one; required precision $\mathcal{I}_{\text{req}} = 1$ ("one effective quote"); gate exponent $\gamma = 1$ except where fig. 10.2(a) varies it. Figure 10.3(a) spreads nine equally-weighted quotes uniformly over $[-W, W]$ and sweeps the half-width W; the wing legs sit at ± 0.18.

The overnight experiment (fig. 10.4). Yesterday's smile is $0.20 - 0.35k + 0.80k^2$; today's truth adds 4 points uniformly. Today's quotes: 15 mids of the truth at equal spacing on $[-0.10, 0.10]$, stated noise 30 bp. Shape baskets: five candidate rows (level; risk reversal and butterfly at leg pairs ± 0.20 and ± 0.30), targets evaluated on yesterday's curve; each row is priced by eq. (10.7), gated by eq. (10.5) ($\mathcal{I}_{\text{req}} = 1$, $\gamma = 1$), and a total budget worth twenty clean quotes is split over the open gates in proportion to their values — the closed level and moderate-butterfly rows (gates 0.00 and 0.00) are dropped. Absolute anchors: yesterday's volatilities imposed at the four wing strikes, the same total budget split equally. Errors are read at $k = -0.30$.

The observation budget (fig. 10.6). A synthetic chain of 21 quotes at equal spacing on $[-0.25, 0.25]$, truth $0.20 - 0.30k + 0.60k^2$, base noise 30 bp, fitted by the quadratic family $\theta_0 + \theta_1 k + \theta_2 k^2$ whose parameters are the handles (curvature $= 2\theta_2$), so eqs. (10.9) and (10.10) are exact. The realized-misfit multiple is $\max(1,\ \chi^2/(m-d))$ capped at 25, with χ^2 the sum of squared whitened quote residuals, m the quote count and d the parameter count. Panel (b) kinks the two adjacent strikes nearest $k = +0.03$ by $\pm\epsilon$ with ϵ swept to three vol points, and evaluates gains against a stated prediction budget of $(60\,\text{bp},\ 0.03,\ 0.15)$ per handle sd — the level entry is a four-day-old state under the walk scale of fig. 10.7 ($\sqrt{4} \times 30$ bp).

The stale-strike vignette (fig. 10.5). Stated moments as printed in section 10.5; the update is eq. (10.8) verbatim, per handle.

The audit world (fig. 10.7). Seed 20260807 (NumPy default generator). Truth: level starts at 20% and takes 500 daily Gaussian steps of sd 30 bp, plus +5-point jumps after days 151 and 320. Observations: truth plus Gaussian noise of sd 30 bp, except every seventh ("wide-spread") day at 100 bp — neither jump day is a wide day — and the filter is always told the true per-day noise. Filters: eq. (10.8) with $\mathcal{V}^- = \mathcal{V}^+_{\text{prev}} + q^2_{\text{walk}}$ per day; the starved run uses $q_{\text{walk}} = 10$ bp, the honest run 30 bp; the surprise-widened run multiplies $\mathcal{V}^-$ by (standardized innovation/3)^2, capped at 25, when the innovation exceeds 3 standardized units. $\mathcal{Z}$ is computed against the known truth from day two on; panel (c) averages absolute errors over the two jumps.

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Bibliography

[1] Brian D. O. Anderson and John B. Moore. *Optimal Filtering.* Prentice-Hall, Englewood Cliffs, 1979.

[2] Jesper Andreasen and Brian Huge. Volatility interpolation. *Risk*, pages 76–79, March 2011.

[3] Thierry Ané and Hélyette Geman. Order flow, transaction clock, and normality of asset returns. *Journal of Finance*, 55(5):2259–2284, 2000.

[4] Shalom Benaim and Peter Friz. Regular variation and smile asymptotics. *Mathematical Finance*, 19(1):1–12, 2009.

[5] Henri Berestycki, Jérôme Busca, and Igor Florent. Asymptotics and calibration of local volatility models. *Quantitative Finance*, 2(1):61–69, 2002.

[6] Lorenzo Bergomi. *Stochastic Volatility Modeling.* Chapman & Hall/CRC, 2016.

[7] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences.* SIAM, Philadelphia, 1994.

[8] Fischer Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3 (1–2):167–179, 1976.

[9] Mary Ann Branch, Thomas F. Coleman, and Yuying Li. A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems. *SIAM Journal on Scientific Computing*, 21(1):1–23, 1999.

[10] Douglas T. Breeden and Robert H. Litzenberger. Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651, 1978.

[11] Olena Burkovska, Kathrin Glau, Mirco Mahlstedt, and Barbara Wohlmuth. Calibration to American options: Numerical investigation of the de-Americanization method. *Quantitative Finance*, 18(7), 2018. arXiv:1611.06181.

[12] Peter K. Clark. A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1):135–155, 1973.

[13] John C. Cox, Stephen A. Ross, and Mark Rubinstein. Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3):229–263, 1979.

[14] Karl E. Dambis. On the decomposition of continuous submartingales. *Theory of Probability and Its Applications*, 10(3):401–410, 1965.

[15] Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. A guide to volatility and variance swaps. *Journal of Derivatives*, 6(4):9–32, 1999.

[16] Emanuel Derman. Regimes of volatility. *Risk*, 12(4):55–59, 1999.

[17] Lester E. Dubins and Gideon Schwarz. On continuous martingales. *Proceedings of the National Academy of Sciences*, 53(5):913–916, 1965.

[18] Bruno Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994.

[19] Nicole El Karoui, Monique Jeanblanc-Picqué, and Steven E. Shreve. Robustness of the Black and Scholes formula. *Mathematical Finance*, 8(2):93–126, 1998.

[20] Matthias R. Fengler. Arbitrage-free smoothing of the implied volatility surface. *Quantitative Finance*, 9(4):417–428, 2009.

[21] Kenneth R. French and Richard Roll. Stock return variances: The arrival of information and the reaction of traders. *Journal of Financial Economics*, 17(1):5–26, 1986.

[22] Frederick N. Fritsch and Ralph E. Carlson. Monotone piecewise cubic interpolation. *SIAM Journal on Numerical Analysis*, 17(2):238–246, 1980.

[23] Jim Gatheral. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation at Global Derivatives & Risk Management, Madrid, 2004.

[24] Jim Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, Hoboken, 2006.

[25] Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.

[26] Warren G. Gilchrist. *Statistical Modelling with Quantile Functions*. Chapman & Hall/CRC, Boca Raton, 2000.

[27] Michael B. Giles and Rebecca Carter. Convergence analysis of Crank–Nicolson and Rannacher time-marching. *Journal of Computational Finance*, 9(4):89–112, 2006.

[28] István Gyöngy. Mimicking the one-dimensional marginal distributions of processes having an Itô differential. *Probability Theory and Related Fields*, 71(4):501–516, 1986.

[29] Patrick S. Hagan, Deep Kumar, Andrew S. Lesniewski, and Diana E. Woodward. Managing smile risk. *Wilmott Magazine*, September:84–108, 2002.

[30] John C. Hull. *Options, Futures, and Other Derivatives*. Pearson, 10th edition, 2018.

[31] R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[32] Thomas W. Keelin. The metalog distributions. *Decision Analysis*, 13(4):243–277, 2016.

[33] Hans G. Kellerer. Markov-Komposition und eine Anwendung auf Martingale. *Mathematische Annalen*, 198:99–122, 1972.

[34] Roger W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.

[35] Erich L. Lehmann and George Casella. *Theory of Point Estimation*. Springer, New York, 2nd edition, 1998.

[36] Claude Martini and Arianna Mingone. No arbitrage SVI. *SIAM Journal on Financial Mathematics*, 13(1):227–261, 2022.

[37] Robert C. Merton. Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

[38] Anthony Neuberger. The log contract. *Journal of Portfolio Management*, 20(2):74–80, 1994.

[39] Emanuel Parzen. Nonparametric statistical data modeling. *Journal of the American Statistical Association*, 74(365):105–121, 1979.

[40] Goran Peskir and Albert Shiryaev. *Optimal Stopping and Free-Boundary Problems*. Birkhäuser, 2006.

[41] Alexander Petersen and Hans-Georg Müller. Functional data analysis for density functions by transformation to a Hilbert space. *Annals of Statistics*, 44(1):183–218, 2016.

[42] Daniel Revuz and Marc Yor. *Continuous Martingales and Brownian Motion*. Springer, 3rd edition, 1999.

[43] L. C. G. Rogers and M. R. Tehranchi. Can the implied volatility surface move by parallel shifts? *Finance and Stochastics*, 14(2):235–248, 2010.

[44] Peter J. Rousseeuw and Annick M. Leroy. *Robust Regression and Outlier Detection*. Wiley, 1987.

[45] Hans R. Stoll. The relationship between put and call option prices. *Journal of Finance*, 24 (5):801–824, 1969.

[46] Volker Strassen. The existence of probability measures with given marginals. *Annals of Mathematical Statistics*, 36(2):423–439, 1965.

[47] Andrey N. Tikhonov and Vasiliy Y. Arsenin. *Solutions of Ill-Posed Problems*. Winston, Washington, D.C., 1977.